WELL RE-ENTRY PROGRAM

AZADEGAN OIL FIELD

WELL RE-ENTRY PROGRAM

AZNS-040 (PAD #50), TYPE: IIF

Drilling Department

JULY 2023

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1. General Information

1.1 Field Introduction

Azadegan South Oil Field is situated in West Karoun province on a large long-axis anticline with N-S trending with approximate dimensions about 20 km width and 60 km length near to Iran-Iraq border, to the east of Yaran Oil Field.

The trap area of the south Azadegan is limited to 43.2x17.2 km. The Azadegan Oil field was discovered in 1999, when Well AZN-1 was drilled vertically down to the Cretaceous Fahliyan Formation with total depth of 4,643m. The results of this well, revealed the reservoir characteristics of the field. Following to the result of the well no. AZN-1, another 5 exploration wells was drilled in different parts of this huge field down to Fahliyan Formation. In the later stage the well No.10 assumed as exploration well for evaluation of deeper formations. This well was drilled in 2008, down to the depth of 5340m (Neyriz Fm.), mainly to evaluate the Jurassic formations.

Based on the exploration wells results, carbonate reservoir of Sarvak Formation is broadly developed in Azadegan Field. This formation was penetrated in all of seven exploration wells (AZN-1, AZN-2, AZN-3, AZN-4, AZN-5, AZN-6 and AZN-10). Clastic oil-bearing zones of Kazhdumi Formation developed in main anticline of Azadegan Field, extending from north to center of South Azadegan. The Clastic oil-bearing zones of Gadvan Formation developed only in crest area and south parts of South Azadegan Filed, which was discovered by AZN-1 in crest and AZN-4 in south parts. The carbonates of Fahliyan Formation also developed in Azadegan field, but the oil reservoir of this formation developed in South part of South Azadegan.

The Azadegan Field is an elongated anticline, trending in N-S direction, belonging to "Arabian Trend".

The stratigraphy sequence of the field includes AghaJari, Gachsaran, Fahliyan, Pabdeh, Gurpi, Ilam, Laffan, Sarvak, Kazhdumi, Dariyan, Gadvan and Fahliyan Formations, from top to bottom. This sequence in well no AZN-10, the deepest up to now drilled well of the field, continues with Garau, Gotnia, Najmah, Sargalu, Alan, Mus and Neyriz formations.

The Dominant lithofacies of Sarvak Formation are Rudist-bearing packstone and partially grainstone, which contains abundant bioclasts. Some other facies like matrix-dominated wackestone and mudstone are also found.

The AZADEGAN Oilfield is Located in Khuzestan Province and is Approximately Located 80 Kilometers West of Ahwaz City, near to Iran-Iraq Border. It is in The Northwest of The YADAVARAN Field and in The West of JUFEYR Field. The Average Monthly Temperature is 45ºc in summer, and Highest Temperature is 52ºc. The Lowest Temperature is -4.4ºc in winter. Maximum Rainfall is 78 mm in February, Winter Season.

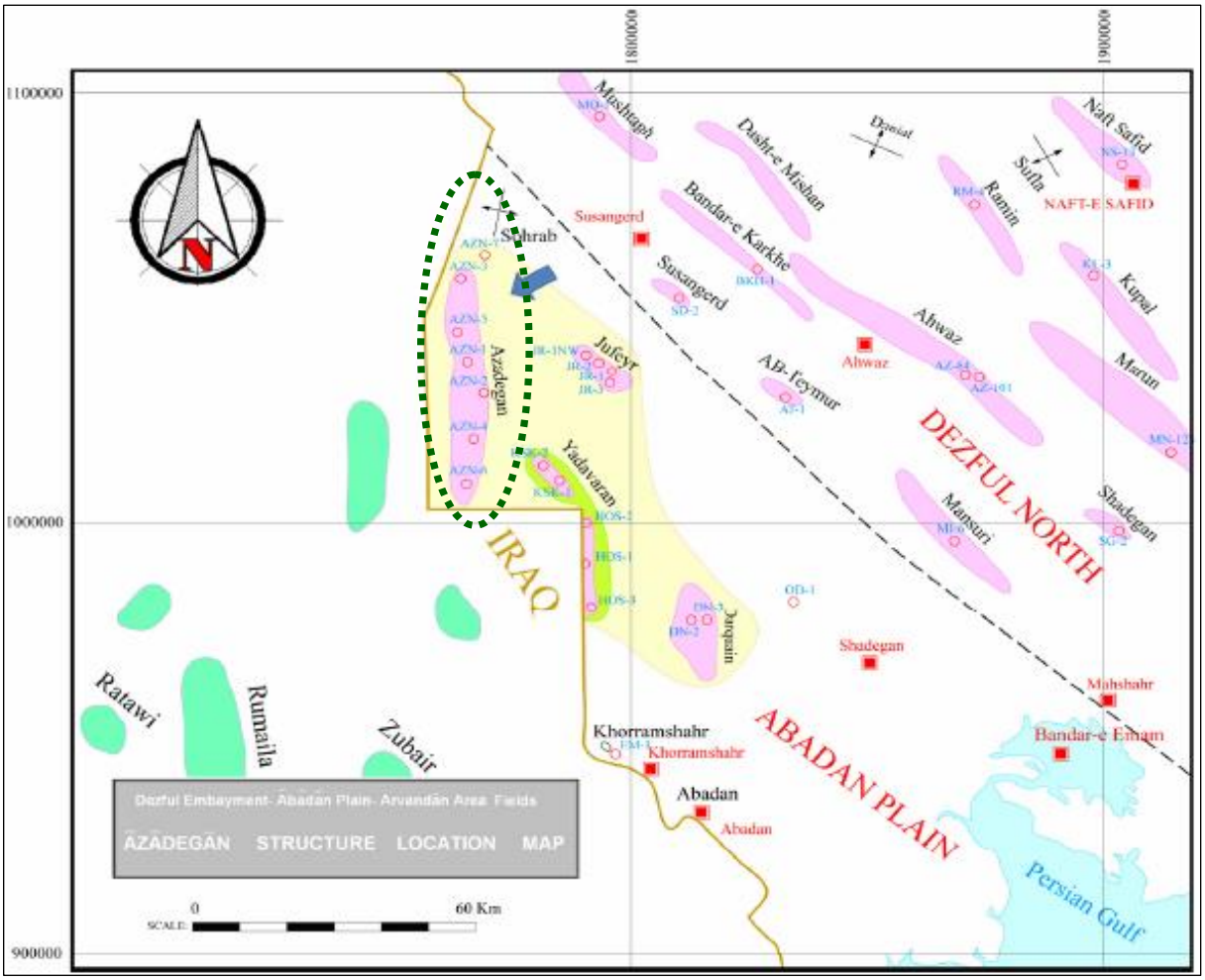


Figure 1-1: AZNS Oilfield Location Map

1.2 Well Information

Table 1-1: AZNS-040 Re-Entry Well Information

AZNS-040 Re-Entry Well Information			
Country / Area	Iran / Khuzestan		
Client:	PPL/PEDEC		
Contractor:	MSA/OEOC		
Field Name:	South Azadegan		
Well Name:	AZNS-040		
Location:	PAD-50 (Area: II)		
Wellhead Co-ordinate (UTM39N, WGS84):	E: 194,235m N: 3,450,715m		
Well Type:	IIF Vertical Well		
Well Objective:	Produce Oil from Fahliyan Reservoir		
RTE:	13.45m	**Est. Moving Time:**	10 Days
GLE:	3.95m	**Spud Date:**	TBA
Rig Name:	OEOC 208	**Est. Well Duration:**	102 Days
Rig Contractor:	OEOC	**Est. Rig Released:**	TBA

Casing / Liner Size				
CSG/LNR Size:	20” CSG	13-3/8” CSG	9-5/8” CSG	7” Liner
Shoe Depth (MD):	152.5	1492	3986	4200

Other Information			
Total Depth:	4200 MD (-4186.55m TVDss)	**Entry Point in Fahliyan Fm.:**	E: 194,235m N: 3,450,715m
Producer Formation:	Fahliyan		
Reservoir Top:	3985.79 MD (-3927.34m TVDss)	**Target Coordinate:**	E: 194,235m N: 3,450,715m
KOP:	--		

Reservoir Data					
Item		**Measure**			**Unit**
		Sarvak	**Kazhdumi**	**Gadvan**	**Fahliyan**
Datum Depth		2750	3590	3920	4050	**M, TVDss**
Datum Pressure (Initial)		4566	5485	5878	8843	**Psi**
Press Gradient		1.14	1.15	0.971	0.912	**Psi/m**
Temp. @ Datum Depth		96	115	124	132	**°C**
Temperature Gradient		2.4	2.4	2.4	N/A	**°C/100m**
H2S (f/ PVT Separator Oil)		0.06	0.0	0.011	0.28	**Mol%**
CO2 (f/ PVT Separator Oil)		1.45	200	1.87	2.68	**Mol%**
Formation Water	**Salt**	200000	180000	185000	143000	**PPM**
	Ca^{++}	9500-11600	11400	9200-10000	8750	**Mg/lit**

1.3 AZNS-040 Well Location Map

This well is located on Pad-50 near wells AZNS-036 and AZNS-017S.T-1.

The pad-50 is located on center of South Azadegan Oil field in Block-II.

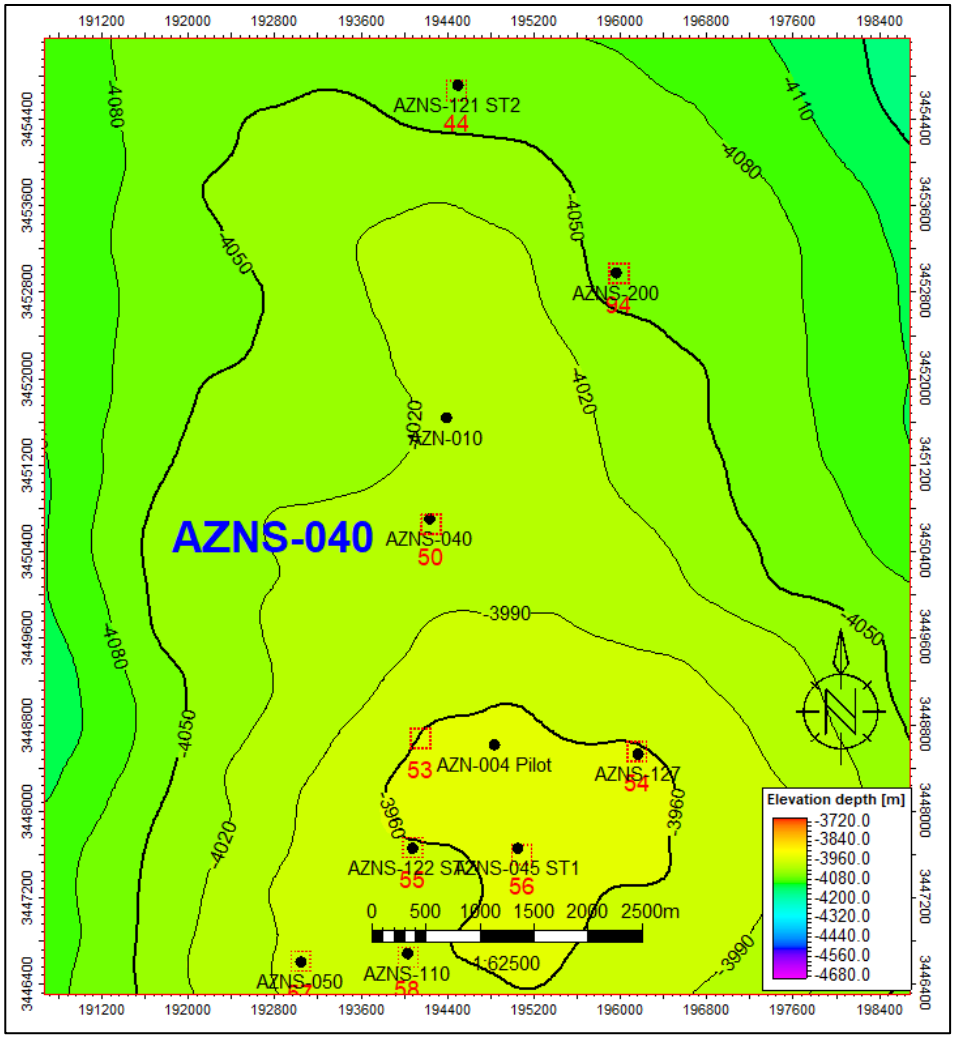


Figure 1-2: AZNS-040 Well Location Map

1.4 Reservoir Information

Key Reservoir Descriptions

Fahliyan Formation: This carbonate to argillaceous carbonate Formation is limited by overlying thick argillaceous successions in the upper part (eaquivalent to the Iraqi Ratawi Formation) and the main part consists of different bioclastic limestones with some layers of shale and argillaceous facies that can be assumed equivalent with Iraqi Yamama Formation. However, in South Azadegan wells 1, 2 and 5 the Fahliyan Formation is characterized by fossiliferous limestone beds and interbedded shales. It should be noted that most of Majnoon wells penetrated into the Sarvak and Fahliyan carbonates as one of the main reservoir targets.

Formation Pore Pressure and Fracture Gradient

Table 1-2: Formation Pressure Prediction in Block II

Formation	Interval	Pore Pressure	Collapse Pressure (Break-Out Pressure)	Fracture Pressure (Break-Down Pressure)
	m	SG	SG	SG
Aghajari	0～1096	0.98～1.18	0.06～1.39	1.01～1.89
Gachsaran	1096～1459	1.13～1.31	0.78～1.36	1.89～2.01
Asmari	1459～1739	1.14～1.26	0.6～1.28	1.43～1.79
Pabdeh	1739～2194	1.20～1.26	0. 82～1.36	1.40～1.78
Gurpi	2194～2538	1.21～1.26	0.66～1.02	1.40～1.76
Ilam	2538～2619	1.18～1.24	0.20～0.9	1.41～1.72
Laffan	2619～2625	1.13～1.18	0.24～1.11	1.41～1.72
Sarvak	2625～3272	1.14～1.26	0.24～1.23	1.44～1.81
Kazhdumi	3272～3475	1.1～1.14	0.46～0.96	1.93～1.94
Darian	3475～3641	1.18～1.25	0.55～1.10	1.96～2.06
Gadvan	3641～3932	1.1～1.25	0.55～1.44	1.97～2.11
Fahliyan	3932～TD	1.60～1.3	1.20～1.75	2.08～2.12

Fahliyan Reservoir Predicted Pressure, Temperature and OWC

Table 1-3: Predicted Pressure and Temperature and TTZ Contact

Reservoir	Sub-layer	Top zone in AZNS_040 (mSSTVD)	Temp (°F)	Temperature Gradient (°F/100ft)	Current Reservoir Pressure (psia)	Basis for Reservoir Pressure	Top of Transition Zone (mTVDSS)	OWC (mTVDSS)	Oil Gradient (psi/ft)
Fahliyan	Fah-1a	-4009	266	1.25	9240	Dynamic Model	-4149	-4163	0.28
	Fah-2a	-4071			9300	Pressure Gradient	-4045	-4060	0.28
	Fah-3a	-4096	271		9050	Dynamic Model	-4095	-4110	0.28
	Fah-4a	-4115			7380	Dynamic Model	-4122	-4127	0.28
	Fah-4b	-4126			7930	Pressure Gradient	-4107	-4112	0.28
	Fah-5a	-4140			8150	Dynamic Model	-4168	-4190	0.28
	Fah-5b	-4163			8200	Dynamic Model	-4168	-4190	0.28

Corrosion Fluid

Table 1-3: South Azadegan Oilfield Reservoirs Corrosion Fluid Content

Reservoir	Sarvak	Kazhdumi	Gadvan	Fahliyan
H_2S Content (ppm)	2436	10	1650	30000
CO_2 Content (ppm)	142929	61107	40684	72769

Note: The corrosive gas concentration is subject to future well test sampling and lab test.

Formation Water Salinity

Table 1-4: South Azadegan Oilfield Formation Water Salinity Content

Reservoir	Sarvak	Kazhdumi	Gadvan	Fahliyan
Salinity (ppm as NaCl)	200000	180000	185000	143000
Calcium Content (mg/lit)	9500-11600	11400	9200-10000	8750

Note: Since there is high corrosive fluid in the reservoirs, during cementing operation, additives to anti-corrosion should be considered.

1.5 AZNS-040 Well History

The AZNS-040 well was drilled from 26-Mar-2014 to 10-May-2014 by CNPCI to total depth 1493m and suspended. Drilling and well suspend sequences are describe in the following.

26” Hole Section Summary

This section was drilled from 15 to 153 m using one rock bit (26" Simth, Type: MSDGH, IADC 135) and bentonite mud w/ 65.5-71.8 pcf weight. Performed ream and back ream prior to each connection. When TD was reached, wiper trip and CBU to clean the hole were done with good condition. Spotted Hi-Vis pill at bottom, dropped single shot to take a survey that resulted in 0.2° at 150 m. 20″ CSG (K-55, 117 ppf, Big Omega) was run to 152.5 m and cemented in place by stinger with 346 bbls cement class G, 117 pcf. 50 bbls slurry returned to surface. N/U 20 3⁄4″ CHH 3K on 20″ CGS supported by a base plate and performed pressure test by 400 psi, 15min. N/U 20 3⁄4″ BOP 3K and performed pressure test. 17 1⁄2″ milled tooth bit was RIH, tagged TOC at 150.5 m, 20″ CGS pressure tested with 1650 psi /15min, then D/O cement & shoe.
Total Duration for this section was 4 days and 2 hrs. Until new formation of next section was started to drill.

17-1/2” Hole Section Summary

Drilled 17 1⁄2" hole section from 153m to 1493m, MW: 71.8-85.5pcf NaCl-KCl polymer Mud &Saturated NaCl Polymer Mud. LOT@156m with 80psi, MW: 71.1pcf, EMD: 93.64 pcf. Took survey and wiper trip every 200m to ensure good well track and hole condition, try to control viscosity and water loss of mud as program, due to claystone & gypsum exist in the interval. From the depth of 1159m to 1385m, we meet very low ROP (min-ROP is 0.1 m/h). total used 5pcs of bits (1 used rock bit, 3 used PDC bits and 1 new PDC bit) when during drilling 17-1/2" hole. total footage 1340m, actual drilling time 687.3Hrs, ROP: 1.95 m/h.

Performed condition hole before run wireline logging (Items: GR, CDR, CALI). After logging performed 1 more time hole condition prior to run casing, Spotted high viscosity mud pill at bottom of the hole. Ran 13 3/8" CSG (68ppf-N80-BTC) to 1492m (casing shoe0.

Performed cementing job: Cement track pumped 25m3 spacer (1.05sg) with FR 317gpm, SPP 200-478psi, dropped BTM plug, pumped 2m3 of spacer(1.05sg), follow by 147.0m3 of lead slurry(average 1.45sg, Max 1.51sg, Min 1.45sg) with FR 528gpm, SPP 1015-345psi, pumped 48m3 tail slurry (average 1.85sg, Max 1.89sg, Min 1.83sg) with FR 528gpm SPP345-0psi. Released top plug, pumped 2m3 of tail slurry (average 1.00sg) with FR 264gpm, SPP 0psi. Pumped 1m3 of fresh water with FR 264gpm, SPP 0psi. Switch to rig pump, displace mud (1.37sg) with FR 740-264gpm, SPP 350-480psi, bumped plug pressure from 732psi to 1083psi. Bleed pressure. 56m3 neat cement slurry returned to surface.

Install 20-3/4"*3K-13-5/8"*5K CHS. P/T of seal of flange connection with 1500psi/ 10min, OK. N/U 13-5/8"*10K BOP single ram.

RIH from surface to 285m with 5"DP to put CMT plug f/50-150 meter below the WH. P/T 13- 3/8" casing with 3000psi/15min, MW 85.49pcf, OK. POOH from 285m to 200m. Pumped 4m3 1.37SG H-Vis mud. POOH from 200m to 150m. Pump 1.0 m3 spacer (fresh water), Pump 8m3 of lead slurry (average 1.85sg), POOH from 150m to surface. WOC, While L/D 5"DPS.

Tagged top of cement plug @55m with 3 Ton.

R/D 13-5/8"*10K BOP single ram. Cover the wellhead with steel plate, and fixed with 4 pieces of bolts and suspend the well.
Total Duration for this section was 41 days and 7 hrs. until release the rig.

1.6 AZNS-040 Re-Entry Well Program Outline

AZNS-40 well, was planned to be drilled as Horizontal well in Sarvak formation on Pad-50. The well was drilled by CNPCI to total depth 1493m in Gachsaran formation as 17-1/2" hole section, 13-3/8" CSG ran and cemented to surface and well has been suspended by CMT plug.

As the Previous exploration/delineation drilling in Azadegan oilfield has proved presence of a very prolific hydrocarbon bearing sedimentary column, particularly within the Cretaceous sequences. The Fahliyan Formation was divided into 6 zones, named as Fahliyan top, Fah-1, Fah-2, Fah-3, Fah-4 and Fah-5.

The objective of Re-Entry this well is to De-suspend the well, vertical drilling 12-1/4″ & 8-1/2″ hole section to new well target in Fahliyan reservoir (**Well target has been changed from Sarvak horizontal well to Fahliyan vertical well**), run 9-5/8" CSG and 7" Liner, perforate 7" Liner in Fahliyan formation and complete the well as Fahliyan vertical oil producer well.

The AZNS-040 Re-Entry sequences summarize as follow:

- **De-Suspend the Well:**

Monitor wellhead pressure and make sure about wellhead integrity. Remove wellhead cover, N/U BOPs. P/U & M/U 12-1/4" D/Out BHA, RIH to top of CMT plug @ 55m, D/Out CMT plug f/ 55-100m with water, RIH to top of 13-3/8" CSG Float collar, D/Out shoe track to 3m above shoe with water, displace well with 75 pcf Salt Saturated/Polymer mud, perform 13-3/8" CIT, D/Out shoe track to 1493m.
Note: if wellhead pressure is not Zero after Bleed off, Contractor shall inform client. Remedial plan to be prepared in advance by operators and will be followed after client's confirmation.

➢ **12-¼" Hole Section:**

Drill 12-1/4" vertical hole section f/ 1493m to 3986m through Asmari, Pabdeh, Gurpi, Ilam, Laffan, Sarvak, Kazhdumi, Dariyan, Gadvan and Fahliyan formations with 75-85 pcf Salt Saturated KCL/Glycol/Polymer mud. Then Full Suite/CDR/GR Logs will be run. The 9-5/8" Casing will be set at ± 3986m and it will be cemented in one stage as primary cementing up to surface. N/U13-5/8", 5K x 11", 10K THS and then N/U 13-5/8", 10K BOPs. RIH with D/Out BHA to TOC, D/Out to 7m above shoe, perform 9-5/8" CIT. Run 9-5/8" Cement quality log. RIH with 8-1/2" PDC Bit BHA to 7m above shoe, D/Out shoe track to 3986m while displace well with 100 pcf Salt Saturated Kcl/Polymer/CaCo3 mud.

➢ **8-½" Hole Section:**

The 8-½" hole section will be drilled vertically from 3986m to 4200m in Fahliyan formation with 100-106 pcf Salt Saturated Kcl/Polymer/CaCo3 mud. Then Full Suite/CDR/GR Logs will be carried out over 8-½" open hole section to evaluate Fahliyan reservoir. The 7" Liner will be run and set at ± 4200m and it will be cemented in one stage as primary cementing up to ± 7 Stands Wet above Liner Lap. Clean out to 7" L.Lap and inside 7" liner to LDC. Perform 7" Liner CIT, L.Lap positive and negative tests and run 7" Liner Cement quality Log. Displace 7" Liner volume with CaBr2 perforation fluid, perform 7" Liner perforation, run CCL/GR log to confirm perforated interval. RIH with 5-7/8" Bit to top of perforation while scrap 7" production PKR setting depth, displace well volume with CaBr2 completion fluid. Run 4-1/2" * 3-1/2", 10K Completion string. N/D BOP, N/U XMT. Perform well Pre-Acid clean up, Acid-Wash, Post-Acid clean up and well performance test operations. Secure the well and release the Rig

1.7 Contractors Involved

Table 1-5: South Azadegan Oilfield Project's Contracts Involved

Item	Service Item	Contractor
1	Drilling Rig, Mobilization and Demobilization the Rig, Rig Move Between Well Pads	OEOC
2	Drilling Fluids and Completion Fluids	VIRA
3	Waste Management	TBA
4	Electrical Logging and Perforation	PANIZ
5	Mud Logging	MAPSA KISH
6	Directional Drilling (MWD)	VIRA
7	H2S Monitoring and Control	MAPSA KISH BONYAN IMEN
8	Casing and Tubing Running	OEOC
9	Cementing	SPAD ENERGY
10	Dry Test, DST and RTTS	OEOC
11	Slick Line	OEOC
12	Stimulation (Coiled Tubing and Acidizing and Nitrogen Lifting)	OEOC
13	Surface Well Testing	OEOC
14	Liner Hanger	PARS POOLAD
15	Well Head	OEOC

1.8 AZNS-040 Well Montage (Before Re-Entry)

Field: South Azadegan | **Pad no:** 50 | **Location:** Iran | **Type:** Horizontal
Rig No: GW262 | **Spude date:** 26-March-2014 | **Release Date:** 10-May-2014 | **Rig Days:** 45.46

South Azadegan Oil Field AZNS-040
as drilled and completed

WH Coordinates: E: 194235 N: 3450715
RTE: 11m
GLE: 4m
Final Well Status: Ran & cemented 13 3/8'' CAG, Set CMT plug 50-150m below the WH.

Formation	Formation Tops Depth m,MD	Formation Tops Thick m	Well Sketch and Completion	Bit Size/Type	ROP m/	Drilling Fluid	Remarks On Hole Condition	Casing	Well Head & BOP	Brief Cementing Report	Logs Tests	Drilling History
Aghajari	15	1120	20" CSG @ 152.5 m; Cement plug from 55-100m; 13-3/8"CSG Shoe @ 1492 m; Shoe track	26" Varel MSDGH/135 Open	4.69	Bent. Mud MW: 66.8-71.8 pcf	Soft formation	20" K-55 117ppf Big Omega	20 ¾" 3K C.H.H Welded on 20" CSG	CMT Class "G" SW: 117.5pcf SV:345bbls 50bbls return to surface	N/A	Drilled f/ 15 to 153 m. Wiper trip, CBU, single shot (0.2° at 150 m). 20" CGS (K-55, 117 ppf, Big Omega) was run to 152.4 m and cemented by stinger with 346 bbls cement class G, 117 pcf. 50 bbls slurry returned to surface. N/U 20 ¾" CHH 3K, pressure test by 400 psi, 15min. N/U 20 ¾" BOP 3K, pressure test.Tagged TOC at 150.5 m, 20" CGS pressure tested with 1650 psi /15min, displaced NaCl/Kcl polymer mud then D/O cement & shoe.
500m				17 1/2" MGSO5V /425/open	1.72	NaCl/KCL Polymer & Salt Saturated MW: 71.8-85.5 pcf	Good Hole Condition NL-NG	13 3/8" N-80 68 ppf BTC	Well Head 20 ¾" 3K* 13 5/8" 5K C.H.S BOP 20 3/4" 3K	Spacer: SW: 65.6 pcf SV: 155 bbls Lead Cmt: Cement class "G" SW: 90.5 pcf, SV:910 bbl Tail Cmt: Cement Class "G" SW:115.5 pcf, SV:297 bbl 347 bbls Return @ Surface	GR-CDR-CAL @ 0.H	Drilled F/153m T/ 1486 m. chloride was increased before entering to Gachsaran formation (1050 m). LOT @ 156 m (2.6m below the shoe) with 93.64 pcf EMW. Drilling was continued to 270m, CBU, single shot survey 0.47° @ 270 m, POOH,change PDC bit & changed the BHA adding drilling Jar. Cont. drilling to 1493m (section TD), CBU, short trip, POOH, Took GR/CDR/CALI log. Change BHA & RIH, CBU, took multi shot survey and POOH to 1404m, FIT. @1493m, surface pressure 500psi, MW 85.49pcf, EMW 100.46pcf (1.61SG). Cont. POOH to surface, Run 13 3/8" CGS (N-80, 68 ppf, BTC) was run to 1492 m, cemented with 157.25 bbls spacer 65.48 pcf, 924.63 bbls lead slurry 90.5 pcf and 301.92 bbls tail slurry 115.44 pcf. Obs. Bumping 1083 psi. 352.24 bbls slurry returned to surface. No loss during the cement job. While WOC, installed CGS hanger and N/U 13 5/8" CHS 5K, pressure test with 2000 psi/15 min, O.K. N/U 13 5/8" 10 K single Ram BOP. RIH from surface to 285m with 5"DP. P/T 13-3/8" casing with 3000psi/15min, MW 85.49pcf , OK. POOH from 285m to 200m. Pump 4m³ 1.37SG H-Vis mud. POOH from 200m to 150m. Pump 1.0 m3 spacer (fresh water) , Pump 8m3 of lead slurry(average 1.85sg) , POOH from 150m to surface. WOC. While L/D 5"DPS. Tag top of cement plug @55m with 3 Ton. R/D 13-5/8"*10K BOP single ram. Cover the wellhead with steel plate, and fixed with 4 pieces of bolts. Abandon well.
1000m				17 1/2" CKS605F /S223/5*15+5*13	3.62							
Gachsaran	1138	357		17 1/2" CKS605F /S223/2*16+5*15+3*13	0.79							
1500m Asmari	1495			17 1/2" SKS76D/ S111/9*18	0.87							
				17 1/2" MS1953SS /S223 4*20+3*16	0.91							
Pabdeh												
2000m Gurpi-1												
Tarbur												
Gurpi-2												
2500m Ilam												
Laffan												
Sarvak-1												
Sarvak-2												
Sarvak-3		-										
3000m											-	

Total Depth @ ...m, ...m TVD Into Sarvak Zone 3

Figure 1-3: AZNS-040 Well Montage (Before Re-Entry)

1.9 AZNS-040 Well Montage (After Re-Entry)

General Information:
Well Name: AZNS-040
Pad No.: 50
Well Type: IIF
Well Progile: Vertical
Block: II
Field: South Azadegan
Country / Area: Iran / Khuzestan
Reservoir: Fahlian

Wellhead Coordinate:
Grid Geodetic System: (UTM39N, WGS84)
E: 194,235
N: 3,450,715

Nearby Wells:
AZN-010
AZNS-045 ST1
AZNS-127
AZNS-017 (KZ-Horizontal)

Parties/Refrence:
Client: PPL/PEDEC
Contractor: MSA/OEOC
Rig Cont: OEOC
Rig: OEOC 208

Well Montage
AZNS-040 Re-Entry
(PAD-50), IIF

Drilling Depth:
TD (MD): 4200m
TD (TVD): 4200m
GLE: 3.94m
RTE: 13.45m

Timing:
Moving Days: 10 Days
Spud Date: TBA
Total Rig Days: 102 Days (With out rigless job)
Rig Released Date: TBA

Field Location/Objective:
Field Location: The AZNS oilfield is located in Khuzestan province, approximately 80 Km to the west of Ahwaz city.
Well Objective: De-suspend the well, vertical drilling to Fahliyan formation, complete well and produce oil from Fahliyan reservoir..

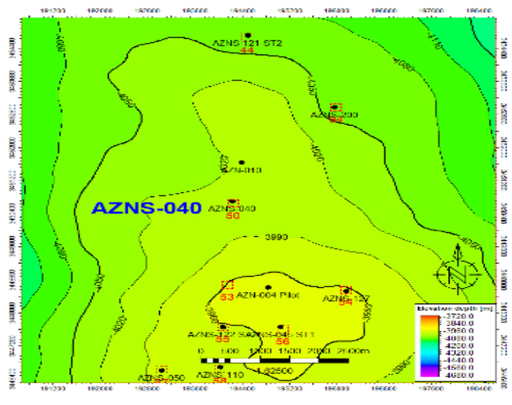
Formation	Formation TOP MD (M)	TVD (M)	TVD ss (M)	Formation Thickness (M)
Aghajari	Surface	Surface	3.95	1123.1
Gachsaran	1138	1138	-1123.1	357.1
Asmari	1493.7	1493.7	-1480.2	282.4
Pabdeh	1776.1	1776.1	-1762.6	421.8
Gurpi	2197.9	2197.9	-2184.4	337.1
Ilam	2435	2435	-2521.5	104.1
Lafan	2639.1	2639.1	-2625.6	6.4
Sarvak	2645.4	1645.4	-2631.9	638
Kazhdumi	3283.5	3283.5	-3270	206.3
Dariyan	3489.7	3489.7	-3476.2	193
Gadvan	3682.7	3682.7	-3669.3	303.2
Fahliyan	3985.79 4200	3986.79 4200	-3972.34 -4186.55	214.2

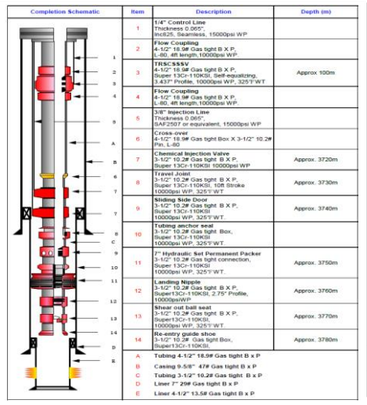
Well Schematic:
- 20" CSG Shoe @ 152.5m
- 13-3/8" CSG Shoe @ 1492m
- 17-1/2" Hole Section TD @ 1493m
- 7" Liner Lap @ 3886m
- 9-5/8" CSG Shoe @ 3986m
- 7" Liner Shoe @4200m
- 8-1/2" Open Hole TD @ 4200m

Hole Size / Bit Type / Nozzle Sizes	BHA	Drilling Parameters	Drilling Fluid	CSG/Liner Design	Cementing	Logging Testing Coring	BOP and Well Head	Remarks (Drilling Issues/Lesson Learned)
26" 26" MT Bit MSDGH 135 (open)		Flow Rate: 800 GPM SPP: 100-350 Psi WOB: 2-8 Kibs RPM: 30-80 ROP: 4.69 m/h	Bentonite Mud 65-72 PCF	0-152.5m: 20", 117 ppf, K-55, Btg Omega	Lead: Class "G" CMT S.V: 345 bbl, SW: 117.5 pcf *50 bbl return @ surface.	NIL	20-3/4", 3K CHH (Weld On 20" CSG)	1. Bit Balled-up with 65 pcf Bentonite Mud in AZNS-045. 2. Cement Losses in AZNS-017 with 115 pcf Sw. 3. Bit balled-up with bentonite mud in AZNS-127.
17-1/2" 17-1/2" TCI Bit MGSO5V 425 (open) 2No. 17-1/2" PDC Bit CKS605F (5x15 + 5 x 13) (2x16 + 5x15 + 3x13) 1No. 17-1/2" PDC Bit SKS76D S111 (9x18) 1No. 17-1/2" PDC Bit MS1953SS S223 (4x20 +3x16)		Flow Rate: 900 GPM SPP: 850-990 Psi WOB: 5-12 Kibs RPM: 60-70 ROP: 1.72 m/h Flow Rate: 890 GPM SPP: 2200/2200/2200/2200 Psi WOB: 10-18/10-18/10-22/10-22 Kibs RPM: 70/70/70/70 ROP: 3.62/0.79/0.87/091 m/h	Salt Saturated Kcl Polymer WBM 71-86 PCF	0-1492m: 13-3/8", 68 ppf, N-80, BTC	Lead: Class "G" CMT S.V: 910 bbl, S.W: 90.5 pcf Tail: Class "G" CMT S.V: 297 bbl, S.W: 115.5 pcf *347 bbl return @ surface.	CDR/GR/Cal	20-3/4", 3K BOPs 20-3/4", 3K x 13-5/8", 5K CHS	1. Bit and Stabilizer balling with 67-86 pcf Kcl/Polymer mud in GS formation in AZNS-045. 2. Calcium ion contamination in GS formation. 3. Frequent Bit/Stb balling and tight hole in AZNS-017 due to low Mw and improper mud properties. 4. Wash down 13-3/8" CSG and set CSG off bottom due to bad hole condition in AZNS-017 (mainly low Mw in compare to hole condition). 5. Obs back flow after Cementing 13-3/8" CSG, apply pressure and close well in several steps in AZNS-045. 6. String stuck while POOH & B.Ream in GS formation with 84 pcf Mw in AZNS-045 well. 7. Frequent tight hole and hole stability problem in GS formation in well AZNS-017 (12-1/4" Hole). 8. WL logging tool held up in GS Fm., in well AZNS-017 (12-1/4" Hole). 9. Bit balled-up in base of GS formation in AZNS-127. 10. Frequent tight hole and string washout in GS in AZN-010 with 75-90 pcf Mw.
12-1/4" 1No. 12-1/4" TCI Bit 537 (3 x 16) 3No. 12-1/4" PDC Bit M323 (5x13 + 3x14) M423 (5x13 + 3x14) M433 (4x13 + 4x14)	D.Out BHA :Bit + Junk Sub + B.Sub + 6 Jts 8-1/2" DC + 8" D. Jar +3 Jts 8-1/2" DC + XOS +6 Jts 5" HWDP Bit + B.Sub + 2 Jt 8-1/2" DC + 12-1/4" S.STB + 7 Jts 8-1/2" DC + 8" D.Jar + 3 Jts 8-1/2" DC + XOS + 9 Jts 5" HWDP	Flow Rate: 500-650 GPM SPP: max 2850 Psi WOB: 5-20 Kibs RPM: 80-110 Flow Rate: 600-750 GPM SPP: max 2850 Psi WOB: 20-30 Kibs RPM: 160-180	Salt Saturated KCL/Glycol/Polymer/CaCo3 WBM 75-85PCF	0-3986m (3986m): 9-5/8", 53.5 ppf (SD), 110SS, Premium Gas Tight Con. ID: 8.535" (SD) Burst: 10900 psi Colapse: 7950 psi Body Yield: 1,710 Kibs B.SF: 1.17 C.SF: 1.99 A.SF: 2.33 TA.SF: 1.42	Lead: Class "G" CMT S.V: bbl, S.W: 81-85 pcf Tail: Class "G" CMT S.V: bbl, S.W: 118pcf	Cased Hole Logs: CBL/VDL from 13-3/8" Casing Open Hole Logs: Full Suite/CDR/GR Cased Hole Logs: SBT or CAST/CCL/CBL/VDL/GR from 9-5/8" Casing	13-5/8", 10K BOPs 13-5/8", 5K x 11", 10K CHH	Main drilling Challenges: Losses in AS/PD/SV/KZ, Chert formation in base of Pabdeh, Tight hole and string stuck, Washout string, Oil/Gas Flow, CMT Losses, Complete Loss, Hole Deviation 1. Frequent tight hole while drilling/tripping through GU (L) in AZNS-045 well. 2. Seepage losses 2-6 BPH with 75-87 pcf Mw in AS/PD/GU formation in AZNS-045. Drilling P/ SV to GD with 0-2 BPH losses w/ 84-87.5 pcf Mw. 3. Differential String stuck in KZ of AZNS-045 w/ 87.5 pcf Mw, free same with HCL Acid and cut MW to 81 pcf. 4. Perform top cement job in 9-5/8" Ann due to failure while displacing cement and remained cement inside casing in AZNS-045. 5. Obs2 BPH losses in KZ Fm., in AZNS-017 and frequent string stuck while drilling horizontal hole in KZ with 76-78 pcf Mw. 6. Obs 2-12 BPH losses while drilling through SV in AZNS-017 with 76-77 pcf Mw. 7. Losses 5-20 BPH losses with 80-81 pcf Mw in AZNS-127, losses inc to 12 BPH in KZ & 15 BPH in GD formation. 8. Frequent string washout and twist off string in AZNS-127 while performance drilling with 80-81 pcf Mw. 9. Nitecsable CMT losses while cementing 9-5/8" CSG w/ 88 pcf Lead CMT Sw in AZNS-127. 10. Low ROP in Cherty Zone (PD) from 1920-1989m in AZNS-127. 11. Obs 40-50 BPH losses while drilling with 80 pcf Mw in top of SV in AZN-010.
8-1/2" 1No. 8-1/2" MT Bit 217 (3 x 14) 1No. 8-1/2" PDC Bit M433 (6 x 14) 1No. 5-7/8" MT Bit 217 (3x15)	D.Out BHA:Bit + J.Sub + B.Sub + 9Jts 6-1/2" DC + 6-1/2" D.Jar + 3Jts 6-1/2" DC + XOS + 6Jts 5" HWDP Bit + B.sub + 1Jt 6-3/4" NMDC + MWD sub + 1Jt 6-3/4" NMDC + 8-1/2" S.STB + 9 JTs 6-1/2" DC + 6-1/2" D.Jar + 3 Jts 6-1/2" DC+ 9 Jts 5" HWDP	Flow Rate: 450-500 GPM SPP: max 2850 psi WOB: 5-15 Kibs RPM: 70-80 Flow Rate: 400-500 GPM SPP: max 2850 Psi WOB: 10-20 Kibs RPM: 150-170	Salt Saturated KCL/Polymer/CaCo3 WBM: 100-106 PCF	3886-4200m (314m): 7", 32ppf, 110SS, Premium Gas Tight Con. ID: 6.094" Burst: 12460 psi Colapse: 10780 psi Body Yield: 1,025 Kibs B.SF: 1.24 C.SF: 1.06 A.SF: 1.51 TA.SF: 1.49	Lead: Class "G" CMT S.V: bbl, S.W: 118pcf	Open Hole Logs: Full Suite/CDR/GR Cased Hole Logs: SBT or CAST/CCL/CBL/VDL/GR from 7" Liner	13-5/8", 10K BOPs 4-1/16", 10K X-Tree	Main drilling Challenges: Tight hole and string stuck, Oil/Gas Flow, Complete Loss, Hole Deviation 1. String stuck in FH formation with 105-106 pcf MW, free same with Overpull/Jarring/Pipe-Lax in AZNS-045. 2. String stuck while coring in FH., LIH and side track well in AZNS-045. 3. 7" L.Lap leakage as Dry Test result in AZNS-127 even with good amount of hard CMT on TOL. Run T.Back Liner and cement same. 4. Seepage losses 2-8 BPH in AZNS-045 with 100-106 pcf Mw, 1-3 BPH w/ 104-105 pcf Mw in AZNS-127 & AZN-010 w/ 100-110 pcf Mw.

Time Breakdown

Hole Size	De-Suspend	12-1/4"	8-1/2"	Completion				Sum (Hrs)	Total %	Total (Days)
Drilling		658.9	72					730.9	32.8%	30.5
Round Trip/BHA		147	19					166	7.5%	6.9
BOP Test		26	24					50	2.2%	2.1
Hole Condition, Reaming, Circ		105	44					149	6.7%	6.2
Deviation Survey		18	3.5					21.5	1.0%	0.9
Directional Drilling								0	0.0%	0.0
Casing/Liner		55	46					101	4.5%	4.2
Cementing		8	5					13	0.6%	0.5
WOC			4					4	0.2%	0.2
N/U&D BOP		18						18	0.8%	0.8
N/U WH		6						6	0.3%	0.3
Drill Out		43	92					135	6.1%	5.6
Logging		71	22					93	4.2%	3.9
Testing (FIT, LOT, CIT, Dry Test, etc.)		6	107					113	5.1%	4.7
DST								0	0.0%	0.0
Coring								0	0.0%	0.0
Plug Back								0	0.0%	0.0
Completion				482				482	21.7%	20.1
Well Testing								0	0.0%	0.0
Rig Repair	1.4	24.4	9.2	13.7				48.7	2.2%	2.0
Slip & Cut		7.5	5					12.5	0.6%	0.5
De-suspend	82							82	3.7%	3.4
Workover								0	0.0%	0.0
Total Hrs per Hole	83.4	1193.8	452.7	495.7	0.0		2225.6			
Total Days per Hole	3.5	49.7	18.9	20.7	0.0		92.7			
Total Days per Hole with 10% Contingency	3.8	54.7	20.7	22.7	0.0		102.0			
Cummulative Time (Days)	3.5	53.2	72.1	92.7	92.7					
Cummulative Time (Days) with 10% Contingency	3.8	58.5	79.3	102.0	102.0					





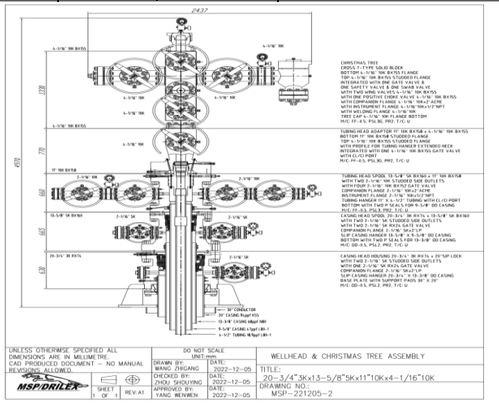


Figure 1-4: AZNS-040 Well Montage (After Re-Entry

1.10 AZNS-040 Basic Well Data

AZNS-040 Well Details Time Breakdown

De-Suspend, Re-Entry and Completion Operation TBD of Well: AZNS-040 Re-Entry	IADC Code	Interval (meters)	Formation	Depth (meters)	ROP Average (MPH)	Time Duration (Hours)	Time Duration (Days)	Total Duration (Days)	Contingency (%) 10	Time Duration w/ contingency (Hours)	Time Duration w/ contingency (Days)	Total Duration w/ contingency (Days)
De-Suspend												
Held PJSM. Spud the Rig. Stand Back required 5" DPs, 5" HWDPs and 8-1/2" DC.	De-Suspend		GS	0		0.0	0.00	0.0	0.0	0.0	0.00	0.00
Record well head pressure, function well head components and fix wellhead if necessary. Bleed off any wellhead/annulus pressure to Zero and monitor same. Remove WH cover. Note: If annulus pressure is not Zero after bleed off, remedial plan will be prepare and follow after Client confirmation.			GS	1493		6.0	0.25	0.3	0.3	6.6	0.28	0.3
N/U 13-5/8", 10M BOPs & related lines.			GS	1493		18.0	0.75	1.0	0.8	19.8	0.83	1.1
Test Accumulator unit, BOPs, choke manifold & WC equipment as per NIOC policy.			GS	1493		8.0	0.33	1.3	0.4	8.8	0.37	1.5
Perform WH pressure integrity test with cup tester.			GS	1493		2.0	0.08	1.4	0.1	2.2	0.09	1.6
P/U & M/U 12-1/4" TCI Bit BHA (including junk sub), RIH to tag CMT plug @ 55m. D/Out CMT plug f/ 55-100m. RIH to tag 13-3/8" CSG Float Collar @ 1468m jt by jt. D/Out plugs, F.C, CMT to 3m above CSG shoe @ 1489m.			GS	1493		36.0	1.50	2.9	1.7	39.6	1.65	3.2
Displace well with 75 pcf Salt Saturted Kcl/Polymer mud, Circ and homogenize mud properties.			GS	1493		4.0	0.17	3.1	0.2	4.4	0.18	3.4
R/U CMT surface line. Perform 13-3/8" CIT. with 1500 psi for 10 min, R/D CMT surface lines.			GS	1493		2.0	0.08	3.2	0.1	2.2	0.09	3.5
D/Out CMT, shoe and rat hole to 1493m.			GS	1493		6.0	0.25	3.4	0.3	6.6	0.28	3.8
Total Duration of De-Suspend the Well:							***3.4***	***3.4***			***3.8***	***3.8***

12-1/4" Hole Section / 9-5/8" Casing

Drilling 12-1/4" Hole to 1498m.	Drilling	5	AS	1498	2.0	2.5	0.10	3.5	0.1	2.8	0.11	3.9
CBU and balance Mw. perform FIT with Required Pressure 410 psi (EMW: 87pcf, should be equal to max. planned Mw for this hole section plus 2 pcf)	FIT		AS	1498		3.0	0.13	3.6	0.1	3.3	0.14	4.0
Cont. Drilling 12-1/4" Hole to 1515m.	Drilling	17	AS	1515	2.7	6.3	0.26	3.9	0.3	6.9	0.29	4.3
POOH to surface, L/D TCI Bit and junk sub.	Trip		AS	1515		7.0	0.29	4.2	0.3	7.7	0.32	4.6
Change Ram, Install shooting nipple, R/U WL Logging Eq. Perform Logging (CBL/VDL logs) from Surface to 1492 m in 13-3/8" CSG, R/D Logging Eq, change Ram.	Log		As	1515		8.0	0.33	4.5	0.4	8.8	0.37	5.0
P/U & M/U 8" DCs and 5" DPs/HWDPs stands for drilling 12-1/4" hole section, R/Back same in derrick.	Trip		AS	1515		32.0	1.33	5.5	1.5	35.2	1.47	6.5
P/U & M/U 12-1/4" PDC Bit (St. STB @ 20m), RIH to Btm @ 1515m.	Trip		AS	1515		8.0	0.33	5.9	0.4	8.8	0.37	6.8
Cont. Drilling 12-1/4" hole f/ 1515m to 1815m.	Drilling	300	AS PD	1815	4.8	62.5	2.60	8.5	2.9	68.8	2.86	9.7
Take single-shot survey every 100m and multi-shot survey every 300m.	Survey		PD	1815		2.0	0.08	8.6	0.1	2.2	0.09	9.8
Perform condition trip every drilled 300m (CBU, POH, RIH, CBU)	Hole Cond/Circ		PD	1815		4.0	0.17	8.7	0.2	4.4	0.18	10.0
Cont. Drilling 12-1/4" hole f/ 1815m to 2115m.	Drilling	300	PD	2115	4.8	62.5	2.60	11.3	2.9	68.8	2.86	12.8
Take single-shot survey every 100m and multi-shot survey every 300m.	Survey		PD	2115		2.0	0.08	11.4	0.1	2.2	0.09	12.9
Perform condition trip every drilled 300m (CBU, POH, RIH, CBU)	Hole Cond/Circ		PD	2115		4.0	0.17	11.6	0.2	4.4	0.18	13.1
Cont. Drilling 12-1/4" hole f/ 2115m to 2365m.	Drilling	250	PD GU	2365	4.4	56.8	2.37	13.9	2.6	62.5	2.60	15.7
Take single-shot survey every 100m and multi-shot survey every 300m.	Survey		GU	2365		2.0	0.08	14.0	0.1	2.2	0.09	15.8
Perform condition trip every drilled 300m (CBU, POH, RIH, CBU)	Hole Cond/Circ		GU	2365		4.0	0.17	14.2	0.2	4.4	0.18	16.0
POOH to surface, L/D PDC Bit.	Trip		GU	2365		12.0	0.50	14.7	0.6	13.2	0.55	16.5
BOP Test.	BOP Test		GU	2365		6.0	0.25	14.9	0.3	6.6	0.28	16.8
P/U & M/U New 12-1/4" PDC Bit and same BHA, RIH to Btm.	Trip		GU	2365		12.0	0.50	15.4	0.6	13.2	0.55	17.4
Cont. Drilling 12-1/4" hole f/ 2365m to 2665m.	Drilling	300	TL LA SV	2665	3.8	78.9	3.29	18.7	3.6	86.8	3.62	21.0
Take single-shot survey every 100m and multi-shot survey every 300m.	Survey		SV	2665		2.0	0.08	18.8	0.1	2.2	0.09	21.1

Perform condition trip every drilled 300m (CBU, POH, RIH, CBU)	Hole Cond/Circ		SV	2665		4.0	0.17	19.0	0.2	4.4	0.18	21.2
Cont. Drilling 12-1/4" hole f/ 2665m to 2965m.	Drilling	300	SV	2965	3.3	90.9	3.79	22.8	4.2	100.0	4.17	25.4
Take single-shot survey every 100m and multi-shot survey every 300m.	Survey		SV	2965		2.0	0.08	22.9	0.1	2.2	0.09	25.5
Perform condition trip every drilled 300m (CBU, POH, RIH, CBU)	Hole Cond/Circ		SV	2965		4.0	0.17	23.0	0.2	4.4	0.18	25.7
Cont. Drilling 12-1/4" hole f/ 2965m to 3200m.	Drilling	235	SV	3200	3.2	73.4	3.06	26.1	3.4	80.8	3.37	29.1
Take single-shot survey every 100m and multi-shot survey every 300m.	Survey		SV	3200		2.0	0.08	26.2	0.1	2.2	0.09	29.1
Perform condition trip every drilled 300m (CBU, POH, RIH, CBU)	Hole Cond/Circ		SV	3200		4.0	0.17	26.3	0.2	4.4	0.18	29.3
POOH to surface, L/D PDC Bit.	Trip		SV	3200		16.0	0.67	27.0	0.7	17.6	0.73	30.1
BOP Test.	BOP Test		SV	3200		6.0	0.25	27.2	0.3	6.6	0.28	30.3
P/U & M/U New 12-1/4" PDC Bit and same BHA, RIH to Btm.	Trip		SV	3200		16.0	0.67	27.9	0.7	17.6	0.73	31.1
Cont. Drilling 12-1/4" hole f/ 3200m to 3500m.	Drilling	300	SV KZ DR	3500	3.5	85.7	3.57	31.5	3.9	94.3	3.93	35.0
Take single-shot survey every 100m and multi-shot survey every 300m.	Survey		DR	3500		2.0	0.08	31.6	0.1	2.2	0.09	35.1
Perform condition trip every drilled 300m (CBU, POH, RIH, CBU)	Hole Cond/Circ		DR	3500		4.0	0.17	31.7	0.2	4.4	0.18	35.3
Cont. Drilling 12-1/4" hole f/ 3500m to 3800m.	Drilling	300	DR GD	3800	3.7	81.1	3.38	35.1	3.7	89.2	3.72	39.0
Take single-shot survey every 100m and multi-shot survey every 300m.	Survey		GD	3800		2.0	0.08	35.2	0.1	2.2	0.09	39.1
Perform condition trip every drilled 300m (CBU, POH, RIH, CBU)	Hole Cond/Circ		GD	3800		4.0	0.17	35.4	0.2	4.4	0.18	39.3
Cont. Drilling 12-1/4" hole f/ 3800m to section TD @ 3986m.	Drilling	186	GD	3986	3.2	58.1	2.42	37.8	2.7	63.9	2.66	41.9
Take single-shot survey every 100m and multi-shot survey every 300m.	Survey		GD	3986		2.0	0.08	37.9	0.1	2.2	0.09	42.0
CBU, Perform condition trip to last CSG Shoe prior to logging, RIH to Btm, CBU, Spot pill.	Hole Cond/Circ		GD	3986		30.0	1.25	39.1	1.4	33.0	1.38	43.4
POOH to surface, L/D PDC Bit.	Trip		GD	3986		19.0	0.79	39.9	0.9	20.9	0.87	44.3
Change Ram, Install shooting nipple, R/U WL Logging Eq. Perform Logging (Full Suite/CDR/GR), R/D Logging Eq, change Ram.	Log		GD	3986		14.0	0.58	40.5	0.6	15.4	0.64	44.9

BOP Test.	BOP Test		GD	3986		6.0	0.25	40.7	0.3	6.6	0.28	45.2
Condition hole prior to run casing w/ last BHA. (RIH to Btm, CBU, Spot pill, POOH to surface)	Hole Cond/Circ		GD	3986		40.0	1.67	42.4	1.8	44.0	1.83	47.0
R/U CSG running gear, Run 9-5/8" casing to Btm, R/D CSG running gear.	CSG		GD	3986		55.0	2.29	44.7	2.5	60.5	2.52	49.5
Perform 9-5/8" casing cementing operation.	CMT		GD	3986		8.0	0.33	45.0	0.4	8.8	0.37	49.9
Set 9-5/8" casing slip and test same w/ 2000 psi. Hang BOPs, Cut & L/D Landing Joint.	N/D BOP		GD	3986		10.0	0.42	45.5	0.5	11.0	0.46	50.4
Dress & bevel CSG neck, Install 13-5/8", 5K * 11", 10K THS. Inject plastic packing & test P-Seal w/ 4000 psi.	N/U WH		GD	3986		6.0	0.25	45.7	0.3	6.6	0.28	50.6
N/U 13-5/8", 10K BOP's & related lines.	N/U BOP		GD	3986		8.0	0.33	46.0	0.4	8.8	0.37	51.0
Test Accumulator unit, BOPs, choke manifold & WC equipments as per NIOC policy.	BOP TEST		GD	3986		8.0	0.33	46.4	0.4	8.8	0.37	51.4
L/D 8-1/2" drill Collars. P/U & M/U 6-1/2" DCs and 5" DPs/HWDPs stands for drilling next hole section, R/Back same in derrick.	Trip		GD	3986		7.0	0.29	46.7	0.3	7.7	0.32	51.7
P/U & M/U 8-1/2" MT Bit BHA, RIH to tag Plugs.	D/Out		GD	3986		17.0	0.71	47.4	0.8	18.7	0.78	52.5
D/Out Plugs, F.Collar & CMT to 7m above CSG Shoe, work on Junk sub.	D/Out		GD	3986		5.0	0.21	47.6	0.2	5.5	0.23	52.7
Circ and balance Mw. R/U CMT surface lines, close well and Perform 9-5/8" CIT. with 1000 psi for 10 min, R/D CMT surface lines.	Testing		GD	3986		3.0	0.13	47.7	0.1	3.3	0.14	52.8
POOH to surface.	D/Out		GD	3986		17.0	0.71	48.4	0.8	18.7	0.78	53.6
RIH 8-1/2" MT Bit with 9-5/8" Scraper to Btm (7m above shoe), CBU, Scrap Casing prior to run CMT quality Log, CBU, Spot pill, POOH.	Log		GD	3986		36.0	1.50	49.9	1.7	39.6	1.65	55.3
Change Ram, Install shooting nipple, R/U WL Logging Eq. Perform Logging (SBT or CAST/GR/CCL/CBL/VDL logs) from Surface to 3986 m in 9-5/8" CSG, R/D Logging	Log		GD	3986		13.0	0.54	50.5	0.6	14.3	0.60	55.9
P/U & M/U 8-1/2" PDC Bit BHA (Including MWD & St. STB @ 20m), RIH to tag CMT (7m above shoe).	Trip		GD	3986		18.0	0.75	51.2	0.8	19.8	0.83	56.7
Displace well volume with 100 pcf Salt Saturated Kcl/Polymer Mud.	D/Out		GD	3986		4.0	0.17	51.4	0.2	4.4	0.18	56.9
Cont' D/Out CMT, Shoe and Rat Hole.	Hole Cond/Circ		GD	3986		3.0	0.13	51.5	0.1	3.3	0.14	57.0
Total Duration of 12-1/4" Hole Section:							***48.4***	***51.5***			***53.3***	***57.0***

8-1/2" Hole Section / 7" Liner

Drilling 8-1/2" hole to 3990m.	Drilling	4	FH	3990	2	2/0	0/08	51/9	0/1	2/2	0/09	57/1
CBU and balance Mw. perform FIT with Required Pressure 728 psi (EMW: 108pcf, should be equal to max. planned Mw for this hole section plus 2 pcf)	FIT		FH	3990		3	0/13	52/0	0/1	3/3	0/14	57/2
Drilling 8-1/2" hole f/ 3990m to well TD @ 4200m.	Drilling	210	FH	4200	3/0	70/0	2/92	55/0	3/2	77/0	3/21	60/4
Take MWD survey every drilled stand.	Survey		FH	4200		3/5	0/15	55/1	0/2	3/9	0/16	60/6
CBU, Perform condition trip to last CSG Shoe prior to Logging, RIH to Btm, CBU, Spot pill.	Hole Cond/Circ		FH	4200		6/0	0/25	55/3	0/3	6/6	0/28	60/9
POOH to surface.	Trip		FH	4200		19/0	0/79	56/1	0/9	20/9	0/87	61/8
BOP Test.	BOP Test		FH	4200		6/0	0/25	56/4	0/3	6/6	0/28	62/0
Change Ram, Install shooting nipple, R/U WL Logging Eq. Perform Logging (Full Suite/CDR/GR Logs), R/D Logging Eq., change Ram.	Log		FH	4200		11/0	0/46	56/8	0/5	12/1	0/50	62/5
RIH w/ 8-1/2" PDC Bit BHA (Without MWD), condition hole prior to 7" Liner running (RIH to Btm, CBU, Spot pill, POOH to surface)	Hole Cond/Circ		FH	4200		38/0	1/58	58/4	1/7	41/8	1/74	64/3
R/U 7" Liner running Eq., Run 7" liner (+/- 314m LNR) to Btm @ 4200m, R/U CMT lines, Set L.Hanger +/- 3886m (100m overlap), release L.Hanger setting tool.	CSG		FH	4200		30/0	1/25	59/7	1/4	33/0	1/38	65/7
Cement 7" Liner as per program.	CMT		FH	4200		5/0	0/21	59/9	0/2	5/5	0/23	65/9
POH w/ L.Hanger setting tool to 3500m, Flush string & CBU, POH to 3300m.	CSG		FH	4200		4/0	0/17	60/1	0/2	4/4	0/18	66/1
Close the well and WOC.	WOC		FH	4200		4/0	0/17	60/2	0/2	4/4	0/18	66/2
Open the Well. Cont' POH to surface & L/D L.Hanger setting tool.	CSG		FH	4200		12/0	0/50	60/7	0/6	13/2	0/55	66/8
BOP Test.	BOP Test		FH	4200		6/0	0/25	61/0	0/3	6/6	0/28	67/1
P/U & M/U 8-1/2" MT Bit BHA, RIH to L.Lap @ +/- 3886m while D/Out cement (If any cement: Considered 7STD wet for CMT Job), Circ & clean the well, POOH.	D/Out		FH	4200		40/0	1/67	62/6	1/8	44/0	1/83	68/9
L/D excess 5" DPs/HWDP.	D/Out		FH	4200		5/0	0/21	62/8	0/2	5/5	0/23	69/1
P/U & M/U 4-3/4" DCs & 3-1/2" DPs/HWDPS, R/Back same in derrick.	D/Out		FH	4200		7/0	0/29	63/1	0/3	7/7	0/32	69/5
P/U & M/U 5-7/8" MT Bit BHA, RIH to tag POB, D/Out POB, RIH to LDC depth @ +/- 4170m while D/Out any cement, work on junk sub and CBU. Perform 7" Liner CIT. (80% of Burst press) with 1500 psi,for 10 min, (If not obs bumping). POOH to top of Liner Lap @ +/- 3886m, CBU, perform Liner lap positive test w/ 2000 psi, POOH to Surface.	D/Out		FH	4200		40/0	1/67	64/8	1/8	44/0	1/83	71/3

RIH with 5-7/8" MT Bit and tandem scraper (7" & 9-5/8") to LDC depth, Scrap inside 7" Liner and 9-5/8" casing before Dry test, CBU, POOH to Surface.	Testing		FH	4200		36.0	1.50	66.3	1.7	39.6	1.65	72.9
BOP Test.	BOP Test		FH	4200		6.0	0.25	66.6	0.3	6.6	0.28	73.2
RIH with 9-5/8" Dry test BHA, RIH to L.Lap @ +/- 3886m, Perform 7" Liner lap Dry Test (based on fahliyan Reservoir Datum Pressure), POOH.	Testing		FH	4200		68.0	2.83	69.4	3.1	74.8	3.12	76.3
Change Ram, Install shooting nipple, R/U WL Logging Eq. Perform CMT quality Logging of 7" Liner from 7" liner shoe @ 4200 m to 7" Liner Lap @ 3886 m (SBT or CAST/CBL/VDL), R/D Logging Eq, change Ram.	Log		FH	4200		11.0	0.46	69.8	0.5	12.1	0.50	76.8
P/U & M/U 5-7/8" MT Bit and C/Out BHA, RIH to LDC depth. Displace 7" Liner volume with CaBr2 Perforation fluid.	Completion		FH	4200		36.0	1.50	71.3	1.7	39.6	1.65	78.5
P/U & M/U DST/TCP Perforation Assy, RIH to perforation depth, correlate depth with WL and perforate Guns as program. POOH DST/TCP Perforation Assy.	Completion		FH	4200		64.0	2.67	74.0	2.9	70.4	2.93	81.4
P/U & M/U 5-7/8" Mill BHA, RIH to top of LDC depth while smoothing Perforated intervals.	Completion		FH	4200		36.0	1.50	75.5	1.7	39.6	1.65	83.1
Change Ram, Install shooting nipple, R/U WL Eq. RIH WL with CCL/GR Log to confirm Perforated intervals, POOH and R/D Logging Eq, change Ram.	Completion		FH	4200		11.0	0.46	76.0	0.5	12.1	0.50	83.6
P/U and M/U 5-7/8" Mill and 7" Scraper. RIH to Perforation depth while Scrap 7" PKR setting depth interval, CBU. Displace well volume with CaBr2 Completion fluid.	Completion		FH	4200		24.0	1.00	77.0	1.1	26.4	1.10	84.7
POOH to surface while L/D all Tubulars.	Completion		FH	4200		56.0	2.33	79.3	2.6	61.6	2.57	87.2
BOP Test.	BOP TEST		FH	4200		6.0	0.25	79.6	0.3	6.6	0.28	87.5
R/U Completion running Eq, Run 4-1/2" * 3-1/2", 10K Completion string as per program, R/D completion running Eq.	Completion		FH	4200		124.0	5.17	84.7	5.7	136.4	5.68	93.2
Install BPV in TBG Hanger, N/D BOPs.	Completion		FH	4200		14.0	0.58	85.3	0.6	15.4	0.64	93.8
Retrieve BPV, set TWCV in TBG Hanger. N/U X-mass Tree, perform required test as per program, Retrieve TWCV.	Completion		FH	4200		8.0	0.33	85.6	0.4	8.8	0.37	94.2
R/U Slick Line, Set protection sleeve in TRSSSV, R/D Slick Line.	Completion		FH	4200		4.0	0.17	85.8	0.2	4.4	0.18	94.4
R/U CTU, Stimulation and Well Testing Eq. RIH w/ CT to Btm, displace the well with Diesel, POOH CT to surface.	Completion		FH	4200		24.0	1.00	86.8	1.1	26.4	1.10	95.5
R/U Slick Line, Retrieve protection sleeve from TRSSSV, R/D Slick Line.	Completion		FH	4200		4.0	0.17	87.0	0.2	4.4	0.18	95.7
Flow the well as Pre-Acid clean up till acceptable BSW and reading stable flow rate on 2 different choke size for 72 hrs.	Completion		FH	4200		77.0	3.21	90.2	3.5	84.7	3.53	99.2
R/U Slick Line, Set protection sleeve in TRSSSV, R/D Slick Line.	Completion		FH	4200		Rig Less	0.00	90.2	0.0	0.0	0.00	99.2
RIH w/ CT and perform Acid Wash as per program, POOH CT to surface.	Completion		FH	4200		Rig Less	0.00	90.2	0.0	0.0	0.00	99.2

R/U Slick Line, Retrieve protection sleeve from TRSSSV, R/D Slick Line.	Completion		FH	4200		Rig Less	0/00	90/2	0/0	0/0	0/00	99/2
Flow the well as Post-Acid clean up till acceptable BSW and reading stable flow rate on 2 different choke size for 96 hrs.	Completion		FH	4200		Rig Less	0/00	90/2	0/0	0/0	0/00	99/2
R/U Slick Line, Set protection sleeve in TRSSSV, R/D Slick Line.	Completion		FH	4200		Rig Less	0/00	90/2	0/0	0/0	0/00	99/2
RIH w/ CT to Btm, displace well with Diesel, POOH CT to surface, R/D CTU and well Testing Eq.	Completion		FH	4200		Rig Less	0/00	90/2	0/0	0/0	0/00	99/2
R/U Slick Line, RIH w/ SL and Install Plug and Prong into NO-GO L.NPL, POOH.	Completion		FH	4200		Rig Less	0/00	90/2	0/0	0/0	0/00	99/2
RIH w/ SL and Retrieve protection sleeve from TRSSSV, R/D Slick Line.	Completion		FH	4200		Rig Less	0/00	90/2	0/0	0/0	0/00	99/2
Secure the well by set BPV in TBG Hanger, close all X-mas Tree valves except BMV, Release the Rig.	Completion		FH	4200		Rig Less	0/00	90/2	0/0	0/0	0/00	99/2
Total Duration of 8-1/2" Hole Section:							***38/4***	***90/2***			***42/2***	***99/2***
NOTE:												
Total Duration for Cut & Slip DL in all hole section.	Cut & Slip		FH	4200		12/5	0/52	90/7	0/6	13/8	0/57	99/8
Total Duration for Rig Service (Consider 0.5hr per day) in all hole section.	Rig Service		FH	4200		48/8	2/03	92/7	2/2	53/7	2/24	102/0
Total Duration of AZNS-040 Re-Entry Well:			**FH**	**4200**				**92/7**				**102/0**

AZNS-040 Re-Entry Well Time vs. Depth

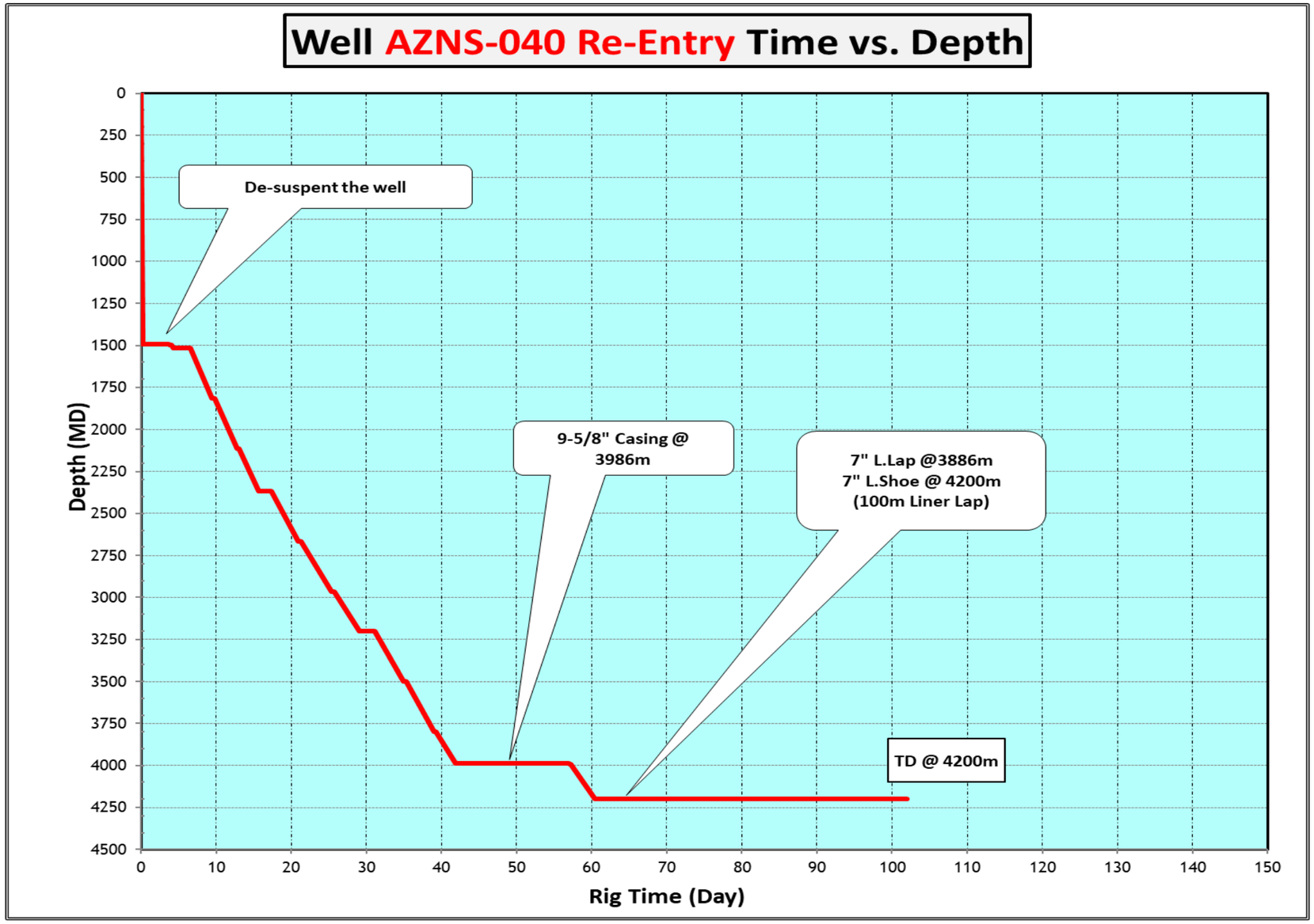


Figure 1-5: AZNS-040 Re-Entry Well Time vs. Depth

AZNS-040 Re-Entry Well Time Table

Time Breakdown								Sum (Hrs)	Total %	Total (Days)
Hole Size	De-Suspend	12-1/4"	8-1/2"	Completion						
Drilling		658.9	72					730.9	32.8%	30.5
Round Trip/BHA		147	19					166	7.5%	6.9
BOP Test		26	24					50	2.2%	2.1
Hole Condition, Reaming, Circ		105	44					149	6.7%	6.2
Deviation Survey		18	3.5					21.5	1.0%	0.9
Directional Drilling								0	0.0%	0.0
Casing/Liner		55	46					101	4.5%	4.2
Cementing		8	5					13	0.6%	0.5
WOC			4					4	0.2%	0.2
N/U&D BOP		18						18	0.8%	0.8
N/U WH		6						6	0.3%	0.3
Drill Out		43	92					135	6.1%	5.6
Logging		71	22					93	4.2%	3.9
Testing (FIT, LOT, CIT, Dry Test, etc.)		6	107					113	5.1%	4.7
DST								0	0.0%	0.0
Coring								0	0.0%	0.0
Plug Back								0	0.0%	0.0
Completion				482				482	21.7%	20.1
Well Testing								0	0.0%	0.0
Rig Repair	1.4	24.4	9.2	13.7				48.7	2.2%	2.0
Slip & Cut		7.5	5					12.5	0.6%	0.5
De-suspend	82							82	3.7%	3.4
Workover								0	0.0%	0.0
Total Hrs per Hole	83.4	1193.8	452.7	495.7	0.0		2225.6			
Total Days per Hole	3.5	49.7	18.9	20.7	0.0		92.7			
Total Days per Hole with 10% Contingency	3.8	54.7	20.7	22.7	0.0		102.0			
Cummulative Time (Days)	3.5	53.2	72.1	92.7	92.7					
Cummulative Time (Days) with 10% Contingency	3.8	58.5	79.3	102.0	102.0					

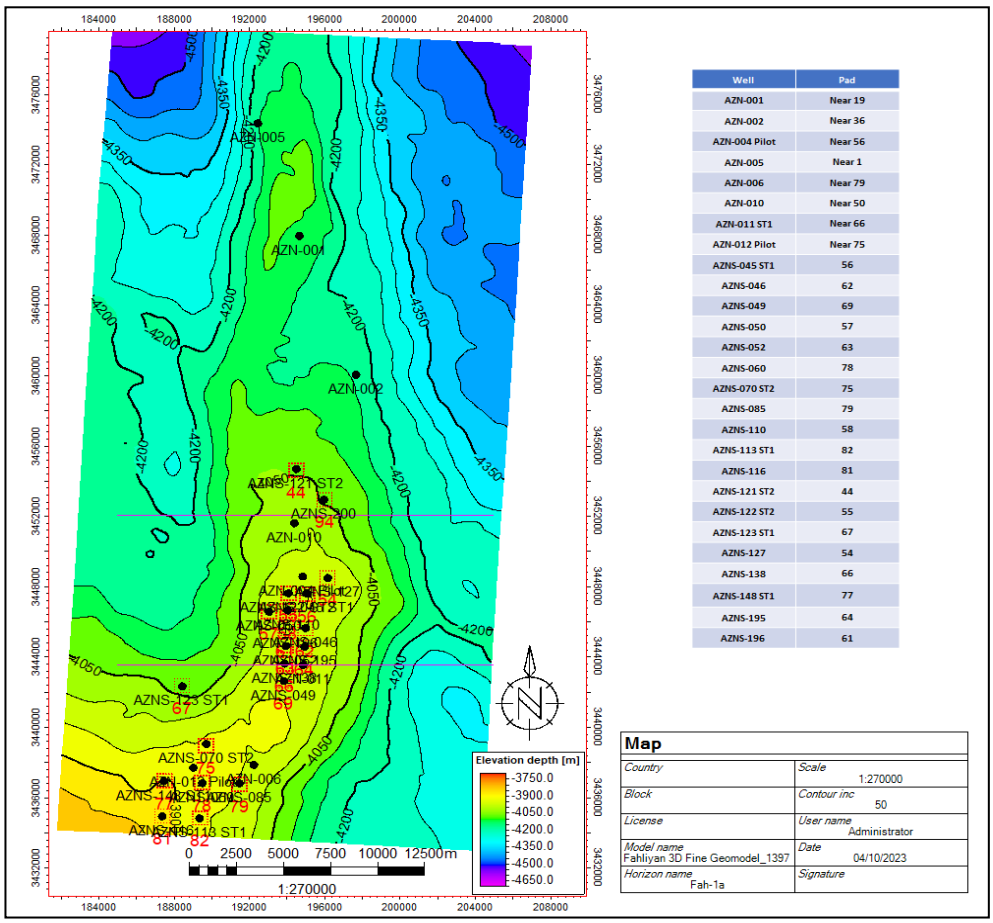
Figure 1-6: AZNS-040 Re-Entry Well Time Table

2. Geological Information

2.1 Structural Geology

The Azadegan oil field is located at 80km west of Ahwaz is an elongated anticline, trending in N-S direction, belongs to Arabian Trend which is different from "Zagros Trend". There is no outcrop for Azadegan structure and it was detected by 2D seismic interpretation. The Previous exploration/delineation drilling in Azadegan oil field has proved presence of a very prolific hydrocarbon bearing sedimentary column, particularly within the Cretaceous sequences. Fahliyan Formation is one of the important hydrocarbon reservoirs in the southern part of Azadegan field, which has been developed in recent years. The Fahliyan Formation was divided into 6 zones, named as Fahliyan top, Fah-1, Fah-2, Fah-3, Fah-4 and Fah-5.

Figure 2-1: Distribution of Fahliyan wells in South Azadegan Oilfield on Top of Fah-1 Zone



2.2 Formation Description

The stratigraphy sequence of this field includes Aghajari, Gachsaran, Asmari, Pabdeh, Gurpi, Ilam, Laffan, Sarvak, Kazhdumi, Dariyan, Gadvan and Fahliyan formations. This sequence in well no. AZN-10, the deepest up to now drilled well of the field, continues with Garau, Gotnia, Najmah, Sargalu, Alan, Mus and Neyriz formations.

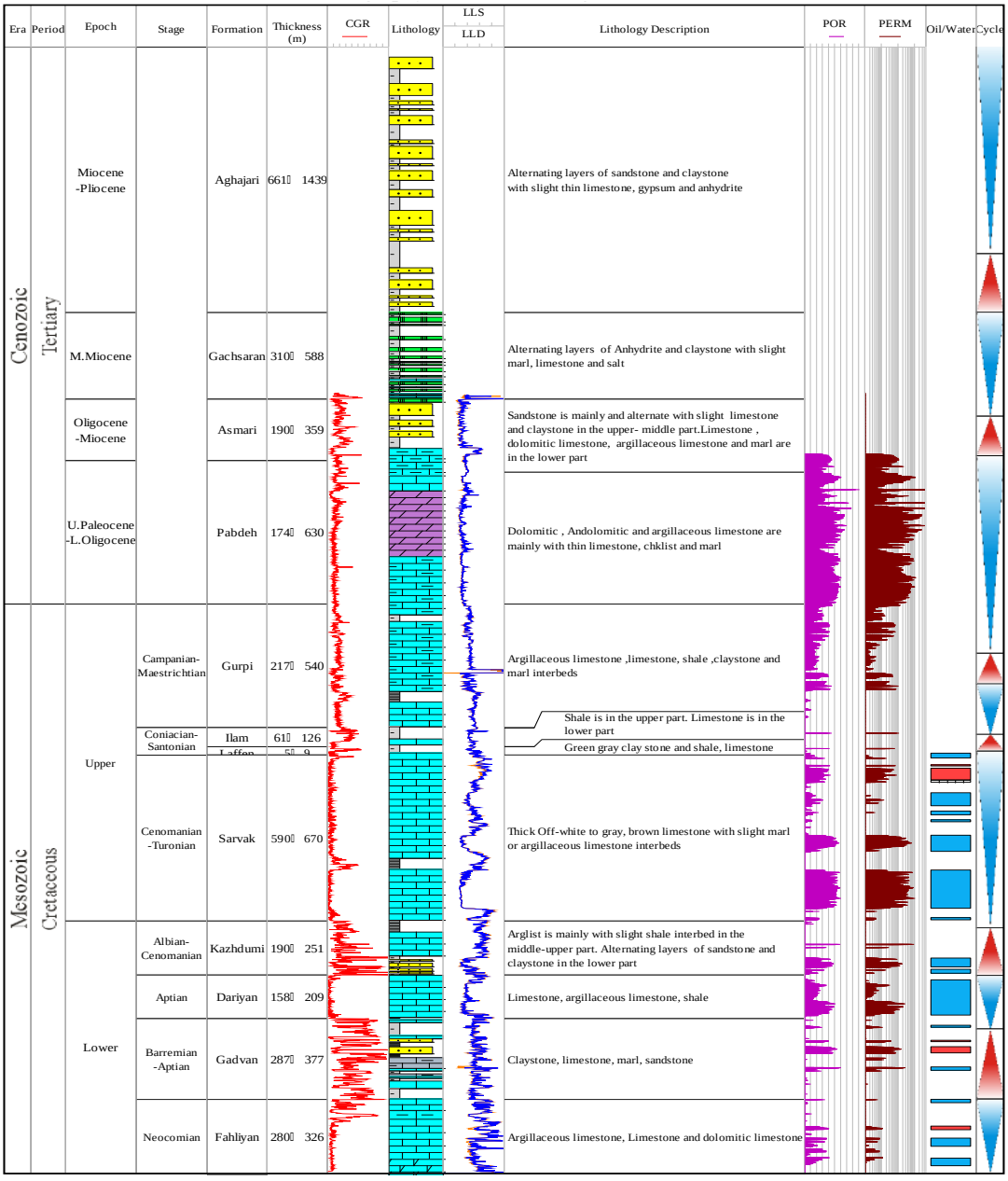


Figure 2-2: Stratigraphy Column of South Azadegan Oilfield

The **AZN-040 Re-Entry** well is carried out inside of Gachsaran to Fahliyan Formations, a brief stratigraphic description of these formations in South Azadegan field has been described as follows:

- **Gachsaran Formation**

This formation begins with white, glassy, soft, medium, hard anhydrite and off white, light gray to gray, soft to medium hard, dolomitic limestone. Comprises of anhydrite interbedded with gray marl and red marl, salt beds of variable thickness and some thin limestone beds. This formation in type section not divided to other member. The Anhydrite is white off white, milky, soft, and partly crystalline to crystalline, medium hard to hard. The Gray marl is gray to light gray, soft. The Red marl is red brown to brown, soft, sticky. The Limestone is gray to light gray, buff to buff gray, hard, and slightly dolomitic.

- **Asmari Formation**

The Asmari Formation in Azadegan Field developed a massive sandstone interval, with a few thin argillaceous limestones in the upper sections. Sandstone are clear, fine to medium-sorted, and poorly consolidated. On the other hand, the lower sections consist of intercalation of sandy, chalky, slightly dolomitic high-energy limestone with sandstone layers.

- **Pabdeh Formation**

Pabdeh Formation is divided into two parts in the Azadegan field. The upper part is made of grey to dark- brown argillaceous limestone and marl. Planktonic fauna dates that this part was deposited in Oligocene. The lower part is composed of dark-brown, slightly argillaceous limestone. Pelagic fauna, mostly Globigerina and Globorotalia, indicates that this part was formed during late Paleocene to late Eocene in age. It seems that no early Paleocene formation exists in the Azadegan area.

- **Gurpi Formation**

This formation consists of three main units: a) Upper Gurpi Member, b) Tarbour Tongue and c) Lower Gurpi Member.

➢ **Upper Gurpi Member**

This member begins with light gray, gray, off white, soft to medium, sticky argillaceous limestone. The Upper Gurpi Member is mostly consisting of gray buff to gray with of trace partly buff gray, dark gray, medium hard to hard argillaceous, high argillaceous graded to marl argillaceous limestone interbedded with anhydrite marl and chalky limestone beds. The anhydritic marl is light gray to gray, soft,

sticky, graded to high argillaceous calcareous. The chalky limestone is light gray to buff gray, medium hard to soft, chalky. II, a slightly argillaceous.

➢ **Tarbur Tongue**

In the western Abadan Plain about 50 meters thick chalky limestone was encountered in the middle part of the Gurpi Formation, which contains abundant fragments of bioclasts, suggesting very shallow depositional environment. This interval is equivalnt to the Tarbur Formation with reservoir quality in other regions.

➢ **Lower Gurpi Member**

This Member begins with off white, cream, soft to medium hard argillaceous limestone and light gray to gray, soft, sticky marl. The Member consisting of dark gray to gray, light gray, soft to medium hard argillaceous limestone and light cream and soft. II, A. chalky limestone beds.

- **Ilam Formation**

This formation is mostly consisting of limestone interbedded with shale and clay stone beds. The Limestone is cream, buff to buff gray, light brown, soft to medium hard, chalky, II, A, oil stained Partly slightly argillaceous. The Shale is green gray to light green gray, mostly brown gray, medium hard, pyritic, slightly calcareous. The Claystone is green gray to light green gray and soft, slightly calcareous.

- **Laffan Formation**

The Laffan Formation is mostly consists of greenish-grey pyritic shale layer with ostracods fauna interbedded with Claystone beds that capping the Sarvak Formation.

- **Sarvak Formation**

The Sarvak Formation developed a carbonate unit, which was gradually overlying the Kazhdumi Formation, and it is abruptly terminated by an erosion phase, covered by a various late Cretaceous sediment. The lithology of Sarvak consists of the dark grey fine grained nodular argillaceous limestone interbedded with thin marls in the bottom. The Limestone graded upward into a white-to-buff massive chalky limestone with siliceous nodules; developed a very massive tan-to-brown limestone containing abundant rudist debris in the middle part. In the upper part is made up of rubbly limestone with red staining and leached zones of breccia and the top of Sarvak Formation is very ferruginous. For the layering scheme of the reservoir section, sequence stratigraphic concept was applied to the Azadegan wells. Judging from the characteristics of gamma ray and other wireline logs, relative sea- level change during the deposition of the reservoir sections were

interpreted in terms of sequence stratigraphic context. Based upon the sea-level change inferred, the sections were interpreted as vertical accumulation of a number of zones, each of which is geologically termed as a depositional sequence, implicitly representing a genetically related body of sedimentary sequence. The formation was divided totally into thirteen zones by interpreting.

- **Kazhdumi Formation**

This formation is start with dark brown to dark green gray, medium hard to hard, soft to medium, high Bituminous, calcareous shale interbedded with argillaceous limestone, sandy claystone, claystone, marl, sandy limestone, limestone beds. the middle part developed light-grey, micritic to chalky limestone, the low part, is around 40-60m in thichness clean channel sands interbeded with argillaceous sands with streaks of shale.

- **Dariyan Formation**

In the Azadegan Field, the average thickness of the Dariyan Formation is 183.5 m; its greatest thickness is 197.5 m in well AZN-1. This formation overlies the Gadvan Formation conformably and is overlain disconformably by the Kazhdumi Formation. Dariyan Formation starts with white to white cream grey clean limestone type I/II, I. Base of Dariyan Formation appears grey, green pyritic shale interbedded with argillaceous limestone. This zone called transition zone and thickness is 16m. after last grey limestone indicates of Gadvan Formation.

- **Gadvan Formation**

In this formation a variety of lithological rock types such as carbonates, rich organic shales and sandstones were deposited in various environments. The lower part of the Gadvan Formation is composed of low- energy, but not very deep, micritic, often argillaceous limestone with echinoids and algal fragments. Two sandstone units were defined in the Gadvan Formation lateral changes and especially discontinuity of the sandstone layers which are consist of fine clastics (shales and siltstones) may be expected The Gadvan formation divided to three part:

➢ **Upper Gadvan**

The upper part appears with occurrence of terrigenous rock. The lithology is composed of grey claystone and some shale.

➢ **Khalij Member**

The Khalij member appear with pure limestone, cream, light brown and in some intervals oil stain.

- **Lower Gadvan**

The lower part consisting of alternating of marl, argillaceous limestone, limestone, claystone, sandy claystone and sandstone beds. The Marl is green gray, soft, sticky, slightly calcareous. The Argillaceous Limestone is gray white gray medium hard to soft, partly pyritic. The Limestone is white, white to light gray, hard. The Claystone is light gray, soft, sticky. The Sandy Claystone is light gray, soft, sticky, fine sand grained. The Sandstone is brown to gray, partly white buff, gray, friable, sorted, sub rounded, medium quartz grained, partly calcareous. This sandstone which occur in Zone-3 and Zone-5 are the reservoir of this formation and in this part is equivalent with Kushk member.

- **Fahliyan Formation**

This formation is start with light gray, light green gray, soft Limestone interbedded with green gray, soft, sticky Marl, green gray, light gray, soft, Argillaceous Limestone and dark green gray, soft, sticky Clay stone beds. This formation at continue is consisting of buff white light gray, light brown, brown, medium hard, hard, densely, slightly argillaceous limestone and off white, white gray, medium hard chalky limestone beds.

2.3 Expected Formation Tops and Thickness

The formation tops and thicknesses that expected to be encountered in AZNS-040 Re-Entry well is shown in below table.

Formation /Member Name		Top		Thickness
		mTVDss (RTE:13.45m)	mMD	mTVD
Aghajari		Surface	Surface	1123.1
Gachsaran		-1123.1	1138.0	357.1
Asmari		-1480.2	1493.7	282.4
Pabdeh		-1762.6	1776.1	421.8
Gurpi	U-Gurpi	-2184.4	2197.9	124.1
	Tarbur	-2308.5	2322.0	87.6
	L-Gurpi	-2396.1	2409.6	125.4
Ilam		-2521.5	2535.0	104.1
Laffan		-2625.6	2639.1	6.3
Sarvak	Sar-1	-2631.9	2645.4	19.6
	Sar-2	-2651.6	2665.0	7.6
	Sar-3	-2659.1	2672.6	39.5
	Sar-4_6	-2698.6	2712.1	86.6
	Sar-7	-2785.2	2798.7	15.0
	Sar-8	-2800.2	2813.7	61.6
	Sar-9	-2861.9	2875.3	46.5
	Sar-10	-2908.4	2921.8	17.2
	Sar-11	-2925.5	2939.0	27.6
	Sar-12	-2953.2	2966.6	82.2
	Sar_Intra	-3035.4	3048.8	234.6
Kazhdumi	Kaz-Top	-3270.0	3283.5	147.3
	Kaz-1-Shale	-3417.3	3430.7	7.7
	Kaz-2-Sand	-3424.9	3438.4	20.6
	Kaz-3-Shale	-3445.6	3459.0	7.3
	Kaz-4-Sand	-3452.9	3466.3	18.7
	Kaz-5-Shale	-3471.6	3485.0	4.7
Dariyan		-3476.2	3489.7	193.0
Gadvan	Gad-1	-3669.3	3682.7	9.2
	Gad-2u	-3678.4	3691.9	55.8
	Khalij	-3734.2	3747.7	11.2
	Gad-2l	-3745.4	3758.8	4.9
	Gad-3	-3750.3	3763.7	8.6
	Gad-4	-3758.9	3772.4	21.1
	Gad-5a	-3780.0	3793.5	3.3
	Gad-5b	-3783.3	3796.7	7.1
	Gad-5c	-3790.4	3803.8	4.8
	Gad-6	-3795.1	3808.6	22.4
	Gad-7	-3817.5	3831.0	26.5
	Gad-8	-3844.1	3857.5	128.3
Fahliyan	Fahliyan	-3972.34	3985.79	37.0
	Fah-1a	-4009.32	4022.77	15.3
	Fah-1b	-4024.66	4038.11	14.7
	Fah-1c	-4039.34	4052.79	31.7
	Fah-2a	-4071.05	4084.5	15.5
	Fah-2b	-4086.6	4100.05	9.8
	Fah-3a	-4096.44	4109.89	9.5
	Fah-3b	-4105.92	4119.37	9.5
	Fah-4a	-4115.44	4128.89	10.2
	Fah-4b	-4125.61	4139.06	14.7
	Fah-5a	-4140.27	4153.72	22.4
	Fah-5b	-4162.65	4176.1	23.9
Total Depth		-4186.55	4200	-

Figure 2-3: Expected Formation Tops and Thickness in AZNS-40 Re-Entry Well

Stratigraphy correlation of AZNS-040 well is shown in below figure:

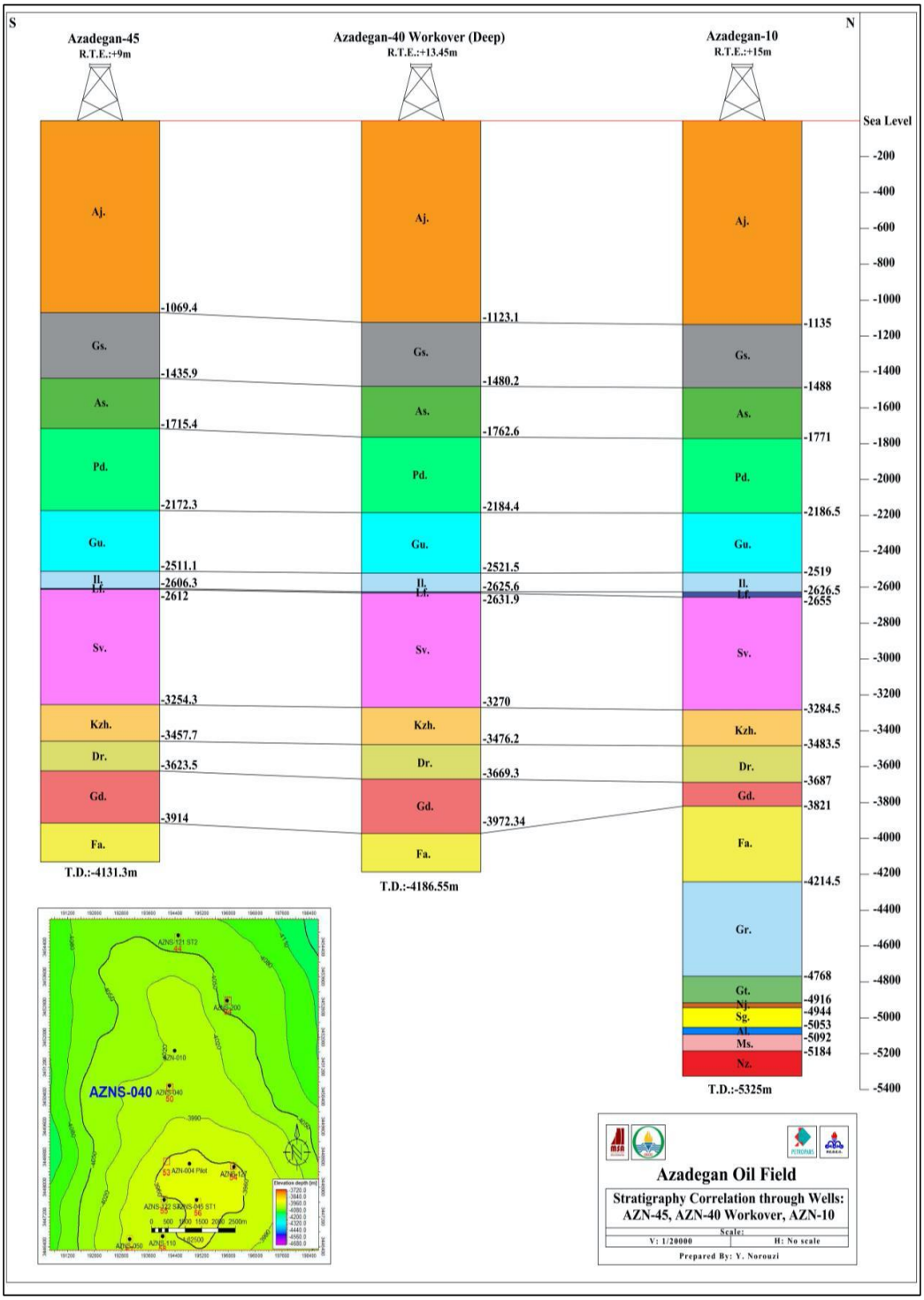
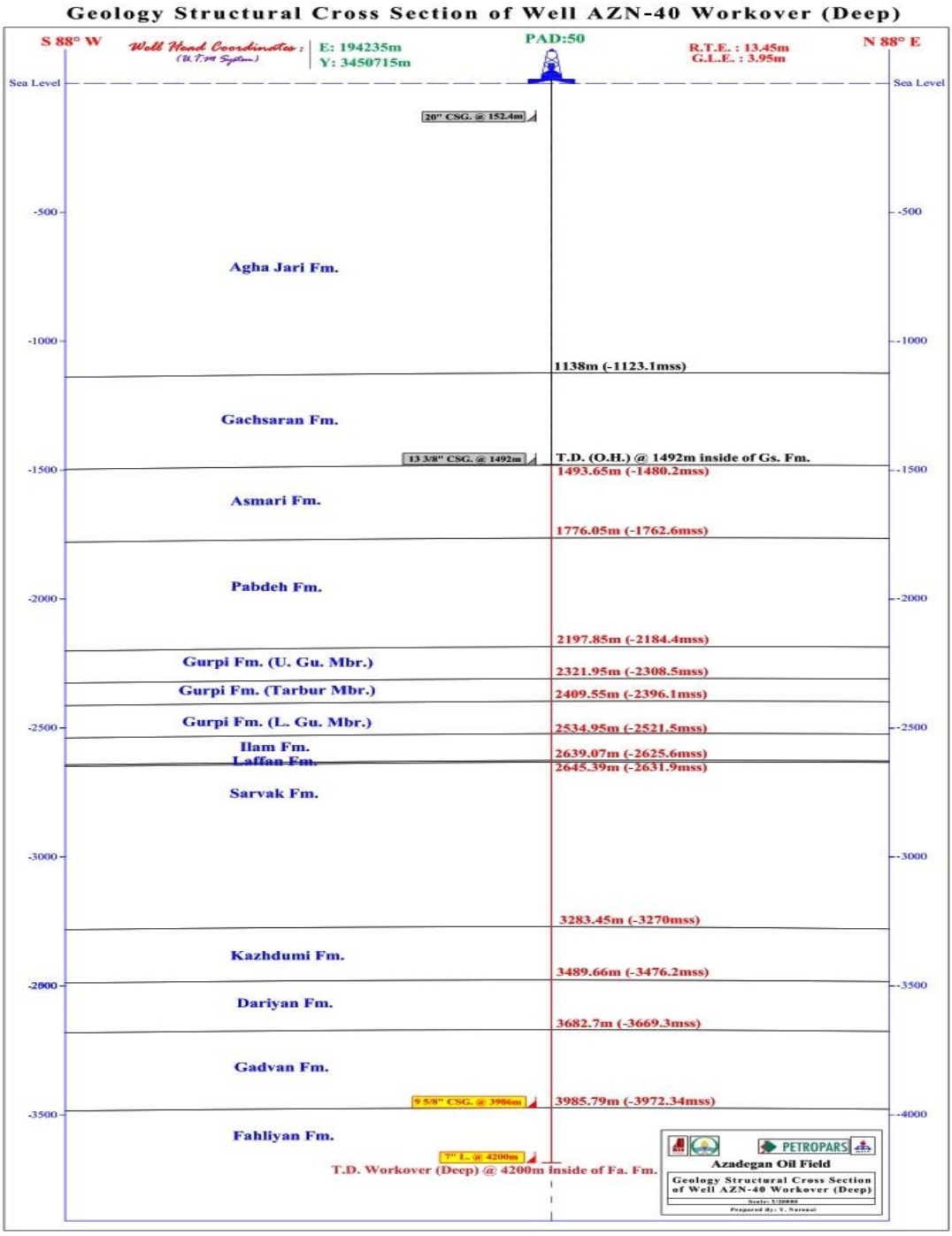


Figure 2-4: Stratigraphy Correlation of AZNS-040 Re-Entry Well

Below figure shows AZNS-040 well geological structural map.

Figure 2-5: AZNS-040 Well Geological Structural Map



2.4 Cutting Sample Program

For identification, description and interpretation of lithology, sampling intervals and preservation of cuttings are required as follow:

Table 2-1: Cutting Sampling Program

Hole Size	Formation/Member	Sampling Interval (m)	Measured Depth (m)
12-1/4" Open Hole	Gachsaran	2	1492m T.D. (O.H.) to 3986
	Asmari		
	Pabdeh		
	Gurpi		
	Ilam		
	Laffan		
	Sarvak		
	Kazhdumi		
	Dariyan		
	Gadvan		
8-1/2" Open Hole	Fahliyan	2	3986-4200m T.D. Workover (Deep)

Note:

- Thin section must be prepared every 10 meters and these intervals could be changed according to the well site geologist.
- For any decision such as top formations or members, casing point depth, K.O.P., core point, logging point and intervals and Total depth and other important issues during drilling of the well, wellsite geologist is fully responsible to determine the frequency of cutting sampling and mud logging crew shall take sample based on wellsite geologist advice. In

deviated interval wellsite geologist should take care of sampling and collecting sample of any cuttings with different characteristics.

2.5 Logging and Testing Program

Logging Program

Well AZNS-040 will undergo wireline logging operation in all required intervals. Since this well logging plan is targeted to evaluate the hydrocarbon potential, the full suite Logs should be run in 12-¼" and 8-½" open hole to collect reservoir and required data.
Logging is limited to GR/CDR logs in 12-¼" and 8-½" open holes and CAST log in 9-5/8" casing and 7" Liner that must be carried out by wire line.
CBL/VDL logs in 13-3/8" casing must be carried out by wire line.
The elaborative Logging and Formation Testing program in Well AZNS-040 are as below:

Table 2-2: Logging Plan for AZNS-040 Re-Entry Well

Hole Size	Casing/Liner Size	Logs		Test	Formation
		Open Hole	Cased Hole		
-	13-3/8"	-	CBL/VDL/CCL/GR	-	Aj.-GS.
12-1/4"	9-5/8"	Full Suite Log	CAST/CBL/VDL/CCL/GR or SBT/VDL/CCL/GR Or USIT/CBL/VDL/CCL/GR	N/A	Gs.-As.-Pd.-Gu.-Il.-Lf.-Sv.-Kzh.-Dr.-Gd.
		CDR/GR Logs			
8-1/2"	7"	Full Suite Log	CAST/CBL/VDL/CCL/GR or SBT/VDL/CCL/GR Or USIT/CBL/VDL/CCL/GR	Positive & Dry Test (7" Liner lap)	Fa.
		CDR/GR Logs			

Note:

- Full Suite Logs is including the GR, Spectral GR, Resistivity, Neutron, Density, and PEF logs.
- The CBL/VDL log will be performed after running and cementing, 9-5/8" casing from casing shoe to 100m above 13-3/8" shoe, and after cementing 7" liner, the CBL/VDL logging will be run from the shoe to liner hanger.

Testing Program

- **FIT:** Perform FIT with EMW equals to planned section Mw plus 2 pcf after drill 3-5m new formation below 13-3/8" CSG shoe and 9-5/8" CSG shoe.
- **Dry Test:** will be carried out from 7" Liner in order to check 7" L.Lap integrity with 2000 psi differential pressure with regard to Fahliyan formation datum pressure.

3. Nearby Wells

The well AZNS-040 is located in Pad#50 and as per well geology program and pad location proximity wells the nearby wells are AZNS-045, AZNS-127, AZN-010 and AZNS-017.

3.1 Nearby Wells Drilling Challenges and Lesson Learned

Drilling challenges and lesson learned for well AZNS-040 are extracted from DDR and EOWR of nearby wells.
Furthermore, based on NBW study, the well AZNS-017 is a Kazhdumi Horizontal well and it's drilling sequences are different from typical Fahliyan well in South Azadegan oilfield, therefore all drilling challenges and lesson learned have been correlated to typical Fahliyan well sequences.

Drilling Challenges and Lesson Learned in 26" Hole Section

Generally, 26" hole section is first step of drilling sequence in AZNS oilfield and it will be drill through Aghajari formation.
This hole section main issue while drilling including Tight hole, Bit/stabilizer balling and hole deviation control.
The best practice to drill this hole section is to commence drilling with low parameters and high viscous spud mud for first drilled stand in order to prevent washout around well location/cellar area, then increase drilling parameters to maximum gradually.
In order to prevent Bit/Stabilizer balling, flow rate and mud properties must be in range.
For preventing hole deviation, use stabilized BHA and drilling with controlled parameters, if hole deviate from vertical, try to decrease hole inclination by reaming and varying drilling parameters such as decrease WOB and increase RPM.
Furthermore, reamer trip is essential to make hole as smooth as possible prior to run casing.
In well AZNS-040 this section already drilled to 153m with 65-71 pcf Mw and 20" CSG has been run to TD and cemented to surface without any problem.
Nearby wells drilling issue in this section are listed as below:

1. Bit balling in well AZNS-045 with 65 pcf Bentonite mud.
2. Noticeable cement losses while cementing 18-5/8" CSG with 115 pcf Sw in well AZNS-017.

3. Bit balling in well AZNS-127.

Drilling Challenges and Lesson Learned in 17-1/2" Hole Section

For typical Fahliyan vertical well in area II, this hole section drill through Gachsaran formation which is not high pressure in this zone.
Based on NBW study and formation description, the main drilling issues in 17-1/2" hole are tight hole, calcium ion contamination, hole deviation from vertical and CMT loss while CSG cementing.
While drilling through Gachsaran formation, drilling fluid must be salt saturated and Mw must be adjust with hole condition and in case of any indication of abnormality while drilling/tripping/reaming such as Hi-Trq, abnormal drag, tight hole, increase in calcium ion value, etc, the Mw should increase in 1 pcf increment to overcome drilling issues and prevent induced losses, also mud should be inhibited with Kcl or any caly inhibition additive to achieve better hole condition.
For good hole condition and reasonable ROP, mud properties must be in range and flow rate must be as maximum acceptable rate especially while drilling with PDC Bit to obtain planned HIS.
Nearby wells drilling issue in this section are listed as below:

1. Frequent Bit and Stabilizer balling in GS formation with 67-86 pcf KCL/Polymer mud due to bad mud properties in well AZNS-045.
2. Drilling fluid contamination with Ca^{++} in Gachsaran formation in well AZNS-045.
3. String stuck in GS formation in well AZNS-045 while back reaming due to bad hole condition with 84 pcf KCL/Polymer mud.
4. Observed back flow after cementing 13-3/8" CSG in well AZNS-045, fix problem after several step apply pressure in well and hold.
5. Frequent tight hole and reaming due to low Mw with respect to hole condition in base of Aghajari formation in well AZNS-017.
6. Bit and Stabilizer balling due to low Mw and improper Mud properties in well AZNS-017. Mud must be inhibited to drill clay formation.
7. Washing down 13-3/8" Casing and set casing off bottom (5.5m) due to bad hole condition in well AZNS-017.
8. Bill balling in well AZNS-127 in base of GS formation.
9. Frequent tight hole and string washout in GS formation in AZN-010 while drilling with 74-90 pcf Mw.

Drilling Challenges and Lesson Learned in 12-1/4" Hole Section

For typical Fahliyan vertical well in area II, this hole section drilling starts from Asmari formation and will be finish to top of Fahliyan formation and pass through Sarvak, Kazhdumi and Gadvan reservoirs which contain H2S and Co2.

The 9-5/8" Production Casing will be set in top of Fahliyan to isolate high pressure Fahliyan formation from other formation.

The key point while drilling this hole section is to adjust proper Mw with respect to hole condition in order to prevent hole problems such as tight hole, wellbore stability problem, string stuck and induced losses.

Also, inhibited drilling fluid system must be used to drill this hole section due to presence of different kind of formation, the normal inhibition additives such as PHPA, KCL, Glycol must be used in system with regard to formation being drilled and mud properties must be controlled as per program.

Consider H2S and Co2 effect on mud while drilling through SV, KZ, GD formation and try to keep PH in range.

As per NBW study, this section is challenging hole section of typical Fahliyan vertical well profile and tight hole, wellbore stability problems, mud contamination, string washout and string stuck have been recorded frequently in this hole section.

The best practice to drill safely is to start drilling with salt saturated polymer mud and change add Kcl into mud before top of Pabdeh and control Mw as low as possible while drilling AS/Pabdeh formations and continue drilling while increase Mw in 1 pcf increment with CaCo3 if observed any abnormal sign while drilling/tripping such as Hi-Trq, abnormal drag, increase in calcium ion value, etc. Consider losses while drilling through SV formation.

Furthermore, while drilling through Sarvak and Kazhdumi formations, mud inhibition must be in range (Glycol/Kcl). In case of any losses in AS, PD and SV formations tried to cure same with proper LCM while keep Mw as low as possible.

Also, due to drilling through different H2S bearing formation in this section, consider H2S safety requirement.

Nearby wells drilling issue in this section are listed as below:

1. Frequent Tight hole in GU formation (Lower) in well AZNS-045 due to low Mw and improper mud inhibition.
2. Drilling through AS/PD/GU with 2-6 BPH losses with 75-84 pcf Mw in well AZNS-045. Drilling to section TD with 0-2 BPH losses with 84-87.5 pcf Mw.

3. Differential String stuck in KZ formation in well AZNS-045, free same with HCL Acid, cut Mw from 87.5 pcf to 81 pcf.
4. Remain cement in casing sue to increase in pressure while displacing cement after release top plug and no more pumping. Perform Top job in 9-5/8 annulus and D/Out cement inside 9-5/8" CSG.
5. Tight hole while trip in GS formation in well AZNS-017 which cured by gradually increase Mw.
6. Bit and Stabilizer balled-up in well AZNS-017 due to improper mud properties and low Mw.
7. Wireline Log tools held up in GS formation due to low Mw and bad mud properties in well AZNS-017.
8. Cement losses while cementing casing with 112 pcf Lead Sw (Mw 100 pcf) in GS formation well AZNS-017.
9. Observed 2-12 BPH losses with 75-77 pcf WBM in Sarvak formation in well AZNS-017 (8-1/2" Hole, KZ-H well).
10. Observed seepage losses 2 BPH and several string stuck while directional drilling in Kazhdumi formation with 78 pcf OBM in well AZNS-017 (8-1/2" Hole, KZ-H well).
11. Observed noticeable cement losses while cementing 7" Liner in AZNS-017. Observed no cement on 7" L.Lap in well AZNS-017 and took injectivity. Cure Lap leakage by remedial cement job with RTTS (8-1/2" Hole, KZ-H well).
12. Low ROP in Cherty Zone (PD) from 1920-1989m in AZNS-127.
13. Observed 3-5 BPH losses with 80-81 pcf Mw in 12-1/4" Hole section which increased to 12 BPH in KZ and 15 BPH in GD formation in well AZNS-127, cure same by decrease flow rate and spot LCM pill.
14. Several washout and string twist off while drilling as performance drilling with 80-81 pcf Mw in Well AZNS-127.
15. Noticeable cement losses while cementing 9-5/8" CSG with 88 pcf Lead CMT Sw in well AZNS-127.
16. Observed 40-50 BPH losses while drilling with 80 pcf Mw in top of SV in well AZN-010.

Drilling Challenges and Lesson Learned in 8-1/2" Hole Section

For typical Fahliyan vertical well in zone-II, this hole section is the last section of drilling sequences and drill through Fahliyan formation, which is high pressure and high temperature and contains Co2 and H2S.

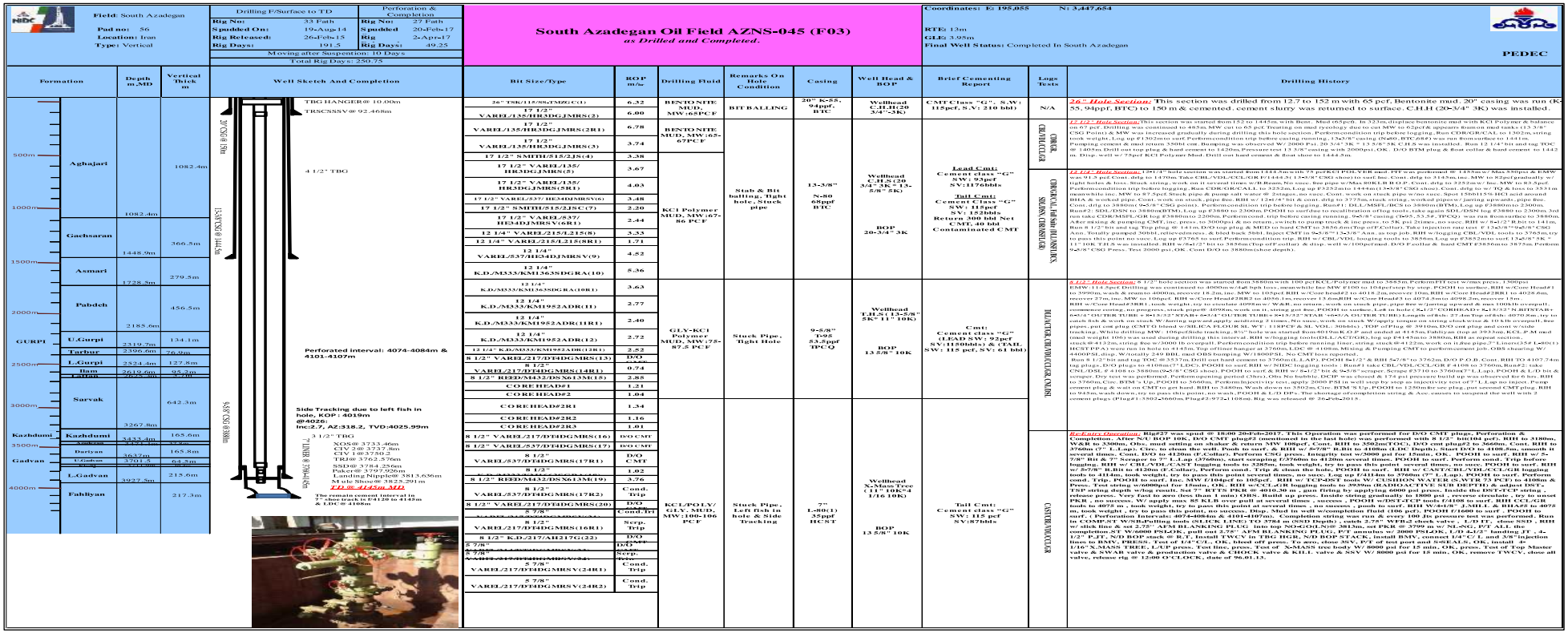
The main issue while drilling this hole section is string stuck and tight hole so best drilling practice must be considered while drilling through Fahliyan and Mud properties must be in proper range. For this section, the KCL/Polymer/Limestone mud is good choice to drill safely and efficiently.

Consider H2S safety requirements while drilling/completion in Fahliyan formation.

Nearby wells drilling issues in this section are listed as below:

1. Drilling with 2-6 BPH seepage losses in FH FM., with 100-106 pcf Mw in well AZNS-045.
2. Several string stuck in FH Fm., in well AZNS-045, with 105-106 pcf MW, free same with Overpull and Pipe-Lax.
3. String stuck while coring in FH Fm., in well AZNS-045 with 106 pcf Mw, Left Fish in hole, set CMT plug and side track well.
4. Observed seepage losses 1-3 BPH with 100-105 pcf Mw in well AZNS-127 & with 100-110 pcf Mw in well AZN-010 while drilling through FH.

Offset Well AZNS-045 (F03) Well Montage


South Azadegan Oil Field AZNS-045 (F03)
as Drilled and Completed.
Field: South Azadegan
Pad no: 56
Location: Iran
Type: Vertical
Drilling F/Surface to TD
Rig No: 33 Fath
Spudded On: 19-Aug-14
Rig Released: 26-Feb-15
Rig Days: 191.5
Perforation & Completion
Rig No: 27 Fath
Spudded 20-Feb-17
Rig 2-Apr-17
Rig Days 49.25
Moving after Suspention: 10 Days
Total Rig Days: 250.75
Coordinates: E: 195,055 N: 3,447,654
RTE: 13m
GLE: 3.95m
Final Well Status: Completed In South Azadegan
PEDEC
Formation
Depth m,MD
Vertical Thickness
Well Sketch And Completion
Bit Size/Type
ROP m/hr
Drilling Fluid
Remarks On Hole Condition
Casing
Well Head & BOP
Brief Cementing Report
Logs Tests
Drilling History
Aghajari
Gachsaran
Asmari
Pabdeh
U.Gurpi
Tarbur
L.Gurpi
Ilam
Sarvak
Kazhdumi
Durjan
L.Gadvan
Fahliyan
GURPI
Gadvan
1082.4m
1448.9m
1728.8m
2185.6m
2319.7m
2396.6m
2524.4m
2614.6m
3267.8m
3433.4m
3637m
3701.5
3927.5m
366.5m
279.5m
456.5m
134.1m
76.9m
127.8m
95.2m
642.3m
165.6m
165.8m
64.5m
218.6m
217.3m
TBG HANGER@ 10.00m
TRSCSSSV@ 92.468m
4 1/2" TBG
Perforated interval: 4074-4084m & 4101-4107m
Side Tracking due to left fish in hole, KOP: 4019m @4026: Inc:2.7; AZ:318.2, TVD:4025.99m
3 1/2" TBG
XOS@ 3733.46m
CIV 2@ 3737.8m
CIV 1@3750.2
TRV@ 3762.576m
SSD@ 3784.256m
Packer@ 3797.926m
Landing Nipple@ 3813.636m
Mule Shoe@ 3825.201m
TD @ 4145m MD
The remain cement interval in 7" shoe track is f/4126 to 4148m & f.D/C @ 4106m
20" K-55, 94ppf, BTC @ 50m
13 3/8" K-55, 68ppf, BTC @ 1445m
9 5/8" T-95 53.5ppf TPCQ @ 3844.1m
7" L-80(1) 35ppf HCST @ 4144.9m
26" TSK/118/SS-EMGC(1) 6.32 BENTONITE MUD, MW:65PCF BIT BALLING 20" K-55, 94ppf, BTC Wellhead C.H.H(20-3/4"-3K) CMT Class "G", S.Wt: 115pcf, S.V.: 210 bbl) N/A 26" Hole Section: This section was drilled from 12.7 to 152 m with 65 pcf, Bentonite mud. 20" casing was run (K-55, 94ppf, BTC) to 150 m & cemented. cement slurry was returned to surface. C.H.H (20-3/4" 3K) was installed.
17 1/2" VAREL/135/HR3DGJMRS(2) 6.00
17 1/2" VAREL/135/HR3DGJMRS(2R1) 6.78
17 1/2" VAREL/135/HR3DGJMRS(3) 3.74
17 1/2" SMITH/S15/2JS(4) 3.38
17 1/2" VAREL/135/HR3DGJMRS(5) 3.67
17 1/2" VAREL/135/HR3DGJMRS(5R1) 4.03
17 1/2" VAREL/837/HE34DJMRSV(6) 3.48
17 1/2" SMITH/S15/2JSC(7) 2.20
17 1/2" VAREL/837/HE34DJMRSV(6R1) 2.44
12 1/4" VAREL/215/L215(8) 3.33
12 1/4" VAREL/215/L215(8R1) 1.71
12 1/4" VAREL/537/HE34DJMRSV(9) 4.52
12 1/4" KD./M333/KM1363SDGRA(10) 5.36
12 1/4" KD./M333/KM1363SDGRA(10R1) 3.63
12 1/4" KD./M333/KM1982ADR(11) 2.77
12 1/4" KD./M333/KM1982ADR(11R1) 2.40
12 1/4" KD./M333/KM1952ADR(12) 2.72
12 1/4" KD./M333/KM1952ADR(12R1) 2.52
8 1/2" VAREL/217/DT6DGMRS(13) D/O CMT
8 1/2" VAREL/217/DT6DGMRS(13R1) 0.74
8 1/2" REED/M432/DSX613M(15) 2.88
CORE HEAD#1 1.21
CORE HEAD#2 1.04
CORE HEAD#2R1 1.34
CORE HEAD#2R2 1.16
CORE HEAD#2R3 1.01
8 1/2" VAREL/217/DT6DGMRS(16) D/O CMT
8 1/2" VAREL/837/DT6DGMRS(17) D/O CMT
8 1/2" VAREL/837/DT6DGMRS(17R1) D/O CMT
8 1/2" 1.02
8 1/2" REED/M432/DSX613M(19) 3.76
8 1/2" VAREL/537/DT6DGMRS(17R2) Cond. Trip
8 1/2" VAREL/217/DT6DGMRS(20) D/O
8 5/8" Cond.trip
8 1/2" VAREL/217/DT6DGMRS(16R1) Scrp. Trip
8 1/2" K.D./217/AH217G(22) D/O
5 7/8" D/O+Scrp.
5 7/8" Scrp.
5 7/8" Cond. Trip
VAREL/217/DT6DGMRSV(24R1)
5 7/8" Cond. Trip
VAREL/217/DT6DGMRSV(24R2)
KCl Polymer MUD, MW:67-86 PCF
KCL/POLY/GLY-MUD, MW:1100-106 PCF
Wellhead C.H.H(20 3/4"-3K" 13-5/8" 5K)
BOP 20-3/4" 3K
Lead Cmt: Cement class "G" SW: 93pcf SV:1176bbls
Tail Cmt: Cement Class "G" SW: 115pcf SV: 152bbls Return 300 bbl Net CMT: 40 bbl Contaminated CMT
17 1/2" Hole Section: 13-1/4" hole section was started from 1141.9m with 73 pcf KCl POLYER mud. PIT was performed @ 1153m w/ Max.330psi & EMW was 15.3 ppg. Drilling to 1445m. Take CBL/VDL/CCL/GR(1444.2)-13-3/8" CSG shoe) to surf Inc: Cont. drlg to 1445m, MW: to 82pcf gradually w/ tight holes & loss. Stuck string, work on it several times w/R.Roam, No succ. free pipe w/Max 80KLB R.O.P. Cont. drlg to 3232m w/ Inc. MW to 83.3pcf. Perform condition trip before logging. Run CDR/OBC/CALL to 3232m.Log up 03232m to 1444m(13-3/8" CSG shoe). Cont. drlg to w/ 82 & loss to 3511m meanwhile inc. MW to 87.5pcf. Stuck pipe & pump sat water @ 2stages, no succ. Cont. work on stuck pipe w/no succ. Spot 156bl15% HCl acid around BHA & worked pipe. Cont. work on stuck pipe free. RIH w/ 12-1/4" bit & cont. drlg to 3775m, stuck string, worked pipe/w/ jarring upwards. pipe free. Cont. drlg to 3880m(9-5/8" CSG point). Perform condition trip before logging. Run#1: DLL/MSFL/BCS to 3880m(BTM).Log up @3880m to 2300m. Run#2: SDL/DSN to 3880m(BTM).Log up @3880 to 2300m. POOH to surf/due to recalibration of log tools, take again SDL/DSN log @3880 to 2300m. 3rd run take CDR/MSFL/GR log @3880m to 2200m. Perform cond. trip before casing running. 9-5/8" casing (O-95, 53.5#, TPCQ) was run 80m surface to 3880m, Rig up & pumping CMT. Inc. pump rate to 8bbl/min & no returns, switch to pump rate 6 bbl/min & no returns, RIH w/ 9-5/8" Stinger to 3844.1

3.1.1.1 Offset Well AZNS-045 (F03) (26" Hole Section Summary)

This section was drilled from 13 to 152m with 65pcf, Bentonite mud. 20" casing was run to 150m & cemented.

26" HOLE SUMMARY REPORT

General Data

FIELD	South Azadegan	WELL NO.	045	LOCATION	3447608	N	
PAD NO	56				195087	E	
RIG NO	33	HOLE DEPTH	152 m MD	MUD WT. Range	65	pcf	
HOLE SIZE	26"		152 m TVD	MUD TYPE	BENTONITE		
FORMATION	AGHAJARI						

Mud Weight Change

MW	Depth (m)	Reason of MW change
		MW wasn't changed.

Drilling Problems

Problem	Depth (m)	Corrective Action
Bit balling	152	POOH, clean the bit & Stabilizer.

Cementing

	VOL (bbl)	WT.(pcf)	CMT (SXS)	Additive
Spacer	70	62.4		No Add.
Lead CMT	210	115	806 SXS "G"-DELIJAN	CACL2 3630 LBS, CD-1B 179 LBS & Antifoam 17 Gal

COMMENTS :R/U CMT LINES TEST TO 3000 PSI/ 10 MIN, OK. HOLD PJSM WITH RELEVANT CREW & PERSONNEL.PUMP 70 BBL FRESH WATERAS SPACER, MIX AND PUMP 806 SXS CMT "G" (S.V= 210 BBL, SW= 115 PCF) AS PER PROGRAM, GET 15 BBL RETURN CMT,DISPLACE W/ 6.5 BBL FRESH WATER(2 BBL SHORT DISPLACE), CHECK BACK FLOW,OK.

Wellhead & Casing

Equipment	Property
26" Casing	K-55, 94 PPF, 150m (Surface to 150m)
C.H.H.	20-3/4" 3K (MSP)
BOP STACK	NO NEED BOP

COMMENTS:

Bit Record

Bit Run.	From (m)	To (m)	Type/IADC/Nozz.	ROP (MPH)	Bit Run.	From(m)	To(m)	Type/IADC/Nozz	ROP (MPH)
1	13	152	115,3*18	6.3	3				
2					4				

COMMENTS:

LOGGING

Logging wasn't performed.

BHA

26" R.Bit*0.56m+ 7 5/8" Bit sub * 0.89m +2 jts 9 9/16" DC *19.46m+ 26" S.Stabliser * 2.38m +4 jts 9 9/16" DC *38.44m.

3.1.1.2 Offset Well AZNS-045 (F03) (17-1/2” Hole Section Summary)

This section was started from 152 to 323m with Bentonite Mud (67 pcf). KCL/Polymer was used from 323 to section T.D. (1445m). While reaming the interval from 350m to 1200m and increasing mud density to 86pcf in several stages to obviated tight hole @ 485m. continue drilling @ 903m stop drilling due to 150PSI press. drop. Continue drilling to 1286m mean while treating mud rheology several times, due to tight hole hard wash & ream f/1280m to 1286m, stuck @ 1284m due to hydraulic up, string free by pumping, POOH to surface, string stuck at several spots, work on them and string got free. 13 3/8" CSG was run and cemented.

General Data

FIELD	South Azadegan	**WELL NO.**	AZNS 045		**LOCATION**	3447608	N
PAD NO.	56					195087	E
RIG NO.	33	**HOLE DEPTH**	1445	m MD	**MUD WT. Range**	65-86	PCF
HOLE SIZE	17-1/2"		1445	m TVD	**MUD TYPE**	BENTONITE/KCL-POLYMER	
FORMATION	AGHAJARI-GACHSARAN						

Mud Weight Change

MW (PCF) From	MW (PCF) To	Depth(m)	Reason of MW change
65	67	323	As per Program
67	67	394	As per Program
62	70	485	Treating on mud ryeology due to cut mud weight to 62 pcf and appear foam on mud tanks
71	74	1004	As per Program
74	78	1173	Tight Hole, increase C++
78	79	1272	Increase C++
79	84	1286	Tight Hole, increase C++
84	86	1445	As per Program

Drilling Problems

Problem	Depth (m)	Corrective Action
Stabilizer & Bit Balling	778	Stabilizer 50 % & Bit 100 % balled up, obs wash out on nozzle seat and lost one 18/32" (Stop drlg due to 400 psi press drop)
Stabilizer & Bit Balling	903	Stabilizer & Bit 100 % balled up (Stop drlg due to 150 psi press drop)
Low ROP	1001.7	Treating mud rheology (Foam on return mud)
Tight Hole	1173	Wash and Ream
Stuck Pipe	1284	Hard wash and ream, pumping after 30 min Ca++:2000 ppm, due to hydraulic up
Stuck pipe	958,881	while POOH, low back ream and pumping
Stabilizer & Bit Balling	1286	Stabilizer & Bit 100 % balled up

COMMENTS: The main problems in this hole were stuck pipe & bit balling.

Cementing

	VOL (bbl)	WT.(PCF)	CMT (SXS)	ADDITIVE
Spacer				80 BBL SALT SATURATED WATER
Lead CMT	1176	93	2537 SXS, CMT "G" DELIJAN	O-BENTONITE 7566 LBS, O-R5 392 LBS & ANTIFOAM 55 GAL
Tail CMT	152	115	583 SXS, CMT "G" DELIJAN	O-R5 76 LBS, ANTIFOAM 14 GAL & CFR 3 190 LBS

COMMENTS: R/U CMT LINE AND TEST TO 3000 PSI, OK.PUMP 80 BBLSALT WATER AS SPACER , MIX AND PUMP 2537 SXS CMT " G "DELEJAN (LEAD SL.VOL = 1176 BBL, SL.WT= 93 PUMP 583 SXS CMT "G" DELEJAN (TAIL SL.VOL= 152 BBL, SL.WT 115 PCF) DROP TOP CMT PLUG, (WHILE AFTER PUMPING 300 BBL CMT STROKE COUNTER OF PUMP TRUCK FAILED CONT. CEMENTING BASE ON CMT SXS),DISPLACE WITH 701 BBL + 25 % OF SHOE TRACK (3 BBL) MUD 86 PCF BASE ON MUD ENG. LEVEL, WITHOUT OBS. BUMPING CHECK BACK FLOW ,OK. RETURN 40 BBL CONTAMINATED AND 300 BBL CMT.AFTER 15 MIN OBS. BACK FLOW, PUMP 10 BBL WATER, CLOSE THE WELL AND HOLD 200 PSI TILL M.N.WAIT ON CEMENT MEANWHILE HOLD 200 PSI.PRESS DECREASE TO 50 PSI,OPEN THE WELL OBS. 9 BPH GAIN (73 PCF FLUID), PUMP 10 BBL WATER, CLOSE THE WELL AND HOLD 200 PSI, PRESS DECREASE TO 50 PSI, OPEN THE WELL OBS. 8.5 BPH GAIN (73 PCF FLUID) @ R.T.CLOSE THE WELL AND CASING , RECORD PRESSURE(WELL:250 PSI , CASING:650 PSI), RELEASE PRESS.(CASING: STATIC , WELL: 8 BPH GAIN (66 PCF FLUID). CLOSE THE WELL HOLD 250 PSI PRESS. WHILE PERFORM INJECTIVITY TEST IN 2 TIMES (PRSS.:400 PSI , 0.5 BPM INJECTION RATE) (PRESS.:500 PSI , 1 BPM INJECTION RATE). WAIT ON REC. DRILLING SUPERVISOR WHILE R/U CMT LINE & TEST W/3000 PSI FOR 15 MIN ,OK, MIX AND PUMP 395 SXS CMT " G "DELEJAN (SL.VOL = 100 BBL, SL.WT= 115 PCF) (MAX RATE:1 BPM , MAX PRESS.:500 PSI) WAIT ON REC. DRILLING SUPERVISOR WHILE PUMP 10 BBL WATER, CLOSE THE WELL AND HOLD 200 PSI, CLOSE THE WELL HOLD 350 PSI PRESS.PRESS DECREASE TO 0 PSI,WAIT ON REC. DRILLING SUPERVISOR WHILE AFTER 17 HRS, OPEN THE WELL, CHECK BACK FLOW, OK.

Wellhead & Casing

Equipment	Property
13.375" Casing	N-80, 68 PPF, BTC & Length:1444.5m
C.H.S	20-3/4"3K* 13-5/8"5K C.H.S(MSP)
BOP Stack	20 3/4" BOP Stack 3K (Hydrill, Blind Rams, Pipe Rams)

COMMENTS:

Bit Record

Bit No.	From (m)	To (m)	Manufacture/IADC/Nozz	ROP (MPH)	Bit No.	From(m)	To(m)	Manufacture/IADC/Nozz	ROP (MPH)
2	150	179	VAREL/135/3*18		6	1143	1286	VAREL/537/3*18	3.48
2R1	179	735	VAREL/135/3*18	6.80	7	1286	1374	SMITH/515/3*18	2.2
3	735	778	VAREL/135/3*18	3.70	6R1	1374	1445	VAREL/537/3*18	2.45
4	778	903	SMITH/515/3*18	3.40					
5	903	1004	VAREL/135/3*18	3.70					
5R1	1004	1143	VAREL/135/3*18	4.02					

COMMENTS: **x*Ry** that means the bit No. x was run for "y'th time.

LOGGING

COMMENTS: RIH W/ CDR/GR TOOLS TO 1445M (BTM), TAKE LOG UP F/1445M TO 150M.TAKE CBL/VDL/CCL/GR LOG F/1444.5(13 3/8" SHOE) TO SURFACE (POUR QUALITY OF CCL LOG).

BHA

17 1/2" R.Bit*0.47m+ 9 1/2" Bit sub * 0.89m +6 jts 9 9/16" DC *57.90m+9 1/8"-8 3/8" XOS*0.74+6 jts 8 1/2" DC *58.11m+9 9/16" XOS*0.88

17 1/2" R.Bit*0.47m+ 9 1/2" Bit sub * 0.89m +2 jts 9 9/16" DC *19.46m+ 17 1/2" S.Stabliser * 1.54m+4 jts 9 9/16" DC *38.44m+7 5/8"-6 5/8" XOS*0.74+6 jts 8 1/2" DC *58.11m+4 1/2"-6 5/8" XOS*0.88+6 jts 5" DPHW*56.07m.

17 1/2" R.Bit*0.46m+ 9 1/2" Bit sub * 0.89m +2 jts 9 9/16" DC *19.46m+ 17 1/2" S.Stabliser * 1.54m+7 jts 9 9/16" DC *67.44m+9 1/2" jar*9.55m+1 jts 9 9/16" DC *9.71m+9 1/8"-8 3/8 XOS*0.74+2 jts 8 1/2" DC *19.42m+7 13/16-6 9/16" XOS*0.88+9 jts 5" DPHW*84.22m.

17 1/2" R.Bit*0.42m+ 9 1/2" Bit sub * 0.89m +2 jts 9 9/16" DC *19.46m+ 17 1/2" S.Stabliser * 1.54m+7 jts 9 9/16" DC *67.44m+9 1/2" jar*9.55m+1 jts 9 9/16" DC *9.71m+9 1/8"-8 3/8 XOS*0.74+2 jts 8 1/2" DC *19.42m+7 13/16-6 9/16" XOS*0.88+9 jts 5" DPHW*84.22m.

17 1/2" R.Bit*0.47m+ 9 1/2" Bit sub * 0.89m +2 jts 9 9/16" DC *19.46m+ 17 1/2" S.Stabliser * 1.54m+7 jts 9 9/16" DC *67.44m+9 1/2" jar*9.55m+1 jts 9 9/16" DC *9.71m+9 1/8"-8 3/8 XOS*0.74+2 jts 8 1/2" DC *19.42m+7 13/16-6 9/16" XOS*0.88+9 jts 5" DPHW*84.22m..

17 1/2" R.Bit*0.48m+ 9 1/2" Bit sub * 0.89m +2 jts 9 9/16" DC *19.46m+ 17 1/2" S.Stabliser * 1.54m+7 jts 9 9/16" DC *67.44m+9 1/2" jar*9.55m+1 jts 9 9/16" DC *9.71m+9 1/8"-8 3/8 XOS*0.74+2 jts 8 1/2" DC *19.42m+7 13/16-6 9/16" XOS*0.88+9 jts 5" DPHW*84.22m..

17 1/2"B. Bit*0.42m+ 9 9/16" Bit sub * 0.89m +2 join 9-3/4" DC *19.46m+9 3/4" S.Stabliser *1.54m +7 join 9-3/4" DC *64.44m+9 1/8" D.JAR *9.55+ 1joint 9-3/4" DC *9.71+ 9 1/8"-8 3/8" X.O.S *0.74m.+2 join 8 1/16" DC *19.42m+8 1/2" +7 13/16" X.O.S *0.88m+9 join 5"H.W *84.22m.

17 1/2"B. Bit*0.48m+ 9 9/16" Bit sub * 0.89m +2 join 9-3/4" DC *19.46m+9 3/4" S.Stabliser *1.54m +7 join 9-3/4" DC *64.44m+9 1/8" D.JAR *9.55+ 1joint 9-3/4" DC *9.71+ 9 1/8"-8 3/8" X.O.S *0.74m.+2 join 8 1/16" DC *19.42m+8 1/2" +7 13/16" X.O.S *0.88m+9 join 5"H.W *84.22m.

3.1.1.3 Offset Well AZNS-045 (F03) (12-1/4” Hole Section Summary)

Drilling of hole section was started from 1444.5m and continued to 1702m with KCl/Polymer type mud (density 75pcf) with2-12bph loss in formation. Then drilling was continued from 1702 to 2170mt with 1-4bbl/hr loss in formation by increased mud Weight from 75 to 77pcf considering to hole conditions to obviate tight holes is several spots. Continue drilling to 2519m w/2-4 bph loss. Continue drilling to 3002m meanwhile washed & ream several times to obviate tight holes, Cont’ drilling to 3145m, stuck pipes, free pipe. Continue drilling to 3252m w/2-4bph loss while increase MW due to tight hole. For prevented loss in formation while drilling used high viscosity pill mixed by mica (LCM). RIH and Continue drilling to 3365m w/2-4bhp loss & Hi TQ=3.5-5K to 3531m, Obs tight hole, stuck pipe, after stuck pipe at 3531m pumped KCl brine and pipe lax in several steps. Continue drilling to 3600m, hard back ream, Continue drilling w/2-4bph loss to 3728m. drilling to 3788 w/4-6bph loss, obs 100psi drop in pressure, wash out on 12jts, RIH, Continue drilling to 3880m (CSG Point) For this type fluid when increased total hardness frequently used starch to control fluid loss. Finally, 9-5/8" CSG was run from surface to 3880m and cemented.

General Data

FIELD	South Azadegan	WELL NO.	AZNS 045	LOCATION	3447608	N	
PAD NO.	56				195087	E	
RIG NO.	33	HOLE DEPTH	3880	m MD	MUD WT. Range	75-87.5	PCF
HOLE SIZE	12-1/4"		3880	m TVD	MUD TYPE	KCl-Polymer	
FORMATION	ASMARI+PABDEH+GURPI+ILAM+LAFAN+SARVAK +KAZHDUMI+DARIYAN+GADVAN						

Mud Weight Change

MW (PCF) From	MW (PCF) To	Depth(m)	Reason of MW change
75	77	1702	as per program
77	78	2170	as per program
78	79	2386	as per program
79	81	2992	as per program
81	82	3135	Due to tight hole on conn.
82	83.5	3191	Due to tight hole on conn.
83.5	84.5	3252	as per program
84.5	86.5	3292	as per program
86.5	87.5	3400	as per program
87.5	81	3531	Work on stuck pipe
81	83	3600	as per program
83	84	3880	as per program

COMMENTS:

Drilling Problems

Problem	Depth (m)	Corrective Action
Tight Hole	2976-2992	Washed and reamed several times and increased MW to 81 pcf
Tight Hole	2992-3002	Washed and reamed several times until clear
Tight Hole	3135	Work on it w/B.Ream several times & pump 50 bbl low vis pill + 50 bbl Hi vis pill
Stuck pipe	3145	Work on it w/B.Ream several times, pump 25bbl low vis pill + 50 bbl Hi vis pill
Tight Hole	3191	Work on it w/B.Ream several times while pump Hi vis pill periodic
Stuck Pipe	3531	Work on stuck pipe downwards several times,spot pipe lax pill and saltwater, cut mw to 81 pcf and contiune work on stuck pipe, spot HCl acid around BHA, pipe free
Stuck Pipe	3775	Worked pipe w/jarring upward, pipe free

COMMENTS Obs several tight spots and stuck pipe during drilling 12 1/4" hole section

Cementing

	VOL (bbl)	WT.(PCF)	CMT (SXS)	ADDITIVE
Pre Flush	70	75		
Spacer				SALT SATURATED WATER (70 bbl)
Lead CMT	1150	92	2240 SXS CMT "G" DELIJAN	O-Micro Silca 30518 LBS, O-bentonite 3370 LBS, O-R5 1260 LBS, O-Micro Block 1020 gal & Antifoam 39 gal
Tail CMT	61	115	181 SXS CMT "G" DELIJAN	Antifoam 7 gal, Silica Flour 6969 LBS, O-Uni FLC5 338 LBS, O-R12 46 LBS, CD-1B 91 LBS

COMMENTS : R/UP CMT LINE& TEST TO 3000PSI, OK, PUMP 70BBL SATURATED SALT WTR AS SPASER, MIX&PUMP 2240SXS CMT "G" DELIJAN (BLEND W/ O-MICRO SILICA) SL.WT=92PCF& SL.VOL=1150BBL AS LEAD, THEN MIX& PUMP 181SXS CMT "G" DELIJAN (BLEND W/ SILICA FLOUR) SL.WT=115 PCF& SL.VOL=61BBL AS TAIL WITHOUT RETURN (ESTIMATED TOP OF CMT 2500m) DROP TOP PLUG, DISP CMT W/30BBL 84PCF MUD, INC PRESS TO 3000PSI& NO RETURN, SWITCH TO PUMP TRUCK& INC PRESS TO 5K PSI TWO TIMES NO SUCC, OBS WELL STATIC.

Wellhead & Casing

Equipment	Property
9.625" Casing	T-95, 53.5 PPF, TPCQ & Length:3880m
T.H.S	T.H.S (13-5/8", 5K*11", 10K; MSP DRILEX)
BOP Stack	13-5/8"-10K BOP

COMMENTS:

Bit Record

Bit No.	From (m)	To (m)	Manufacture/IADC/Nozz	ROP (MPH)	Bit No.	From(m)	To(m)	Manufacture/IADC/Nozz	ROP (MPH)
8	1445	1470	VAREL/215/3*16	3.33					
8R1	1470	1702	VAREL/215/3*16	1.71					
9	1702	2170	VAREL/537/3*16	4.52					
10	2170	2519	KINGDREAM/M333/8*12	5.36					
10R1	2519	3252	KINGDREAM/M333/8*12	3.63					
11	3252	3531	KINGDREAM/M333/7*14	2.78					
11R1	3531	3728	KINGDREAM/M333/7*14	2.4					
12	3728	3788	KINGDREAM/M333/8*12	2.72					
12R1	3788	3880	KINGDREAM/M333/8*12	2.5					

COMMENTS: **x*Ry** that means the bit No. x was run for "y 'th time.

LOGGING

RIH W/CDR/GR/CAL TOOLS TO 3252 MT (BTM) & LOG UP F/ 3252 TO 1444 (13 3/8" CSG SHOE).RUN#1:TAKE DLL/MSFL/BCS LOG F/3880M TO 2300M.RUN#2: TAKE SDL, DSN LOG F/3880M (BTM) TO 2300M.RECALIBRATE TOOLS, TAKE AGAIN SDL,DSN LOG F/3880M (BTM) TO 2300M.RUN#3: TAKE CDR,MSFL,GR LOG F/3880M (BTM) TO 2200M.

BHA

12 1/4""R. Bit*0.32m+ J.sub * 1.13m +j.sub 0.87m+12 join DC 8 1/2"*116.19m+8" D.JAR *5.8+ 3 joint 8 1/2" DC *28.94m.

12 1/4""R.BIT#8R*0.32m + B.SUB*1.08m + 4STD 8-1/2 "DC+ST STAB *118.58m + D.JAR *5.8m+ 1STD 8-1/2" DC*28.97m +4.5"XOS *0.88+3STD 5" HWDP*84.22m

12 1/4""R.BIT*0.32m + B.SUB*1.08m + 4STD 8-1/2 "DC+ST STAB *118.58m + D.JAR *5.8m+ 1STD 8-1/2" DC*28.97m +4.5"XOS *0.88+3STD 5" HWDP*84.22m

12 1/4""R.BIT*0.32m + B.SUB*1.08m + 4STD 8-1/2 "DC+ST STAB *120.08m + D.JAR *9.56m+ 1STD 8-1/2" DC*28.97m +4.5"XOS *0.88+3STD 5" HWDP*84.22m

12 1/4""R.BIT*0.32m + B.SUB*1.08m + 4STD 8-1/2 "DC+ST STAB *118.58m + D.JAR *5.7m+ 1STD 8-1/2" DC*28.97m +4.5"XOS *0.88+3STD 5" HWDP*84.22m

12 1/4""R.BIT*0.32m + B.SUB*1.08m + 4STD 8-1/2 "DC+ST STAB *120.08m + D.JAR *9.56m+ 1STD 8-1/2" DC*28.97m +4.5"XOS *0.88+3STD 5" HWDP*84.22m

12 1/4""R.BIT*0.32m + B.SUB*1.08m + 4STD 8-1/2 "DC+ST STAB *120.08m + D.JAR *9.56m+ 1STD 8-1/2" DC*28.97m +4.5"XOS *0.99+3STD 5" HWDP*84.22m

12 1/4""R.BIT*0.40m + B.SUB*1.08m + 4STD 8-1/2 "DC+ST STAB *119.64m + D.JAR *9.56m+ 1STD 8-1/2" DC*28.97m +4.5"XOS *0.99+3STD 5" HWDP*84.22m

3.1.1.4 Offset Well AZNS-045 (F03) (8-1/2” Hole Section Summary)

The 8 1/2" hole was started from 3880m & Continue to 4000m w/4-8 bph loss. Wash & ream drilled interval while increase MW to 104pcf due to tight hole. KCL.P.M mud (mud weight 100 to 106pcf) was used during drilling this phase. In this phase, mud loss and stuck pipe happened, stuck pipe @ 4098m, work on it, string got free, POOH to surface, Left in hole Length of fish: 27.4m Top of fish: 4070.8m, try to catch fish & work on stuck W/Jarring upward, apply acidizing 2 times, No succ., work on stuck W/apply torque on string clockwise & 10 klb over pull, free pipes, put cmt plug, top of plug @ 3910m, D/O cmt plug and cont w/side tracking, Side tracking, 8 1/2" hole was started from 4019m K.O.P. and ended at 4145m. In this phase following formations were drilled, Fahliyan (top at 3933m), KCL.P.M mud (mud weight 106pcf) was used during drilling this phase. 7" Liners (35# L-80 HCST) were run in hole to 4145m. Top of liner hanger at 3760m. In this phase, mud loss happened. While drilling this phase we have 1.5bbl/hr losses. MW: 106pcf.

General Data

FIELD	South Azadegan	WELL NO.	AZNS 045		LOCATION	3447608	N
PAD NO.	56					195087	E
RIG NO.	33	HOLE DEPTH	4145	m MD	MUD WT. Range	100-106	PCF
HOLE SIZE	8-1/2"		4145	m TVD	MUD TYPE	KCl.P.M	
FORMATION	GADVAN+FAHLIYAN						

Mud Weight Change

MW (PCF) From	To	Depth(m)	Reason of MW change
100	103	3905	Due to tight hole on connection
103	104	4000	Due to tight hole on connection
104	105	4010	Due to tight hole on connection
105	106	4056,1	Due to obs tight hole

COMMENTS:

Drilling Problems

Problem	Depth (m)	Corrective Action
STUCK PIPE	4080	PIPE FREE W/JARRING UPWARD & MAX 100KLB O.PULL
STUCK PIPE	4098	PIPE FREE W/JARRING UPWARD & MAX 100KLB O.PULL
LEFT IN HOLE (8-1/2" CORHEAD+ 8-15/32" N.BITSTAB+ 6-3/4" OUTER TUBE + 8-15/32" STAB+ 6-3/4" OUTER TUBE+ 8-15/32" STAB '+6-3/A OUTER TUBE) LENGTH OF FISH: 27.4m TOP OF FISH: 4070.8m	4098	CATCH FISH & WORK ON STUCK W/JARRING UPWARD, WORK ON STUCK W/APPLY TORQUE ON STRING CLOCKWISE & 10 KLB OVERPULL, FREE PIPES, PUT CMT PLUG, TOP OF PLUG @ 3910M, D/O CMT PLUG AND CONT SIDE TRACKING
STUCK PIPE	4122	WORK ON STUCK PIPE UPWARD & DOWNWARD, SPOT PIPE LAX PILL, WORK ON STUCK PIPE UPWARD W/160KLB OVERPULL, PIPE FREE

COMMENTS: *OBS TIGHT HOLE IN SEVERAL INTERVALS.*

Cementing

	VOL (bbl)	WT.(PCF)	CMT (SXS)	ADDITIVE
Pre Flush	15	75		
Spacer				Salt Saturated Water (15 bbl), O-Chemical Wash (188 LBS), O-SURFACTANT (23 GAL), LIME STONE POWDER (570 LBS) & ANTIFOAM (5 GAL)
Lead CMT				
Tail CMT	87	115	136 SXS (1.36 CUFT/SXS) CMT "G" DELIJAN	Silica Flour 9856 LBS, CFL-350 470 LBS, O-R12 72.5 LBS, O-Gas Block 92.6 gal, CFL-387 26.4 gal & Anti Foam 5 gal

COMMENTS: R/UP CMT LINE & TEST TO 3500PSI OK, DROP BALL, SET LNR HGR W/ 2400PSI & BLOW LDC W/ 3400PSI, RELEASE S.TOOLS AS MECHANICAL (F.SHOE@4145m F.COLLAR @4120m LDC@ 4108m P.O.B@3762m L.LAP@3760m) CIRC MUD WELL STATIC,PUMP 15BBL S.WTR w/CH.WASH AS SPACER, MIX & PUMP 256 SXS CMT "G" DELIJAN SL.W=115PCF& SL.V=87BBL) DICPL W/ 209 BBL MUD OBS SHEARING W/ 4400PSI CONT DISPL W/TOTALLY 249 BBL MUD OBS BUMPING W/1800PSI, RELEASE PRESS FLOAT OK, NO CMT LOSS REPORTED, POH 13 STD WET TO 3350m FLUSH DPS WELL STATIC, POH TO 3180m OBS WELL STATIC CLOSE THE WELL APPLY 100 PSI IN WELL & WOC TO GET HARD MONITOR PRESS , DROP PRESS TO ZERO. WAIT ON CMT TO GET HARD AFTER 8HRS SURFACE SAMPLE SOFT,OPEN WELL STATIC.

Wellhead & Casing

Equipment	Property
7" Liner	L-80(1), 35 PPF, HCST PPA & Length: 385m
3-1/2" Tubing	Tubing, 3-1/2"-9.2# CRA, G3-110KSI, V.TOP (91.6m)
4-1/2" Tubing	Tubing, 4-1/2-18.9# TPCQ-EX,L80(1), B X P (3734m)
BOP Stack	13-5/8" 10K

COMMENTS:

Bit Record

Bit No.	From (m)	To (m)	Manufacture/IADC/Nozz	ROP (MPH)	Bit No.	From(m)	To(m)	Manufacture/IADC/Nozz	ROP (MPH)
13	DRILL OUT		VAREL/217/3*16		19	4107.4	4145	REED/M432/5*14+15	3.76
14R1	3880	3888.6	VAREL/217/OPEN	0.74	17R2	4145	4145	VAREL/537/3*16	COND. TRIP
15	3888.6	4000	REED/M432/3*14+3*15	2.85	20	4145	4145	217/3*18	D/O CMT
CORE HEAD#1	4000	4018.2	VAREL	1.21	21	4145	4145	VAREL/217/3*18	COND. TRIP
CORE HEAD#2	4018.2	4028.6	VAREL	1.04	16R1	4145	4145	VAREL/217/3*16	SCRAPING TRIP
CORE HEAD#2R1	4028.6	4056.1	VAREL	1.34	22	4145	4145	KINGDREAM/217/3*18	D/O CMT
CORE HEAD#2R2	4056.1	4070.68	VAREL	1.17	23	4145	4145	VAREL/214/OPEN	D/O CMT
CORE HEAD#2R3	4070.68	4098.2	VAREL	1.01	24	4145	4145	5-7/8, VAREL/214/OPEN	SCRAPING TRIP
16	3912	4010	VAREL/217/3*16	D/O CMT	24R1	4145	4145	5-7/8, VAREL/214/OPEN	COND. TRIP
17	4010	4014.6	VAREL/537/3*16	D/O CMT	24R2	4145	4145	5-7/8, VAREL/214/OPEN	COND. TRIP
17R1	4014.6	4032	VAREL/537/3*16	D/O CMT					
18	4098.2	4107.4	KINGDREAM/M323/6*14	1.02					

COMMENTS: **x*Ry** that means the bit No. x was run for "y"th time.

LOGGING

RUN#1:(DLL/ACT/GR),RIH W/LOGGING TOOLS TO 4145M(BTM).LOG UP F/4145M TO 3880M.RIH AS REPEAT SECTION TO 4145M.LOG UP, LOGGING TOOLS STUCK AT 4132M, STRING FREE W/3000 LB OVERPULL. POOH LOGGING TOOLS TO SURF L/D LOGGING EQUIPMENT. RUN#1:RIH W/CBL-VDL-GR-CCL TOOLS TO 4108MT (LDC) & LOG UP F/3760 MT (7" L.LAP). RUN#2: TAKE CNL/DSL LOG F/4108M (LDC) TO 9 5/8" SHOE (3880m). R/UP LOGGING EQUIPMENT (NIDC), RIH W/ CBL/VDL/CAST WIRELINE LOG TOOLS TO 3285M, TOOK WEIGHT, TRY TO PASS THIS POINT AT SEVERAL TIMES, NO SUCCESS, PULL OUT TO SURFACE, AGAIN RIH W/ CBL/VDL WIRELINE LOG TOOLS TO 3200M, TOOK WEIGHT F/ 3200 TO 3240M, TRY TO PASS AT SEVERAL TIMES, NO SUCCESS, PULL OUT TO SURFACE. R/UP CAST/CBL/VDL/CCL/GR LOG EQUIPMENT (NIDC), RIH W/ CAST/CBL/VDL/CCL/GR LOG TOOLS TO 4114M, TOOK WEIGHT, TRY TO PASS THIS POINT AT SEVERAL TIMES, NO SUCCESS, LOG UP F/ 4114 TO 3760M (7"LINER LAP), CONT. PULL OUT THE LOGGING TOOLS TO SURFACE, L/D IT.

BHA

8 1/2"R.BIT*0.25m + J.SUB*1.2m+B.SUB* 0.98m + 6STD 6-1/2" DC *167.44m + D.JAR *5.41m + 1STD 6-1/2" DC*28.13m + 3STD 5" HWDP*84.22m

8 1/2" R.BIT* 0.32m + B.SUB* 0.98m + 6STD 6-1/2"DC + ST STAB *169.28m + D.JAR *5.41m + 1STD 6-1/2" DC*28.13m + 3STD 5" HWDP*84.22m

8 1/2" CORE HD#2 *0,3m + 8-15/32" N.BITSUB* 0.92m + 8-3/4" OUTER TUBE* 7.92m + 8-15/32" ST STAB* 1.22m + 6-1/2" OUTER TUBE* 7.92m + 8-15/32" ST STAB* 1.22m + 8-3/4" OUTER TUBE* 7.92m + 6-3/4" SLICK* 1.22m + 6-3/4" SAFETY JOINT* 0.77m + 6-3/4" DOUBLE PIN* 0.32m + 6STD 6-1/2" DC*167.44m + D.JAR* 5.17m + 1STD 6-1/2" DC* 28.13m

8-1/2" CORE HD#1 (CK-58) + 8-15/32" N.BIT STAB + 6-3/4" OUTER TUBE + 8-15/32" STAB + 6-3/4" OUTER TUBE + 6-3/4" SLICK + 6-3/4" SAFETY JT + 6-3/4" DBL PIN + 6STD 6-1/2"DC + 6-1/2"DRLG JAR + 1STD 6-1/2"DC +3STD 5"HWDP

8-1/2" CORE HD#2 (CK-58) + 8-15/32" N.BIT STAB + 6-3/4" OUTER TUBE + 8-15/32" STAB + 6-3/4" OUTER TUBE + 6-3/4" SLICK + 6-3/4" SAFETY JT + 6-3/4" DBL PIN + 6STD 6-1/2"DC + 6-1/2"DRLG JAR + 1STD 6-1/2"DC +3STD 5"HWDP

8-1/2" CORE HD#2RR1 (CK-58) + 8-15/32" N.BIT STAB + 6-3/4" OUTER TUBE + 8-15/32" STAB + 6-3/4" OUTER TUBE + 6-3/4" SLICK + 6-3/4" SAFETY JT + 6-3/4" DBL PIN + 6STD 6-1/2"DC + 6-1/2"DRLG JAR + 1STD 6-1/2"DC +3STD 5"HWDP

8-1/2" CORE HD#2RR2 (CK-58) + 8-15/32" N.BIT STAB + 6-3/4" OUTER TUBE + 8-15/32" STAB + 6-3/4" OUTER TUBE + 6-3/4" SLICK + 6-3/4" SAFETY JT + 6-3/4" DBL PIN + 6STD 6-1/2"DC + 6-1/2"DRLG JAR + 1STD 6-1/2"DC +3STD 5"HWDP

8-1/2" CORE HD#3 (CK-58) + 8-15/32" N.BIT STAB + 6-3/4" OUTER TUBE + 8-15/32" STAB + 6-3/4" OUTER TUBE + 6-3/4" SLICK + 6-3/4" SAFETY JT + 6-3/4" DBL PIN + 6STD 6-1/2"DC + 6-1/2"DRLG JAR + 1STD 6-1/2"DC +3STD 5"HWDP

8-1/2" CORE HD#3RR1 (CK-58) + 8-15/32" N.BIT STAB + 6-3/4" OUTER TUBE + 8-15/32" STAB + 6-3/4" OUTER TUBE + 6-3/4" SLICK + 6-3/4" SAFETY JT + 6-3/4" DBL PIN + 6STD 6-1/2"DC + 6-1/2"DRLG JAR + 1STD 6-1/2"DC +3STD 5"HWDP

8-1/2" M.T BIT#16 (USED,217&3*16NZL)+ BIT SUB + 4 STD 6-1/2" DCS + 6-1/2" D.JAR + 1 STD 6-1/2" DCS + 3 STD 5" HWDP

8-1/2" B.BIT#17 (537.3*16NZL) + 6-3/4" LUHAI MOTOR (AKO=2.12) + 6-1/2" FLOAT SUB + 6-1/2' U.B.H.O.SUB + 1 JT 6+1/4" NMDC + 5" SHORT FLEX MONEL + 3 STD 5" HWDP+ 15 JTS 5" HWDP JT BY JT + 6-1/2" D.JAR + 6 JTS 5" HWDP JT BY JT

BIT#17R,(00-NO-A-E-I-NO-DMF) & MOTOR AKO=1.83 + 6-1/2" FLOAT SUB + 6-3/16" NMDC + 6-3/4"T.M.SUB + 5" SHORT FLX MONEL + 5STD 5" HWDP

8-1/2" PDC BIT#18(KING DREAM, S/N:14010029, 5%USED, 1-2-WT-A-X-I-BT-TD, 6*14) SAME MOTOR(AKO=1.15) &SAME BHA

8-1/2" PDC BIT#18 RR1(KING DREAM, S/N:14010029, 5%USED, 1-2-WT-A-X-I-BT-TD, 6*14), MOTOR(AKO=1.15) &SAME BHA

8-1/2" PDC BIT #19(SN:NC208461, TYPE:DSX613M, IADC CODE:M432 REED) W/5*14+1*15 NOZZLES. + B.SUB + 2*DC 6-1/2" + STABILIZER 8-15/32" + 13*D.C 6-1/2" + JAR 6-1/2"+ 3*D.C + HWDP 5"

8 ½" Rock.Bit*0.24cm+6 1/2" bit sub*0.99+12 jnts 6 ½" DC*114+D.JAR 6 ½"*5.18+3 jnts 6 ½"*28.63+9 jnts 5" HWDP*84.35

8 ½" Rock.Bit*0.24cm+6 1/2" bit sub*0.99+12 jnts 6 ½" DC*114+D.JAR 6 ½"*5.18+3 jnts 6 ½"*28.63+9 jnts 5" HWDP*84.35

Offset Well AZNS-017 Well Montage

Well Montage

Field : South Azadegan | Spudded: 18:00 O'Clock od date 87/12/07.
Well No. AZN-017 | Rig Released: 18:00 o'clock date 1388-04-08.
Well Type : Horizontal well | Well Objectives: Produce Oil from Kazhdumi reservoir
Pad #: 50 | Rig : Fath-86
GLE : 7 m | RTE: 15 m | Max Inc: 90.39 Deg
Coordinates: E: 1765998 N: 1019996

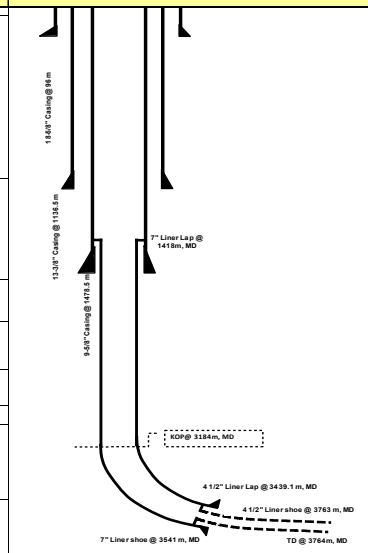
Formation	Formation Tops MD (m)	Well Sketch	Bit Size/Code	Drilling Fluid	Drilling History	Drilling Problems/Remarks
	13		26"/ SDS	Water Bentonite Mud MW=65-66 pcf FV=70-77 PV=30 YP=28 PH=10.5	**26" Hole section:** RIH w/26" bit to 16m, tag formation. Drilled 26" hole section f/16 to 97m with 65-66 pcf Bentonaite mud w/0-1 BPH losses. RIH w/26" Reamer to30m, ream to bottom, POOH. Run 18 5/8" casing (87.5#, J-55, BTS) to 95m. Mix & pump cement "A" (193 bbl, 115 pcf), displace w/80 bbl mud, during cement job 33 bbl mud & 54 bbl cement losses and 20 bbl cement return. Install 21 ¼"-2K C.H.H. N/U 20"-2K Hydrill.	Partial losses Tight hole
Aghajari	1122		17-1/2"/ GTXG1 GTX1	S. Water Mud MW=70-82 pcf FV=43-64 PV=-23 YP=12-16 PH=8.3-8.6	**17 ½" Hole Section:** RIH w/17 ½" bit to 80m, tag cement. Test C.H.H w/400 psi, ok. D/O hard cement & shoe f/80 to 95m, displace mud in hole with 70 pcf salt water. Drilled 17 ½" hole section f/95 to218m with 70 pcf salt water mud. Obs 600 psi dnped & slow pump rate increased, POOH to200m, obs wash out on 5" HWDP, Cont POOH to surface. RIH w/17 ½" bit to 218m. Drilled 17 ½" hole section f/218 to 980m with 70 pcf salt water mud w/0-2 BPH losses. Increase MW to 73 pcf. Drilled 17 ½" hole section f/980 to 1050m with 73-75 pcf salt water mud w/0-2 BPH losses. Due to decrease ROP, POOH w/back ream to 900m, cont POOH pump out to 790m, while increase MW to 77 pcf w/0-2 BPH losses. Cont POOH pump out (with back ream in some interval) to surface. Obs bit & stabilizer fully baled up. RIH w/17 ½" bit to 725m, W&R f/725 to 1050m, while increase MW to 82 pcf. Drilled 17 ½" hole section f/1050 to 1140m (CSG point) with 82 pcf salt water mud w/1-2 BPH losses. Due to Torque & tight in connection increase MW to 83 pcf, CBU. POOH for condition well with tight hole f/1140 to 500m, while increase MW to 85 pcf due to tight hole. RIH to 710m, W&R @ several times (710-730 & 880-900 & 1030-1140) to bottom. POOH to surface. Run 13 3/8" casing ((72#, N-80, BTS (13jts) & 68#, J-55, BTS (83 jts)) to 1134.5m (5.5m off bottom & wash f/116 to 1134.5m and try to more progress, no success). Mix & pump cement "G" (Lead: 863 bbl, 90 pcf Tail: 185 bbl, 115 pcf), displace w/548 bbl mud, obs bumping w/1500 psi, 200 bbl cement return. Instal 120"-2K x 12"-3K on C.H.H. N/U BOP.	Partial losses Tight hole Bit & stabilizer balling
Gachsaran	1491		12-1/4" SV MK566DG1 (PDC)	S. Water Mud MW=85-100pcf FV=43-48 PV=15-24 YP=7-13 PH=8.8-9	**12 ¼" Hole section:** RIH w/12 ¼" bit to 1107m, tag plug, D/O plug, D.F.Collar, hard cement and shoe to 1134.5m. Harded W&R f/1134.5m to bottom. Drilled 12 ¼" hole section f/1140 to 1152m with 85 pcf salt water mud. Work on junk sub several times, POOH. RIH w/12 ¼" PDC bit to 1152m. Drilled 12 ¼" hole section f/1152 to 1225m with 85 pcf salt water mud w/NL-NG. Continue Drilled 12 ¼" hole section f/1225 to 1479m (CSG Point) with 85-88 pcf salt water mud w/2-4 BPH losses. Performed hole condition prior to run casing (POOH w/B.reamf/1479 to 1180m, RIH w/W&R f/1195 to bottom), while increase MW f/88 to 100 pcf at two stage, NL-NG. POOH to surface, obs bit & stabilizer baled up. RIH w/logging tools (CDR-GR) to 1333m, try to pass no success, log up to 930m (angle@1330m 0.5 deg), POOH. Run 9 5/8" casing ((47#, L-80, PJE(137jts) & (53.5#, L-80, Vam(2jts)) to 1478.5m. Mix& pump cement "G" (Lead: 337 bbl, 112 pcf Tail: 62 bbl, 113 pcf), displace w/350 bbl mud (40 bbl loss while displace and return 56 bbl cement). Obs bumping w/1800 psi. Install 13 5/8"-3K x 11"-5K T.H.S. Install BOP.	Tight hole Bit & stabilizer balling
Asmari	1820		8-1/2"/ F2 MF20 MKS66DG1 (PDC) MKS66DG (PDC)	S. Water Mud MW=75-77 pcf FV=32-42 PV=5-8 YP=7-10 PH=8.5-9 OBM MW=77-78 & 76 pcf FV=61-68 PV=26-38 YP=9-19 PH=4.2-4.5	**8 ½" hole section:** RIH w/8 ½" B.Bit to 1456m, D/O plugs, F.Clollar and cement to 1474m, displace well with 75 pcf mud, NL-NG. D/O cement, shoe and formation f/1456 to 1500m with 75 pcf salt water w/NL-NG. work on junk sub several times, POOH. RIH w/8 ½" bit to 1500m, Drilled 8 ½" hole section f/1500 to 1970m with 75 pcf salt water w/2-12 BPH losses. Cont drilled 8 ½" hole section f/1970 to 2010m with 75 pcf salt water w/NL-NG. due to tight hole in connection, increase MW to 77 pcf. Cont drilled 8 ½" hole section f/2010 to 2600m with 77 pcf salt water w/2-4 BPH losses. Displace mud in hole w/77 pcf oil base mud. Cont drilled 8 ½" hole section f/2600 to 2709m with 77 pcf O.B.M w/2-6 BPH losses. Due to tight hole while drilling increase MW to 78 pcf. Cont drilled 8 ½" hole section f/2709 to 3184m (KOP) with 78 pcf O.B.M w/2-6 BPH losses. Cont dir drilled 8 ½" hole section f/3184to 3314m with 78 pcf O.B.M w/2-4BPH losses. Obs 500 psi press droped, POOH to surface, no wash out founded. RIH w/8 ½" bit to 3314m, Cont dir drilled 8 ½" hole section f/3314 to 3534m with 78 pcf O.B.M w/1-2 BPH losses. While drilling string got stuck@ 3512, 3520, 3529, 3533m. work on stuck up & down ward string got free. Decrease MW to 76 pcf, Cont dir drilled 8 ½" hole section f/3314 to 3622m with 78 pcf O.B.M w/0-2 BPH losses. Performed hole condition prior to run log. RIH w/ full suit logging tools (AIT,BHC, TLD, CNL, HNGS) to 3622m, log up f/3622 to 3559m. RIH w/open ended DP to 3610m. mix & pump magneset (50 bbl, 100 pcf), displace w/182 bbl mud as balance plug w/NL-NG, POOH. RIH w/8 ½" bit to 3370m, tag magneset plug, D/O magneset f/3370 to 3541.4m (liner point) with 76 pcf O.B.M w/NL-NG. Performed hole condition prior to run 7" liner. Run 7" liner (29#, C-75,N.Vam) f/1418 to 3541m. mix & pump cement "G" (213 bbl, 115 pcf) displace w/335 bbl mud, didn't obd shearing & bumping, 12 bbl losses while displace cement. RIH w/8 ½" bit to 1418m (L.Lap), CBU, did not obs cement on L.Lap, POOH. RIH w/6 1/8" bit ,took injectivity w/1 BPM rate & max press 1200 psi (inject 17 bbl mud), pump off drop pressure to 1050 psi, open the well return 13 bbl mud, obs well static, POOH. RIH w/9 5/8" RTTS (53.5#) and set same@1280m (138m above L.Lap), took injectivity w/5 BPM rate & max press 1500 psi (inj 40 bbl mud), pump off drop pressure to 1000 psi. Unset packer, POOH and set same@1225m, took injectivity w/5 BPM rate & max press 1500 psi (inj 25 bbl mud), Mix & pump cement "G" (110 bbl 115 pcf), inject w/94 bbl mud w/6-1 BPM rate & max press 1300 psi, WOC to get hard. Unset RTTS, POOH. RIH w/8 ½" bit to 1322m, tag & D/O hard cement f/1322 to 1418m (L.Lap), POOH. RIH w/6 1/8" B.Bit to 1418m, D/O P.Of bushing & hard cement f/1418 to 1450m, cont RIH to 2560m, W&R f/2560 to 2680m, cont RIH to 3337m, D/O plugs , hard cement, shoe & magneset f/3337 to 3570m, obs formation cutting f/3563 to 3570m (100% Fm.) POOH to surface. RIH w/2 7/8" open ended DP to 3570m, mix & pump cement (20 bbl, 120 pcf), as balance plug w/NL-NG. POOH. RIH w/6 1/8" bit to 3394m, tag cement, D/O hard cement f/3394 to 3543m, POOH. RIH w/6 1/8" bit and directional BHA to 3552m (9.59m deeper), try to side track hole f/3552 to 3565.7m w/76 pcf O.B.M w/NL-NG, no success. (3562.5-3565m: 50% Fm, 50% CMT), POOH to surface. RIH w/2 7/8" open ended DP to bottom, mix & pump cement "E" (12 bbl, 120 pcf), displace w/122 bbl mud, POOH.	Losses Tight hole Stuck pipe
Pabdeh	2178					
Gurpi	2532					
Ilam	2622					
Sarvak	3291					
Kazhdumi			6-1/8"/ F2 XR20T STR445	OBM MW=76-73 pcf FV=-54-73 PV=27-35 YP=9-14 PH=4.5	**6 ¾" hole section:** RIH w/6 1/8" bit to 3435m, tag cement, D/O soft cement f/3435 to 3440m, cont D/O hard cement f/3440 to 3542m w/76 pcf O.B.M w/NL-NG. POOH. RIH w/6 1/8" bit and directional BHA to 3542m, side track hole f/3542 to 3560.2m w/76 pcf O.B.M w/NL-NG. RIH w/6 1/8" bit to 3560.2m, dir drilled 6 1/8" hole section f/3560.2 to 3595.5m with 76 pcf O.B.M w/NL-NG, pipe got stuck, work on stuck with over pull, pipe got free. Cont dir drilled 6 1/8" hole section f/3595.5 to 3650m with 76 pcf O.B.M w/NL-NG. cut MW to 73 pcf. Cont dir drilled 6 1/8" hole section f/3650to 3764m (TD) with 73 pcf O.B.M w/NL-NG. Performed hole condition prior to run log. RIH w/ full suit logging tools (TLD/CNL/AIT/BHC/NGS) to 3480m, try to latch DWCH sub several times, no success. POOH to surface. RIH w/6 1/8" bit to TD, performed hole condition prior to run log. R/U coil tubing equipment and test same, ok. Performed logging with CT in two run. First run AIT/BHC/GR to 3764m (TD), log up f/3764 to 3541m (7" Liner shoe), POOH to surface, Second run TLD/CNL/NGS to 3764m (TD), log up f/3764 to 7" Liner shoe@ 3541 (CNL/NGS f/TD to 2500m), POOH to surface. RIH w/6 1/8" bit to 3764m (TD), performed hole condition prior to run 4 ½" liner. Run 4 ½" liner ((13.5#, L-80, HFJ (19 jts slotted liner & 8 jts blank liner)) f/3439.1 to 3763m. RIH w/3 ¾" E.Mill and 7" casing scraper to 3762m, displace mud in hole w/73 pcf salt water, POOH to surface. Run completion string to 1615m. (Packer depth@ 1601.16). N/D BOP, N/U x-mass tree. Complete well head, secure the well. Rig release @18:00 o'clock date 1388-04-08.	Stuck pipe
Total Depth (TD), m	3764 m, MD					

Figure 3-2: AZNS-017 Well Montage

3.1.1.5 Offset Well AZNS-017 (26” Hole Section Summary)

RIH w/ 26” Bit to 16m, tag formation. Drilled 26” hole section f/16 to 97m with 65-66 pcf Bentonite mud w/0-1 BPH losses. RIH w/26” Reamer to 30m, ream to bottom, POOH. Run 18-5/8” casing (87.5#, J-55, BTS) to 95m. Mix & pump cement “A” (193 bbl, 115 pcf), displace w/ 80 bbl mud, during cement job 33 bbl mud & 54 bbl cement losses and 20 bbl cement return. Install 21 ¼”-2K C.H.H. N/U 20”-2K Annular Prev.

3.1.1.6 Offset Well AZNS-017 (17-1/2” Hole Section Summary)

RIH w/17 ½” Bit to 80m, tag cement. Test C.H.H w/400 psi, ok. D/O hard cement & shoe f/80 to 95m, displace mud in hole with 70 pcf salt water. Drilled 17 ½” hole section f/95 to218m with 70 pcf salt water mud. Obs 600 psi Press dropped & slow pump rate increased, POOH to 200m, obs wash out on 5” HWDP, Cont’ POOH to surface.
RIH w/17 ½” Bit to 218m. Drilled 17 ½” hole section f/218 to 980m with 70 pcf salt water mud w/0-2 BPH losses. Increase MW to 73 pcf. Drilled 17 ½” hole section f/980 to 1050m with 73-75 pcf salt water mud w/0-2 BPH losses. Due to decrease ROP, POOH w/back ream to 900m, Cont’ POOH pump out to 790m, while increase MW to 77 pcf w/0-2 BPH losses. Cont’ POOH pump out (with back ream in some interval) to surface. Obs Bit & stabilizer fully baled up. RIH w/17 ½” Bit to 725m, W&R f/725 to 1050m, while increase MW to 82 pcf. Drilled 17 ½” hole section f/1050 to 1140m (CSG point) with 82 pcf salt water mud w/1-2 BPH losses. Due to Torque & tight in connection increase MW to 83 pcf, CBU. POOH for condition well with tight hole f/1140 to 500m, while increase MW to 85 pcf due to tight hole. RIH to 710m, W&R @ several times (710-730 & 880-900 & 1030-1140) to bottom. POOH to surface. Run 13 3/8” casing ((72#, N-80, BTS (13jts) & 68#, J-55, BTS (83 jts)) to 1134.5m (5.5m off bottom & wash f/116 to 1134.5m and try to more progress, no success). Mix & pump cement ”G” (Lead: 863 bbl, 90 pcf Tail: 185 bbl, 115 pcf), displace w/548 bbl mud, obs bumping w/1500 psi, 200 bbl cement return. Instal l20”-2K x 12”-3K on C.H.H. N/U BOP.

3.1.1.7 Offset Well AZNS-017 (12-1/4" Hole Section Summary)

RIH w/12 ¼" Bit to 1107m, tag plug, D/O plug, D.F.Collar, hard cement and shoe to 1134.5m. Herd W&R f/1134.5m to bottom. Drilled 12 ¼" hole section f/1140 to 1152m with 85 pcf salt water mud. Work on junk sub several times, POOH. RIH w/12 ¼" PDC Bit to 1152m. Drilled 12 ¼" hole section f/1152 to 1225m with 85 pcf salt water mud w/ NL-NG. Continue Drilled 12 ¼" hole section f/1225 to 1479m (CSG Point) with 85-88 pcf salt water mud w/2-4 BPH losses. Performed hole condition prior to run casing (POOH w/ B.Ream f/1479 to 1180m, RIH w/W&R f/1195 to bottom), while increase MW f/88 to 100 pcf at two stage, NL-NG. POOH to surface, obs Bit & stabilizer baled up. RIH w/logging tools (CDR-GR) to 1333m, try to pass no success, log up to 930m (angle@1330m 0.5 deg), POOH. Run 9 5/8" casing ((47#, L-80, PJE(137jts) & (53.5#, L-80, Vam(2jts)) to 1478.5m. Mix& pump cement "G" (Lead: 337 bbl, 112 pcf Tail: 62 bbl, 113 pcf), displace w/350 bbl mud (40 bbl loss while displace and return 56 bbl cement). Obs bumping w/1800 psi. Install 13 5/8"-3K x 11"-5K T.H.S. Install BOP.

3.1.1.8 Offset Well AZNS-017 (8-1/2" Hole Section Summary)

RIH w/8 ½" B.Bit to 1456m, D/O plugs, F.Clollar and cement to 1474m, displace well with 75 pcf mud, NL-NG. D/O cement, shoe and formation f/1456 to 1500m with 75 pcf salt water w/NL-NG. work on junk sub several times, POOH. RIH w/8 ½" Bit to 1500m, Drilled 8 ½" hole section f/1500 to 1970m with 75 pcf salt water w/2-12 BPH losses. Cont drilled 8 ½" hole section f/1970 to 2010m with 75 pcf salt water w/NL-NG. due to tight hole in connection, increase MW to 77 pcf. Cont drilled 8 ½" hole section f/2010 to 2600m with 77 pcf salt water w/2-4 BPH losses. Displace mud in hole w/77 pcf oil base mud. Cont drilled 8 ½" hole section f/2600 to 2709m with 77 pcf O.B.M w/2-6 BPH losses. Due to tight hole while drilling increase MW to 78 pcf. Cont' drilled 8 ½" hole section f/2709 to 3184m (KOP) with 78 pcf O.B.M w/2-6 BPH losses. Cont dir drilled 8 ½" hole section f/3184to 3314m with 78 pcf O.B.M w/2-4BPH losses. Obs 500 psi press droped, POOH to surface, no wash out founded. RIH w/8 ½" Bit to 3314m, Cont dir drilled 8 ½" hole section f/3314 to 3534m with 78 pcf O.B.M w/1-2 BPH losses. While drilling string got stuck@ 3512, 3520, 3529, 3533m. work on stuck up & down ward string got free. Decrease MW to 76 pcf, Cont' dir drilled 8 ½" hole section f/3314 to 3622m with 78 pcf O.B.M w/0-2 BPH losses. Performed hole condition prior to run log. RIH w/ full suit logging tools (AIT, BHC, TLD, CNL, HNGS) to 3622m, log up f/3622 to 3559m. RIH w/open ended DP to 3610m. mix & pump magneset (50 bbl, 100 pcf), displace w/182 bbl mud as balance plug w/NL-NG, POOH. RIH w/8 ½" Bit to

3370m, tag magneset plug, D/O magneset f/3370 to 3541.4m (liner point) with 76 pcf O.B.M w/NL-NG. Performed hole condition prior to run 7" liner. Run 7" liner (29#, C-75,N.Vam) f/1418 to 3541m. mix & pump cement "G" (213 bbl, 115 pcf) displace w/335 bbl mud, didn't obs shearing & bumping, 12 bbl losses while displace cement. RIH w/8 ½" Bit to 1418m (L.Lap), CBU, didn't obs cement on L.Lap, POOH. RIH w/6 1/8" Bit , took injectivity w/1 BPM rate & max press 1200 psi (inject 17 bbl mud), pump off drop pressure to 1050 psi, open the well return 13 bbl mud, obs well static, POOH. RIH w/9 5/8" RTTS (53.5#) and set same@1280m (138m above L.Lap), took injectivity w/5 BPM rate & max press 1500 psi (inj 40 bbl mud), pump off drop pressure to 1000 psi. Unset packer, POOH and set same@1225m, took injectivity w/5 BPM rate & max press 1500 psi (inj 25 bbl mud). Mix & pump cement "G" (110 bbl 115 pcf), inject w/94 bbl mud w/6-1 BPM rate & max press 1300 psi, WOC to get hard. Unset RTTS, POOH. RIH w/8 ½" Bit to 1322m, tag & D/O hard cement f/1322 to 1418m (L.Lap), POOH. RIH w/6 1/8" B.Bit to 1418m, D/O P.Off bushing & hard cement f/1418 to 1450m, cont RIH to 2560m, W&R f/2560 to 2680m, cont RIH to 3337m, D/O plugs , hard cement, shoe & magneset f/3337 to 3570m, obs formation cutting f/3563 to 3570m (100% Fm.) POOH to surface. RIH w/2 7/8" open ended DP to 3570m, mix & pump cement (20 bbl, 120 pcf), as balance plug w/NL-NG. POOH.

3.1.1.9 Offset Well AZNS-017 (6-1/8" Hole Section Summary)

RIH w/ 6-1/8" Bit to 3394m, tag cement, D/O hard cement f/3394 to 3543m, POOH. RIH w/6 1/8" Bit and directional BHA to 3552m (9.59m deeper), try to side track hole f/3552 to 3565.7m w/76 pcf O.B.M w/NL-NG, no success. (3562.5-3565m: 50% Fm., 50% CMT), POOH to surface. RIH w/ 2-7/8" open ended DP to bottom, mix & pump cement "E" (12 bbl, 120 pcf), displace w/122 bbl mud, POOH. RIH w/6 1/8" Bit to 3435m, tag cement, D/O soft cement f/3435 to 3440m, cont D/O hard cement f/3440 to 3542m w/76 pcf O.B.M w/NL-NG. POOH. RIH w/6 1/8" Bit and directional BHA to 3542m, side track hole f/3542 to 3560.2m w/76 pcf O.B.M w/NL-NG. RIH w/6 1/8" Bit to 3560.2m, dir drilled 6 1/8" hole section f/3560.2 to 3595.5m with 76 pcf O.B.M w/NL-NG, pipe got stuck, work on stuck with over pull, pipe got free. Cont' dir drilled 6 1/8" hole section f/3595.5 to 3650m with 76 pcf O.B.M w/NL-NG. cut MW to 73 pcf. Cont dir drilled 6 1/8" hole section f/3650to 3764m (TD) with 73 pcf O.B.M w/NL-NG. Performed hole condition prior to run log. RIH w/ full suit logging tools (TLD/CNL/AIT/BHC/NGS) to 3480m, try to latch DWCH sub several times, no success. POOH to surface. RIH w/6 1/8" Bit to TD, performed hole condition prior to run log. R/U coil tubing equipment and test same, ok. Performed logging with CT in two runs. First run AIT/BHC/GR to 3764m (TD), log up f/3764 to 3541m (7" Liner shoe),

POOH to surface, second run TLD/CNL/NGS to 3764m (TD), log up f/3764 to 7" Liner shoe@ 3541 (CNL/NGS f/TD to 2500m), POOH to surface. RIH w/6 1/8" Bit to 3764m (TD), performed hole condition prior to run 4 ½" liner. Run 4 ½" liner ((13.5#, L-80, HFJ (19 jts slotted liner & 8 jts blank liner)) f/3439.1 to 3763m. RIH w/3 ¾" E.Mill and 7" casing scraper to 3762m, displace mud in hole w/73 pcf salt water, POOH to surface. Run completion string to 1615m. (Packer depth@ 1601.16). N/D BOP, N/U x-mass tree. Complete well head, secure the well. Rig release @18:00 o'clock date 1388-04-08.

Offset Well AZNS-127 Well Montage

Well Name: AZNS-127
RIG Name: 304 TDDC
Well Type: IIF
Final Well Status: Suspended

Location : F01, pad 54
RTE /GLE: 14.45 / 3.95
Pay Zone: Fahliyan

Surface Coordinate
Latitude: 3,448,561
Longitude: 196,167

Target Coordinate
Latitude: 196,163.4
Longitude: 344,853,8.7

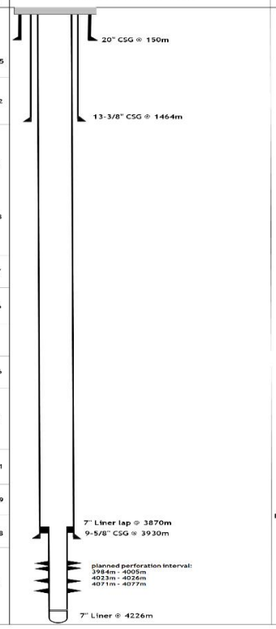
AZNS-127, F-01, PAD54, Type: IIF

Spud Date: 1396/01/05
Actual Rig Days: 152 days
Actual Rig Releas Date: 1396/06/01 at 00:00

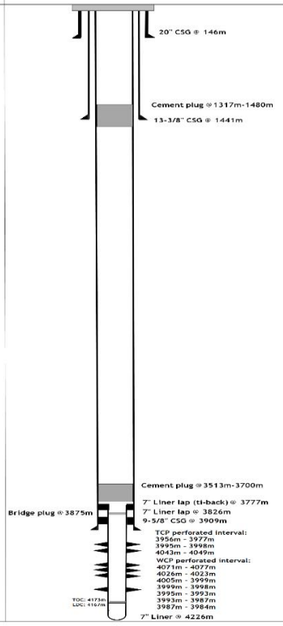
objective
To evaluate characteristics of Gurpi Formation to Lower Fahliyan reservoir. This well will be completed as a Lower Fahliyan oil producer as vertical hole.

Geology Forecast			Actual Geology	
Formation	Top (m,MD)	Top TVD (M, SS)	Top (m,MD)	Top TVD (M, SS)
Aghajari	Surface	14.45	Surface	14.45
Gachsaran	1107.45	-1093	1105	-1090.55
Asmari	1470.45	-1456	1448.84	-1434.2
Pabdeh	1747.45	-1733	1740.68	-1726
U.Gurpi	2203.45	-2189	2182.49	-2167.8
Tarbur	2335.45	-2321	2310.38	-2295.7
L.Gurpi	2407.45	-2393	2390.33	-2375.6
Ilam	2537.45	-2523	2516.7	-2502
Laffan	2635.45	-2621	2618.34	-2603.6
Sarvak	2641.45	-2627	2624.7	-2610
Kazhdumi	3286.45	-3272	3273.82	-3259.1
Darlyan	3495.45	-3481	3478.62	-3463.9
Gadvan	3664.45	-3650	3650.57	-3635.8
Fahliyan	3951.45	-3937	3937.73	-3923

Planned Well Sketch



Actual Well Sketch



Hole size/Bit		Mud Type	MW (pcf)	Casing	Cement Program	Log	Survey	WH	Drilling Problems
26"	115	BENT-MUD	66-68	• 20" slab in shoe 94ppf, k-55, BTC (KTK Co.) • 2 joints 20" casing 94ppf, k-55, BTC to 25m • Circ'd mud and check float shoe-ok • Cont'd run with 10 joints 20" casing 94ppf, K-55, BTC to 146m	• Sting in, Pumped 20bbl fresh water as spacer • 689sxs cement class "G" Kerman (SW: 118pcf, SV: 170bbl) • Displaced with /bbl 68pcf mud (1 /bbl short displace) while got 40bbl cement slurry on surface (10bbl contaminate, 30bbl 118pcf cement slurry).	-	Well deviation at 150m: 0.5deg	20-3/4" CHH 3K (MSP DRILEX)	• Bit balled up- pumped brine periodically • Low ROP
17-1/2	115 / 415 PDC MDI 619	BENT-MUD / KCL POLYMER MUD	68-90 At 502m Displaced w /KCL POLYMER mud	• 13-3/8" float shoe KTK, 68ppf, N-80, BTC • 2 Joints 13-3/8" casing 68ppf, N-80, BTC as shoe track • 13-3/8" float collar KTK, 68ppf, N-80, BTC • 2 Joints 13-3/8" casing 68ppf, N-80, BTC • Circulated mud & tested f.shoe & f. collar-ok. • Continued RIH with 13-3/8" casing 68ppf, N-80, BTC (Totally:127 jts) to bottom at 1441m.	• 40bbl chemical wash. • 80bbl 93pcf spacer. • 2004sxs CMT "G" Kerman as lead (SV:780bbl, SW:98pcf). • 582sxs CMT "G" Kerman as tail (SV:142bbl, SW:118pcf). • Dropped top plug (KTK). • Displaced w/696 bbl mud, bumped plug w/ 1500psi for 15 min-ok, while got 150bbl cement slurry return on surface with 120bbl contaminate and recorded 40bbl loss during cement job.	**Open hole:** RIH Sonic/Caliper/CDR/GR to 1438m (3m off bottom), held up, tried to pass this point-not succeed, logged up from 1438 to 1280m, due to failure of Sonic tools, stopped log up and pulled out of hole to surface. changed sonic tools and RIH to 1438m (3m off bottom), tried to pass this point-not succeed, logged up from 1438 to 146m, continue logged up with GR to surface.	Totco @ 500m, well dev.:0.75deg Totco @ 800m, well dev.:0.5deg Totco @ 1100m, well dev.:0.5deg Totco @ 1354m, well dev.:0.5deg Multi shot: @1421.4m dev.:0.94, @1200m dev.:0.95, @1000m dev.:0.74, @800m dev.: 0.66, @600m dev.: 0.61, @400m dev.:0.54, @146m dev.: 0.48	20-3/4" 3K x 13-5/8" 3K CH5 (MSP DRILEX) 20-3/4"x13-3/8" non-auto casing slips (MSP DRILEX)	• Bit balled up
12-1/4"	215 / 517 / 437 PDC	GLYCO-DRILL FLUID	80-81	• 9-5/8" float shoe KTK, 53.5ppf, C-110, TSH Blue • 2 joints 9-5/8" casing 53.5ppf, T-95, TSH Blue as shoe track. • 9-5/8" float collar KTK, 53.5ppf, C-110, TSH Blue. • 2 joints 9-5/8" CSG 53.5#, T-95, TSH Blue. • Circulated mud & function test float shoe & float collar-ok. • Continued RIH with 99 joints 9 5/8" casing 53.5ppf, T-95, TSH blue totally 103 joints) to 1163.6m. • 9-5/8" XOS TSH blue pin x TPCQ box. • Continued run with 9-5/8" casing to bottom at 3910m with good hole condition while washed down from 3900m to 3910m while and circulated mud at 1441m & checked casing fill up every 10 jts- ok. • Ran totally: 103 joints 9-5/8" CSG 53.5ppf, T-95, TSH Blue - 9-5/8" XOS TSH Blue pin x TPCQ box + 226 joints 9-5/8" CSG 53.5ppf, T-95, TPCQ.	• 20bbl 85pcf spacer. • Lead: 2282sxs CMT "G" deltjan w/cenosphere, micro silica (SV: 1133bbl, SW: 88pcf). • Tail: mixed & pumped 363sxs cement "G" deltjan with silica flour (slurry volume: 116bbl, slurry weight: 118pcf). • Dropped top plug (KTK), displaced with 902bbl 81pcf mud, bumped plug with 3000psi and held pressure for 15 min-ok. • After 702bbl displacement, observed cement slurry on surface (got 200bbl slurry cement & 22bbl contaminate-on surface), recorded 280bbl loss during job.	**Open hole:** • **1st full suite toolstring as:** RIH MSFL/DLL/SONIC/CALIPER/GR/CDR tools to 3900m, string held up, tried to pass this point w/ no success, logged up from 3900m to 15-3/8" csg shoe at 1441m, POOH to surf and L/D logging tools while decided to repeat DLL log. • **2nd full suite logging toolstring as:** ZDL/CNC/DSL/WTS/ TTRM to 3859m, logged up f/ 3859m to 13-3/8" csg shoe at 1441m •**3rd logging toolstring as:** DLL/GR, ran in hole to 3630m, again recorded resistivity log data as client request f/3630m to 1441m and logged up logged up f/3630m to 2150m **Cased hole:** VDL/SBT/CCL/GR, string held up at 3905m, tried to pass several times with no success, logged up f/ 3905m to 100m above of 13-3/8" CSG shoe at 1341m, cont'd POOH to surf. and rigged down logging equip.	**Gyro Survey** f/1850-1400m (Dev @1850m, 0.35deg, max Dev @1400m:1.45deg) **Gyro Survey** f/2570-1850m (Dev @2570m, 0.26deg, max Dev @2370m:0.88deg) **Gyro Survey** f/ 3165- 2570m (Dev@3165m, 0.89deg, max dev @3065m, 1.07deg) **Gyro Survey** (Dev @3570m: 0.39deg, _Dev @3520m: 0.44deg, Dev @3470m: 0.27deg , dev @3420m: 0.37deg,dev @3370m: 0.16 deg, dev @3320m: 0.91 deg, dev @3270m: 0.41 deg) **Gyro Survey** (Dev @3570m: 0.39deg ,dev @3520m: 0.44 deg,dev @3470m: 0.27deg , dev @3420m: 0.37deg, dev @3370m: 0.16 deg, dev @3320m: 0.91 deg, dev @3270m: 0.41 deg, dev @3220m:0.88 deg, dev @3170m: 0.64 deg) At section TD at 3910m, dropped **multi shot**, survey shot each 30m f/3868 to 3500m as per program	13-5/8" 5K x 11" 10K THS (MSP Drillex),13-5/8" x 5-5/8" casing slips (non-auto & auto, MGP Drilex)	• PDM failure at 1612m • Due to change in inh., low ROP from 1750m to 1797m, pumped brinc periodically • Low ROP f/ 1920m to 1989m, chorty zone. • Wash out on XOS (6-5/8" Reg. pin x 4-1/2" IF box) bet 5" HWDP and 8-1/2" DC at 1989m • PDM twist off at 2354m • Low ROP from 2371 to 2383m • Stuck pipe at 3040m while POOH. • Wash out F. sub above 8" PDM at 3377m. • Loss increased f/ 6 to 15bph at 3327m, spotted 100bbl 20ppb hi-vis LCM pill 20ppb (10ppb shell fish med+10 ppb limestone coarse)- Loss decreased to 5-6bph. • At 3705m loss increased to 12bph, decreased flow rate from 630gpm to 550gpm & pumped 20ppb hi vis bentonite LCM pill (10ppb shell fish med+10 ppb limestone coarse). Continued drilling 12-1/4" hole size from 3786m to 3910m while periodically pumped 15ppb hi-vis bentonite LCM pill (5ppb shell fish med+5 ppb limestone coarse) every 90 min, observed flow rate from 630gpm to 550gpm & pumped 20ppb hi vis bentonite LCM pill (10ppb shell fish med+10

3.1.1.10 Offset Well AZNS-127 (26” Hole Section Summary)

Drilled 26” hole section from 14m to 150m while experienced low ROP and bit balled up several times. Ran in hole with same rock bit and 26" reamer to 150m and reamed from surface to 150m.
Ran 20” casing to 146m and cemented same in place with no major problem.

3.1.1.11 Offset Well MB-011 (17-1/2” Hole Section Summary)

Drilled 17-1/2” hole section from 150m to 13-3/8” casing point at 1441m while to prevent balled up bit pumped salt saturated water several times, displaced hole with 74 pcf Kcl polymer mud at 502m, experienced 3-6bph mud loss.
Ran Sonic/Caliper/CDR/GR log and then ran in hole with 13-3/8” casing to 1441m and performed cement job with no major problem.

3.1.1.12 Offset Well AZNS-127 (12-1/4” Hole Section Summary)

Drilled 12-1/4” hole section as performance drilling from 1441m to 2371m while at 1474m displaced hole with 80pcf clay-bond drilling fluid and periodically pumped tandem pill with 3-5bph loss. Experienced PDM twist-off at 2354m which was recovered from hole.
Continued drilling 12-1/4” with B. bit from 2371m to 2383m and from 2383m to 3090m as performance drilling while periodically pumped hi-vis pill with 2-6 bph loss. Spotted LCM pill due to increase in mud loss rate. Continued performance drilling 12-1/4” hole from 3090m to 3117m, with 9-12 bph loss, due to PDM failure pulled out of hole, laid down PDM and continued drilling with PDC bit from 3117m to planned 9-5/8” casing point at 3910m with 6-15 bph loss, spotted LCM pill several times.
Ran Full suit log in three stage and then ran 9-5/8” casing to bottom at 3910m and cemented in place with no major problem.

3.1.1.13 Offset Well AZNS-127 (8-1/2” Hole Section Summary)

Drilled 8-1/2” hole section from 3910m to 3913m, displaced hole with 104pcf ultra-carb drill fluid at 2m above 9-5/8" shoe at 3907m. Ran VDL/SBT/CCL/GR and logged 9-5/8” casing

Continued drilling 8-1/2” hole section vertically from 3913m to planned hole TD at 4226m with 0-3bph loss and no major problem. Ran full suit log in three stage.

Ran in hole with 7” liner shoe to bottom at 4226m and cemented same with no major problem. Ran VDL/SBT/WTC/GR/CCL and logged 7” liner.

Due to negative result of 7” liner lap inflow test, ran 7” liner tie-back assembly and cemented same with no major problem. Ran dry test assembly to inflow test 7” liner lap which test result was positive.

To perforate 7” liner as under balance, ran in hole with TCP-DST to planned depth at 4077m, ran CCL/GR and correlated depth, set DST packer at 3954m and fired guns.

To check perforated interval depth, ran in hole CCL/GR logging toolstring, according to CCL/GR log result, perforation was done approximately 28m above planned interval.

To determine exact perforated depth, ran MFC/CCL/GR log and confirmed perforated interval as: 3956 to 3977m, 3995 to 3998m, 4043 to 4049m.

As remedial action perforated planned perforation interval as WCP in 6 step which this time CCL/GR log confirmed correct perforated intervals.

Rigged up Completion running equipment, ran 1st completion, set packer at 3906m, during pressure test observe return from annulus, several times worked on same with no success, to distinguish leakage reason, ran PLT, the results showed leakage at 3-1/2" anchor seal assembly and 7" permanent production packer, unlatch anchor seal and pulled out to surface.

Ran 2nd completion to 3837m, pressure tested completion string, pressure dropped, worked on same several times with no success, pulled out to surface and on surface observed rupture disks were broken.

Ran 3rd completion, packer pre-set at 3890m, tried to pull out with completion string and worked on same several times with no success, unlatched anchor seal and pulled out to surface. Ran in hole with anchor seal on drill pipes and tried to retrieved completion tail with no success.

Milled both upper and lower packer by 6-1/8” metal muncher mill and retrieved packer and completion tail by taper tap.

To temporary suspend the well due to unavailability of completion string, set bridge plug at 3875m and dry test same with positive result and also set two cement plug 1st from: 3513m-3700m and 2nd from: 1317m-1480m.

Landed tubing hanger in THS and tight tie down screws, back off 5" landing joint, installed 4" two way in tubing hanger and tested it with 1000 psi-ok.

Secured well and released rig on 96/06/12 at 2400hrs.

Offset Well AZN-010 Well Schematic

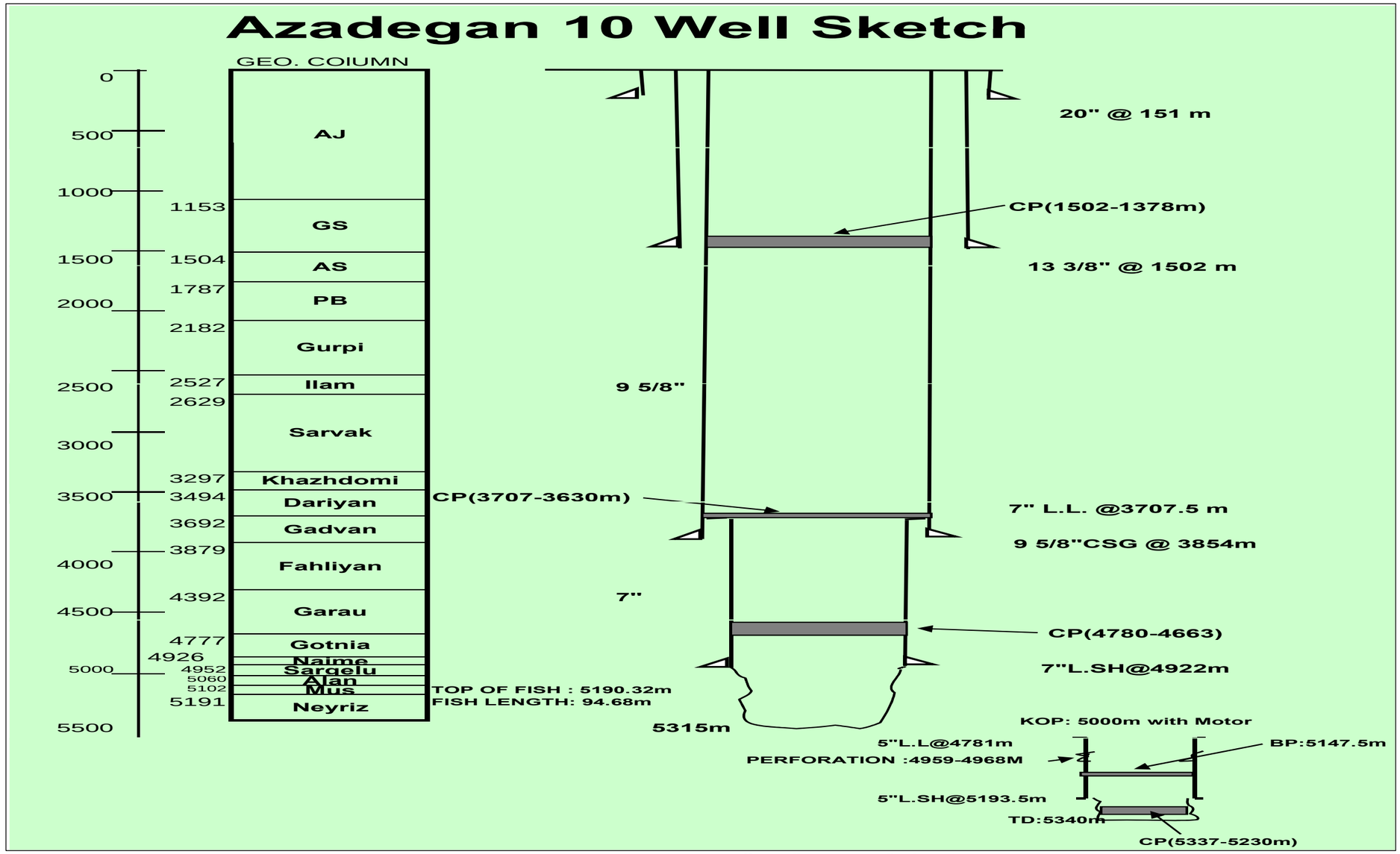


Figure 3-4: Offset Well AZN-010 Well Schematic

3.1.1.14 Offset Well AZN-010 (26" Hole Section Summary)

Drilling started with 26" Bit size and 65 pcf bentonite mud to 152.5m then 20" casing was run and cemented at 151m with cement return to surface.

3.1.1.15 Offset Well AZN-010 (17-1/2" Hole Section Summary)

Drilling 17-1⁄2" hole with 74-90 pcf Mw to 1502m then 13-3/8" casing was run to 1502 m (end of Gachsaran) and cemented to surface.

3.1.1.16 Offset Well AZN-010 (12" Hole Section Summary)

Drilling 12" hole size with 85 pcf Mw to top of Sarvak with 0-8 BPH losses, drilling with 40-50 BPH losses in top of Sarvak and 6-8 BPH to base of Sarvak and drilling with 0-8 BPH while increased Mw to 85 pcf to 3857m, conducted logs in hole in between and at the end of section. 9-5/8" casing shoe was set in 3854m and cemented in base of Gadvan Formation with return contaminated cement to surface.

3.1.1.17 Offset Well AZN-010 (8-3/8" Hole Section Summary)

Hole 8-3/8" drilled with 100-110 PCF polymer mud from Fahliyan formation. Sonic log was run in open hole and then after MDT was run and get 30 pressure points from 4025 to 4221 m in Fahliyan formation. Drilling continued with several problems in surface and one stuck of string and fishing in hole. In depth of 4536 m the string stucked again and released. In Gotnia formation, loss was about 2-4 barrel per hour and 20-80 percent of gas. Also, in the end of Gotnia loss increased to 110 bbl./hr which controlled with LCM Pill and soon after they applied 135 PCF mud weight to control well. Some log conducted (details are noted in logging table). 7" liner run in hole and cemented from 3707.5 to 4922m.

3.1.1.18 Offset Well AZN-010 (5-7/8" Hole Section Summary)

Hole 5 7/8" drilled with 110-120 PCF mud weight and there were several tightness and stuck and one time they applied FPIT to detect the stuck point and used electrical pack off to release string. Left a fish of 94.68 m length in hole while their final TD was 5315 mdd. They could take two well testing in intervals of 4922-5315 and 4922-5120 m. A

cement plug was set in 4996 m and testing done in interval 4922-4996 m with F.B. DST. All the testing had a result of gas, condensate and H2S.

Side track hole of 5 7/8" was conducted from KOP of 5000 m with 112 PCF mud weight and drilled to TD 5340m in Neyriz formation. Some logs were run in hole and VSP log conducted in almost whole well length. 5" liner run in hole and set from 4781 to 5193.5m. A F.B DST was done in Neyriz open hole section in interval of 5193.5- 5340m.

Two cement plugs were set in 5260 and 5230m. Two DST performed in open hole section of Neyriz 5193.5-5260 and 5193.5- 52302m which produced gas and condensate.

A bridge plug set in 5147.5 m and a test was run in perforated interval of 4959-4968m in Sargelu formation.

4. Directional Program

The AZNS-040 Re-Entry well is a vertical Fahliyan well.

5. Sequence of Operation

5.1 Rig Move

➢ Hold pre-move meeting on location. Discuss contingency plans for:

- Safe arrangements for R/U and R/D.
- Safe standby areas for Rig.
- In case move have to be suspended.
- Loss of trucks in case of repair.
- Moving at night.

➢ Hold pre-move safety meeting. Ensure everyone understands individual responsibilities:

➢ Commence rigging down. The ordering of trucks will be done by rig manager.

➢ Commence Rig move to AZNS-40, PAD NO. 50

- Position Rig for optimal operations.
- Mud tanks should be empty during move. Fill water tanks and keep as full as possible

➢ Rig up on cellar

- Final rig up and commence operational procedure for (AZNS-40)
- Before leaving previous location, all equipment should be checked.

➢ Test all manifold and surface safety valves

5.2 Pre-spud

➢ **Certificate**

1. All Rig equipment is on location, properly rigged up and function-tested. Check availability and content of Rig equipment inspection & maintenance records according to Client and API standards.

➢ **Safety Device**

1. Test draw-works crown block saver and record results on daily drilling report; test derrick man escape device (Geronimo line) and record time required for test weight on daily drilling report; test rig intercom system, especially for Driller-Derrick Man and Driller-Mud Logger communication link.
2. Check that all rig safety equipment, e.g., fire extinguishers, eye-wash stations etc., is installed at the proper locations as per rig layout diagrams.
3. Verify that substructure is level and foundation adequate to support drilling unit during whole well period and that mast is centered above rotary table.

4. Ensure rig alignment with Top Drive System. Make sure rig tongs and rig floor hoisting equipment are adequate for handling 8-1/2″ drill collars. Make sure that mechanical tong torque gauges for drill pipe/drill collar are properly functioning and have been calibrated; re-calibrated as required.

➢ **Rig Equipment**

1. Verify that the De-suspension phase and 12-1/4" phase equipment is available in Rig site.
2. 12-1/4" Tri-cone Bit is ready to be picked up, all nozzles installed and 12-1/4″ Bit breaker are on the rig floor.
3. Make sure single-shot or multi-shot survey tools available for vertical section survey.
4. Make sure all solids control equipment are correctly installed and have been function-tested with charging pressure gauges in place where applicable. Make sure adequate screen inventory for the shale shakers is available in the sizes required for the well. Make sure water tanks are completely filled and transfer lines from water tanks to mud system are hooked up. Make sure all mud materials and chemicals required to drill 12-1/4″ hole section is on location, in sound condition and suitably stored. Make sure that De-suspension fluid and kill mud has been made in sufficient volume.
5. Make sure that all mud pumps have been fitted with new consumable parts: liners, pistons, valves, seats etc. Make sure that pulsation dampeners are charged with Nitrogen to the proper pre-charge pressure, 30% of expected standpipe pressure during drilling of the 12-1/4" hole section; check if sufficient Nitrogen bottles are available on location to adjust pulsation dampener pre-charge pressure as required. Make sure that mud pump pop-off valves are set at the proper pressure for the liner sizes installed and that the mud discharge lines from the pop-off valves to the mud tanks are properly secured. Check that all high-pressure hoses are properly secured with safety chains. P/T should be done to high-pressure surface line of circulating system.

➢ **Wellhead & BOP**

1. 20-3/4" 3K CHH and 20-3/4", 3K x 13-5/8", 5K CHS. Have cover on CHS.
2. Record well head and annulus pressure under Client supervision.
3. Overall check of well head components and function same, verify all are in good working order under Cliet supervision.
4. 13-5/8", 10K BOPs and related well control equipment must be ready in Rig site and pressure tested on test stump and ready for N/U on well head.
5. Make sure proper Cup tester, plug tester and wear bushing are available in Rig site.

➢ **Others**

1. Make sure that adequate supply of tools and equipment for De-suspension and drilling 12-1/4" hole section is available.

2. Ensure mud logging unit is properly rigged up and equipment are calibrated and tested properly.
3. Verify that the Rig's fuel tanks are full to meet minimum of 3 days Rig consumption available on site.
4. Make sure all waste treatment equipment available and commissioning test is ok.
5. Hold pre-spud meeting with relevant service Company personnel on location. Discuss procedures involved throughout De-suspension phase and 12-1/4″ hole section drilling, reporting system and critical safety operation measures.
6. Prepared a pre-spud checklist as per Client standard and distributed it in Rig.
7. Check & observe the alignment of the rig. This check will be performed in four stages; rig levelling, derrick verticality, derrick alignment with rotary table, rotary table alignment with conductor pipe. To do this pick up 3 joints of 9-3/4" DC and check the verticality of the derrick with well head.
8. Record Rotary Table-GL and Rotary Table-CHS elevation.
9. Mechanical facilities for mud treatment such as:

a. Desander, Desilter, mud cleaner, proper size of shaker–screens, hoppers, mud guns & mixers, should be all available on Rig and functioning properly.

b. All mud tanks, mud transfer lines & connection should be checked by filling them with water & cross wellhead circulation.

c. Mud engineer laboratory should be equipped with a complete set of testing equipment. The calibration should be done under supervision of well site drilling supervisor before starting the operation.

d. Weight indicator, rotary speed gauge, any smart drilling instrumentation, Top Drive, Stroke meter, pressure gauges on manifold and in the drilling console, loss / gain gauge and any other gauges should be checked and calibrated.

e. Stand pipe manifold valves & connection must be checked & pressure tested.

f. All motors should be checked under load prior to start drilling

g. Check & control mud stock & cement stock every day & if there is any necessity to request mud material, cement or any drilling tool, it should be prepared & requested at proper time.

h. P/U & M/U required 5" DP, 5" HWDP and 8-1/2" DC.

Note: Fill up following forms and send to drilling office:

- Pre-spud check list
- Location and Well Hand Over Form (Including wellhead diagram with dimensions and serial numbers and visual condition)

5.3 De-Suspend the Well

Sequence of Operation

1. Record well head and annulus pressure. If any pressure, bleed off and make sure pressure is zero and does not charge again.

NOTE:

- If there is any pressure after bleed off, call town for instruction.

2. Remove well head cover. Visually check well head, 13-3/8" CSG slip and CHS ring groove, make sure all are in good working order.
3. N/U 13-5/8", 10M BOPs (13-5/8", 5M Annular preventer, 13-5/8″, 10M Double Ram preventer including Pipe Ram and Shear/Blind Ram & 13-5/8", 10M Single Pipe Ram preventer) and related lines. N/U bell nipple and flow line.

NOTE:

- N/U all BOPs component on BOP stump as Rig-less, test same and be ready for final N/U on WH.

4. Function test Accumulator unit. Function & P/T BOPs and related lines as per NIOC standard.
5. Perform WH pressure integrity test by Cup tester.

NOTE:

- Make sure proper Cup test is available at well site.
- Limit P/T to 80% of WH RWP.

6. P/U and M/U 12-1/4" TCI Bit (**Bit#1**, 517 w/ 3x15/32" Jet Nozzles) and D/Out BHA (Including Junk sub), RIH to tag top of Cement plug @ 55m.
7. D/Out Cement plug f/ 55-100m with water while sweeping well with Hi-vis pill intermittently and dump remained fluid in the well.

D/Out Parameters:

- Flow Rate: 400-500 GPM
- RPM: 60
- WOB: 5-20 Klbs

NOTE:

- Start D/Out CMT plug with 5 Jts 8-1/2" DC and add BHA components as D/Out progress.
- Be careful for probable trapped pressure below CMT plug.

8. Cont' RIH joint by joint to tag top of 13-3/8" Float Collar (Cementing Plugs) @ 1468m.
9. D/Out Plugs, F.Collar and CMT to 3m above 13-3/8" Shoe @ 1489m. Work on junk sub.

D/Out Parameters:

- Flow Rate: 500-600 GPM
- RPM: 60-80
- WOB: 5-20 Klbs

10. Displace well with 75 pcf Salt Saturated Polymer mud, Circ and homogenize mud properties.

11. R/U CMT surface lines, P/T same to 3000 psi. Perform 13-3/8" Casing Integrity Test, R/D CMT surface lines.

NOTE:

- Limit CIT pressure to max 80% of 13-3/8" CSG Burst pressure with 1500 psi for 10 min (Consider current Mw).

12. Resume D/Out CMT, Shoe and Rat hole to 1493m. Work on junk sub.

D/Out Parameters:

- Flow Rate: 500-600 GPM
- RPM: 60-70
- WOB: 5-10 Klb

NOTE:

- D/Out with controlled parameters and watch carefully for any indication of abnormal torque, since there might be a possibility of shoe being backed off.

13. Fill up the following forms and send them to drilling office:

- BOP's Schematics with all related dimensions
- BOP's Test Report
- 13-3/8" CIT Report

NOTE:

- Use low parameters while D/Out CMT plug.
- Perform BOP drill and Trip drill before D/Out CMT plug.
- Perform BOP's test on each 14 days.
- Choke manifold and choke line must be lined up for **Hard SHUT-IN**.
- Ensure that all drilling tools and equipment are on well site and are in serviceable condition.
- Ensure that adequate fishing tools are available.
- Ensure that enough Cement and Mud Material specially weighting agent and LCM are available at Rig site.
- Ensure that proper water supply and enough water storage is available on Rig site and water supply never cease.

5.4 12-1/4” Hole Section

General

Table 5-1: 12-1/4” Hole Section General Data of AZNS-040 Re-Entry Well

Casing Point	9-5/8” Casing shoe @ 3986m (Top of Fahliyan)	
Formation	Asmari, Pabdeh, Gurpi, Ilam, Laffan, Sarvak, Kazhdumi, Dariyan, Gadvan	
Risks & Hole Problems	Losses, Tight Hole & Pipe Stuck, Oil/Gas Flow, Shale Instability,Hole Deviation , Complete Loss	
Deviation Control	Single-shot every 100m and Multi-shot every 300m	
Bit Type & Nozzle	TCI, 537 (3x15/32”) & PDC, M323 (4x13 & 4x14) & PDC, M423 (4x13 & 4x14) & PDC, M433 (3x13 & 5x14)	
Logging	**Cased Hole: CBL/VDL/CCL/GR from 13-3/8” Casing** **Open Hole:** Full Suite/CDR/GR **Cased Hole:** SBT or CAST/CCL//VDL/GR from 9-5/8” Casing	
Testing	Nil	
Drilling Fluid Parameters	**Salt Saturated Kcl/Glycol/Polymer/CaCo3 WBM**	
	MW (PCF)	75-85
	FV (Seq/qt)	32-52
	PV (CP)	6-22
	YP (Lbs./100ft^2)	15-25
	HPHT FL (CC/30min)	<7
	Gel 10 sec/10 min (Lbs./100ft^2)	6> / <18
	PH	9-10.5
Drilling Parameters	GPM	650-750 / 600-750
	WOB (Klbs)	20-25 / 20-30
	RPM	90-100 / 160-180

❖ Formation Sampling

Sample would be taken from Gachsaran to Gadvan every 2-5 meters

NOTE: In addition to this general rule, geologist may decide to take more samples.

Sequence of Operation

❖ Drilling

1. Drilling 12-¼″ hole (5m new formation) from 1493m to 1498m.
2. Work on junk sub, CBU and balance Mw. Perform FIT with Required Pressure 410 psi .

NOTE:

- EMW (**87 pcf**) should be equal to max. planned Mw in this hole section (**85 pcf**) plus 2 pcf.

3. Cont' drilling f/ 1498m to 1515m.
4. POOH to surface, L/D TCI Bit and Junk sub.
5. Change Ram, Install shooting nipple, R/U WL Logging Eq.Perform Logging (CBL/VDL /CCL/GR logs) from Surface to 1492 m in 13-3/8" CSG, R/D Logging Eq, and change Ram.
6. P/U & M/U 12-¼″ PDC Bit (**Bit#2**, M323 w/ 4x13/32" & 3x14/32" Jet Nozzles) BHA (including St. STB @ 20m), RIH to bottom @ 1515m.
7. Resume drilling f/ 1515m to 2365m. POOH to surface for change Bit.
8. P/U & M/U 12-¼″ PDC Bit (**Bit#3**, M423 w/ 4x13/32" & 4x14/32" Jet Nozzles) BHA (including St. STB @ 20m), RIH to bottom @ 2365m.
9. Resume drilling f/ 2365m to 3200m. POOH to surface for change Bit.
10. P/U & M/U 12-¼″ PDC Bit (**Bit#4**, M433 w/ 3x13/32" & 5x14/32" Jet Nozzles) BHA (including St. STB @ 20m), RIH to bottom @ 3200m.
11. Resume drilling f/ 3200m to section TD @ 3986m.

NOTE:

- ✓ 12-1/4" hole section in this well contains H2S bearing formation. H2S service is required to drill this hole section.
- ✓ Ensure that all drilling tools and equipment are on well site and are in serviceable condition.
- ✓ Take multi-shot survey every 300m.Perform BOP's test on each 14 days.
- ✓ Perform BOP drill, Trip drill and Pit drill regularly.
- ✓ Choke manifold and choke line must be lined up for **SOFT SHUT-IN**.
- ✓ The TD will be based on formation tops detected by the well site geologist. TD should be adjusted in such a way to have rat hole as minimum as possible for probable fillings.

- ✓ Adjust Mw according to NBW and with respect to the hole condition. If tight hole encountered, Mw should be increased in 1-2 pcf increment. In addition, the shale inhibitor, lubricant and Torque-less additives should be added into mud system with regard to formation type and drilling challenges.
- ✓ The mud engineer should check the mud properties in a pre-planned and agreed upon time schedule for (Mw, FV, GS, FL, PH, Ca^{++}, TH, Cl^{-})
- ✓ Ensure that enough amount and type of LCM are available at Rig site and keep ready to mix and pump LCM Pills to cure possible losses for different losses scenario. If loss couldn't be cured by LCM pill use cement plug to cure same. Heavy losses shall be cured prior to drilling ahead. Use LCM pill as per recommendation in drilling fluid section of well program.
- ✓ Ensure that adequate fishing tools for this hole section are available.
- ✓ Ensure that proper water supply and enough water storage is available on Rig site and make sure water supply never cease.
- ✓ Perform condition trip every drilled 300m (CBU, POH, RIH, CBU).
- ✓ Use single-shot before each trip to surface and multi-shot each 300m to control well deviation.
- ✓ Flow rate in above table are calculated based on maximum anticipated Mw. Keep maximum acceptable flow rate (650-800 GPM) while drilling with PDC Bit.
- ✓ Pump Truck must be as standby in wellsite.
- ✓ Add proper additives to mud system based on formation being drilled and hole condition as per mud program.

12. At section TD, Circulate and clean hole. Perform condition trip to last casing shoe prior to logging, rest hole for 0.5 hr. and flow check, RIH to bottom and check the hole for any fill at bottom, CBU, check Btm's up sample for any influx, Spot pill, POOH.

NOTE:

- Strap pipe while POOH.
- To either spot the pill or not prior to logging consult with Drilling Manager.
- If observed any indication of drag, run back to bottom, wash and ream to make sure that the hole condition is good prior to logging operation.

❖ Logging (Full Suite/CDR/GR)

13. R/U wire line logging equipment. Run WL with "Full Suite /CDR/GR log Tools", take log f/1493-3986m. POOH & R/D wire line.

14. RIH with last BHA to bottom, CBU and check for any influx, if required increase MW beore run CSG, spot pill, POOH.

NOTE:

- Strap pipe while POOH.
- To either spot the pill or not prior to CSG running consult with Drilling Manager.
- If observed any indication of drag, run back to bottom, wash and ream to make sure that the hole condition is good prior to run casing.
- Do not forget to change Top Ram before/after Logging.

❖ Casing Running

15. Held PJSM. R/U 9-5/8" casing running gear.

16. Run 9-5/8" casing as per program to bottom at 3986m (Pump mud as 1.2 casing annulus volume @ 13-3/8" casing shoe).

NOTE:

- Function test floating equipment.
- Casing shoe should be adjusted in such a way to have minimum Rat hole and enough stick up (1.5m) on rig floor.
- Use thread lock compound to M/U all shoe track connection.
- Use centralizer as per stand-off report in final CMT program. In shale and sticky Marl layers use bow types on each 10 jts.
- Torque-Turn system (Jam system) must be used while running Casing to make sure applying right M/U Trq.
- Take care when running casing not to set down excessive weight if tight hole or ledges are encountered. Continuously monitor WOH during Casing running.
- Fill casing with Non-LCM fresh mud every 10 joints (or every jt with F/U tool).
- Do not forget to change Top Ram before/after CSG running.

❖ Cementing

17. R/U cementing head with related lines (DSV shall witness the loading of plugs). P/T the lines to 4000 psi for 15 minutes. Pump mud as 1.2 casing annulus volume.

18. Cement casing as per program. Displace with mud. Bump the plug with 500 psi over the final circulating pressure or maximum 3000 psi pressure. Bleed off pressure, check for back flow. R/D cementing head.

NOTE:

- Make sure that the cement head is capable of holding pressure while displacing and testing cement (Visually).
- Any minor adjustment for Cementing program regarding the corrected depth will be done in due time.
- Rig pump efficiency should be check in trip-tank before cement job.

- Closely monitor cementing pressure. Record and report the final pressure before bump the plug.
- Cementing Truck must be equipped with DAS to record job.
- If cement returns to surface and observe bumping with 3000 psi while displacing cement slurry, hold pressure for 15 minutes and assume bumping pressure as Casing integrity test and note in DDR.
- If bumping was not observed, CIT must be performed after D/Out CMT to 7m above shoe.
- If any back flow observed after Bleed off pressure, hold the pressure inside till cement to get hard.

❖ Wellhead and BOP

19. Flush BOPs. Hold on casing string weight. Drop 13-5/8" x 9-5/8" casing slip, slack off 80% casing weight, Test CSG Slip with 2000 psi (or as per WH specialist instruction) while keep open CHS side outlet valves.

Note: Consider well control and HSE policy while this operation.

20. Hang BOP's. Rough cut 9-5/8" landing joint, L/D same. Dress and bevel 9-5/8" casing neck.

21. Install 13-5/8", 5K x 11", 10K Tubing Head Spool. Pack P-seal and test THS and cavity as per WH specialist instruction (Limit test pressure to max. 80% of CSG collapse pressure).

NOTE:

- Before installing the casing head spool, open all the test ports and clean inside it with water and air.

22. Install 11", 10K x 13-5/8", 10K DSA.

23. N/U 13-5/8", 10K BOPs (13-5/8", 5K Annular preventer, 13-5/8″, 10K Double Ram preventer including Pipe Ram and Shear/Blind Ram & 13-5/8", 10K Single Pipe Ram preventer) w/ Related lines. N/U Bell nipple and Flow line.

Note: Do not forget to check the wear bushing, Plug Tester and tubing Hanger with THS and tie down flange before installing.

24. Perform Accumulator unit function test, BOPs function test and BOPs Pressure test as per NIOC standards.

❖ D/Out

25. L/D 8-1/2" Drill collars, P/U & M/U required 6-1/2" DCs and 5" DPs/HWDPs to drill next hole section.

26. P/U & M/U 8-1/2" Milled Tooth Bit (**Bit#5**, 217 w/ 3x16/32" Jet Nozzles) BHA including Junk sub, RIH to tag plugs.

27. D/Out Plugs, F.Collar and CMT to 7m above 9-5/8" Casing shoe with controlled parameters. Work on junk sub.

D/Out Parameters:

- ❖ Flow Rate: 450-550 GPM
- ❖ RPM: 60-80
- ❖ WOB: 5-15 Klbs

28. Circ and balance Mw. R/U CMT surface lines and P/T to 4000 psi. Perform 9-5/8" Casing Integrity Test, R/D CMT surface lines. POOH.

NOTE:

- ↳ Limit CIT pressure to max 80% of 9-5/8" CSG Burst pressure which is 1000 psi for 10 min (Consider current Mw).
- ↳ If bumping not observed while cementing proceed to CIT, otherwise consider bumping pressure as CIT.

❖ Logging (9-5/8" Casing Cement Quality Log)

29. RIH with 8-1/2" Tri-Cone Milled Tooth Bit (**Bit#5RR**, 217 w/ 3x16/32" Jet Nozzles) with 9-5/8" Scraper to 7m above CSG shoe while scrap 9-5/8" Casing prior to run CMT quality Log, CBU, POOH.

30. R/U wireline logging equipment. Run 9-5/8" Casing cement quality log (SBT or CAST/GR/CBL/VDL/CCL log), take Log, POOH & R/D wire line.

NOTE:

- ↳ Do not forget to change Top Ram before/after running logs.

31. P/U & M/U 8-1/2" PDC Bit (**Bit#6**, M433 w/ 6x14/32" Jet Nozzles) BHA (Including **MWD** and **St. STB @ 20m**), RIH to tag TOC (7m above CSG shoe).

32. D/Out CMT and F.Shoe with controlled parameters. Work on junk sub.

D/Out Parameters:

- ❖ Flow Rate: 450-550 GPM
- ❖ RPM: 50-70
- ❖ WOB: 5-10 Klbs

NOTE:

- ↳ Watch carefully for any indication of abnormal torque, since there might be a possibility of shoe being backed off. In case, report immediately.

33. Displace mud in well with 100 pcf Salt saturated/KCL/Polymer/CaCo3 Mud.

34. Fill up the following forms and send them to the drilling office:

- ↳ 12-1/4" Hole Summary
- ↳ 12-1/4" Bit Record
- ↳ 9-5/8" Casing Tally
- ↳ 9-5/8" Casing Report (w/ TMS Graph)
- ↳ 9-5/8" Cement Report (w/ CDAS Graph)
- ↳ Wellhead diagram with dimensions and serial number
- ↳ BOP's Schematics with all related dimensions
- ↳ BOP's Test Report
- ↳ 9-5/8" CIT Report
- ↳ FIT Report

Logging Program

Cased Hole: CBL/VDL /CCL/GR from 13-3/8" Casing.
Open Hole Logs: Full Suite/CDR/GR Log
Cased Hole Logs: SBT or CAST/VDL/GR Logs from 9-5/8" Casing

Hole Deviation

Single-shot every 100m and Multi-shot every 300m

5.5 8-1/2” Hole Section

General

Table 5-2: 8-1/2” Hole Section General Data of AZNS-040 Re-Entry Well

Casing/Liner Point	7” Liner shoe @ 4200m (7” L.Lap @ 3886m)	
Formation	Fahliyan	
Risks & Hole Problem	Oil Flow, Tight Hole and Differential Sticking, H2S,Complete Loss, Hole Deviation	
Deviation Control	Single/Multi shot Survey	
Bit Type & Nozzle	MT, 217 (3x16/32”) & PDC, M433 (6x14/32”)	
Logging	**Open Hole:** Full Suite/CDR/GR **Cased Hole:** SBT or CAST/CCL/VDL/GR from 7” Liner	
Testing	Dry Test from 7” L.Lap	
Drilling Fluid Parameters	**Salt Saturated Kcl/Polymer/CaCo3 WBM**	
	MW (PCF)	100-106
	FV (Seq/qt)	40-56
	PV (CP)	26-48
	YP (Lbs./100ft^2)	15-27
	API FL(CC/30min)	< 5-3
	Gel 10 sec/10 min (Lbs./100ft^2)	>8 / <18
	KCL (Lb/bbl)	10-15
Drilling Parameters	GPM	400-500
	WOB	15-20
	RPM	150-170

Sequence of Operation

❖ Formation Sampling

Sample would be taken from Fahliyan every 2 meters

NOTE: In addition to this general rule, geologist may decide to take more samples.

❖ Drilling

1. Drilling 8-1/2" hole (4m new formation) from 3986m to 3990m.
2. CBU and homogenize Mw. Perform FIT with Required Pressure 728 psi.

NOTE:

- EMW (**108pcf**) should be equal to max. planned Mw in this hole section (**106pcf**) plus 2 pcf.

3. Cont' drilling 8-1/2" hole from 3990m to well TD @ 4200m.

NOTE:

- Fahliyan reservoir of AZNS oilfield contains H2S. H2S service is required to drill/complete this hole section.
- Ensure that all drilling tools and equipment are on well site and are in serviceable condition.
- Perform H2S drill, BOP drill, Trip drill and Pit drill regularly.
- Perform BOP's test on each 14 days.
- Choke manifold and choke line must be lined up for **HARD SHUT-IN**.
- The Mud engineer should check the mud properties in a pre-planned and agreed upon time schedule.
- The section TD maybe changed based on formation tops detected by the well site geologist.
- Ensure that enough amount and type of LCM (Acid soluble) are available at Rig site and keep ready to mix and pump LCM Pills to cure losses. If loss couldn't be cured by LCM pill use Magneset plug(s) to cure same. Heavy losses shall be cured prior to drilling ahead.
- Make sure proper water supply and enough storage of water is available in case of encountering heavy losses/complete losses, fill the well with water. Water supply must never cease.
- Ensure that adequate fishing tools for this hole section are available.
- Pump Truck must be as standby in wellsite.

4. CBU, condition hole to last CSG shoe prior to logging operation, RIH to bottom, Spot pill, POOH.

NOTE:

- To either spot the pill or not prior to logging, consult with Drilling Manager.
- If observed any indication of drag, run back to bottom, wash and ream to make sure that the hole condition is good prior to logging operation.

❖ Logging (Full Suite, CDR/GR)

5. R/U wireline Eq., Run WL with "Full Suite Log Tools" to bottom, take log f/ 3986-4200m, POOH & R/D wireline Eq.

NOTE:

- Do not forget to change Top Ram before/after running logs.

6. RIH with last 8-1/2" PDC Bit (**Bit#6 RR**, M433 w/ 6x14/32" Jet Nozzles) BHA (**Without MWD**) to bottom while condition hole prior to run 7" Liner, CBU, Spot pill, POOH.

NOTE:

- Strap pipe while POOH.
- To either spot the pill or not prior to Liner running consult with Drilling Manager.
- If observed any indication of drag, run back to bottom, wash and ream to make sure that the hole condition is good prior to run casing.
- Do not forget to change Top Ram before/after CSG running.

❖ 7" Liner Running

7. Retrieve wear bushing.

8. R/U 7" Liner running gear, Run ±314m 7" Liner, M/U Liner Hanger (record Liner Up/Dn wt), Circ as 1.2 Liner Capacity. R/D Liner running gear.

9. Continue run 7" Liner on 5" DP stands down to 9-5/8" CSG shoe, Circ as 1.2 string capacity. Cont' RIH and tag Btm @ 4200m and confirm well TD (Record Up/Dn wt.). P/U string off-Btm and space out for proper R/U of cementing head.

NOTE:

- Function test floating equipment.
- Use thread lock compound to M/U all shoe track connection.
- Use centralizer as per stand-off report in final CMT program.
- Torque-Turn system (Jam System) must be used while running Liner to make sure applying right M/U Trq.

- Liner Hanger Specialist supervise making up the liner hanger as per the installation procedure. Install the liner wiper plug in presence of WSS.
- Liner Hanger Specialist is to be on Rig floor at all times while running the liner.
- While running Liner, break circulation every 500m. Define max rate/ pressure as per LH specialist.
- Once in open hole, beware the string hanging up on ledges or washed out zones. Pump and reciprocate the string as required to ease the shoe to pass through any difficult zones.
- Wash down last stand to Btm. Take care not to plug the shoe.
- Make sure all Tubular (DP, Pup joint, XOS, etc.) for running liner have been drifted with proper drift size as per job requirement.
- Fill 7" Liner every 5 joints (or every joint with F/U tool) and 5" DP stands every 7 stands with Non-LCM fresh mud.

❖ Cementing

10. PJSM. R/U Cementing Head, connect CMT surface lines, Pressure test with 5000 psi. Meanwhile prepare CMT solution.

11. Circ 1.2 string capacity or till clean the hole and monitor loss/gain while prepare Cement solution.

12. Set Liner Hanger @ ± 3886m (100m above 9-5/8" CSG shoe) and release LH setting tool as per LH Specialist instruction, pull out setting tool 1m and confirm releasing.

13. Circulate mud and prepare for CMT job.

14. Cementing Liner as per program, pull out setting tool and POH to 3500m.

15. Flush string and CBU. POH to 3300m.

16. Close the well and monitor pressure for 4 hrs (or till surface cement sample get hard)

17. Open the well, POOH and L/D Liner Hanger setting tool.

NOTE:

- If it is not possible to set liner hanger, set liner shoe in TD then release liner hanger setting tool.
- Mud engineer should confirm mud properties as per the cementing program and precisely monitor and record any losses prior to job.
- While Circ. at TD before set Hanger, define max rate/ pressure as per LH specialist.
- Any minor adjustment for Cementing program regarding the corrected depth will be done in due time.
- Gauge the OD of pump down plug and ensure it is compatible with drill pipe tools joint ID.
- Closely monitor cementing pressure. Record and report the final pressure before shearing and bump the plug.
- Pump **2 bbl** cement slurry behind the pump down plug as a part of displacement.

- Cementing Truck must be equipped with DAS to record job.
- If observed bumping to 3000 psi while displacing cement, hold pressure for 15min and assume bumping pressure as 7" CIT and note in DDR.
- If any back flow observed after Bleed off pressure, hold the pressure inside for the cement to get hard.
- Record and report losses prior, while and after the cementing job.
- Check the Mud Pump efficiency if use rig pump as contingency in case of pump truck failure.
- Keep one CMT sample in Mud logging oven and set oven temperature as per latest recorded static temperature @ L.Lap depth (From Full suite log data)

❖ **Clean Out (Above 7" L.Lap & Inside Liner)**

18. RIH with 8-1/2" Milled Tooth Bit (Bit#5 RR2, 217 with 3x14/32" Jet Nozzles) to top of Liner Lap @ ± 3886m while D/Out any cement (Considered 7 stands wet while CMT job), CBU, POOH while L/D excess 5" DPs/HWDPs.

D/Out Parameters:

- ❖ Flow Rate: 400-450 GPM
- ❖ RPM: 60-80
- ❖ WOB: 5-15 Klbs

19. P/U & M/U required 3-1/2" DP/HWDP and 4-3/4" DC, stand back same in derrick.

20. RIH with 5-7/8" Milled Tooth Bit (Bit#7, 217 with 3x15/32" Jet Nozzles) with Junk sub to tag pack-off bushing, D/Out pack-off bushing and smooth same, work on Junk sub. Continue RIH to LDC depth @ ±4170m while D/Out any cement, CBU and work on junk sub.

D/Out Parameters:

- ❖ Flow Rate: 350 GPM
- ❖ RPM: 50-70
- ❖ WOB: 5-10 Klbs

21. R/U CMT surface lines and pressure test same to 4000 psi. Perform 7" CIT, R/D CMT surface lines. POOH.

NOTE:

- Limit CIT pressure to max 80% of 7" Liner Burst pressure which is 1500 psi,for 10 min, (Consider current Mw).
- If bumping not observed while cementing proceed to CIT, otherwise consider bumping pressure as CIT.

22. POH to 7” L.Lap, CBU. R/U CMT surface lines, pressure test same with 4000 psi, perform 7” Liner lap positive test with 2000 psi. POOH.
23. RIH with 5-7/8” Milled Tooth Bit (Bit#7RR, 217 with 3x15/32” Jet Nozzles) and tandem scraper (7" & 9-5/8") to LDC depth, Scrap inside 7" Liner and 9-5/8" casing before Dry test, CBU, spot pill, POOH to Surface.
24. RIH 9-5/8” Dry test BHA to ± 20m above 7” L.Lap, set packer, perform 7” L.Lap Dry test as per program, unset packer, POOH and L/D Dry test tools.

NOTE:

- Dry test Liner Lap with 2000 psi differential press between water cushion and Fahliyan datum pressure. Detailed Dry Test program will be issued in due time.
- Before performing L.Lap Dry test, make sure there is hard CMT inside 7” Liner.
- If Dry test result was not satisfying, next program will be issued in due time.
- Perform BOP test before run Dry Test operation.

❖ Logging (7” Liner Cement Quality Log)

25. R/U wireline logging equipment. Run WL with 7” Liner cement quality log tools (SBT or CAST/CCL/VDL/GR), take Log from 7” Liner shoe @ 4200 m to 7" Liner Lap @ 3886 m POOH & R/D wire line.

NOTE:

- Do not forget to change Top Ram before/after running logs.

❖ Completion Phase

NOTE:

- Detailed Completion program will be issued in due time.

26. RIH with 5-7/8” Milled Tooth Bit (Bit#7RR2, 217 with Open Jet Nozzles) to LDC @ 4170m. Displace 7” Liner volume with **CaBr2 Perforation Fluid**, POOH.

NOTE:

- Perforation fluid type and weight might be changed and will be issued to client in due time with well completion program.

27. P/U & M/U DST/TCP Perforation Assy. RIH to perforation depth, correlate depth with CCL/GR, perforate guns and POOH.

NOTE:

- If perforation interval is short and there is not any limitation to perform perforation with WL, consider WL perforation to save Rig time.

28. P/U & M/U 5-7/8" Mill (Junk Mill or E.Mill) BHA, RIH to tag LDC while clear and smooth perforated intervals.

29. R/U WL Eq. RIH WL with CCL/GR Log to confirm Perforated interval, POOH and R/D Logging Eq.

NOTE:

- Do not forget to change Top Ram before/after running WL.

30. RIH with 5-7/8" Milled Tooth Bit (Bit#7RR3, 217 with Open Jet Nozzles) and 7" Liner Scraper to top of Perforations. Displace well volume with **CaBr2 Completion Fluid**.

31. POOH while L/D all drilling tubulars.

NOTE:

- If there is possibility, L/D drilling tubulars after complete the well when make sure about well integrity.

32. R/U Completion running gear. Run 4-1/2" * 3-1/2", 10K Completion string as per program, R/D Completion running gear.

33. Install BPV in TBG Hanger, N/D BOPs.

34. Retrieve BPV, set TWCV in TBG Hanger. N/U X-mass Tree, perform required test as per program, Retrieve TWCV.

35. R/U Slick Line, Set protection sleeve in TRSSSV, R/D Slick Line.

36. R/U CTU, Stimulation and Well Testing Eq. RIH w/ CT to Btm, displace the well with Diesel, POOH CT to surface.

37. R/U Slick Line, Retrieve protection sleeve from TRSSSV, R/D Slick Line.

38. Flow the well as Pre-Acid clean up till acceptable BSW and reading stable flow rate on 2 different choke size for 72 hrs.

39. R/U Slick Line, Set protection sleeve in TRSSSV, R/D Slick Line.

40. RIH w/ CT and perform Acid Wash as per program, POOH CT to surface.

41. R/U Slick Line, Retrieve protection sleeve from TRSSSV, R/D Slick Line.

42. Flow the well as Post-Acid clean up till acceptable BSW and reading stable flow rate on 2 different choke size for 96 hrs.

43. R/U Slick Line, Set protection sleeve in TRSSSV, R/D Slick Line.

44. RIH w/ CT to Btm, displace well with Diesel, POOH CT to surface, R/D CTU and well Testing Eq.

45. R/U Slick Line, RIH w/ SL and Install Plug and Prong into NO-GO L.NPL, POOH.

46. RIH w/ SL and Retrieve protection sleeve from TRSSSV, R/D Slick Line.

47. N/D BOPs and N/U X-Tree.

Note: Consider well control and HSE policy while nipple down BOPs.

48. Secure the well and release the Rig.

49. Fill up the following forms and send them to the drilling office:

- 8-1/2" Hole summary
- 8-1/2" Bit Record
- 5-7/8" Bit Record
- 7" Liner Tally
- 7" Liner Report (w/ TMS Graph)
- 7" Cement Report (w/ CDAS Graph)
- Wellhead diagram with Dimensions and Serial Number
- BOP's Schematics with All Related Dimensions
- BOP's Test Report
- 7" Positive Test and Dry Test Reports
- 7" CIT Report
- FIT Report

Logging Program

Open Hole Logs: Full Suite Log/CDR/GR Log.
Cased Hole Logs: SBT or CAST/CCL/VDL/GR Logs from 7" Liner.

Testing Program

Dry Test: From 7" L.Lap with 2000 psi differential press between water cushion and Fahliyan reservoir datum pressure.

Hole Deviation

MWD

6. BHA Program

6.1 12-1/4” Hole Section

❖ 13-3/8” CSG D/Out BHA and 22m of 12-1/4” Hole (from 1493m to 1515m)

BIT SIZE (inch)	12-1/4"	BHA Configuration to Drill CMT Plug, Shoe Track and 12-1/4" Hole Section				DC	OD
Depth In/Out (m)	1493-1515					DC-1	8-1/2"
HOLE INCLINATION (deg)	0						
MUD WEIGHT (pcf)	75						
B.FACTOR	0.847						
Weight in Air (lb)	54,877						
Bouyed Weight (lb)	46,476						
Bouyed Weight Below Jar (lb)	25,738						

SCHEMATIC	DESCRIPTION	QUANTITY (JT)	OD (in)	ID (in)	CONNECTION	LENGTH (m)	
	5" Drill Pipe	-	5	4.276	4-1/2" IF P × B	-	
	5" Heavy Weight Drill Pipe	6	5	3	4-1/2" IF P × B	54.84	151.15
	XOS	1	8	3	6-5/8" Reg P × 4-1/2" IF B	1	96.31
	8-1/2" Drill Collar	3	8-1/2	2-13/16	6-5/8" Reg P × B	27.42	95.31
	8" D.Jar	1	8	3	6-5/8" Reg P × B	9.75	67.89
	8-1/2" Drill Collar	6	8-1/2	2-13/16	6-5/8" Reg P × B	54.84	58.14
	Bit Sub W/ Float Valve	1	8-1/2	3	6-5/8" Reg B. × 6-5/8" Reg B	1	3.3
	Junk Sub	2	7-3/4	3.5	6-5/8" Reg B × 6-5/8" Reg P	2	2.3
	12-1/4" TCI Bit	1	12-1/4	-	6-5/8" Reg.Pin	0.3	0.3

Figure 6-1: 13-3/8” CSG D/Out BHA and 22m of 12-1/4” Hole (from 1493m to 1515m)

❖ **12-1/4” PDC Bit BHA (from 1515m to 3986m)**

BIT SIZE (inch)	12-1/4"	BHA Configuration to Drill 12-1/4" Hole Section (PDC Bit)				DC	OD
Depth In/Out (m)	1515-3986					DC-1	8-1/2"
HOLE INCLINATION (deg)	0						
MUD WEIGHT (pcf)	87						
B.FACTOR	0.822						
Weight in Air (lb)	74,718						
Bouyed Weight (lb)	61,450						
Bouyed Weight Below Jar (lb)	37,491						

SCHEMATIC	DESCRIPTION	QUANTITY (JT)	OD (in)	ID (in)	CONNECTION	LENGTH (m)	
	5" Drill Pipe	-	5	4.276	4-1/2" IF P × B	-	
	5" Heavy Weight Drill Pipe	9	5	3	4-1/2" IF P × B	82.26	205.99
	XOS	1	8-1/2	3	6-5/8" Reg P × 4-1/2" IF B	1	123.73
	8-1/2" Drill Collar	3	8-1/2	2-13/16	6-5/8" Reg P × B	27.42	122.73
	8" D.Jar	1	8	3	6-5/8" Reg P × B	9.75	95.31
	8-1/2" Drill Collar	7	8-1/2	2-13/16	6-5/8" Reg P × B	63.98	85.56
	12-1/4" S.Stabilizer	1	8-1/2	3	6-5/8" Reg P × B	2	21.58
	8-1/2" Drill Collar	2	8-1/2	2-13/16	6-5/8" Reg P × 6-5/8" Reg B	18.28	19.58
	Bit Sub W/ Float Valve	1	8-1/2	3	6-5/8" Reg B. × 6-5/8" Reg B	1	1.3
	12-1/4" PDC Bit	1	12-1/4	-	6-5/8" Reg.Pin	0.3	0.3

Figure 6-2: 12-1/4” PDC Bit BHA (from 1515m to 3986m)

6.2 8-1/2” Hole Section

❖ 9-5/8” CSG D/Out BHA

BIT SIZE (inch)	8-1/2"	BHA Configuration to Drill 8-1/2" Hole Section (Shoe Track Drilling)				DC	OD
Depth In/Out (m)	3986					DC-1	6-1/2"
HOLE INCLINATION (deg)	0						
MUD WEIGHT (pcf)	87						
B.FACTOR	0.822						
Weight in Air (lb)	42,291						
Bouyed Weight (lb)	34,781						
Bouyed Weight Below Jar (lb)	20,355						

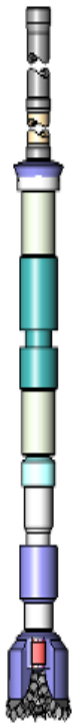
SCHEMATIC	DESCRIPTION	QUANTITY (JT)	OD (in)	ID (in)	CONNECTION	LENGTH (m)	
	5" Drill Pipe	-	5	4.276	4-1/2" IF Pin x Box	-	
	5" Heavy Weight Drill Pipe	6	5	3	4-1/2" IF Pin x Box	54.84	177.32
	XOS	1	6-1/2	3	4-1/2" Reg. P x 4-1/2" IF B	1	122.48
	6-1/2" Drill Collar	3	6-1/2	2-13/16	4-1/2" Reg. P x B	27.42	121.48
	6-1/2" D.Jar	1	6-1/2	2.75	4-1/2" Reg. P x B	9.5	94.06
	6-1/2" Drill Collar	9	6-1/2	2-13/16	4-1/2" Reg. P x B	82.26	84.56
	Bit Sub W/ Float Valve	1	6-1/2	3	4-1/2" Reg. P x B	1	2.3
	Junk Sub	1	5-1/2	2-1/4	4-1/2" Reg. B x P	1	1.3
	8-1/2" MT Bit	1	8-1/2	-	4-1/2" Reg. P x B	0.3	0.3

Figure 6-3: 9-5/8” CSG D/Out BHA

❖ 8-1/2” PDC Bit BHA (from 3986m to 4200m)

			DC	OD
BIT SIZE (inch)	8-1/2"	BHA Configuration to Drill 8-1/2" Hole Section (PDC Bit)	DC-1	6-1/2"
Depth In/Out (m)	3986-4200			
HOLE INCLINATION (deg)	0			
MUD WEIGHT (pcf)	106			
B.FACTOR	0.784			
Weight in Air (lb)	53,044			
Bouyed Weight (lb)	41,568			
Bouyed Weight Below Jar (lb)	24,181			

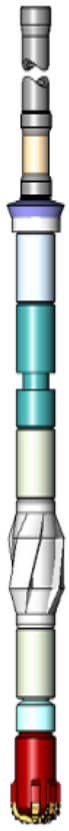
SCHEMATIC	DESCRIPTION	QUANTITY (JT)	OD (in)	ID (in)	CONNECTION	LENGTH (m)	
	5" Drill Pipe	-	5	4.276	4-1/2" IF Pin x Box	-	
	5" Heavy Weight Drill Pipe	9	5	3	4-1/2" IF Pin x Box	82.26	226.04
	XOS	1	6-1/2	3	4-1/2" Reg. P x 4-1/2" IF B	1	143.78
	6-1/2" Drill Collar	3	6-1/2	2-13/16	4-1/2" Reg. P x B	27.42	142.78
	6-1/2" D.Jar	1	6-1/2	2.75	4-1/2" Reg. P x B	9.5	115.36
	6-1/2" Drill Collar	9	6-1/2	2-13/16	4-1/2" Reg. P x B	82.26	105.86
	8-1/2" S.Stabilizer	1	6-1/2	3	4-1/2" Reg. P x B	2	23.6
	6-1/2" NMDC	1	6-3/4"	3.25	4-1/2" Reg P x B	9.4	21.6
	6-3/4" MWD	1	6-3/4"	3.25	4-1/2" Reg P x B	1.5	12.2
	6-3/4" NMDC	1	6-3/4"	2-13/16	4-1/2" Reg. P x B	9.4	10.7
	Bit Sub W/ Float Valve	1	6-3/4"	3	4-1/2" Reg. B x B	1	1.3
	8-1/2" PDC Bit	1	8-1/2	-	4-1/2" Reg. P x B	0.3	0.3

Figure 6-4: 8-1/2” PDC Bit BHA (from 3986m to 4200m)

❖ 7” Liner D/Out BHA

BIT SIZE (inch)	5-7/8"	BHA Configuration to C/Out Inside 7" Liner	Collar	OD
Depth In/Out (m)	4170		DC-1	4-3/4"
HOLE INCLINATION	0			
MUD WEIGHT (pcf)	106			
B.FACTOR	0/784			
Weight in Air (lb)	29,642			
Bouyed Weight (lb)	23,229			
Bouyed Weight Below Jar (lb)	16,462			

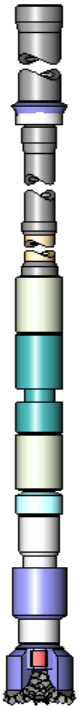
SCHEMATIC	DESCRIPTION	QUANTITY (JT)	OD (in)	ID (in)	CONNECTION	LENGTH (m)	
	5" Drill Pipe	-	5	4/276	4-1/2" IF Pin x Box	-	
	XOS	1	4-3/4	2.25	3-1/2" IF P x 4-1/2" IF B	1	345/61
	3-1/2" Drill Pipe	12	3-1/2	2-7/16	3-1/2" IF P x B	115/2	344/61
	3-1/2" Heavy Weight Drill Pipe	6	3-1/2	2-1/4	3-1/2" IF P x B	54/84	229/41
	4-3/4" Drill Collar	3	4-3/4	2.25	3-1/2" IF P x B	27/42	174/57
	4-3/4" Jar	1	4-3/4	2.25	3-1/2" IF P x B	8	147/15
	4-3/4" Drill Collar	15	4-3/4	2.25	3-1/2" IF P x B	137/1	139/15
	4-3/4" Bit Sub W/ Float Valve	1	4-3/4	2.25	3-1/2" Reg. P x 3-1/2" IF B	0/75	2/05
	Junk Sub	1	4-1/4	1-1/2	3-1/2" Reg. B x P	1	1/3
	5-7/8" MT Bit	1	5-7/8	-	3-1/2" Reg. P x B	0/3	0/3

Figure 6-5: 7” Liner D/Out BHA

7. Mud & Cement Program

7.1 Drilling Fluid Summary

Table 7-1: AZNS-040 Well Drilling Fluid Summary

Hole Size (in)	Casing Size (in)	Casing Program	Depth (m)	Drilling Fluid System	Drilling Fluid Properties
12-1/4”	9-5/8”		1493-3986	Salt Saturated KCL/Glycol/Polymer/ CaCo3 mud	**MW:** 75-85 PCF **F.Vis:** 32-52 SPQ **PV:** 6-22 **YP:** 15-25 Lbs./100ft^2 **FL:** <7 CC/30 Min **GS:** >6 / <18 Lbs./100ft^2 **PH:** 9-10.5 **KCL:** 15-25 PPB
8-1/2"	7" Liner		3986-4200	Salt Saturated KCL/Polymer/CaCo3 mud	**MW:** 100-106 PCF **F.Vis:** 40-56 SPQ **PV:** 26-48 **YP:** 15-27 Lbs./100ft^2 **FL:** <5-3 CC/30 Min **GS:** >8 / <18 Lbs./100ft^2 **PH:** 9.5-10.5 **KCL:** 10-15 PPB

7.2 12-1/4” Hole Section

12-¼ " Hole Interval to Set 9-⅝" Casing

From: 1493 m	**To:** 3986 m	2493 m to drill	**Last Casing Depth:** 1492 m

Fluid System: Salt/Kcl/Glycol/Sulfonated Asphalt/Polymer/$CaCO_3$ Mud

Interval Threats / Barriers

Table 7-2: 12-1/4” Interval Threats/Barriers

Threat	Barriers
HSE	Ensure that all PPE as specified in MSDS is worn when mixing/ handling Mud Chemicals.
Downhole Loss	No action to be taken for losses lower than 10 bbl/hr. Close monitoring shall be made. For larger losses the procedure. Losses can be experienced in whole section; seepage to moderate in Kazhdumi, Dariyan and Gadvan. Close monitoring shall be made. Sufficient quantities of acid soluble LCM should be on hand. Have LCM pill ready. Losses should be cured for the rates above 10 lb/hr.
Tight Hole / Stuck Pipe	Tighten HPHT Fluid Loss lower than 7 ml. Apply correct mud weight. Improve Inhibition and Hole Cleaning.
Influx / Gain	Apply correct mud weight. Close monitoring of indications; drop of return MW, changes in mud properties and/or appearance, Mud Logging Flow Paddle.

Potential Problem

- Downhole Losses
- Complete loss
- Gas Cut / Gain
- Tight Hole / Pipe Stuck
- Hole Deviation

Formation to drill

- **ASMARI:** SANDSTONE, LIMESTONE/DOLOSTONE AND MARL
- **PABDEH:** ARGILLACEOUS LIMESTONE, SHALE, LIMESTONE, MARL AND CHERTY LIMESTONE
- **L.GURPI, TARBUR, U.GURPI:** LIMESTONE, ARGILLACEOUS LIMESTONE AND MARL
- **ILAM:** LIMESTONE, MARL, SHALE
- **LAFFAN:** CLAYSTONE, ANHYDRIDE, SHALE, LIMESTONE
- **SARVAK:** LIMESTONE, ARGILLACEOUS LIMESTONE, MARL, SHALE, DOLOMITIC LIMESTONE AND CHERTY LIMESTONE.
- **KAZHDUMI:** ARGILLACEOUS LIMESTONE, LIMESTONE, SANDSTONE, MARL AND SHALE.
- **DARIYAN:** LIMESTONE, ARGILLACEOUS LIMESTONE, DOLOMITIC LIMESTONE AND SANDY LIMESTONE.
- **GADVAN:** ARGILLACEOUS LIMESTONE, LIMESTONE, SHALE AND SANDSTONE.

Discussion

12-1/4" Interval will be drilled through formations of Asmari, Pabdeh, Gurpi, Ilam, Laffan, Sarvak, Kazhdumi, Darian and Gadvan. Salt/ KCl/ Glycol/ Sulfonated Asphalt/ CaCo3/Polymer Mud is designed for drilling this section. Shoe of 9⅝" casing will be set at 3986m in top of Fahliyan. Different rates of downhole losses, from seepage to complete, may be experienced through this section. Tight hole leading to pipe stuck is the other risk of drilling here.

Based on the lithology to be drilled in 12-1/4” hole section, it is necessary to utilize shale inhibitors in active system in case of need. The primary shale inhibitor in the system is Potassium ion for cation exchange in the clays structure, to reduce their capacity of hydration and swelling. This will be provided by KCl concentration of ≥15 ppb, but to be added only 100 m prior to reach Pabdeh formation (due to probable losses in Asmari formation).

Inhibition of hydratable lithology, especially in Kazhdumi, is going to be provided by existence of 15 pbb KCl in mud system and minimum 2-3% by volume of High Temperature Cloud Point Glycol. it is recommended to have initial sufficient concentration of KCl (along with formulated 2-3% Glycol) in the system to commence a safe and smooth drilling through Kazhdumi formation, Since KCl of the mud is a disturbing factor for the quality of the Gamma log, the KCl will be allowed to be depleted and not maintained in the course of onward drilling, so that the concentration of disturbing potassium ion drops down. Glycol might play the role of primary shale inhibitor for the intervals exposed to low-KCl mud.

To minimize the risk of polymer fermentation whilst drilling, Oxygen Scavenger S is formulated at concentration of 0.3 ppb while having routine daily treatment as well. A very commonly used type of oxygen scavenger is a Sodium Sulphate-based product (If required). It reacts with and eliminates dissolved oxygen as a factor of increasing fermentation rate of natural-sourced polymers and also as a potential source of metal corrosion in mud systems.

KCl will be allowed to be depleted gradually in the course of drilling. It will be maintained only if situation dictates. Use of 3-4 ppb Sulfonated Asphalt while drilling through Laffan formation may be helpful against the threat of Laffan Shale hydration in case of hole dictation. This will also improve the quality of wellbore when we are close to running casing.

NaCl (72 ppb) should be solved in base water to get 72.5 pcf brine. Starch HT at 5-6 ppb and PAC LV at 1.5-2 ppb will be used to control the HPHT Fluid Loss below 5 ml/30min and keep the filter cake as thin as possible on bore hole. 1-2 ppb Xanthan Gum provides both suspension capacity for adding $CaCO_3$-F as weight agent to reach desired MW and required cutting carrying capacity to clean the hole. Use of Barite should be avoided as it can damage Ilam and Sarvak reservoirs. Although the intervals of these two formations are going to be cased with 9⅝" casing, they are subject to be perforated and their productivities are subject to be evaluated as secondary targets after Fahliyan reservoir tested.

Based on well bore condition and stability and Glycol, as shale inhibitor is going to be mixed equivalent of 2-3% by volume 100 m before we reach the top of Pabdeh formation. It should be a Medium-Cloud Point Glycol with the primary function of shale inhibition for water-based muds. It can be used while drilling Pabdeh and Laffan shales without further maintaining to let it be depleted from the system gradually.

Centrifuges and hydrocyclones should be kept running to keep Low Gravity Solids content below 7% in order to minimize reactive solids and improve ROP.

At section TD, minimum 1.2 times of annular volume circulation should be done prior to make a wiper trip to surface. Monitoring the shakers for being clear of cutting helps in deciding when to stop circulation.

Solids Control Management

- Shakers to be dressed initially with proper mesh screens (175/200 mesh size) to minimize the increment of fine solids in mud system.
- There may be a requirement to dress the shakers with coarser screens for a number of circulations in order to minimize over flow of mud at the beginning of this section and to allow full dispersion of hydratable polymers.

- Constant monitoring of the underflow should determine the efficiency of this equipment. All efforts should be made to maintain the LGS content below 7%.

Interval Key Issues

- Drill the cement inside the casing to one or two meters before 13⅜" casing shoe, then pump Hi-Vis pill as a sweep.
- Maintain the Yield Point in the range 15-25 lb/100ft² to ensure adequate hole cleaning. Cuttings should be circulated up far above the BHA prior to connections. It should also be considered to ream at each connection. This will minimize the potential of cuttings settling during connections, causing packing off and stuck pipe. Prior any trips hole should be circulated to be cleaned and at least 1.2 times of bottoms-up. This will prevent tight hole on the way out and bridges, fill etc. on the way back in.
- Total Hardness content in the filtrate should be controlled less than 600 mg/lit by using Soda Ash and Caustic Soda

Fluid Properties

Table 7-3: 12-1/4" Hole Section Drilling Fluid Properties

Property	Specification
Mud Type	Salt Saturated/ KCl/ Glycol/ Polymer
Density (pcf)	75 - 85
FV (sec/qt)	32 - 52
PV (Cps.)	6-22
YP (lb/100 ft²)	15 - 25
10 Sec / 10 Min Gel	>6 / <18
API Fluid Loss (ml/30min)	< 5
pH	9 - 10.5
MBT (lb/bbl)	< 7.5
Hardness (mg/lit)	< 600
Ca++(mg/lit)	< 500
KCl (lb/bbl)	15-25
Chloride (mg/lit)	> 115,000
Med-Cloud Point Glycol (%Vol) [if required]	2-3
Lubricant (%Vol.) [if required]	1 – 2
Corr. Inhibitor (%Vol.) [if required]	0.5 - 1.0

Drilling Fluid Formulations

Table 7-4: 12-1/4" Hole Section Drilling Fluid Formulation

Component	Concentration (ppb)	Component Description
Drill Water	0.8812	Base Fluid
Soda Ash	0.3	Hardness Remover and pH Modifier
Caustic Soda	0.3	pH Controller
Starch NF	5	Fluids Loss Controller
PAC LV	2	Fluids Loss Controller
Salt	77	Non-Solid Weight Agent
KCl	13.41	Shale Inhibitor (prior to Pabdeh)
Xanthan Gum	2	Rheology Modifier
CaCO3 F	As Required	Bridging Agent, Weighting Material
Med-Cloud Point Glycol	2-3%-Vol.	Shale inhibitor (prior to Pabdeh) [if required]
Sulfonated Asphalt	Contingency	Shale Inhibitor (in Laffan) [if required]
Water-Disperse Lubricant	1-2%-Vol.	Biodegradable Lubricant, T&D Reducer) [if required]
Oxygen Scavenger S	As Required	Oxygen Scavenger, Polymer Extender
pH Buffer Agent	If Required	Polymer Temperature Stabilizer
Barite	Contingency	Weight Agent
Corrosion Inhibitor	Contingency	Corrosion Inhibitor
Zinc Carbonate	Contingency	H_2S Scavenger
Lime	Contingency	CO_2 Treatment

Mixing Procedure

1. Fill tanks with calculated amount of Seawater 0.881 bbl/bbl
2. Treat Drill water with Caustic soda and Soda Ash
3. Mix 5-6 ppb Starch NF and 1.5-2 ppb PAC LV at rate of 1 sack per 7 min
4. Mix NaCl at 77 ppb
5. Xanthan Gum at range of 1-2 ppb at rate of 1 sack per 15 min
6. Adjust mud weight as per hole condition by adding $CaCO_3$-Fine

Note: As it was mentioned above, KCl, Glycol and Sulfonated Asphalt are not planned to be mixed in initial preparation of mud. They are supposed to be added in case of hole dictation in the system as per below plan:

- **KCl**

  To be added 100 m prior to reach Pabdeh formation due to probable losses of Asmari. It will be mixed at initial concentration of 15 ppb, but is supposed to be maintained in range 10-15 ppb to section TD, unless hole condition dictates otherwise.

- **Glycol**

  Based on hole condition can be added while drilling through Pabdeh formation and Laffan formation (Optional). In each of these formations, it will be mixed at initial concentration of 2-3% by volume without further maintaining to let it be depleted gradually, unless hole condition dictates otherwise.

Properties Discussion

Mud Weight

Mud weigh has been planned for this section to be at range of 75-85 pcf. Controlling of MW could be done by running hydrodyclones and centrifuges and dressing the shakers with finest mesh screens and dumping mud from sand trap and dilution instead.

Plastic Viscosity

Plastic viscosity should be maintained as low as possible by extensive use of solids control equipment, coupled with whole new mud dilution.(Predicted Optimum range is 6-22Cp but definite range shall be obtained by Pilot test) .

Fluid Loss

It is recommended that API Fluid Loss be maintained below 7 ml/30min during the section. This is be achieved by the addition of a combination of Starch NF at 5 ppb and PAC LV at 2 ppb.

Rheology

Maintain the Yield Point in the range of 15 to 25 lb/100ft² to ensure having adequate hole cleaning, making adjustment with fresh mud or water as required. Thixotropic characteristic of the fluid is supposed to be provided by Xanthan Gum for an optimized rheological profile with elevated low-shear-rate-viscosity and high shear-thinning characteristics which result in fluid with yield point greater than plastic viscosity.

Total Hardness

Total hardness should be maintained below 600 mg/lit for optimum polymer performance with the addition of Soda Ash and Caustic Soda.

PH and Alkalinity

PH should be maintained in the range of 9 to 10 using Caustic Soda. This will maximize chemicals performance and minimize corrosion problems.

MBT

Measuring of the MBT is the direct indication of formation reactivity. MBT of range of below 7 is acceptable. Using fine screens and high speed centrifuges will help in maintaining lower values of MBT by discarding the fine solids and consequently minimizes the dilution rate.

Shale and Clay Inhibition

Shale and Clay inhibition will be achieved by utilizing KCl at range of 15-25 lb/bbl and Glycol at range of 2-3% by volume.

Drilling Issues

Stuck Pipe

Evidences of tight hole and stuck pipe occurred in some wells while drilling this Section. Mud must be inhibited to avoid the reaction with the shale and proper mud weight must be controlled. The main actions to take in order to avoid differential sticking are tight fluid loss control, and continuous movement of pipe and controlling mud weight. In the event of stuck pipe, environment-friendly pipe-freeing pill can be spotted around the stuck area. The other candidate in this case can be using HCl

10-15%. Because of undesired side effect damage to metallic parts and hole wash out, using acid is only considered for highly dramatic situations.

Hole Cleaning

Xanthan Gum is the base for Rheology in the KCl Polymer system. If additional Rheology is required, increase Xanthan Gum concentration to enhance hole cleaning. Xanthan Gum will be used to provide an optimized rheological profile with elevated low-shear-rate-viscosity and high shear-thinning characteristics which result in fluid with yield point greater than plastic viscosity.

Carbon Dioxide

Although it is unlikely to detect Carbon Dioxide in this section, based on an established protocol, the mud should be monitored continuously for any trace of gas, here specifically CO_2. In case of Carbon Dioxide contamination, Lime should be added at 0.5-2 ppb in order to remove CO_2 in the form of precipitating Calcium Carbonate**.**

Hydrogen Sulphide

Regardless that the interval has or has not the potential of emitting any gas, this is an established protocol that the mud should be monitored continuously for any trace of gas, here specifically H_2S. If any trace of H_2S is detected, treat the mud with 1-2 ppb Zinc Carbonate and increase pH by Caustic Soda to maintain it around 10. Corrosion Inhibitor should be used in this case at range of minimum 1% by volume.

Lubricity

In case of high torque and drag, Lubricant can be used at range of 1-2% by volume. This lubricant should be of the type of Water-Dispersible Readily Biodegradable which provides the characteristics string wet-ability by reduction of surface tension. This results in less chance of bit and BHA balling and enhances ROP.

Downhole Losses

Losses/Differential sticking will be minimized/avoided by the planned use of properly sized bridging/weight material (and Polymer) in conjunction with appropriate drilling practices. If seepage losses are noted on the well no specific action will be taken but close monitoring of the loss rate is to be made. For moderate loss rates addition of $CaCO_3$ 50 micron and 150 micron is recommended to ensure that there is adequate correctly sized material to act as bridging agents.

$CaCO_3$ will minimize filtrate/whole mud invasion of the reservoir and reduce potential problems of losses and differential sticking. The fluid loss and cake quality are two of the most important mud

properties to prevent or alleviate potential stuck pipe. Note that, the maintenance of the correct amount and size of bridging material cannot be over-stressed.

❖ **Seepage losses Less than 10 bbl/hr.**

No specific action will be taken, but close monitoring of the probable loss rate is made.

❖ **Losses 10-30 bbl/hr:**

- Load CaCO3-M in active system up to 5-7 ppb
- Load CaCO3-C in active system up to 5-7 ppb

If no success, then:

- Pump 50 bbl LCM pill containing:

- 7 ppb CaCO3-M
- 7 ppb CaCO3-C
- 5 ppb Mica Medium
- 5 ppb Walnut Shells Fine
- 5 ppb Kwik-Seal Fine
- 5 ppb CaCO3-M

Note: System is loaded gradually from low concentration to maximum mentioned ranges.

❖ **Losses 30-100 bbl/hr:**

- Load CaCO3-M in active system up to 7 ppb
- Load CaCO3-C in active system up to 7 ppb
- Load Kwik-Seal F in system up to 5 ppb
- Shaker-man should be ready for managing a part of the return flow toward the pit-room by-pass shakers to avoid screens get blinded and LCM get wasted.

If no success, then:

- Stop Drilling
- Pull out string above the loss zone
- Spot 60-90 bbl pill containing:

- Pre-Hydrated Bentonite 5-10 ppb
- 30 lb/bbl CaCO3-F,M
- 10 lb/bbl Kwik-Seal F

- 10 lb/bbl Kwik-Seal M
- 10 lb/bbl Walnut-Shells F
- 10 lb/bbl Mica M
- Repeat the pill twice
  - If LCM pill at above concentration cannot pass through the string (bit nuzzles, etc.), we need to pump the pill through a Mule-Shoe while following the procedure of the next state.

❖ **Losses more than 100 bbl/hr to Total Losses:**

In the event of severe losses, Squeeze Pills, Barite or CaCO3 or Cement Plugs, Silicate Pill or Cross-Linking Pills should be pumped to minimize losses.

❖ **LCM Pill Application Procedure**

1. After spotting LCM pill over thief zone, pull the bit above the calculated top of the pill. Wait 30 minutes and monitor the annulus. After 30 minutes, attempt to fill the annulus, if not already full.
2. If the hole can be filled, close the BOP and squeeze 10 bbl of LCM pill into the formation without exceeding 200 psi.
3. If the hole remains stable, continue to drill ahead.
4. If the LCM pill failed to reduce the losses or the hole cannot be filled, then mix and spot another 50 bbl of LCM pill.
5. Attempt to fill the annulus. If successful, close the BOP and squeeze 20 bbl of LCM pill into the formation without exceeding 200 psi.
6. If a squeeze pressure develops, hold pressure for 1 hour before running the string back to bottom.

Notes:

- All LCM pills must be thoroughly mixed and homogenized and enough viscous before pumping.
- Remove filter screens from the drill pipe and if necessary, remove the strainers from the mud pumps suction lines.

Casing Running & Cementing

It is important that the hole is fully clean prior to running casing. Any adjustments to mud properties before cementing will be discussed and, if necessary, made after running the casing. Make pit management plan for dealing with returns from running and cementing casing, with contingencies or lost circulation and dealing with spacer returns and contaminated mud.

Estimated Contingency Material

During all the phase keep a minimum safety stock of below items

Estimated Requirement Volume and Chemicals for 12-1/4” Hole Section

Table 7-5: Estimated Contingency Material for 12-1/4" Hole Section

Chemical Type	Unit	Qty	Function
Barite	1MT BB	30	Weight agent
PHPA [If Required]	25 KG	31	Shale encapsulation
Glycol CP [If Required]	55 GA	106	Shale inhibitor
$CaCO_3$ M & C	1 MT BB	10 of each	Loss Circulation Material
Pipe-Freeing Agent	55 Gal DM	8	Environment-Friendly Stuck Pipe Agent
Lime	25 Kg Sx	80	CO_2 Contamination
Zinc Carbonate	25 Kg Sx	120	H_2S Scavenger
Sodium Bicarbonate	25 Kg Sx	160	Cement Contamination
Citric Acid	25 Kg Sx	160	Cement Contamination
Defoamer A	55 Gal DM	4	Anti-Foam
Bio-Cide	55 Gal DM	4	Anti-Bacteria
Oyster shell F/M/C	25 Kg Sxs	160each	Loss circulation material
Kwik Seal F/M/C	25 Kg Sxs	160each	Loss circulation material
Mica F/M/C	25 Kg Sxs	160each	Loss circulation material
Wallnut Shell F/M/C	25 Kg Sxs	160each	Loss circulation material
Drilling Detergent	55Gal	4	Bit Balling Reducer
Oxygen Scavenger S	25Kg	30	Oxygen Scavenger, Polymer Extender
pH Buffer Agent	55Gal	4	Polymer Temperature Stabilizer

Corrosion Inhibitor	55Gal	4	Corrosion Inhibitor
Lubricant Biodegradable	55Gal	54	lubricator

7.2.1.1 Estimated Chemical Consumption and Material Requirement for 75 pcf Mud System

Table 7-6: Estimated Chemical Consumption and Material Requirement for 75 pcf Mud system

12-1/4" Hole Interval					
Mud Density (pcf)	75	-	85	**Mud System**	**Salt KCl Glycol Polymer CaCO3**
Hole Size (inch)	12 1/4				
Section TD (m)	3986			**Solids Control Equip Eff.%**	75
Casing OD/ ID (inch)	13 3/8	/	12.415	**Hole Washout%**	15
Casing Depth (m)	1492			**Dilution Factor**	3.32
LGS below (%vol)	7			**Meters Drilled (m)**	2,458

Volumes Information	**Qty (bbl)**
Surface System [Considering dead volumes of pits and lines while transferring, active pits+ Reserve pits] [Considering dead volumes of pits and lines while transferring+ active pit+ reserve pits in case of severe losses]	1000
Cased Hole Volume	733
New Hole Volume	1176
Washout	176
Sub Total	3085
Dilution	3905
Sub Total	6990
Volume Carried Over from Previous Section	
Total New Volume Required (bbl)	6990

Estimated Chemical Consumption and Cost based on 75 pcf MW

PRODUCT	**UNIT SIZE**	**UNIT PRICE (EURO)**	**CONC'n (lb/bbl)**	**Expected Consump. for Mixing**	**Expected Consump. for Treating**	**Total Qty Expected To Be Used**	**TOTAL COST (EURO)**
Drill Water	1 bbl		0.8812	6,160		6160	
Caustic Soda	25 KG		0.3	38	160	198	
Soda Ash	50 KG		0.3	19	80	99	
Starch- HT	25 KG		5	634	320	954	
Pac-Lv	25 KG		2	254	80	334	
Sodium Chloride	1 MT		77	244	213	377	

Potassium Chloride	1 MT		13.41	43	28	71	
Xanthangum	25 KG		2	254	121	375	
Calcium Carbonate	1 MT		7.4	23	354	377	
PHPA [If Required]	25 KG		0.25	Contingency			
Glycol CPG If Required]	55 GA		7	Contingency			
Lubricant Biodegradable [If Required]	55 Gal		3.5	Contingency			
Materials Required for Probable Contingency Situations							
Barite	1.5MT BB			Contingency	30	30	
PHPA [If Required]	25 KG		0.25	Contingency	31	31	
Glycol CPG [If Required]	55 GA		7	Contingency	106	106	
Lubricant Biodegradable [If Required]	55 Gal		3.5	Contingency	54	54	
CaCO3 M & C	1 MT			Contingency	10 of each	10 of each	
Pipe-Freeing Agent	55 Gal			Contingency	8	8	
Lime	25 Kg			Contingency	80	80	
Zinc Carbonate	25 Kg Sx			Contingency	120	120	
Sodium Bicarbonate	25 Kg Sx			Contingency	160	160	
Citric Acid	25 Kg Sx			Contingency	160	160	
Defoamer A	55 Gal DM			Contingency	4	4	
Bio-Cide	55 Gal DM			Cement Drill Out	4	4	
Oyster shell F/M/C	25 Kg Sxs			Cement Drill Out	160each	160each	
Drilling Detergent	55 Gal DM			Contingency	4	4	
Oxygen Scavenger S	25 Kg Sx			Contingency	30	30	
pH Buffer Agent	55 Gal DM			Contingency	4	4	
Corrosion Inhibitor	55 Gal DM			Contingency	4	4	

NOTE:

- Above calculation are based on normal hole condition
- Above Calculated is Based on for **75 pcf Mw.**

7.2.1.2 Estimated Chemical Consumption and Material Requirement for 85 pcf Mud System

Table 7-7: Estimated Chemical Consumption and Material Requirement for 85 pcf Mud system

12-1/4" Hole Interval					
Mud Density (pcf)	75 - 85		**Mud System**	**Salt KCl Glycol Polymer CaCO3**	
Hole Size (inch)	12-1/4				
Section TD (m)	3950		**Solids Control Equip Eff.%**	75	
Casing OD/ ID (inch)	13-3/8 / 12.415		**Hole Washout%**	15	
Casing Depth (m)	1492		**Dilution Factor**	3.32	
LGS below (%vol)	7		**Meters Drilled (m)**	2,493	

Volumes Information	Qty (bbl)
Surface System [Considering dead volumes of pits and lines while transferring, active pits+ Reserve pits] [Considering dead volumes of pits and lines while transferring+ active pit+ reserve pits in case of severe losses]	1000
Cased Hole Volume	733
New Hole Volume	1176
Washout	176
Sub Total	3085
Dilution	3905
Sub Total	6990
Volume Carried Over from Previous Section	
Total New Volume Required (bbl)	6990

Estimated Chemical Consumption and Cost based on 85 pcf MW							
PRODUCT	**UNIT SIZE**	**UNIT PRICE (EURO)**	**CONC'n (lb/bbl)**	**Expected Consump. for Mixing**	**Expected Consump. for Treating**	**Total Qty Expected To Be Used**	**TOTAL COST (EURO)**
Drill Water	1 bbl		0.78	5,452		5452	
Caustic Soda	25 KG		0.3	38	160	198	
Soda Ash	50 KG		0.3	19	80	99	
Starch- HT	25 KG		5	634	320	954	
Pac-Lv	25 KG		1	127	80	207	
Sodium Chloride	1 MT		80.1	254	133	387	
Potassium Chloride	1 MT		11.9	38	28	66	
Xanthangum	25 KG		1	127	121	248	
Calcium Carbonate	1 MT		98	311		311	
PHPA [If Required]	25 KG		0.25	If required	31	31	

Glycol CPG [If Required]	55 GA		7	If required	106	106	
Lubricant Biodegradable [If Required]	55 Gal		3.5	If required	54	54	
Materials Required for Probable Contingency Situations							
Barite	1.5MT BB			Contingency	30	30	
PHPA [If Required]	25 KG		0.25	Contingency	31	31	
Glycol CPG [If Required]	55 GA		7	Contingency	106	106	
Lubricant Biodegradable [If Required]	55 Gal		3.5	Contingency	54	54	
CaCO3 M & C	1 MT			Contingency	10 of each	10 of each	
Pipe-Freeing Agent	55 Gal			Contingency	8	8	
Lime	25 Kg			Contingency	80	80	
Zinc Carbonate	25 Kg Sx			Contingency	120	120	
Sodium Bicarbonate	25 Kg Sx			Contingency	160	160	
Citric Acid	25 Kg Sx			Contingency	160	160	
Defoamer A	55 Gal DM			Contingency	4	4	
Bio-Cide	55 Gal DM			Cement Drill Out	4	4	
Oyster shell F/M/C	25 Kg Sxs			Cement Drill Out	160each	160each	
Drilling Detergent	55 Gal DM			Contingency	4	4	
Oxygen Scavenger S	25 Kg Sx			Contingency	30	30	
pH Buffer Agent	55 Gal DM			Contingency	4	4	
Corrosion Inhibitor	55 Gal DM			Contingency	4	4	
Lubricant Biodegradable	55 Gal DM			Contingency	20	20	

NOTE:

- Above calculation are based on normal hole condition
- Above Calculated is Based on for **85 pcf Mw.**

7.3 8-1/2” Hole Section

From: 3986 m	**To:** 4200 m	214 m to drill	**Last Casing Depth:** 3986 m

Fluid System Salt Saturated/ KCl/ Glycol/ Polymer/ $CaCO_3$/ *
(*subject to offer alternative. Subject to pilot testing.)

Interval Threats / Barriers

Threat	Barriers
Personnel Safety using Chemicals	Ensure that all PPE as specified in MSDS is worn when mixing/ handling Mud Chemicals.
Downhole Loss	Losses can be experienced in whole section; Close monitoring shall be made. Sufficient quantities of acid soluble LCM should be on hand. Have LCM pill ready. Losses should be cured for the rates above 10 lb/hr.
Gas Influx	Maintain proper mud weight.
Stuck Pipe	The potential of differential and/or mechanical stuck pipe is expected. Maintaining mud weight close to pore pressure, tightening fluid loss and improving hole cleaning and hole condition are recommended.
Tight Hole / High Torque	Maintain proper mud weight. Improve Hole Cleaning. Tighten fluid loss for having thin filter cake. Apply adequate wash reaming and back reaming.

Potential problems

- Downhole Losses
- Complete Loss
- Differential Stuck Pipes
- Gas Kick and Oil Flow

Formation to Drill

- **Upper Fahlian:** Marl, Limestone, Arg. Limestone
- **Lower Fahlian:** Limestone

Discussion

This section will be drilled at mud weight 100 to 106 pcf and Salt Saturated/ KCl/ Glycol/ Polymer/ $CaCO_3$/ Mud system will be used. The possible risks of this section are downhole losses, flow of gas and/or oil and tight hole which may lead to stuck pipe. Severe losses were experienced in porous layers, especially in Fahlian formation.

Starch HT at 5-6 ppb will be used to control the API Fluid Loss below 5-3 ml/30min and keep the filter cake as thin as possible. Differential Stuck pipes can be encountered in this section therefore it is vital to keep thin filter cake.

Xanthan Gum is used as the base of a thixotropic Rheology to provide adequate YP and high LSRV (Low Shear Rate Viscosity) and high-shear-thinning-viscosity which are the key factors to ensure adequate hole cleaning.

Mud Weight

1. **Mud Density vs. Reservoir Consideration**

Density of base water will be increased by combination of Salt (65 ppb) and KCl (9.5 ppb) to get brine at density of 75.5 pcf, then for increasing the density up to 82 pcf, $CaCO_3$-F will be utilized.

Rheology

Thixotropic characteristic of the fluid is supposed to be provided by Xanthan Gum for an optimized rheological profile with elevated low-shear rate-viscosity and high-shear-thinning characteristics which result in fluid with yield point greater than plastic viscosity. Plastic viscosity should be maintained as low as possible by extensive use of solids control equipment, coupled with new mud dilution.

Water-Dispersible Lubricant

It should be an environmentally friendly, biodegradable and non-damaging lubricant which increases the lubricity of the system and reduces the torque and drag, compatible with most water-based muds. It is formulated at range of 1-2% by volume in this interval.

Clay-based Lithology Inhibition; a matter of caring reservoir as well

Potassium Chloride is used as a Clay/Shale Inhibitor where clay can be existing in limestone layered. The concentration required will be based on the extent of the clay but will generally vary

from 3-5% by weight in the system. Use of KCl in reservoir mud has also the good effect of minimizing the hydration and consequently swelling of the clay-based lithology of the pore's walls. This helps in less damage to the reservoir caused by inevitable mud invasion.

Bio-Cide

Starch HT, PAC LV and Xanthan Gum are subject to bacterial degradation even in saturated brines, therefore, a biocide should be used in the system to retard this happening and extend lift-time of the said products.

Solids Control Management

- Shakers to be dressed initially with proper mesh screens (200/230 mesh size). There may be a requirement to dress the shakers with coarser screens for a number of circulations in order to minimize surface loss of mud.
- Combination of linear motion shale shakers with centrifuges proved best results. If centrifuges are planned to run it should be noted that there is the risk discarding $CaCO_3$. Running retort test on samples of centrifuged fluids gives ideas on how justifiable the plan is.
- Constant monitoring of the underflow should determine the efficiency of this equipment. All efforts should be made to maintain the LGS below 7%.

Interval Key Issues

- Mud weight in this interval is critical for wellbore stability and successful drilling and hole condition will dictate the mud density.
- Cuttings should be circulated up above the BHA prior to connections. It should also be considered to ream at each connection. This will minimize the potential of cuttings settling during connections, causing packing off and stuck pipe. Prior to trips the hole should be circulated for at least 1.5 times of annular volume circulation should be done prior to make a wiper trip to surface. Monitoring the shakers for being clear of cutting helps in deciding when to stop circulation. This will prevent tight hole on the way out and bridges, fill etc. on the way back in.
- Maintain a contingency stock of Caustic Soda, Zinc Carbonate, Lime, Citric Acid, Barite, Hematite and fine and medium size of acid-soluble LCM on board.
- Total Hardness content in the filtrate should be controlled less than 600 mg/lit by using Soda Ash and Caustic Soda for better performance of polymers.

- Use pre-mixes to maintain system volumes and to make necessary dilutions. The pre-mixes will contain sufficient polymers to replace polymers lost by either degradation or adsorption to drilled cuttings.

Fluid Properties

Table 7-8: 8-1/2" Hole Section Drilling Fluid Properties

Property	Specification
Mud Type	Salt Saturated/ KCl/ Polymer/ $CaCO_3$/
Density (pcf)	100-106
FV (sec/qt)	40 - 56
PV (Cps.)	26-48
YP (lb/100 ft²)	15 - 27
10 Sec / 10 Min Gel	>8 / <18
API Fluid Loss (ml)	< 5
pH	9.5 - 10.5
MBT (lb/bbl)	< 5
Hardness (mg/lit)	< 600
Ca++(mg/lit)	< 500
KCl (lb/bbl)	10 - 15
Chloride (mg/lit)	Saturated
Biodegradable Lubricant (%-vol.)	1 - 2
Corrosion Inhibitor (%-vol.)	0.5 - 1.0

*Subject to offer alternative. Subject to pilot testing.

Drilling Fluid Formulations

Table 7-9: 8-1/2" Hole Section Drilling Fluid Formulations

Component	Conc. (ppb)	Component Description
Drill Water	0.63	Base Fluid
Soda Ash	0.63	Hardness Remover and pH Modifier
Caustic Soda	0.3	pH Control
Starch HT	0.3	High Temperature Fluid Loss Controller
Sodium Chloride	5	Non-Solid weighting Agent

Component	Conc. (ppb)	Component Description
KCl	65	Clay/ Shale Inhibitor
Xanthan Gum	9.5	Rheology Provider (As Required 1-2pbb)
$CaCO_3$ Fine	0.5	Acid Soluble Weight Agent
Biodegradable Lube	1-2%-vol.	T&D Reducer, ROP Enhancer (Contingency)

NOTE:

- Above calculation is based on mud weight 100 pcf.
- Pilot Test is required.

Mixing Procedure

1. Fill tanks with calculated amount of Drill Water 0.722 bbl/bbl
2. Treat Drill Water with Caustic soda and Soda Ash
3. Mix 5-6 ppb Starch HT, at rate of 1 sack per 7 min
4. Mix KCl at 9.5 ppb and NaCl at 65 ppb
5. In case of what hole dictates, Addition of pH-Buffer, Oxygen Scavenger S, Corrosion Inhibitor, $ZnCO_3$, Bio-Cide is while drilling as required.
6. Add $CaCO_3$-F (256ppb) to reach MW of ±100 pcf
   - In case of foaming use Alcohol-Based Defoamer.
   - In case of excessive torque use Lubricant at range of 1-2% by volume.

Properties Discussion

Rheology

Thixotropic characteristic of the fluid is supposed to be provided by Xanthan Gum for an optimized rheological profile with elevated low-shear rate-viscosity and high-shear-thinning characteristics which result in fluid with yield point greater than plastic viscosity. Plastic viscosity should be maintained as low as possible by extensive use of solids control equipment, coupled with new mud dilution.

Fluid Loss

It is recommended that the API Fluid Loss be maintained at <5-3 ml/30min. This may best be achieved by the addition of 6 ppb Starch HT. Offering other High-Temp Drill-In Fluid Loss controllers is welcome and usage is subject to be investigated.

Total Hardness

Total hardness should be maintained below 600 mg/lit for optimum polymer performance with the addition of Soda Ash and/or Caustic Soda.

pH and Alkalinity

PH should be maintained in the range of 9 to 10.5 using Soda Ash and/or Caustic Soda. This will maximize chemicals performance and minimize corrosion problems.

Drilling Issues

Stuck Pipe

The main actions to take in order to avoid differential sticking are tight fluid loss control, and continuous movement of pipe and controlling mud weight. In the event of stuck pipe, environment-friendly pipe-freeing pill can be spotted around the stuck area. The other candidate in this case can be using HCl 10-15%. Because of undesired side effect damage to metallic parts and hole wash out, using acid is only considered for highly dramatic situations.

Hole Cleaning

The driving parameter that shows the hole cleaning, are VG-meter 6 and 3 dial readings. Prior to connections, the cuttings should be circulated well above the bottom hole assembly to avoid packing off after connections. Hole should be circulated clean before starting to POOH or back reaming.

- **Parameters affecting Hole Cleaning:**

1. Monitoring shakers to become clear of cutting is a key factor to judge if the hole is clean.
2. Pump as fast as possible.
3. Control ROP's must be considered to avoid heavy load of cuttings in annulus.

4. In laminar flow the hole cleaning capacity, especially at low shear rates in large diameter sections, can be improved by raising the low shear-rate viscosity & gels. The best indicators will be the 6 & 3 rpm and initial gel.

Lubricity

In case of high torque and drag, Lubricant can be used at range of 1-2% by volume. This lubricant should be of the type of environment-friendly water-dispersible readily biodegradable which provides the characteristics of string wet-ability by reduction of surface tension. This results in less chance of bit and BHA balling and enhances ROP.

Hydrogen Sulphide

Regardless that the interval has or has not the potential of emitting any gas, this is an established protocol that the mud should be monitored continuously for any trace of gas, here specifically H_2S. If any trace of H_2S is detected, treat the mud with 1-2 ppb Zinc Carbonate and increase pH by Caustic Soda or Lime to maintain it around 10. Corrosion Inhibitor should be used in this case at range of minimum 1% by volume.

Carbon Dioxide

Although it is unlikely to detect Carbon Dioxide in this section, based on an established protocol, the mud should be monitored continuously for any trace of gas, here specifically CO_2. In case of Carbon Dioxide contamination, Lime should be added at 0.5-2 ppb in order to remove CO_2 in the form of precipitating Calcium Carbonate.

Downhole Losses

Losses/Differential sticking will be minimized/avoided by the planned use of properly sized bridging/weight material (and Polymer) in conjunction with appropriate drilling practices. If seepage losses are noted on the well no specific action will be taken but close monitoring of the loss rate is to be made. For moderate loss rates addition of $CaCO_3$ 50 micron and 150 micron is recommended to ensure that there is adequate correctly sized material to act as bridging agents.

$CaCO_3$ will minimize filtrate/whole mud invasion of the reservoir and reduce potential problems of losses and differential sticking. The fluid loss and cake quality are two of the most important mud properties to prevent or alleviate potential stuck pipe. Note that, the maintenance of the correct amount and size of bridging material cannot be over-stressed.

❖ Seepage losses Less than 10 bbl/hr

No specific action will be taken, but close monitoring of the probable loss rate is made.

❖ Losses from 10-30 bbl/hr

- Load $CaCO_3$-M (150-300 micron) in system up to 5 ppb
- Since some of the $CaCO_3$-M will be discarded over the shakers, add 100 kg per 10 meter drilled to compensate the separated amount on the shakers.

If loss not cured and even tends to increase, then:

- Pump 40-60 bbl of pill made by 7-10 ppb $CaCO_3$ 150-300µ & 7-10 ppb $CaCO_3$ 600µ

❖ Losses from 30-50 bbl/hr

- Load $CaCO_3$-M (150-300 micron) in system up to 5 ppb
- Since some of the $CaCO_3$-M will be discarded over the shakers, add 100 kg per 10 meter drilled to compensate the separated amount on the shakers.

If loss not cured and even tends to increase, then:

- Pump 40-60 bbl of pill made by
- 10-20 ppb $CaCO_3$ 200-300µ
- 10-20 ppb $CaCO_3$ 600 µ
- 10pbb Oyster Shell-F
- 10pbb Oyster Shell-M

❖ Losses from 50-100 bbl/hr

- Load $CaCO_3$-M (150-300 micron) in system up to 5 ppb
- Load $CaCO_3$-C (600-100 micron) in system up to 5 ppb
- Since some of the $CaCO_3$-M&C will be discarded over the shakers, add 100 kg per 10 meter drilled to compensate the separated amount on the shakers.

If loss not cured and even tends to increase, then:

- Pump 40-60 bbl of the following pill:
  - 10-20 ppb $CaCO_3$ 200-300 micron

- 10-20 ppb $CaCO_3$ 600 micron
- 7-10 ppb of each $CaCO_3$ 1000 micron & $CaCO_3$ 2000 micron
- 5-10 pbb Oyster Shell -F
- 5-10 pbb Oyster Shell -M

❖ Losses more than 100 bbl/hr

High concentrated LCM pill of various sizes of Calcium Carbonate and Oyster Shell-F&M&C should be pumped and spotted and gently squeezed over the loss zone.

❖ Spotting Procedure

1. After spotting LCM pill, pull the bit above the calculated top of the pill. Wait 30 minutes and monitor the annulus.
2. After 30 minutes, attempt to fill the annulus, if not already full. If the hole can be filled, close BOP and squeeze 10 bbl LCM pill into formation without exceeding 200 psi.
3. If the hole remains stable, continue to drill ahead.
4. Should the LCM pill fail to reduce the losses or the hole cannot be filled, then mix and spot another 50 bbl of LCM pill.
5. Attempt to fill the annulus. If successful, close the BOP and squeeze 20 bbl of LCM pill into the formation without exceeding 200 psi.
6. If a squeeze pressure develops, hold the pressure for 1 hour before running the drill string back to bottom.

Notes:

- LCM pills must be thoroughly homogenised and viscous enough before being pumped.
- Pumping high-concentrated LCM pill is subject to use of open-end DP or Mule-Shoe. Use of PBL Sub is a good practice too.
- Remove filter screens from the drill pipe (if there is any) and if necessary, remove strainers from the mud pump suction lines.

Logging, Casing Running & Cementing

It is most importance that the hole is clean prior to logging and running liner. Any adjustments to mud properties, such as reduction of gel strength, before logging and before running liner will be discussed. Make pit management plan for dealing with returns from running and cementing casing, with contingencies or lost circulation and dealing with spacer returns and contaminated mud. Reduce yield point to minimum level (if practical) prior to cement job.

Estimated Contingency Material

During all the phase keep a minimum safety stock of below items:

Table 7-10: 8-1/2" Hole Section Estimated Contingency Material

Chemical Type	Unit	Qty	Function
$CaCO_3$ M & C	1 MT BB	10 of each	Loss Circulation Material
Pipe-Freeing Agent	55 Gal DM	8	Environment-Friendly Stuck Pipe Agent
Lime	25 Kg Sx	80	CO_2 Contamination
Zinc Carbonate	25 Kg Sx	120	H_2S Scavenger
Sodium Bicarbonate	25 Kg Sx	160	Cement Contamination
Citric Acid	25 Kg Sx	160	Cement Contamination
Defoamer A	55 Gal DM	4	Anti-Foam
Bio-Cide	55 Gal DM	4	Anti-Bacteria
Oyster shell F/M/C	25 Kg Sxs	160each	Loss circulation material
Drilling Detergent	55Gal	4	Bit Balling Reducer
Oxygen Scavenger S	25Kg	30	Oxygen Scavenger, Polymer Extender
pH Buffer Agent	55Gal	4	Polymer Temperature Stabilizer
Corrosion Inhibitor	55Gal	4	Corrosion Inhibitor
Glycol CPG [If Require]	55Gal	32	Shale inhibitor

Lubricant	55Gal	24	Biodegradable lubricator

Estimated Requirement Volume and Chemicals for 8-1/2” Hole Section

7.3.1.1 Estimated Chemical Consumption and Material Requirement for 100 pcf Mud System

Table 7-11: Estimated Chemical Consumption and Material Requirement for 100 pcf Mud System

8½" Hole Interval					
Mud Density (pcf)	100 - 106		**Mud System**	**Salt Saturated KCl Polymer CaCO3**	
Hole Size (inch)	8-1/2				
Section TD (m)	4200		**Solids Control Equip Eff.%**	75	
Casing OD/ ID (inch)	9-5/8 / 8.535		**Hole Washout%**	20	
Casing Depth (m)	3986		**Dilution Factor**	3.32	
LGS below (%vol)	7		**Meters Drilled (m)**	214	

Volumes Information	**Qty (bbl)**
Surface System [Considering dead volumes of pits and lines while transferring, active pits+ Reserve pits] [Considering dead volumes of pits and lines while transferring+ active pit+ reserve pits in case of severe losses]	950
Cased Hole Volume	931
New Hole Volume	58
Washout	12
Sub Total	1951
Dilution	193
Sub Total	2144
Volume Carried Over from Previous Section	
Total New Volume Required (bbl)	**2144**

Estimated Chemical Consumption and Cost based on 100 pcf MW							
PRODUCT	**UNIT SIZE**	**UNIT PRICE (EURO)**	**CONC'n (lb/bbl)**	**Expected Consump. for Mixing**	**Expected Consump. for Treating**	**Total Qty Expected To Be Used**	**TOTAL COST (EURO)**
Drill Water	1 bbl		0.63	1,351		1351	
Caustic Soda	25 Kg		0.3	12	80	92	
Soda Ash	50 Kg		0.3	6	74	80	
Starch HT	25 Kg		5	195	125	320	
Salt	1 MT		65	63	30	93	
KCl	1 MT		9.5	9	46	55	

Xanthan gum	25 Kg		0.5	19	9	28	
CaCO3-F	1 MT		256	249	40	289	
Lubricant Biodegradable [if required]	55 Gal		7	If required	32	32	
Materials Required for Probable Contingency Situations							
CaCO3 M & C	1 MT BB			Contingency	10 of each	10 of each	
Pipe-Freeing Agent	55 Gal DM			Contingency	8	8	
Lime	25 Kg Sx			Contingency	80	80	
Zinc Carbonate	25 Kg Sx			Contingency	120	120	
Sodium Bicarbonate	25 Kg Sx			Contingency	160	160	
Citric Acid	25 Kg Sx			Contingency	160	160	
Defoamer A	55 Gal DM			Contingency	4	4	
Bio-Cide	55 Gal DM			Cement Drill Out	4	4	
Oyster shell F/M/C	25 Kg Sxs			Cement Drill Out	160each	160each	
Drilling Detergent	55 Gal DM			Contingency	4	4	
Oxygen Scavenger S	25 Kg Sx			Contingency	30	30	
pH Buffer Agent	55 Gal DM			Contingency	4	4	
Corrosion Inhibitor	55 Gal DM			Contingency	4	4	
Lubricant	55 Gal DM			Contingency	32	32	

NOTE:

- Above calculation are based on normal hole condition.
- Above calculated is based on for **100 pcf MW.**

7.4 Completion Phase

Note: Completion fluid specification will be prepared by Mahan Sanat Arya in completion program and after client approval will be implemented.

Fluid#1: Completion Fluid

Clear, solids-free brines are the most commonly used fluids in completion and work over operations to minimize formation damage whilst perforation job and completion phase to improve production rate. Completion and work over fluids are specialized fluids used during well completion operations and remedial workover procedures. These fluids must control not only subsurface pressure with density, but also must minimize formation damage during completion and workover operations. The use of fluids that cause minimal formation damage can result in improved production. Most reservoirs are sensitive to any fluids other than those contained in them naturally.

At the beginning, all surface pits and all involved lines need to be cleaned and flushed thoroughly, then hole is supposed to be displaced with inhibited $CaBr_2$ Brine at density of 105 pcf (last mud weight applied in drilling 8 ½" section), to be ready for completing the well.

Density:

Density of this completion fluid is to be provided by readily commercial liquid Calcium Bromide with initial density of 14.2 ppg (= 106 pcf = 1702 kg/m^3). It is used for blending clear brine, workover brine and completion fluids. It is supplied in 55 gal drums. Note that, when consulting salt tables for formulating the fluid, the inevitable drop of density due to addition of corrosion inhibitor should be considered.

Quality and specifications of $CaBr_2$ shall meet the criteria of a standard regulation to be announced and conformity of the product to be supplied with the regulation is subject to be confirmed through an approved third-party quality testing evidence. Some very common and must-meet specs of an approved sample of commercial Calcium Bromide are as below:

- Identity: Calcium Bromide Liquid, 14.2 SG
- Structural Formula: Br-Ca-Br;HOH
- Molecular Formula: $CaBr_2$;H20
- Molecular Weight: 199.9
- $CaBr_2$ %: 52% by weight
- Boiling Point: 129°C (265°F)

- Bulk Density: ~14.2 lb/gal (1.70 - 1.73 SG) at 25°F
- Solubility in Water: Miscible
- Appearance and Odor: Clear fluid, Colorless liquid with no odor
- Turbidity: 6 NTU maximum
- pH Value: 7 ±1
- Crystallization Temperature: -12.2°C (10°F) Approx.
- Corrosion Rate: < 5 MPY maximum

Corrosion:

Salt solutions are often highly corrosive. Dissolved oxygen and massive concentration of ions are the primary corrosive agents in sodium-, potassium-, calcium-chloride- or bromide-brine base completion fluids. The solubility of oxygen in these brines decreases as saturation with the salt is approached. Anyhow corrosion inhibitor at concentration of 1% by volume and oxygen scavenger (special type compatible with Calcium brines) at concentration of 0.5 ppb should be added in completion fluid to minimize the risk of corrosion.

Appearance of some corrosion inhibitors available in market is amber in color and slightly dark (not 100% clear). Using these types of product may cause reading the turbidity somehow more than what recorded before adding it. Selecting a type of corrosion inhibitor that besides providing the expected metal protection characteristics causes excess value of NTU as less as possible is a point which needs to be considered.

Dissolved Oxygen:

In order to eliminate the threat of dissolved oxygen as a possible source of corrosion, use of Oxygen Scavenger CA is formulated in completion fluid at concentration of 0.5 ppb. Oxygen Scavenger CA is a catalyzed Sodium-based chemical used to scavenge oxygen dissolved in Calcium Brines and also drilling fluids having more than 300 ppm filtrate Calcium. It lowers the risk of corrosion and extends the life of metallic parts including production tubing. It is a stable solid and does not deteriorate when exposed to the atmosphere for extended periods of time.

Oxygen Scavengers must be added directly through the hatch from the top of suction pit, not using/running mixing pump and just prior to pump the fluid into the well. Although, the

concentration is very little, since this stage is after filtering the brine, it is recommended to do a pilot test in lab to ensure turbidity remains in acceptable range after adding this material.

H2S Scavenger:

Use of liquid H_2S scavenger, as a preventive safety factor, is recommended to be added at concentration of 0.5%by volume completion fluid. It should be of the type designed specifically for brine-base fluid systems to remove H_2S.

7.4.1.1 Mixing Procedure; Completion Fluid (based on 105 pcf using only $CaBr_2$)

1. Make sure that all pits and lines involved with completion fluid are thoroughly cleaned, flushed and free of any fluid and solid.
   - Make sure that at least one large pit is kept empty for transferring ready brine into after given time for settling. After transferring one pit to empty pit, the first one is to be flushed and cleaned for accommodating the brine of next ready pit, and so on to the last pit.
   - Depends on the rig facilities and structure, mud engineer on board may find it better to prepare the fluids pit-by-pit.
2. Transfer the content of $CaBr_2$ drums into designated pits.
   - Record the number of drums poured in each tank for accurate volume in it as the base of calculation for adding other additives into; do not rely on pit gauge, sounding or mud logging sensor, but have a look on them for comparison and/or correlation if found useful.
3. Mix Corrosion Inhibitor into pits at concentration of 3.5pbb (1% by volume).
4. Mix H2S Scavenger Liquid into pits at concentration of 0.5% by Volume.
5. Do not add Oxygen Scavenger CA at this stage as it should be added directly from the top of the pit, just prior to pump the fluid into the well.
6. Shut down all agitators and mixing pumps, and leave the mixed fluid in the pits for at least 12 hours to let any solid particle inside settles down.
7. Start transferring the clear fluid of the first pit to the empty pit by means of an air-powered Wilden pump.
   - Make sure that the suction hose of the Wilden pump is not put deep close to the bottom of the pit, otherwise, settled solids are sucked in.

8. After finishing the first pit, stop Wilden pump, wash, flush and clean the pit, use it as the empty pit for accommodating the fluid of second pit using the and same above method.

TSS (Total Suspended Solids) and Turbidity (NTU: Nephelomertric Turbidity Units) of a fluid are the parameters that are used to express the extent to which the fluid is free of solids. These factors are measured on wellsite using a HACH® Test Kit. Most of the operations, require TSS and NTU ranging up to 40. Here, we need TSS<30 and NTU<40 for the fluid pumping in the well and both TSS and NTU less than 100 for the returned fluid.

For the fluids we are going to work with, i.e. packer fluid and CaBr2 completion fluid, typically, there is a high chance of getting desired low value of TSS and NTU easily, because they are made with clear liquid CaBr2, not powder, therefore, there is no impurity in the form of solid

Fluid#2: Packer Fluid

Packer Fluid, here, is suggested to be basically CaBr2 brine same as completion fluid introduced earlier.

Fluid#3: Non-Solid Hi-Vis Pill

Generally, we prepare and use different types of Hi-Viscous pills based on the specifications of the two fluids behind and ahead of the pills and the probable limitations applicable, e.g. the concern of di-valent/mono-valent salts, or remarkably different densities.

If one of the two fluids is $CaBr_2$ brine, the suggestion for the Hi-Vis pill is to make it using a batch of saturated $CaCl_2$ brine (86 pcf = 11.5 ppg) in which Hydroxy Ethyl Cellulose (HEC) is mixed as the viscosifier.

HEC is a modified high-molecular-weight. Non-ionic natural polymer especially suited for the viscosity of brine. This non-fermenting polymer increases viscosity and carrying capacity of workover and completion fluids and heavy brines.

HEC is used to viscosify single salt CaCl2 brines and all mono-valent-salt brines such as NaCl, NaBr, KCl, etc. While mixing, it shows no tendency to form fisheyes and rapidly and smoothly get viscous without the need for high shear mixing. The H-Vis pill made by HEC has a short lifetime and overtime the polymer may become separated from the carrier fluid, therefore, it should be only prepared upon requirement and for the needed volume. Routine usage of HEC is at concentration of ±2 ppb, at pH range of 7 to 9.

Fluid#4: Non-Solid Spacer

Due to having a sensitive completion fluid, in terms of cost, density and being non-solid, use of spacer between every two batches of different fluids to be pumped is a must. Here, the spacer is designed to be again saturated $CaCl_2$ brine (86 pcf = 11.5 ppg). This is heaviest cost-effective non-solid fluid which can be prepared to play the role of a spacer on one end exposed to $CaBr_2$ completion fluid. In the other side, since both $CaBr_2$ and $CaCl_2$ are di-valent salts of the same cation (Calcium) there is no risk of forming precipitation as a result of touching two brines of di-valent and mono-valent salts. This $CaCl_2$ Brine spacer is simply made of 0.823 bbl/bbl fresh water in which 193 ppb of $CaCl_2$ 95% purity is solved. No any other additive needs to be used.

Estimated Requirement Volume and Chemicals for Completion Phase

7.4.1.2 Estimated Chemical Consumption and Material Requirement for Completion Phase (Fluid#1&2)

Table 7-12: Estimated Chemical Consumption and Material Requirement for Completion Phase (Fluid#1&2)

Completion Phase (Fluids#1&2)			
Fluid Density (pcf)	105	**Fluid System**	**COMPLETION & PACKER FLUID $CaBr_2$ Brine**
Hole Size (inch)	Completion Phase		
Casing Depth (m)	3986	**Solids Control Equip Eff.%**	
Casing OD/ ID (inch)	9-5/8 / 8.681	**Well TD (m)**	4200
Liner Depth (m)	4200	**Liner Lap (m)**	3886
Liner OD / ID (inch)	7 / 6.094	**Liner Length (m)**	314

Volumes Information	Qty (bbl)
Surface Volume [Considering dead volumes of pits and lines while transferring]	250
Cased Hole Volume [Capacity of 3886m 9⅝" Casing + Capacity of 314m 7" Liner]	939
Total New Volume Required (bbl)	**1189**

Estimated Chemical Consumption and Cost based on 105 pcf MW							
PRODUCT	**UNIT SIZE**	**UNIT PRICE (EURO)**	**CONC'n**	**Expected Consump. for Mixing**	**Expected Consump. for Treating**	**Total Qty Expected To Be Used**	**TOTAL COST (EURO)**
Calcium Bromide (14.2 ppg)	55 Gal	N/A	0.928 bbl/bbl	888	62	950	N/A
Corrosion Inhibitor	55 Gal	N/A	1% Vol	3	1	10	N/A
H2S Scavenger Liquid	55 Gal	N/A	0.5% Vol	5	3	8	N/A
Oxygen Scavenger CA	25 Kg	N/A	0.5 ppb	9	6	16	N/A

NOTE:

- Above calculation are based on normal hole condition
- Above Calculated is based on **105 pcf Mw.**

7.4.1.3 Estimated Chemical Consumption and Material Requirement for Completion Phase (Fluid#3)

Table 7-13: Estimated Chemical Consumption and Material Requirement for Completion Phase (Fluid#3)

Completion Phase (Fluid#3)					
Fluid Density (pcf)	86		**Fluid System**	**HI-VIS PILL CaCl2 Brine 86 pcf + HEC**	
Hole Size (inch)	Completion Phase				
Casing Depth (m)	3986		**Solids Control Equip Eff.%**		
Casing OD/ ID (inch)	9-5/8	/ 8.535	**Well TD (m)**	4200	
Liner Depth (m)	4200		**Liner Lap (m)**	3886	
Liner OD / ID (inch)	7	/ 6.094	**Liner Length (m)**	314	

Volumes Information	Qty (bbl)
Hi-Vis Brine	**150**
Total New Volume Required (bbl)	**150**

Estimated Chemical Consumption and Cost based on 86 pcf MW							
PRODUCT	**UNIT SIZE**	**UNIT PRICE (EURO)**	**CONC'n**	**Expected Consump. for Mixing**	**Expected Consump. for Treating**	**Total Qty Expected To Be Used**	**TOTAL COST (EURO)**
Fresh Water	55 Gal	N/A	0.823 bbl/bbl	124		124	N/A
CaCl2 95%	25 Kg	N/A	195 ppb	531	80	611	N/A
HEC	25 Kg	N/A	1 ppb	3	3	6	N/A

NOTE:

- Above calculation are based on normal hole condition.
- Above Calculated is based on **86 pcf Mw.**

7.4.1.4 Estimated Chemical Consumption and Material Requirement for Completion Phase (Fluid#4)

Table 7-14: Estimated Chemical Consumption and Material Requirement for Completion Phase (Fluid#4)

Completion Phase (Fluids#4)				
Fluid Density (pcf)	86		**Fluid System**	**SPACER CaCl2 Brine 86 pcf**
Hole Size (inch)	Completion Phase			
Casing Depth (m)	3986		**Solids Control Equip Eff.%**	
Casing OD/ ID (inch)	9-5/8	8.535	**Well TD (m)**	4200
Liner Depth (m)	4200		**Liner Lap (m)**	3886
Liner OD / ID (inch)	7	6.094	**Liner Length (m)**	314

Volumes Information	Qty (bbl)
CaCl2 Brine 86 pcf	**150**
Total New Volume Required (bbl)	**150**

Estimated Chemical Consumption and Cost based on 86 pcf MW							
PRODUCT	**UNIT SIZE**	**UNIT PRICE (EURO)**	**CONC'n**	**Expected Consump. for Mixing**	**Expected Consump. for Treating**	**Total Qty Expected To Be Used**	**TOTAL COST (EURO)**
Fresh Water	55 Gal	N/A	0.823 bbl/bbl	124		124	N/A
CaCl2 95%	25 Kg	N/A	195 ppb	531	80	611	N/A

NOTE:

- Above calculation are based on normal hole condition.
- Above Calculated is based on **86 pcf Mw.**

7.5 Cement Program

NOTE: Final CMT program and Stand-off design will be issued by service company in due time.

General Guideline for Cementing in AZNS Oilfield

Table 7-15: General Guideline for Cementing in AZNS Oilfield

Casing/liner		20″ Surface Casing	13-3/8″ Intermediate Casing in Block I	13-3/8″ Intermediate Casing in Block II & III	9-5/8″ Production Casing in Block I	9-5/8″ Production Casing in Block II & III	7″ Production Liner in Block I	7″ Production Liner in Block II & III
Cementing Method		Stab-in cementing	Double Plugs	Double Plugs	Double Plugs	Double plugs	Rotating/ Non-Rotating Liner	Rotating/ Non-Rotating Liner
Top of Cement		Surface	Surface				Top of Liner	
Slurry Weight (sg)	**Lead**	NA	1.30-1.50		1.9-2.08	1.30-1.35	1.35	1.9
	Tail	1.85-1.89	1.9		1.9		1.9	
Slurry Excess		80%-100% OH volume or return to surface	50%-100% OH volume depending on hole condition & 20% for cased hole annular volume				30% over caliper or 60% OH for tail and 40% for lead slurry & 20% for cased hole annular volume	
Thickening time (min)		≥Cementing time plus 40min	Lead ≥Cementing time plus 6s0min Tail ≥Pumping tail time + Displacement time plus 60min				Cementing time plus 90min	
Fluid loss(ml)		NA	LEAD/TAIL≤150		LEAD ≤80 TAIL≤50		LEAD/TAIL≤50	
Free water (ml/250ml)		NA	LEAD/TAIL : 0					
Initial consistency (BC)		≤30	LEAD/TAIL : ≤30					
Compressing strength (MPa/24h)		≥3.5	LEAD : ≥7 TAIL : ≥14					≥14

Cementing Program Overview in AZNS-040

Table 7-16:Cementing Program Overview in AZNS-040

Casing/Liner	20"	13-3/8"	9-5/8"	7"
CMT Class	Already done w/ return to surface by CNPCI	Already done w/ return to surface by CNPCI	"G"	"G"
Excess (OH-CSG)			50-100%	40-60%
Excess (CSG-CSG)			20%	20%+7Std Wet
Lead Cement (PCF)			85-90	-
Tail Cement (PCF)			115-118	115-118
CMT Additives	It will be issued in due time			
Thickening Time (min)	It will be issued in due time			

Cementing Procedure

Table 7-17: Cementing Procedure

Casing	Cementing Method	TOC
20”	Already done w/ return to surface by CNPCI	Surface
13-3/8”	Already done w/ return to surface by CNPCI	Surface
9-5/8”	Single Stage-Double Plugs	Surface
7”	Single Stage-Rotating Liner Hanger	7 Stand Wet

8. Bit & Hydraulic Program

8.1 12 1/4" HJT217GL – 311.15 mm

- **Metal face sealed journal bearing**

HJ series bit uses journal bearing and metal face seal; it can perform consistently in high speed drilling application.

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- **Special gage protection on cone**

A row of trimmers is added between heel row and gage row compacts. As a result, not only cone shell is protected, but also the borehole wall is properly trimmed.

- **Head OD reinforcement**

Head OD are designed with carbide inserts to protect the shirttail and seal to effectively improve the bit's ability to prevent diameter shrinkage.



- **Hardfacing on shirttail**

This feature can provide better protection for seal and bearing to extend the bit's service life.

- **L -Head OD stabilization**

Ovoid inserts are set on head OD, which near compensator recess. Better protection of bit head and reservoir hole can be achieved in directional and horizontal well drilling applications.

Technical specifications

IADC	2-1-7
Bearing and seal	"O" ring sealed journal bearing
Number of rows	4—3—3
Number of cutters	130
Main cutter shape / Cutter exposure	Hardfaced tooth // 22.8*mm*
Gage row cutter shape	
Layout of diamond enhanced inserts	
Type of nozzle	S4
Type of center nozzle	None
Shank	**6 5/8 REG**
Recommended makeup torque	**37.9~43.3 (KN. m)**

Recommended operating parameters

WOB: 160~ 370 (**KN**)

Ideal WOB range: 180~280 (**KN**)

Rotary speed: 50 ~ 100 (**rpm**)

Ideal Rotary speed range: 60~80 (**rpm**)

The above parameters are only for reference. Optimized parameters are determined by drilling process and lithology of formations on site.

Applicable formations

Medium hard with relatively higher compressive strength and hard stringers, such as shale, sandstone and limestone, etc.

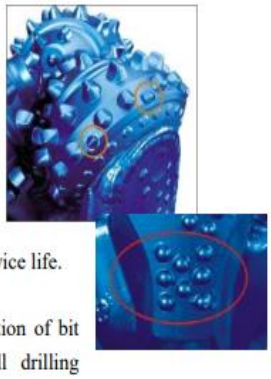
8.2 12 1/4" HJT517GL – 311.15 mm

- **Metal face sealed journal bearing**

HJ series bit uses journal bearing and metal face seal; it can perform consistently in high speed drilling application.

- **Reinforced bit head OD**

Cutter arrangement on head OD area is utilized to protect the shirttail and seal to effectively improve the bit's ability to prevent diameter shrinkage.



- **Special gage protection for cone**

Add one row of trimmers between heel row and gage row, which can repair wall of well as well as protect cone shell.

- **Hardfacing on shirttail**

This feature can provide better protection for seal and bearing to extend the bit's service life.

- **L -Head OD stabilization**

Ovoid inserts are set on head OD, which near compensator recess. Better protection of bit head and reservoir hole can be achieved in directional and horizontal well drilling applications.

Technical specification

IADC	5-1-7
Bearing and seal	Metal face sealed journal bearing
Number of rows	4—4—4
Number of cutters	143
Main cutter shape / Cutter exposure	Scoop chisel compac//14.28mm
Gage row cutter shape	offset wedge
Layout of diamond enhanced inserts	/
Type of nozzle	S4
Type of center nozzle	None
Shank	**6 5/8 REG**
Recommended makeup torque	**37.9~43.3 (KN.m)**

Recommended operating parameters

WOB： 110 ～ 330（**KN**）

Ideal WOB range：140~300（**KN**）

Rotary speed： 50 ～ 240 （**rpm**）

Ideal Rotary speed range：80~200（**rpm**）

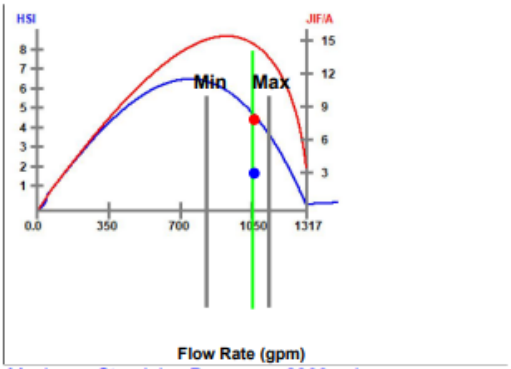
The above parameters are only for reference. Optimized parameters are determined by drilling process and lithology of formations on site.

Applicable formations

Soft with low compressive strength and high drillability, such as shale ,clay ,sandstone and soft limestone, etc .

8.3 12 1/4" Drill Out

Hydraulics Optimization Bit Run Report



Maximum Standpipe Pressure: 2800 psi

Hydraulic Details @ 1515 m

Flow Rate / Hole Diameter	53.1 gpm/in
System Pressure Loss	793 psi
Bit Nozzle Pressure Drop	1455 psi
Annular Pressure Drop	41 psi
Pump Input Hydraulic Power (M.E.=85%)	853 hp
Bit Hydraulic Horsepower	552 hp
Jet Impact Force	1359 lbs
Impact Force/ Hole Area	11.53 lbs/in²
Hydrostatic Head	2586 psi
Equivalent Circulating Density	76.19 lbs/ft³
Cuttings Slip Velocity (Chip Size = .30 in)	15.51 m/min
Surface Equipment Length	46 m X 76.2 mm
Buoyant Collar Weight	56610 lbs

Well Information

Well Name:
AZNS-040
Drilling Contractor:

Operator:

Prepared For:

Location:

Prepared By:

Bit Run Data

Bit Number	1
Bit Size	12 1/4 in
Bit Type	HJT217G

Recommended Hydraulics

Nozzles	15\15\15 32nds
Total Flow Area	0.51 in²

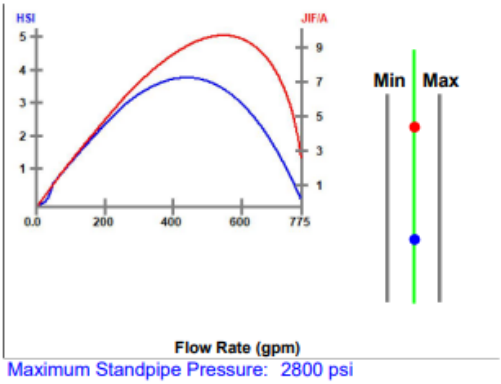
	IN	OUT
BIT HSI	4.68	4.68
JIF/Area	11.53	11.53 lbs/in²
Nozzle Velocity	125	125 m/sec
Flow Rate	650	650 gpm
SPP	2222	2248 psi
% Pressure @ Bit	65.5%	64.7%
Measured Depth	1493	1515 m
TVD	1493	1515 m
Mud Weight	75	75 lbs/ft³
Plastic Viscosity	10	10 cp
Yield Point	15	25 lbs/100 ft²

Drill String And Annular Velocity Data @ 1515 m

Depth m	OD X ID in	Length m	Hole Size in	Bore Loss psi	Ann. Loss psi	Ann. Vel. m/min	Est. Chip Vel. m/min	Crit Vel. m/min	Flow Type
0-1373	DP 5 x 4.276	1373	12.415	402.2	33.3	38	22	97	LAM
1373-1427	HWDP 5 x 3	54	12.415	87.3	1.3	38	22	97	LAM
1427-1428	DC 8 x 3	1	12.415	1.6	.1	54	38	99	LAM
1428-1456	DC 8.5 x 2.813	28	12.415	61.7	1.9	59	44	99	LAM
1456-1457	DC 8 x 3	1	12.415	1.6	.1	54	38	99	LAM
1457-1492	DC 8.5 x 2.813	35	12.415	77.2	2.4	59	44	99	LAM
1492-1511	DC 8.5 x 2.813	19	12.250	41.9	1.4	62	47	100	LAM
1511-1512	DC 8.5 x 3	1	12.250	1.6	.1	62	47	100	LAM
1512-1515	DC 7.75 x 3.5	3	12.250	2.3	.2	54	38	99	LAM

8.4 12 1/4" PDC

Hydraulics Optimization Bit Run Report



Maximum Standpipe Pressure: 2800 psi

Well Information

Well Name:
AZNS-040
Drilling Contractor:
O
Operator:

Prepared For:
Mahan Sanat Arya
Location:

Prepared By:
/Petro Ray

Hydraulic Details @ 3986 m

Flow Rate / Hole Diameter	57.1 gpm/in
System Pressure Loss	2310 psi
Bit Nozzle Pressure Drop	386 psi
Annular Pressure Drop	269 psi
Pump Input Hydraulic Power (M.E.=85%)	1101 hp
Bit Hydraulic Horsepower	158 hp
Jet Impact Force	811 lbs
Impact Force/ Hole Area	6.88 lbs/in²
Hydrostatic Head	7857 psi
Equivalent Circulating Density	89.98 lbs/ft³
Cuttings Slip Velocity (Chip Size = .30 in)	13.37 m/min
Surface Equipment Length	46 m X 76.2 mm
Buoyant Collar Weight	67321 lbs

Bit Run Data

Bit Number	1
Bit Size	12 1/4 in
Bit Type	B616YCM

Recommended Hydraulics

Nozzles	13\13\13\13\13\13\13\13\13 32nds
Total Flow Area	1.16 in²

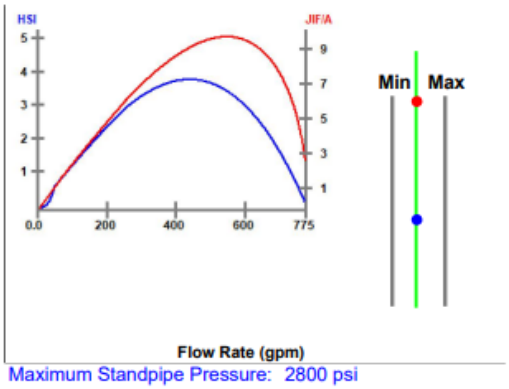
	IN	OUT
BIT HSI	1.34	1.34
JIF/Area	6.88	6.88 lbs/in²
Nozzle Velocity	59	59 m/sec
Flow Rate	700	700 gpm
SPP	1542	2696 psi
% Pressure @ Bit	25.0%	14.3%
Measured Depth	1515	3986 m
TVD	1515	3968 m
Mud Weight	87	87 lbs/ft³
Plastic Viscosity	5	10 cp
Yield Point	15	25 lbs/100 ft²

Drill String And Annular Velocity Data @ 3986 m

Depth m	OD X ID in	Length m	Hole Size in	Bore Loss psi	Ann. Loss psi	Ann. Vel. m/min	Est. Chip Vel. m/min	Crit Vel. m/min	Flow Type
0-1492	DP 5 x 4.276	1492	12.415	564.9	37.1	40	27	90	LAM
1492-3780	DP 5 x 4.276	2288	12.250	866.3	59.4	42	28	90	LAM
3780-3862	HWDP 5 x 3	82	12.250	171.4	2.1	42	28	90	LAM
3862-3863	DC 8.5 x 3	1	12.250	2.1	.1	67	54	92	LAM
3863-3891	DC 8.5 x 2.813	28	12.250	79.8	2.1	67	54	92	LAM
3891-3901	DC 8 x 3	10	12.250	20.9	.6	61	47	92	LAM
3901-3965	DC 8.5 x 2.813	64	12.250	182.4	4.8	67	54	92	LAM
3965-3967	DC 12 x 3	2	12.250	4.2	161.0	863	849	178	TURB
3967-3985	DC 8.5 x 2.813	18	12.250	51.3	1.4	67	54	92	LAM
3985-3986	DC 8.5 x 3	1	12.250	2.1	.1	67	54	92	LAM

8.5 12 1/4" PDC (Nozzle 12/32")

Hydraulics Optimization Bit Run Report



Maximum Standpipe Pressure: 2800 psi

Hydraulic Details @ 3986 m

Flow Rate / Hole Diameter	56.7 gpm/in
System Pressure Loss	2276 psi
Bit Nozzle Pressure Drop	522 psi
Annular Pressure Drop	266 psi
Pump Input Hydraulic Power (M.E.=85%)	1133 hp
Bit Hydraulic Horsepower	211 hp
Jet Impact Force	936 lbs
Impact Force/ Hole Area	7.94 lbs/in²
Hydrostatic Head	7857 psi
Equivalent Circulating Density	89.95 lbs/ft³
Cuttings Slip Velocity (Chip Size = .30 in)	13.37 m/min
Surface Equipment Length	46 m X 76.2 mm
Buoyant Collar Weight	67321 lbs

Well Information

Well Name:
AZNS-040
Drilling Contractor:
O
Operator:

Prepared For:
Mahan Sanat Arya
Location:

Prepared By:
/Petro Ray

Bit Run Data

Bit Number	1
Bit Size	12 1/4 in
Bit Type	B616YCM

Recommended Hydraulics

Nozzles	12\12\12\12\12\12\12\12\12 32nds
Total Flow Area	0.99 in²

	IN	OUT
BIT HSI	1.79	1.79
JIF/Area	7.94	7.94 lbs/in²
Nozzle Velocity	69	69 m/sec
Flow Rate	694	694 gpm
SPP	1661	2798 psi
% Pressure @ Bit	31.4%	18.7%
Measured Depth	1515	3986 m
TVD	1515	3968 m
Mud Weight	87	87 lbs/ft³
Plastic Viscosity	5	10 cp
Yield Point	15	25 lbs/100 ft²

Drill String And Annular Velocity Data @ 3986 m

Depth m	OD X ID in	Length m	Hole Size in	Bore Loss psi	Ann. Loss psi	Ann. Vel. m/min	Est. Chip Vel. m/min	Crit Vel. m/min	Flow Type
0-1492	DP 5 x 4.276	1492	12.415	556.2	37.0	40	27	90	LAM
1492-3780	DP 5 x 4.276	2288	12.250	852.9	59.2	41	28	90	LAM
3780-3862	HWDP 5 x 3	82	12.250	168.7	2.1	41	28	90	LAM
3862-3863	DC 8.5 x 3	1	12.250	2.1	.1	67	53	92	LAM
3863-3891	DC 8.5 x 2.813	28	12.250	78.6	2.1	67	53	92	LAM
3891-3901	DC 8 x 3	10	12.250	20.6	.6	60	47	92	LAM
3901-3965	DC 8.5 x 2.813	64	12.250	179.6	4.8	67	53	92	LAM
3965-3967	DC 12 x 3	2	12.250	4.1	158.5	855	842	178	TURB
3967-3985	DC 8.5 x 2.813	18	12.250	50.5	1.4	67	53	92	LAM
3985-3986	DC 8.5 x 3	1	12.250	2.1	.1	67	53	92	LAM

8.6 8 1/2" HJT517GL – 215.9 mm

- **Metal face sealed journal bearing**

HJT series bit uses journal bearing and metal face seal; it can perform consistently in high speed drilling application.

- **Head OD enhancement**

Adopt teeth layout on head OD to protect the shirttail and seal to effectively improve the bit's ability to prevent diameter shrinkage.

- **Hardfacing on shirttail**

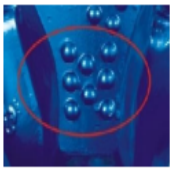
This feature can provide better protection for seal and bearing to extend the bit's service life.

- **Special gage protection for cone**

Add one row of trimmers between heel row and gage row, which can repair wall of well as well as protect cone shell.

- **L -Head OD stabilization**

Ovoid inserts are set on head OD, which near compensator recess. Better protection of bit head and reservoir hole can be achieved in directional and horizontal well drilling applications.



Technical specifications

IADC	5-1-7
Bearing and seal	Journal bearing and Metal face seal
Number of rows	4-4-4
Number of cutters	121
Main cutter shape / Cutter exposure	Scoop chisel compact / 9.93mm
Gage row cutter shape	Offset wedge compact
Type of nozzle	S4
Type of center nozzle	N/A
Shank	4 1/2 REG
Recommended makeup torque	16.3~21.7 (KN.m)

Recommended operating parameters

WOB：75~226（KN）

Ideal WOB range：100~200（KN）

Rotary speed：50~ 240（rpm）

Ideal rotary speed range：70~220（rpm）

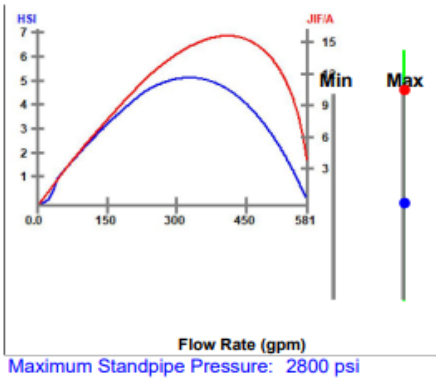
The above parameters are only for reference. Optimized parameters are determined by drilling process and lithology of formations on site.

Applicable formations

Soft with low compressive strength and high drillability, such as shale, clay, salt bed and soft limestone, etc

8.7 8 1/2" PDC

Hydraulics Optimization Bit Run Report



Maximum Standpipe Pressure: 2800 psi

Well Information

Well Name:
AZNS-040
Drilling Contractor:
MSA TBA
Operator:
OEOC
Prepared For:
Mahan Sanat Arya
Location:

Prepared By:
/Petro Ray

Hydraulic Details @ 4200 m

Flow Rate / Hole Diameter	58.8 gpm/in
System Pressure Loss	2212 psi
Downhole Motor / MWD Pressure Drop	200 psi
Bit Nozzle Pressure Drop	539 psi
Annular Pressure Drop	529 psi
Pump Input Hydraulic Power (M.E.=85%)	803 hp
Bit Hydraulic Horsepower	157 hp
Jet Impact Force	756 lbs
Impact Force/ Hole Area	13.33 lbs/in²
Hydrostatic Head	10133 psi
Equivalent Circulating Density	111.53 lbs/ft³
Cuttings Slip Velocity (Chip Size = .30 in)	10.45 m/min
Surface Equipment Length	46 m X 76.2 mm
Buoyant Collar Weight	43126 lbs

Bit Run Data

Bit Number	1
Bit Size	8 1/2 in
Bit Type	VTD616

Recommended Hydraulics

Nozzles	13\13\13\13\13\13 32nds
Total Flow Area	0.77 in²

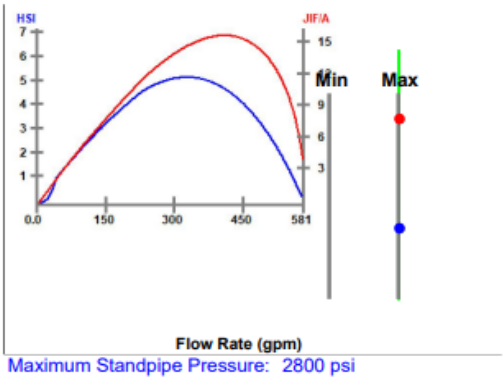
	IN	OUT
BIT HSI	2.62	2.77
JIF/Area	12.58	13.33 lbs/in²
Nozzle Velocity	64	64 m/sec
Flow Rate	500	500 gpm
SPP	2284	2751 psi
% Pressure @ Bit	22.3%	19.6%
Measured Depth	3986	4200 m
TVD	3896	4200 m
Mud Weight	100	106 lbs/ft³
Plastic Viscosity	10	15 cp
Yield Point	15	27 lbs/100 ft²

Drill String And Annular Velocity Data @ 4200 m

Depth m	OD X ID in	Length m	Hole Size in	Bore Loss psi	Ann. Loss psi	Ann. Vel. m/min	Est. Chip Vel. m/min	Crit Vel. m/min	Flow Type
0-1492	DP 5 x 4.276	1492	12.415	387.3	31.3	29	18	85	LAM
1492-3976	DP 5 x 4.276	2484	8.535	644.9	234.6	78	68	88	LAM
3976-3986	HWDP 5 x 3	10	8.535	14.3	.9	78	68	88	LAM
3986-4058	HWDP 5 x 3	72	8.500	103.2	6.9	79	69	89	LAM
4058-4059	DC 6.5 x 3	1	8.500	1.4	.3	125	114	93	TRANS
4059-4086	DC 6.5 x 2.813	27	8.500	52.8	7.1	125	114	93	TRANS
4086-4095	DC 6.5 x 2.75	9	8.500	19.6	2.4	125	114	93	TRANS
4095-4178	DC 6.5 x 2.813	83	8.500	162.2	21.9	125	114	93	TRANS
4178-4180	DC 8.25 x 3	2	8.500	2.9	216.5	892	882	201	TURB
4180-4191	DC 6.75 x 3.25	11	8.500	10.7	4.1	140	130	95	TRANS
4191-4200	DC 6.75 x 2.813	9	8.500	17.6	3.4	140	130	95	TRANS

8.8 8 1/2" PDC (Nozzle 14/32 ")

Hydraulics Optimization Bit Run Report



Maximum Standpipe Pressure: 2800 psi

Well Information

Well Name:
AZNS-040
Drilling Contractor:
MSA TBA
Operator:
OEOC
Prepared For:
Mahan Sanat Arya
Location:

Prepared By:
/Petro Ray

Bit Run Data

Bit Number	1
Bit Size	8 1/2 in
Bit Type	VTD616

Hydraulic Details @ 4200 m

Flow Rate / Hole Diameter	58.8 gpm/in
System Pressure Loss	2212 psi
Downhole Motor / MWD Pressure Drop	200 psi
Bit Nozzle Pressure Drop	401 psi
Annular Pressure Drop	529 psi
Pump Input Hydraulic Power (M.E.=85%)	762 hp
Bit Hydraulic Horsepower	117 hp
Jet Impact Force	652 lbs
Impact Force/ Hole Area	11.49 lbs/in²
Hydrostatic Head	10133 psi
Equivalent Circulating Density	111.53 lbs/ft³
Cuttings Slip Velocity (Chip Size = .30 in)	10.45 m/min
Surface Equipment Length	46 m X 76.2 mm
Buoyant Collar Weight	43126 lbs

Recommended Hydraulics

Nozzles	14\14\14\14\14\14 32nds
Total Flow Area	0.9 in²

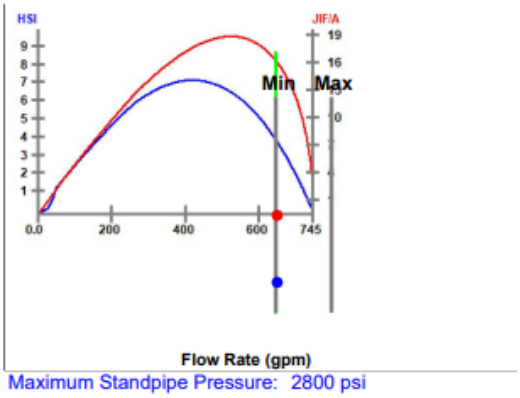
	IN	OUT
BIT HSI	1.94	2.06
JIF/Area	10.84	11.49 lbs/in²
Nozzle Velocity	54	54 m/sec
Flow Rate	500	500 gpm
SPP	2153	2613 psi
% Pressure @ Bit	17.6%	15.3%
Measured Depth	3986	4200 m
TVD	3896	4200 m
Mud Weight	100	106 lbs/ft³
Plastic Viscosity	10	15 cp
Yield Point	15	27 lbs/100 ft²

Drill String And Annular Velocity Data @ 4200 m

Depth m	OD X ID in	Length m	Hole Size in	Bore Loss psi	Ann. Loss psi	Ann. Vel. m/min	Est. Chip Vel. m/min	Crit Vel. m/min	Flow Type
0-1492	DP 5 x 4.276	1492	12.415	387.3	31.3	29	18	85	LAM
1492-3976	DP 5 x 4.276	2484	8.535	644.9	234.6	78	68	88	LAM
3976-3986	HWDP 5 x 3	10	8.535	14.3	.9	78	68	88	LAM
3986-4058	HWDP 5 x 3	72	8.500	103.2	6.9	79	69	89	LAM
4058-4059	DC 6.5 x 3	1	8.500	1.4	.3	125	114	93	TRANS
4059-4086	DC 6.5 x 2.813	27	8.500	52.8	7.1	125	114	93	TRANS
4086-4095	DC 6.5 x 2.75	9	8.500	19.6	2.4	125	114	93	TRANS
4095-4178	DC 6.5 x 2.813	83	8.500	162.2	21.9	125	114	93	TRANS
4178-4180	DC 8.25 x 3	2	8.500	2.9	216.5	892	882	201	TURB
4180-4191	DC 6.75 x 3.25	11	8.500	10.7	4.1	140	130	95	TRANS
4191-4200	DC 6.75 x 2.813	9	8.500	17.6	3.4	140	130	95	TRANS

8.9 8 1/2" PDC (Nozzle 18/32")

Hydraulics Optimization Bit Run Report



Maximum Standpipe Pressure: 2800 psi

Hydraulic Details @ 4000 m

Flow Rate / Hole Diameter	47.1 gpm/in
System Pressure Loss	1043 psi
Bit Nozzle Pressure Drop	308 psi
Annular Pressure Drop	261 psi
Pump Input Hydraulic Power (M.E.=85%)	315 hp
Bit Hydraulic Horsepower	72 hp
Jet Impact Force	414 lbs
Impact Force/ Hole Area	7.3 lbs/in²
Hydrostatic Head	7921 psi
Equivalent Circulating Density	89.87 lbs/ft³
Cuttings Slip Velocity (Chip Size = .30 in)	13.37 m/min
Surface Equipment Length	46 m X 76.2 mm
Buoyant Collar Weight	36497 lbs

Well Information

Well Name:
AZNS-040
Drilling Contractor:
MSA TBA
Operator:
OEOC
Prepared For:
Mahan Sanat Arya
Location:

Prepared By:
/Petro Ray

Bit Run Data

Bit Number	1
Bit Size	8 1/2 in
Bit Type	HAT217G

Recommended Hydraulics

Nozzles	18\18\18 32nds
Total Flow Area	0.74 in²

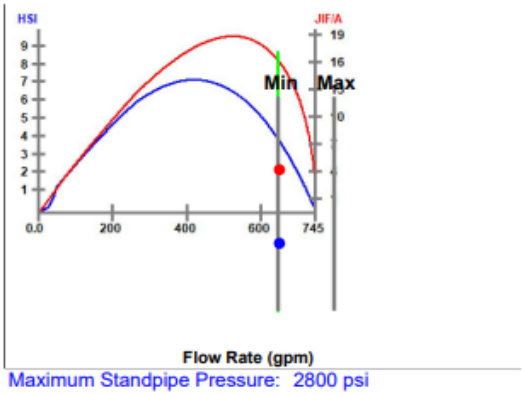
	IN	OUT
BIT HSI	1.27	1.27
JIF/Area	7.3	7.3 lbs/in²
Nozzle Velocity	53	53 m/sec
Flow Rate	400	400 gpm
SPP	1180	1351 psi
% Pressure @ Bit	26.1%	22.8%
Measured Depth	3986	4000 m
TVD	3896	4000 m
Mud Weight	87	87 lbs/ft³
Plastic Viscosity	10	15 cp
Yield Point	15	27 lbs/100 ft²

Drill String And Annular Velocity Data @ 4000 m

Depth m	OD X ID in	Length m	Hole Size in	Bore Loss psi	Ann. Loss psi	Ann. Vel. m/min	Est. Chip Vel. m/min	Crit Vel. m/min	Flow Type
0-1492	DP 5 x 4.276	1492	12.415	219.5	28.4	23	10	94	LAM
1492-3823	DP 5 x 4.276	2331	8.535	342.9	199.5	62	49	98	LAM
3823-3878	HWDP 5 x 3	55	8.535	44.7	4.7	62	49	98	LAM
3878-3879	DC 6.5 x 3	1	8.535	.8	.2	98	84	104	LAM
3879-3906	DC 6.5 x 2.813	27	8.535	29.9	6.2	98	84	104	LAM
3906-3915	DC 6.5 x 2.75	9	8.535	11.1	2.1	98	84	104	LAM
3915-3986	DC 6.5 x 2.813	71	8.535	78.6	16.4	98	84	104	LAM
3986-3998	DC 6.5 x 2.813	12	8.500	13.3	2.9	100	86	105	LAM
3998-3999	DC 6.5 x 3	1	8.500	.8	.2	100	86	105	LAM
3999-4000	DC 5.5 x 2.25	1	8.500	3.2	.1	71	58	100	LAM

8.10 8 1/2" RC BIT (Nozzle 15/32")

Hydraulics Optimization Bit Run Report



Maximum Standpipe Pressure: 2800 psi

Hydraulic Details @ 4000 m

Flow Rate / Hole Diameter	47.1 gpm/in
System Pressure Loss	1043 psi
Bit Nozzle Pressure Drop	639 psi
Annular Pressure Drop	261 psi
Pump Input Hydraulic Power (M.E.=85%)	393 hp
Bit Hydraulic Horsepower	149 hp
Jet Impact Force	597 lbs
Impact Force/ Hole Area	10.52 lbs/in²
Hydrostatic Head	7921 psi
Equivalent Circulating Density	89.87 lbs/ft³
Cuttings Slip Velocity (Chip Size = .30 in)	13.37 m/min
Surface Equipment Length	46 m X 76.2 mm
Buoyant Collar Weight	36497 lbs

Well Information

Well Name:
AZNS-040
Drilling Contractor:
MSA TBA
Operator:
OEOC
Prepared For:
Mahan Sanat Arya
Location:

Prepared By:
/Petro Ray

Bit Run Data

Bit Number	1
Bit Size	8 1/2 in
Bit Type	HAT217G

Recommended Hydraulics

Nozzles	15\15\15 32nds
Total Flow Area	0.51 in²

	IN	OUT
BIT HSI	2.63	2.63
JIF/Area	10.52	10.52 lbs/in²
Nozzle Velocity	77	77 m/sec
Flow Rate	400	400 gpm
SPP	1511	1682 psi
% Pressure @ Bit	42.3%	38.0%
Measured Depth	3986	4000 m
TVD	3896	4000 m
Mud Weight	87	87 lbs/ft³
Plastic Viscosity	10	15 cp
Yield Point	15	27 lbs/100 ft²

Drill String And Annular Velocity Data @ 4000 m

Depth m	OD X ID in	Length m	Hole Size in	Bore Loss psi	Ann. Loss psi	Ann. Vel. m/min	Est. Chip Vel. m/min	Crit Vel. m/min	Flow Type
0-1492	DP 5 x 4.276	1492	12.415	219.5	28.4	23	10	94	LAM
1492-3823	DP 5 x 4.276	2331	8.535	342.9	199.5	62	49	98	LAM
3823-3878	HWDP 5 x 3	55	8.535	44.7	4.7	62	49	98	LAM
3878-3879	DC 6.5 x 3	1	8.535	.8	.2	98	84	104	LAM
3879-3906	DC 6.5 x 2.813	27	8.535	29.9	6.2	98	84	104	LAM
3906-3915	DC 6.5 x 2.75	9	8.535	11.1	2.1	98	84	104	LAM
3915-3986	DC 6.5 x 2.813	71	8.535	78.6	16.4	98	84	104	LAM
3986-3998	DC 6.5 x 2.813	12	8.500	13.3	2.9	100	86	105	LAM
3998-3999	DC 6.5 x 3	1	8.500	.8	.2	100	86	105	LAM
3999-4000	DC 5.5 x 2.25	1	8.500	3.2	.1	71	58	100	LAM

9. Casing Program

9.1 Material Selection

Casing, Liner and Tubing Material selection for South Azadegan oilfield will be describe in the following.

Corrosive Environment Analysis and Strength Check

Based on the data of offset wells in this oilfield, there is a certain degree of corrosive problems in the 4 layers to be produced. Corrosive medium is principally composed of 2 parts: one is the corrosive gases like H2S and CO2 existing in the produced fluid; and the other is the high salinity formation water. The contents of main corrosive fluids in offset wells are shown in below Table:

Well No	*Reservoir formation*	*H₂S from PVT*	*CO₂ from PVT (mol%)*	*H₂S from Seperator*	*CO₂ from Separator*
AZN2	Sarvak	0.13	6.71	7.93/150psi	0.06/150psi
AZN4	Sarvak	0.06	5.44	0.02/150	6.32
	Kazhdumi	0	2.62	0	2.99/150psi
	Gadvan		2.64		
AZN6	Gadvan		2.47		2.566/400psi

1) Corrosion in this oilfield is a result of combined action of H2S, CO2 and high salinity formation water. The corrosive mechanism is complicated, and the corrosive rate on the casing is difficult to determine. Based on the NACE standard and the data from the Table, the producers result in severe corrosive environment. However, by the research on development conditions and technical indices, it can be found that although the corrosive fluid contents are high in the oilfield, the corrosive environment is not extremely serious as the following reasons:

2) In general, the temperature that cause the SSC/SCC is concentrated in the range from 20℃ to 30℃. The average depth of the horizontal section is 2,730m with the reservoir temperature of 96℃, and tensional stress on the casing does not exist in the horizontal section. Therefore, the influence of SSC/SCC may not be considered for the casing/liner.

3) Based on the RMDP, the production GOR in the first development phase should not be higher than the initial GOR, therefore, the flowing pressure is higher than the bubble point pressure all the time, that is, degassing problems do not exist in the horizontal section and some vertical section, the H2S and CO2 have no chance of separation out of oil phase, and it is lack of prerequisite for casing corrosion.

4) Based on above all, the corrosive environment of the casing string is not severe. The optimal selection for the casing string can be stated briefly as follows.

Material Selection of Casing String

In this oilfield, the corrosion of production string obviously occurs in 2 regions. One is the region where the pressure is above the bubble point; in which degassing problems do not exist, and the corrosion environment is relatively weak. Another one is the region where the pressure is below the bubble point; because of the decline of pressure, the concentration of corrosive gas will increase and the corrosion is relatively serious. Considering the corrosive environment is not serious in the low water cut production stage and the production practice in this oilfield has proved the L80 is fit for it, So, the L80 should be taken as the material of 9-5/8″ casing, 7″, 5 1/2″ and 4-1/2″ liner. In order to prevent the hazard of gas leakage after production degassing occurs, 9-5/8″ casing, 7″ and 4-1/2″ liner should be Gas tight thread. 2) The 110S is selected as the material of 9-5/8″, 7″ and4-1/2″ casing rather than L-80 due to the high strength

requirement in Fahliyan wells, what's more the 110S is fit for sour gas environment

9.2 Casing/Liner Design Factors

Table 9-1: Casing/Liner Design Factors

Design Factors			
Pipe Body		**Connection**	
Burst	1.1	Burst	1.1
Tension	1.6	Tension	1.6
Compression	1.300	Compression	1.3
Collapse	1.1		
Triaxial	1.25		

9.3 Casing/Liner Characterize Overview

Table 9-2: Casing/Liner Characterize Overview

Casing Size (in)	Depth (m)		Weight (PPF)	Grade	Connection	Drift Dia. (in)	B.S.F	C.S.F	A.S.F	Triaxial	Formation Remark
	From	To									
20”	0	150	133	K-55	BTS	18.543	Already run and set by CNPCI				AJ
13-3/8”	0	1492	68	N-80	BTS	12.259	Already run and set by CNPCI				GS
9-5/8”	0	3986	53.5 SD	110SS	Premium Gas Tight	8.535	1.17	1.99	2.33	1.42	GD
7”	3886	4200	32	110SS	Premium Gas Tight	6.094	1.24	1.06	1.51	1.49	FH

9-5/8” Casing (12-1/4” Hole Section)

String Design

Table 9-3: 9-5/8" Casing String Design

No.	Item	Description	Length (m)
1	Float Shoe	9-5/8”, 53.5 PPF (SD), 110ss, Premium Gas Tight Connection	0.7
2	Casing	9-5/8”, 53.5 PPF (SD), 110ss, Premium Gas Tight Connection	24
3	Float Collar	9-5/8”, 53.5 PPF (SD), 110ss, Premium Gas Tight Connection	0.7
6	Casing	9-5/8”, 53.5 PPF (SD), 110ss, Premium Gas Tight Connection	3960.6

Casing Specification

Pipe OD (in)	9-5/8
Weight (lb/ft)	53.5
Grade	110SS
Connection	Premium Gas Tight Conn.
Pipe ID (in)	8.535
Collapse Resistance (psi)	7,950
Pipe Body Internal Yield (psi)	10,900
Pipe Body Yield (Klbs)	1,710
Make Up Torque (ft-lb), optimum (for V. Top Conn)	19,080

Minimum Safety Factor

Minimum Safety Factors (9 5/8" Production Casing)						
Depth (MD) (m)	OD/Weight/Grade	Connection	Minimum Safety Factor (Abs)			
			Burst	Collapse	Axial	Triaxial
0	9 5/8", 53.500 ppf, 110ss	N.Vam	1.77 B6 C	2.98 C2	2.33 B8	2.18 B6
10			1.77 B6 C	2.99 C2	2.33 B8	2.18 B6
1328			1.51 B6 C	2.75 C1	3.41 B8	1.74 B6
1328			1.51 B6 C	2.72 C1	2.83 B8	1.82 B6
3836			1.18 B6 C	2.03 C1	3.10 B8	1.44 B6
3920			1.18 B6 C	2.01 C1	3.10 B8	1.43 B6
3986			1.17 B6 C	1.99 C1	3.10 B8	1.42 B6
C	Connection Critical					
B6	Tubing Leak					
B8	Injection Casing					
C1	Full/Partial Evacuation					
C2	Cementing					

Maximum Allowable Overpull

Maximum Allowable Overpull (9 5/8" Production Casing)	
Running Depth (MD) (m)	Max. Overpull (lbf)
0	1315472
10	547493
77	549562
153	547782
230	550144
305	552450
383	554868
460	547918
537	550280
610	552531
613	552643
690	554367
767	557367
843	559729
914	548469
920	548636
997	550999
1073	552405
1150	555723
1219	557861
1303	560447
1328	547759
1328	547760
1380	549355
1456	550442
1524	553799
1533	554079
1610	547354
1686	558803
1763	547711
1829	549737
1840	548480
1916	548611
1993	550974
2070	553336
2134	547906
2146	555698
2223	548429
2300	550791
2376	553153
2438	555068
2453	555516
2530	555647
2606	560240
2683	549147
2743	551006
2760	551509
2836	553872
2913	553685
2990	558596
3048	550130
3066	547503
3143	549866
3219	552228
3296	551722
3353	556337
3373	556952
3449	559314
3526	548222
3603	550584
3658	548770
3679	549441
3756	551803
3833	554165
3909	556527
3962	548213
3986	548940

7” Liner (8-1/2” Hole Section)

String Design

Table 9-4: 7" Liner String Design

No.	Item	Description	Length (m)
1	Float Shoe	7”, 32 PPF, 110ss, Premium Gas Tight Connection	0.5
2	Liner	7”, 32 PPF, 110ss, Premium Gas Tight Connection	22
3	Float Collar	7”, 32 PPF, 110ss, Premium Gas Tight Connection	0.5
4	Liner	7”, 32 PPF, 110ss, Premium Gas Tight Connection	11
5	LDC	7”, 32 PPF, 110ss, Premium Gas Tight Connection	0.5
6	Liner	7”, 32 PPF, 110ss, Premium Gas Tight Connection	277
7	Rot. Liner Hanger Assy.	7”, 32 PPF, 110ss, Premium Gas Tight Connection	1.5

Liner Specification

Pipe OD (in)	7”
Weight (lb/ft)	32
Grade	110SS
Connection	Premium Gas Tight Conn.
Pipe ID (in)	6.094
Wall Thickness (in)	0.453
Collapse Resistance (psi)	10,780
Pipe Body Internal Yield (psi)	12,460
Pipe Body Yield (Klbs)	1,025
Make Up Torque (ft-lb), optimum (for V. Top Conn)	13,860

Minimum Safety Factor

Minimum Safety Factors (7" Production Liner)						
Depth (MD) (m)	OD/Weight/Grade	Connection	Minimum Safety Factor (Abs)			
			Burst	Collapse	Axial	Triaxial
3886	7", 32.000 ppf, 110ss	N.Vam	1.27 B5 C	1.15 C5	(1.59) C5 C	1.54 B5
3986			1.26 B5 C	1.12 C5	(1.56) C5 C	1.53 B5
4123			1.24 B5 C	1.08 C5	(1.53) C5 C	1.50 B5
4200			1.24 B5 C	1.06 C5	(1.51) C5 C	1.49 B5
4200			1.24 B5 C	1.06 C5	(7.40) A5 C	1.49 B5
4200			1.24 B5 C	1.06 C5	(4.31) A5 C	1.49 B5
C	Connection Critical					
B5	Pressure Test					
C5	Full Evacuation Production					
A5	Green Cement Pressure Test(Axial)					
( )	Compression					

- **Maximum Allowable Overpull**

Maximum Allowable Overpull (7" Production Liner)	
Running Depth (MD) (m)	Max. Overpull (lbf)
0	328000
10	328213
76	328752
229	332189
305	333893
305	333907
314	334100
382	* 328350
458	* 330068
610	* 333476
611	* 333505
687	* 335223
840	* 330419
914	* 332093
916	* 332137
993	* 333855
1145	* 329050
1219	* 330710
1222	* 330769
1298	* 332487
1375	* 334205
1451	* 328579
1524	* 329327
1527	* 329401
1680	* 332837
1756	* 334556
1829	* 336186
1909	* 337992
1985	* 339711
2062	* 341429
2138	* 343147
2215	* 344866
2291	* 346584
2367	* 348302
2438	* 349903
2520	* 351739
2596	* 353457
2673	* 355175
2749	* 356894
2825	* 358612
2902	* 360330
2978	* 362049
3055	* 363767
3131	* 365485
3207	* 367204
3284	* 368922
3353	* 370478
3436	* 372359
3513	* 374077
3589	* 375795
3665	* 377514
3742	* 379232
3895	* 382669
3962	* 384196
3971	* 384387
4124	* 387824
4200	* 389542
* Based on Casing Strength only. Running string not included	

9.4 Centralizer Placement Plan

Table 9-5: Centralizer Placement Plan

Casing/Liner Size	Centralizer Type	Installation Patten
9-5/8”	Bow	1No. on First 5 Jts Above Shoe 1No. Each 5 Jts
	Rigid	1No. on Each 8 Jts
7”	Bow	1No. on First 5 Jts Above Shoe 1No. on Each 5 Jts
	Rigid	1No. on Second and Forth Jt below L.Hanger

NOTE: Final Centralizer placement will be issued in due time as stand-off report in cementing program based on real well data.

10. BOPs, Wellhead and Completion Schematic

10.1BOPs Specification and Configuration

NOTE: 26” and 17-1/2” Hole section already drilled by CNPCI.

Table 10-1: AZNS-040 Well BOP Specification

Casing String	Wellhead	BOP	Hole Size
13-3/8" Intermediate casing	Section “B” Casing Head Spool 20-3/4", 3K BTM x 13-5/8", 5K TOP	13-5/8", 5K Annular BOP 13-5/8", 10K Double Rams (Pipe Ram & Shear Ram) 13-5/8", 10K Single Ram 13-5/8", 10K Drilling Spool 13-5/8", 10K Spacer Spool Adapter Flange 13-5/8", 5K BTM x 10K Top	12-1/4″
9-5/8" Production casing	Section “C” Tubing Head Spool 13-5/8", 5K BTM x 11”, 10K TOP	13-5/8", 5K Annular BOP 13-5/8", 10K Double Rams (Pipe Ram & Shear Ram) 13-5/8", 10K Single Ram 13-5/8", 10K Drilling Spool 13-5/8", 10K Spacer Spool Adaptor Flange 11", 10K BTM x 13-5/8" 10K TOP	8-1/2”

12-1/4” Hole Section BOPs Arrangement

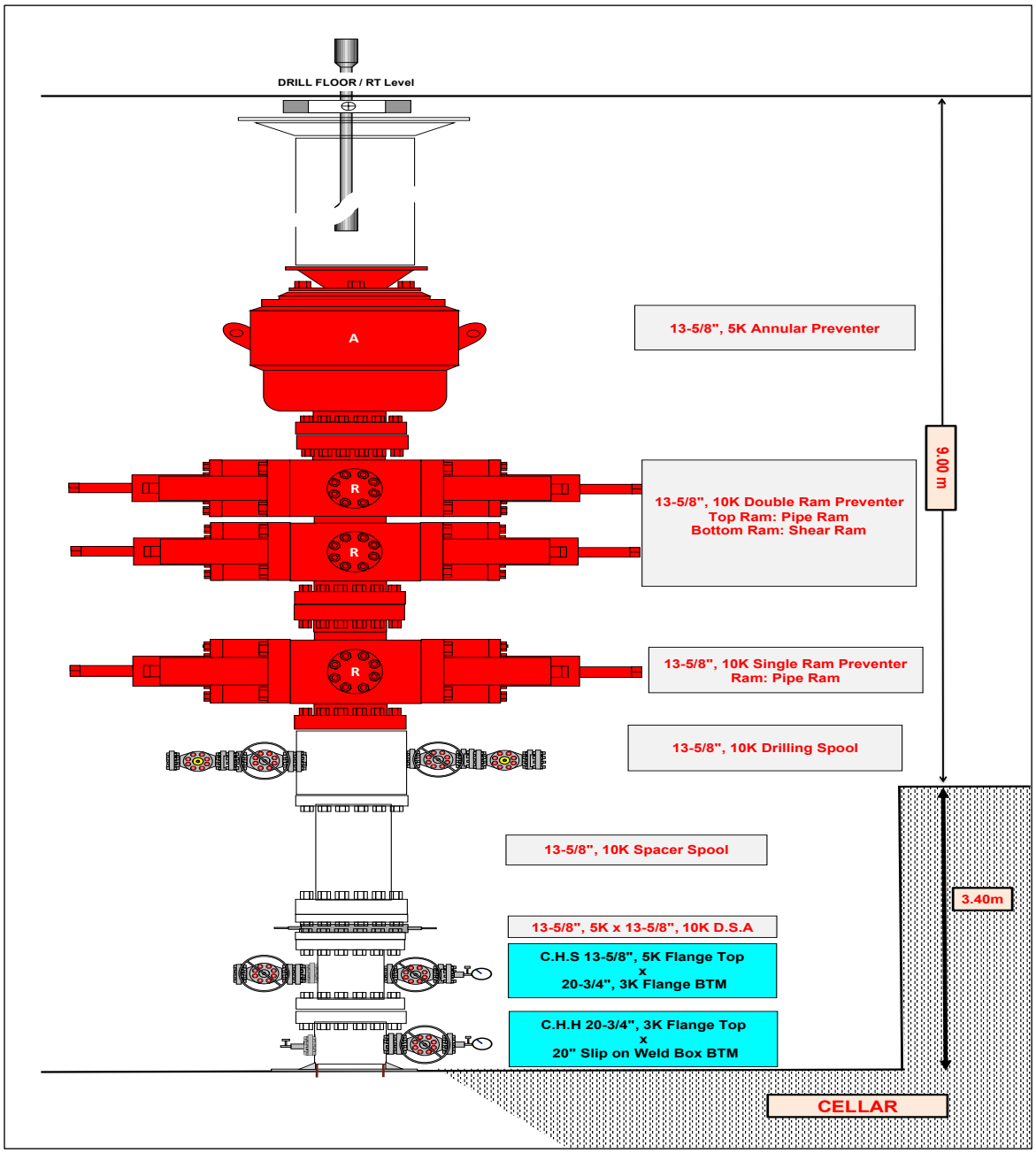


Figure 10-1: 12-1/4" Hole Section BOPs

8-1/2” Hole Section BOPs Arrangement

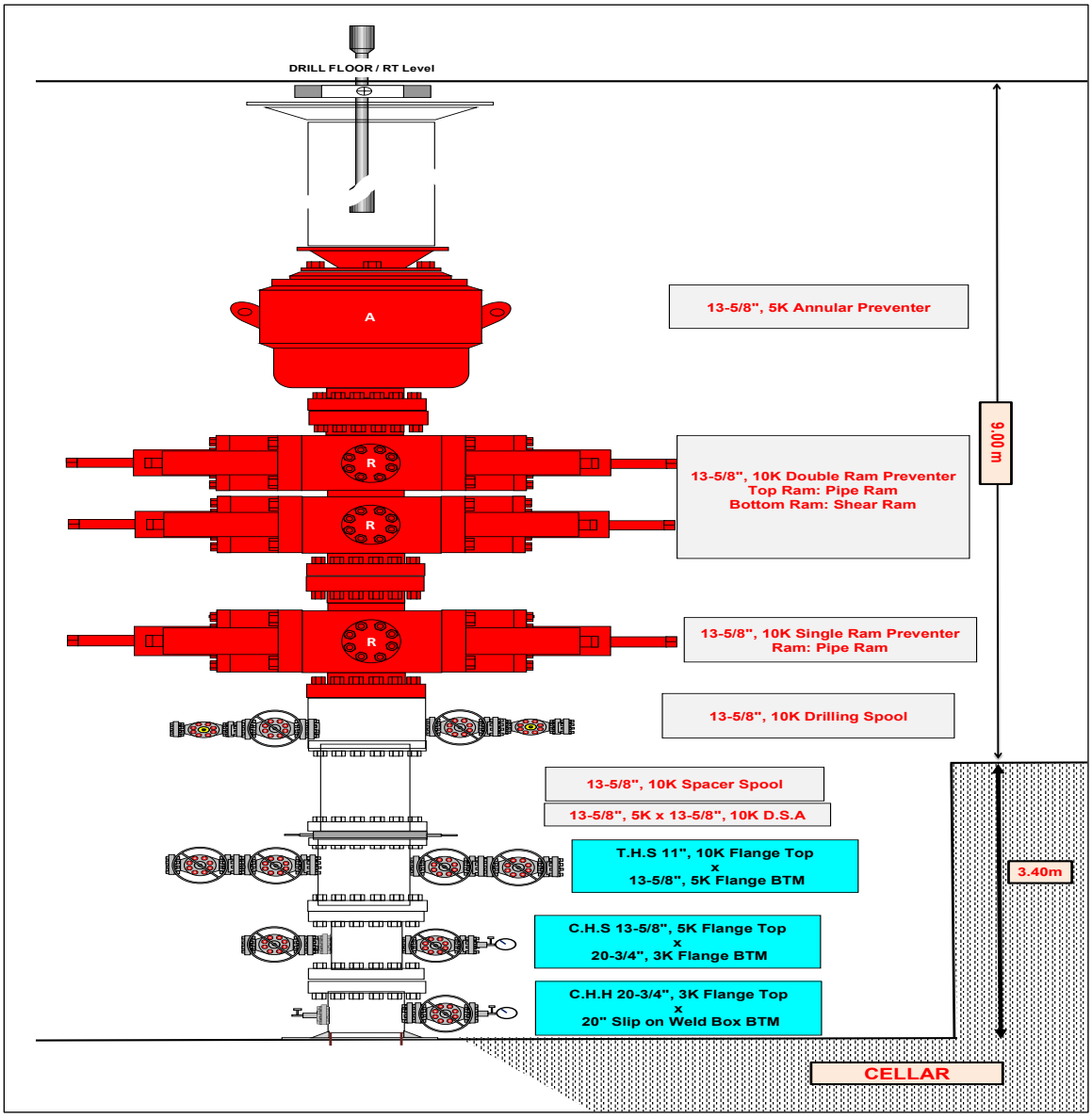


Figure 10-2: 8-1/2" Hole Section BOPs

Kill and Choke Manifold Arrangement

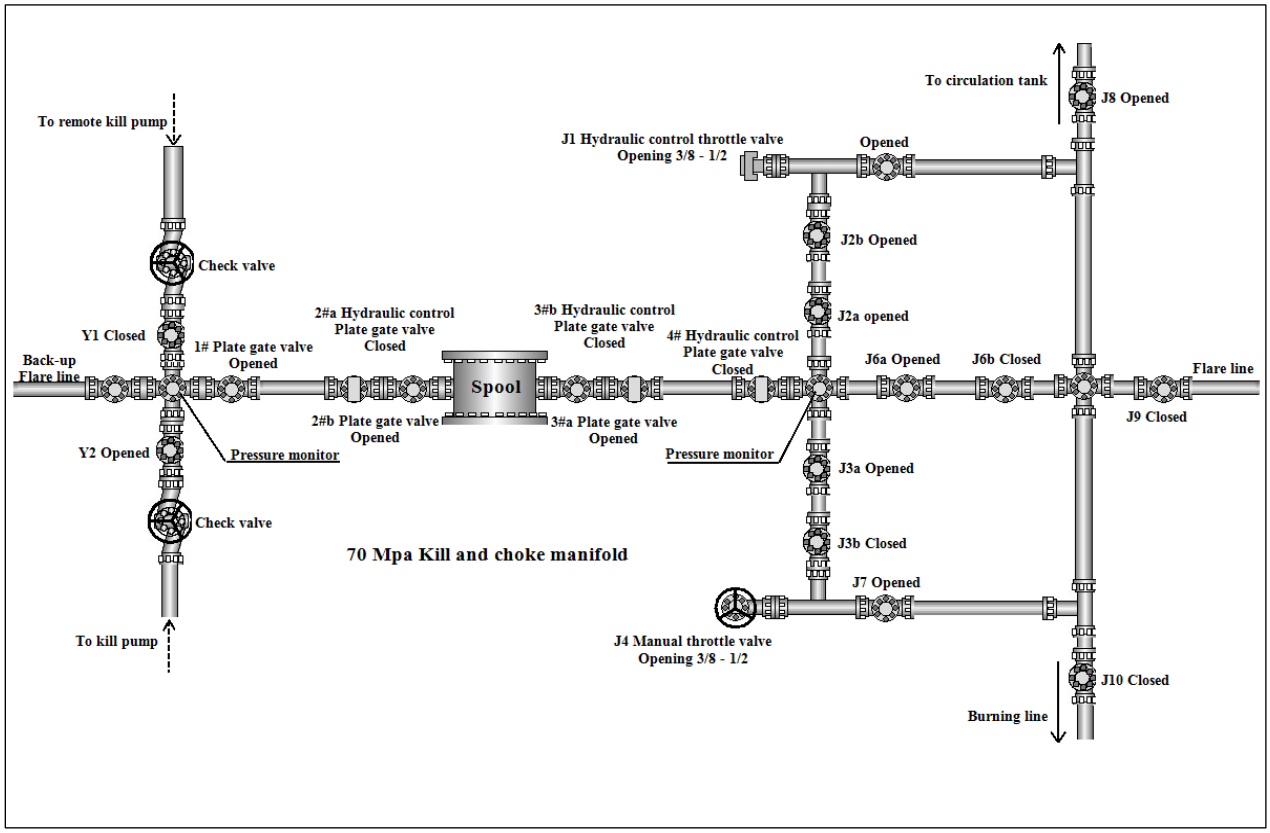


Figure 10-3: Kill and Choke Manifold Arrangement

10.2Wellhead

NOTE: Section "A" & "B" already installed by CNPCI.

Wellhead Material Selection Criteria

A. Selection of material class of Wellhead & X-mas Tree:

Table 10-2: H2S & CO2 Content in AZNS Offset Wells

Well No	Reservoir formation	H2S from PVT (mol%)	CO2 from PVT (mol%)
AZN2	Sarvak	0.13	6.71
AZN4	Sarvak	0.06	5.44
	Kuzhdumi	0	2.62
	Gadvan		2.64
AZN6	Gadvan		2.47

Considering the information above, API 6A 19th & NACE MR-0175 (2003) /ISO 1516, and Cl-content which is being most concerned, the South AZADEGA X-mas tree material and the tubing hanger should be FF0.5 class. As to the casing head, it's barely exposed to resident fluid, or only for quite a short time, when it comes to desert working condition, according to API 6A19th & NACE-MR-0175 (2003) /ISO 1516, the tubing spool should be EE0.5, the casing head material class should be DD0.5 (ANSI 4130) , stainless steel cladding in ring groove

B. Selection of PSL (product specification level):

According to API 6A 19th, The final PSL for wellhead and X-mas tree are as below:

Table 10-3: PSL for Wellhead and X-mas Tree

Item	Specification
Casing Head Housing assembly	PSL-1
Casing spool assembly	PSL-1
Tubing spool assembly	PSL-3
Tubing hanger	PSL-3G
X-Tree assembly	PSL-3G

C. Selection of PR (Performance Requirement)

According to API Specification 6A 19th edition, performance requirements are specific and unique to the product in the as-shipped condition. All products shall be designed to perform according to the requirements of API 6A clauses 4 and clauses 10 while in the pressure and temperature ranges and corrosion environment. Other requirements include load capability, cycles, and operating force or torque. There are two performance requirement levels: PR1 and PR2. Valves operating as safety valves shall be of the PR2 performance level and meet the requirements of Annex I in API 6A For SA wells, the guideline for PR selection is that all parts not contacting well fluids in normal production period are PR1, all parts contacting well fluids are PR2 as follow, Casing head housing and casing head spool assembly - PR1. Tubing head spool, tubing hanger and X-mas tree - PR2.

D. Temperature Class

Environment temp:
Well temp: 0.0244×4100+27.384=127℃
According to API 6A 19th, temp should be U (-18℃-121℃)

Table 10-4: Temperature Class for Wellhead and X-mas Tree in AZNS Oilfield

API TEMP. CLASSIFICATION	OPERATING RANGE
U	-18 TO 121

E. Designed Pressure

Base on wells data of this project, and API spec 6A 19TH, considering the planning casing program, Wellhead and X-mas tree system for the project is designed as followed:

Table 10-5: Wellhead and X-mas tree Designed Pressure in AZNS Oilfield

Item Well type	Casing Head Housing	Casing Spool	Tubing Spool	X-mas Tree
Savark	20-3/4" 3K	13-5/8" 5 K × 20 3/4" 3K	11" 5K × 13 5/8" 5 K	5K
Gadvan / Fahliyan	20-3/4" 3K	13-5/8" 5 K × 20 3/4" 3K	11" 10K × 13 5/8" 5 K	10K
Water disposal wells	20-3/4" 3K	13-5/8" 5 K × 20 3/4" 3K	11" 5K × 13 5/8" 5K	5K
Item	**Specification**			
Casing Head Housing assembly	MC:DD-0.5 3000PSI PSL-2 PR-2, TC: U			
Casing Spool assembly	MC:DD-0.5 5000PSI PSL-2 PR2, TC: U			
Tubing Spool assembly	MC: FF-05 5000PSI PSL-3, PR-2, TC: U (Savark & Water disposal wells)			
	MC: FF-0.5 10000PSI PSL-3G, PR-2, TC: U (K&G&F wells)			
Tubing hanger	MC: FF 0.5 disposal wells 5000PSI PSL-3G, PR2, TC: U (Savark & water)			
	MC: FF 0.5 10000PSI PSL-3G, PR2, TC: U (K&G&F wells)			
X-Tree assembly	MC: FF 0.5 5000PSI PSL-3G, PR-2, TC: U (Savark & Water disposal wells)			
	MC: FF 0.5 10000PSI PSL-3G, PR-2, TC: U (K&G&F wells)			

Wellhead Schematic

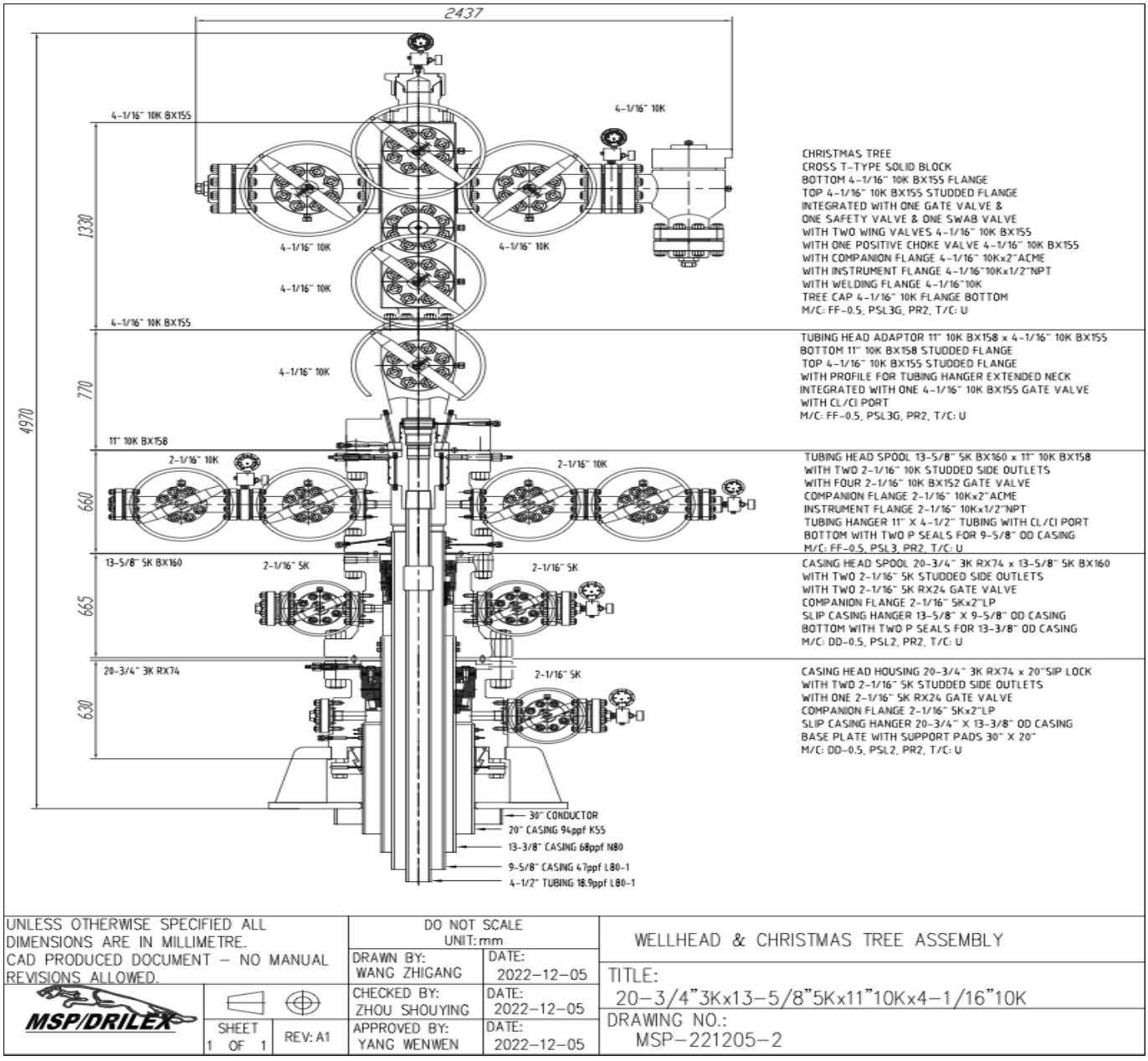


Figure 10-4: AZNS-040 Re-Entry Well Head Schematic

Wellhead Specification

NOTE: Section "A" and "B" already installed by CNPCI.

Table 10-6: AZNS-040 Wellhead Specification

20-3/4" x 13-5/8" x 11" x 4-1/16" x 10,000 psi WP for Fahliyan wells	
Section A	**Casing head housing** for 20" (94lb/ft) OD casing x 20 3/4" 3,000 psi API flange **has hold down bolts for wear bushing,** with 2 x 2- 1/16" 3000 psi studded side outlets, test ports and seals. One side outlet has **one** 2 1/16"x 3,000 psi manual gate valve, 2" LP threaded companion flange and 2" bull plug threaded for ½" NPT and fitted with 0-3,000 psi gauge and needle valve and, the other side to have 2 1/16" x 3,000 psi x 2" LP threaded flange and 2" bull plug. Casing slip and seal assembly self-energizing to hang 13 3/8" (68lb/ft) OD casing. This section is installed on 20" casing stub by way of **slip lock** with landing plate on 30" conductor pipe. **Material DD-0.5, PSL-2, PR-2, Temp U. Height: 630mm** Material: Forged alloy steel 4130 Automatic casing slip material: Low alloy steel 4130
Section B	**Casing head spool** with 13 5/8" 5,000 psi top flange x 20 3/4" 3,000 psi bottom flange has **hold down bolts for wear bushing** and bottom preparation to seal on 13 3/8" OD casing with test ports to test annular space and casing seals. One side outlet has **one** 2 1/16"x 5,000psi manual gate valve, 2" LP threaded companion flange and 2" bull plug on one side and, the other side has **one** 2 1/16"x 5,000psi manual gate valve, 2" LP threaded companion flange, 2" bull plug threaded for ½" NPT and fitted with 0-5,000psi gauge and needle valve. Casing slip and seal assembly self-energizing to hang 9 5/8" (47lb/ft) OD casing. **Material DD-0.5, PSL-2, PR-2, Temp U. Height: 665mm** Material: Forged low alloy steel 4130 13 5/8"x 9 5/8" automatic casing slip material: Low alloy steel 4130

Section C	**Tubing head spool** with 11" 10,000 psi top flange x 13 5/8" 5,000 psi bottom flange has **hold down bolts for wear bushing** and bottom preparation to seal on 9 5/8" OD casing with test ports to test annular space and casing seals C/W T.D.Screws with 410SS material Side outlets have 1.900" sharp vee VR threads on both sides, to have **two** 2 1/16"x 10,000psi manual gate valve, 2"ACME 10K female thread companion flange, 2" bull plug threaded for ½"NPT and fitted with 0-10,000psi gauge and needle valve on one side and, the other side to have the first 2 1/16"x 10,000psi manual gate valve, 2 1/16" 10,000 psi instrument flange installed with 0ne ½"NPT and fitted with 0-10,000psi gauge and needle valve, and the second 2 1/16" x 10,000 psi manual gate valve, 2" ACME 10K female thread companion flange and 2" bull plug.(manual G.valve to be non-rising stem w/back seat capability). Clear bore for passing 8-1/2" BHA. The 11" x 4 1/2" **tubing hanger** has 4 1/2" 18.9ppf premium conn. Top & bottom thread one continuous 1/4" NPT control line exit for SSSV, two channels for **3/8"** chemical injection line exit, one 1/4" line exit for DHPG. 4" type "H" BPV thread preparation, one metal seal & two S-SEALS to pack off the upper neck and two S-SEAL to pack off the body of hanger, and tonging preparation on the lower neck. Tubing hanger shall have factory-torqued double pin tubing pup joint 4 1/2" 18.9 ppf, L80, New Vam approximate length 1600mm to facilitate makeup. **Material FF-0.5, PSL-3, PR-2 with exception to tubing hanger: FF 0.5, PSL-3G, PR-2, Temp U. Height: 850mm** Material of body: Forged low alloy steel 410SS G. valves, material of seats/gates & stem: Forged alloy steel AISI 410SS Seats & gates coated by tungsten carbide by HVOF method.
Section D	**Tubing head adaptor (w/one integral manual G. valve)** 11" 10,000psi studded flange x 4 1/16" 10,000psi full bore manual gate valve with bottom preparation to seal on tubing hanger (one metal seal & 2 S. seal) and with test ports to test annular space and seals, top prepared with one continuous 1/4" NPT control line exit, one 1/4" line exit for DHPG and two channels for **3/8"** chemical injection line exit. (Manual G. valve to be non-rising stem w/back seat capability) The T-TYPE **Christmas tree** consists of the following: (1) **Solid block valve (T-Type):** 4 1/16" 10000psi bottom flange for tubing adaptor. 4 1/16" 10000psi lower manual gate valve, 4 1/16" 10000psi upper gate valve with SSV w/ hydraulic actuator, including wire line cutting function & lock open cap, (2) 4 1/16" 10000psi manual swab valve. With two 4 1/16" 10000psi studded outside for wing valves. With one 4 1/16" 10000psi studded top flange for tree cap. (3) one wing valve 4 1/16" 10,000 psi full-bore manual gate valve, one **instrument flange** with one 1/2" NPT ports c/w one 0-10,000 psi pressure gauge and needle valve installed and one 4 1/16"10,000psi Cage type positive choke with 90 deg outlet.(With 2 no of each Wrench): positive choke with 2 sets of bean sizes 8, 12, 16, 20, 24, 28, 32, 36, 40, 44, 48, 52, 56, 60, 64, 128/64 inch and with welding flange prepared for 4" pipe(specification to be confirmed). (4) The other wing with one 4 1/16" 10,000psi full-bore manual gate valve and one companion flange w/2" ACME 10K female thread. (5) Quick union type Tree Cap above, with 4 1/16" 10,000psi flange at bottom, ACME thread union on top,

	including 1/2"NPT port for needle valve and pressure gauge. **Material FF-0.5, PSL-3G, PR-2, Temp U. Height of integral tubing head adaptor and gate valve=800mm**. Material of body: 410ss G. valves, material of seats/gates & stem: Forged alloy steel, AISI 410SS Seats & gates coated by tungsten carbide by HVOF method.

10.3 Wellhead Specification and Configuration Section "A"

NOTE: This section WH already installed on well by CNPCI.

Casing Head Housing
20″ Slip On Lock x 20-3/4″ 3k FLG with two 2-1/16″ 3K Studded Side Outlet

List of Section A

Item	Description (Equipment and ancillary tools)	P/N & S/N	M/C	T/C	PSL	Qty.
1	20-3/4" 3K Casing head housing	P/N: 2891253 S/N: N130020	DD-0.5	U	PSL1	1 set
2	Casing hanger. Nom. 20-3/4" x 13-3/8"	P/N: 111005 S/N: N130014	DD-0.5	U	PSL1	1 set
3	Gate valve assembly 2-1/16" 3000psi	P/N: 002108A S/N: N120630	DD-0.5	U	PSL1	1 set

Drawing of Section A

A	B	C	D	E
13275mm	885 mm	3260 mm	511 mm	482 mm

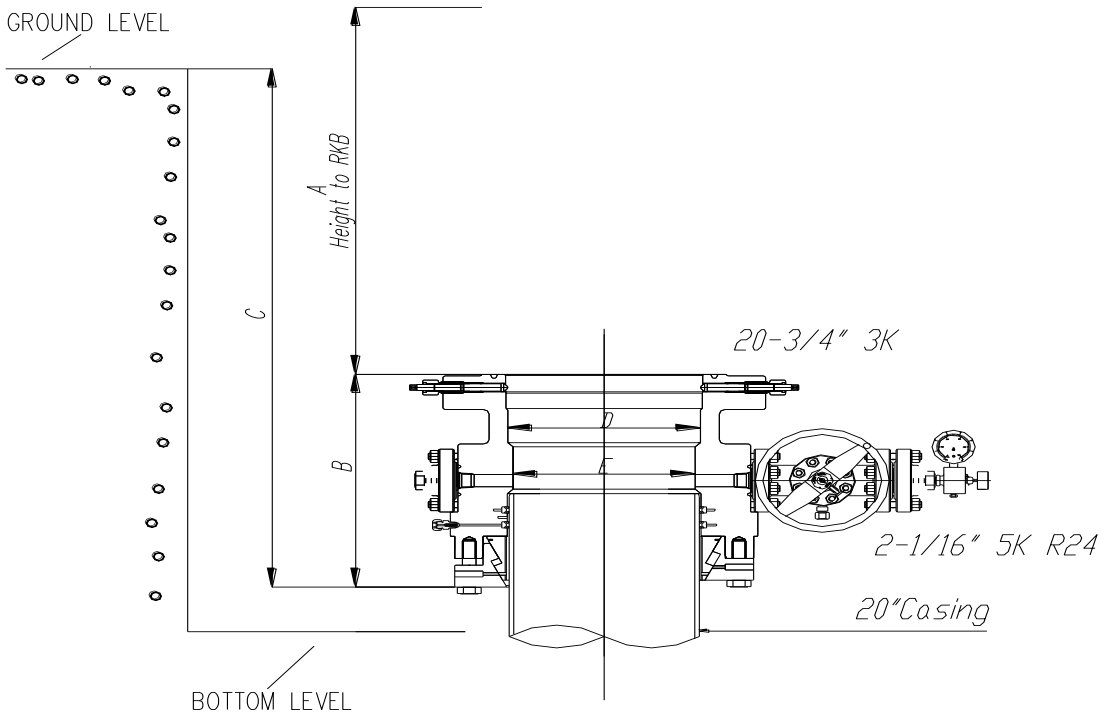


Figure 10-5: AZNS-040 Wellhead Section "A"

10.4 Wellhead Specification and Configuration Section "B"

NOTE: This section WH already installed on well by CNPCI.

Casing Head Spool
20-3/4″ 3k FLG x 13-5/8″ 5k FLG with two 2-1/16″5K Studded Side Outlet

List of Section B

Item	Description (Equipment and ancillary tools)	P/N & S/N	M/C	T/C	PSL	Qty
1	Casing head spool 20-3/4" 3000psi x 13-5/8" 5000psi	P/N: 2891254 S/N: N130011	DD-0.5	U	PSL1	1 set
2	Casing hanger Nom. 13-5/8" x 9-5/8"	P/N: 111001 S/N:	DD-0.5	U	PSL1	1 set
3	Gate valve assembly 2-1/16" 5000psi	P/N: 002108A S/N: N130285	DD-0.5	U	PSL1	1 set
4	Gate valve assembly 2-1/16" 5000psi	P/N: 002108A S/N: N120038	DD-0.5	U	PSL1	1 set

Drawing of Section A

A	B	C	D	E	F
12610mm	665 mm	234 mm	134.1 mm	334 mm	320 mm

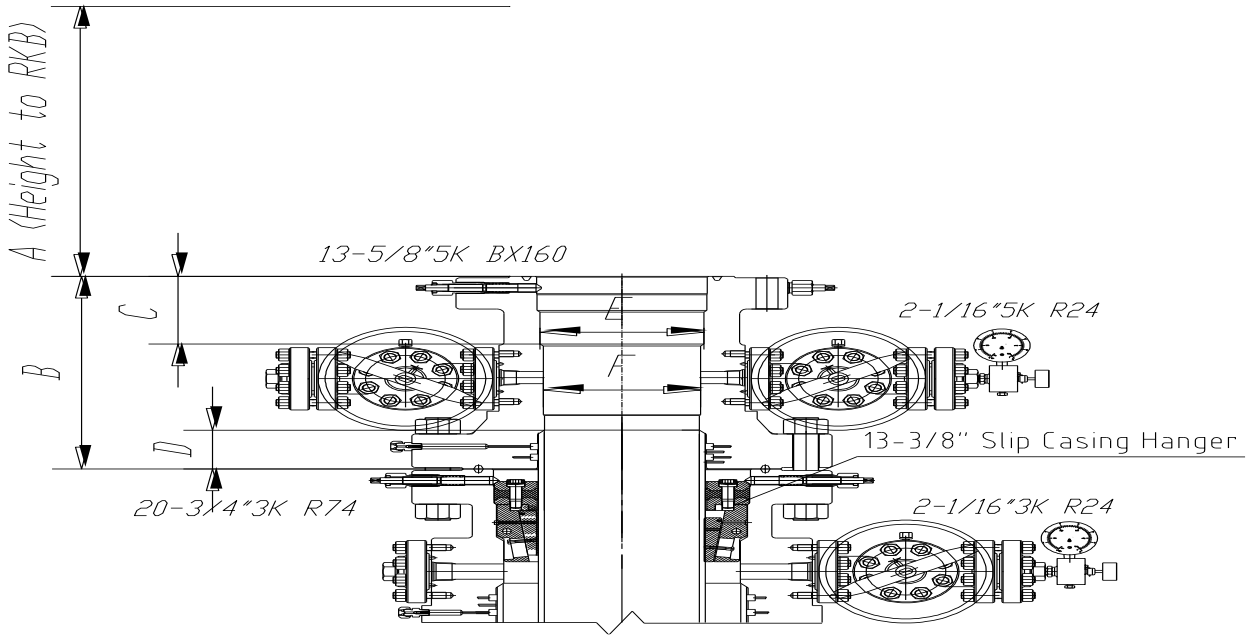


Figure 10-6: Wellhead Section "B"

10.5 Completion String Specification and Schematic

Typical Fahliyan Vertical Well Completion String Specification

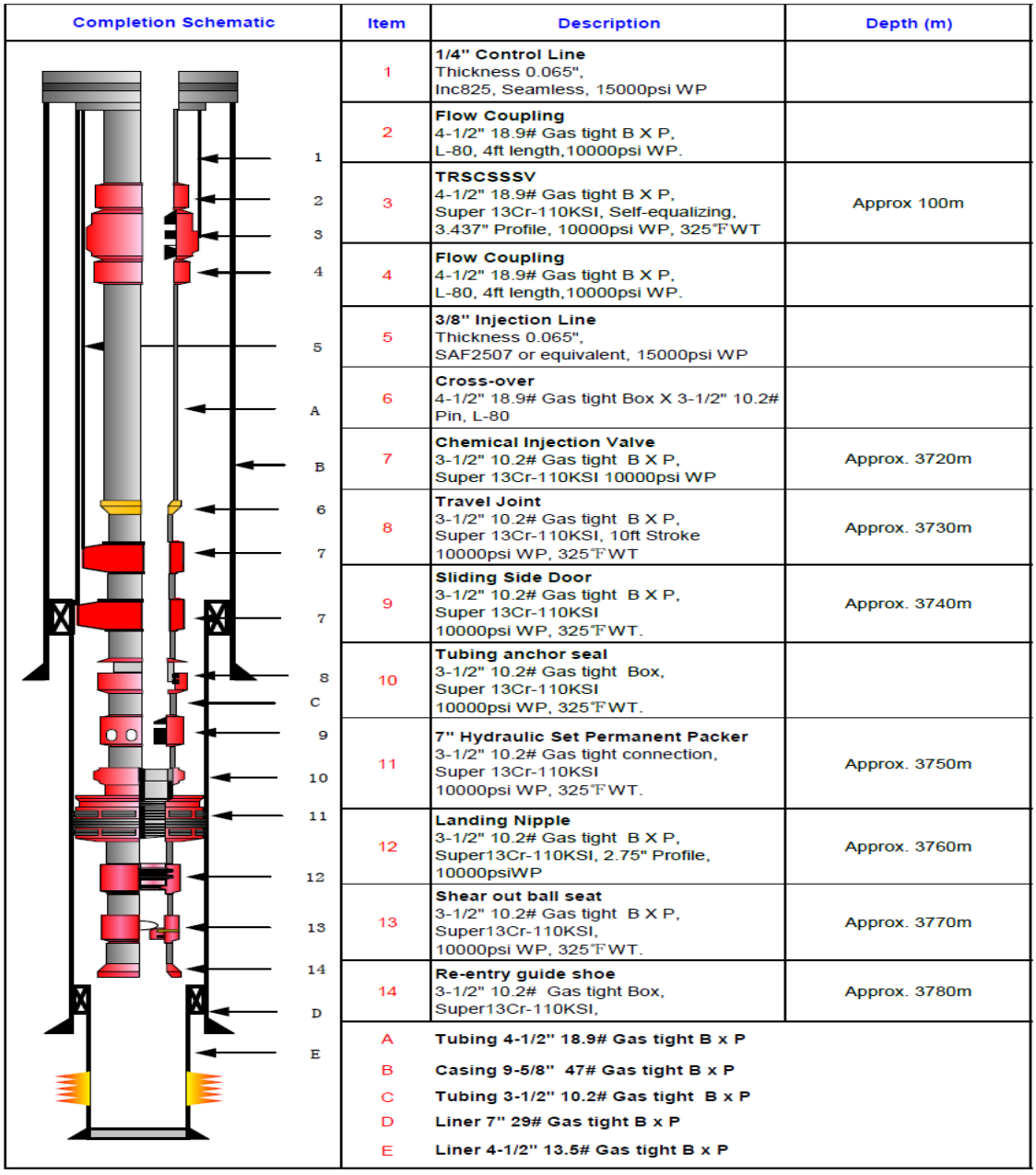
NOTE: Exact Completion string specification and schematic will be issued in due time with Completion program.

Table 10-7: Typical Fahliyan well Completion Specification

No.	Fahliyan wells (10,000 psi)	
1	1/4" Control Line	Thickness: 0.049", INC 825, 120m/Roll, 15,000Psi WP, type: Seamless
2	3/8" Injection Line. encapsulated	Thickness: 0.049", SAF2507, 4000m/Reel, 15,000Psi WP
3	32# Hydraulic Oil	200 Liter Per Tank
4	Fluted Adaptor	4-1/2" 18.9ppf, PREMIUM Pin X Pin, S13CR-110, 10,000 Psi WP
5	4-1/2” Flow Coupling	4-1/2" 18.9ppf, PREMIUM Box X Pin, S13CR-110, 6 FT, 10,000 Psi WP
6	4-/2” Tubing Retrievable Safety Valve. self-equalizing, flapper type	4-1/2" 18.9ppf PREMIUM Box x Pin, S13CR-110, Min ID: 3.437", 10,000 Psi WP, 325 F WT
7	4-1/2” Flow Coupling	4-1/2" 18.9 ppf, PREMIUM Box X Pin, S13CR-110, Min ID: 3.4", Length: 4 FT, WP: 10,000 Psi
8	Cross-over	4-1/2" 18.9#, premium Box X 3-1/2" 10.2# PREMIUM Pin, S13CR-110, Length:24"
9	3-1/2" Travel Joint	3-1/2" 10.2ppf PREMIUM B x P, Stroke Length: 10ft， S13CR-110, WP: 10,000 Psi, WT: 300 F
10	3-1/2” Chemical Injection Nipple, with permanent dual check valve	3-1/2" 10.2ppf PREMIUM B x P, S13CR-110， WP: 15,000Psi
11	Chemical Injection dual check Valve	S13CR-110, WP: 15,000Psi, WT: 325 F
12	3-1/2" Sliding side door, with integral profile	3-1/2" 10.2ppf PREMIUM B x P, S13CR-110, Min. ID: 2.75" Selective Profile, Open Down, WP:10,000 Psi, WT:300 F
13	7" x 3 ½” Hydraulic set Permanent Packer. With anchor latch and MOE	7” 26-29 PPF casing, 3-1/2" 10.2ppf PREMIUM B x P, S13CR-110, WP:10,000 Psi, WT:300 F
14	2.75"No-Go Landing nipple	3-1/2" 10.2ppf PREMIUM B x P, S13CR-110, 2.75” No-Go Profile, WP:10,000 Psi
15	3-1/2" Re-entry guide	3-1/2" 10.2ppf PREMIUM Box Up, S13CR-110, WP 10,000 psi

No.	Fahliyan wells (10,000 psi)	
16	Fitting for safety valve	Size 1/4” male × 1/4” Tube OD, WP 10000Psi, Material: INC 825
17	Fitting for injection nipple	Size 3/8” male × 3/8” Tube, WP 15,000Psi, Material: SAF2507
18	4-1/2" Cross-coupling tubing protector	Matched 4-1/2” 18.9ppf tubing, fit for 9-5/8" 47 to 53.5lb/ft Casing, including 4 slots, two for 1/4" control line, two for 3/8" injection line, Carbon steel.
19	3-1/2” Cross-coupling tubing protector	Matched 3-1/2” 10.2ppf PREMIUM, fit for 7” 29 to 35lb/ft Casing, including 4 slots, two for 1/4” control line, two for 3/8” injection line, Carbon steel.

Typical Fahliyan Vertical Well Completion Schematic



Item	Description	Depth (m)
1	**1/4" Control Line** Thickness 0.065", Inc825, Seamless, 15000psi WP	
2	**Flow Coupling** 4-1/2" 18.9# Gas tight B X P, L-80, 4ft length,10000psi WP.	
3	**TRSCSSSV** 4-1/2" 18.9# Gas tight B X P, Super 13Cr-110KSI, Self-equalizing, 3.437" Profile, 10000psi WP, 325°F WT	Approx 100m
4	**Flow Coupling** 4-1/2" 18.9# Gas tight B X P, L-80, 4ft length,10000psi WP.	
5	**3/8" Injection Line** Thickness 0.065", SAF2507 or equivalent, 15000psi WP	
6	**Cross-over** 4-1/2" 18.9# Gas tight Box X 3-1/2" 10.2# Pin, L-80	
7	**Chemical Injection Valve** 3-1/2" 10.2# Gas tight B X P, Super 13Cr-110KSI 10000psi WP	Approx. 3720m
8	**Travel Joint** 3-1/2" 10.2# Gas tight B X P, Super 13Cr-110KSI, 10ft Stroke 10000psi WP, 325°F WT	Approx. 3730m
9	**Sliding Side Door** 3-1/2" 10.2# Gas tight B X P, Super 13Cr-110KSI 10000psi WP, 325°F WT.	Approx. 3740m
10	**Tubing anchor seal** 3-1/2" 10.2# Gas tight Box, Super 13Cr-110KSI 10000psi WP, 325°F WT.	
11	**7" Hydraulic Set Permanent Packer** 3-1/2" 10.2# Gas tight connection, Super 13Cr-110KSI 10000psi WP, 325°F WT.	Approx. 3750m
12	**Landing Nipple** 3-1/2" 10.2# Gas tight B X P, Super13Cr-110KSI, 2.75" Profile, 10000psiWP	Approx. 3760m
13	**Shear out ball seat** 3-1/2" 10.2# Gas tight B X P, Super13Cr-110KSI, 10000psi WP, 325°F WT.	Approx. 3770m
14	**Re-entry guide shoe** 3-1/2" 10.2# Gas tight Box, Super13Cr-110KSI,	Approx. 3780m
A	**Tubing 4-1/2" 18.9# Gas tight B x P**	
B	**Casing 9-5/8" 47# Gas tight B x P**	
C	**Tubing 3-1/2" 10.2# Gas tight B x P**	
D	**Liner 7" 29# Gas tight B x P**	
E	**Liner 4-1/2" 13.5# Gas tight B x P**	

Figure 10-7: Typical Fahliyan Vertical Well Completion String Schematic

11. Waste Management Program

11.1Drilling Waste Management Design

The contractor is responsible for designing the optimum practices for treating the produced waste during the operation. The initial technical waste management proposal should be included the practices for minimizing, treating and recovering of the wastes.
According to the below categories:

- Water based mud containing minerals and natural synthetic polymers, and
- Cuttings contaminated with WBM.

The contractor is encouraged to offer possible practices to minimize the drilling fluids waste and its regeneration.

11.1.1 Water based mud containing minerals, naturals and synthetic polymers

- **Primary operation**

The proposed primary solids control package should be able to control drilling fluid properties, reducing of mud materials usage against conventional solid control equipment's, and significantly reduce the number of excess fluids generated throughout drilling each phase of the well.

- **Secondary Operation**

All excess mud and spillage (water based) should be collected in rig site pits and free surface fluid should be treated using a built for propose flocculation station (has to join with centrifuge) consisting of integral dosing and pumps along with a centrifuge for efficient, quick and controlled separation of the solids before the cleaned fluids is returned to active system or buffer tanks.

This system should be able to remove colloidal sized particles that cannot be removed with primary solids control system.

This system may use to clean up the remaining water-based mud or decrease weight of oil base mud at the end of the drilling fluids programmed.

11.1.2 Cuttings contaminated with WBM

Spud mud and water-based mud cuttings should be fixed and be ready for transport to disposal zone. For fixing the cutting, 100-150 kg cement shall be used for each cubic meters of cutting as well as 3-4-liter Sodium Silicate for each cubic meter of cutting. (Consumption of cement and sodium silicate varies in different formations). Gypsum is not allowed for cutting fixation. This method is physically Bonds Contaminants to Prevent Leaching to Environment. Final product will pass can test and in case of approval by company man it'll send to landfill.

11.2 Waste discharge and handling

The Company is responsible for designing of required pits and The Contractor is responsible for procedure for burring of waste cuttings in disposal zone; supervision and managing of requirement operation for deposit oil waste cutting.
The contractor is responsible for treating all the generated waste in site without any discharges in environment or damages. It is also the responsibility of the contractor for handling/discharge of generated waste after treatment.
The contractor manages and supervises the solids and fluid waste streams loading and off-loading on/from the designated areas and responsible for the loader along with the responsibility of the storage pits, tanks, etc.

11.3 Laboratory equipment's and analysis

The contractor should prepare suitable laboratory analysis equipment's to perform complete analysis services of waste matter. The provided equipment's shall make this opportunity to study the mentioned environmental policy of waste analysis.

11.4 Reports and comments

The contractor provides the following reports and documentations on contract duration:

- Daly and monthly waste management report.
- Final waste management report, including:
- Summarized amount of generated waste on each section,
- Total treated waste,
- Total chemical consumption,

. Cost monitoring report, and
. Special activities.

11.5 The equipment's installation and commission

. First location

The Maximum days for installation and commissioning of all equipment and necessary piping will be 7 days after rig spud. Contractor is responsible for any delay to installation and commissioning of equipment.

. Other locations

The Installation of all equipment should be completed during rig spud. Contractor is responsible for any delay to installation and commissioning of equipment.

11.6 Clean up process

After drilling operations and drilling rig removal, the well area, waste pit, and corral should be completely cleaned. The contractor shall remove the fixed cuttings to pre-determine locations (disposal) which can protect the environment is very effective.

11.7 Supplying services

Waste treatment package . Dewatering unit . Filter press
Transfer pump package . Air diaphragm pump . Submersible pump . Centrifugal pump
Storage tank . Waste water storage tank
Cutting treatment . Loader
Waste transportation . Cutting transport truck . Tunker truck for waste water transfer
Other . Wash gun . Laboratory and cabin

11.8 Waste management policy

Waste management services should be utilized to minimize generated waste and contaminated discharged cuttings during drilling operations. This helps keep the environment clean and ensures that our operations are in compliance with DOE (Department of Environment) legislations.

We provide cost-effective and optimum solutions from environmentally acceptable procedures to achieve zero-discharge plans and improve fluids recovery. The outcome reduces operational costs and enables recycling.

The waste is analyzed to study various parameters such as chloride, nitrate, and phosphate. Other analysis is carried out to monitor treatment processes on a regular basis during waste generation. Our waste management personnel are fully trained experienced and have the competencies in place to provide the best reliable services with concentrated support and the least liabilities.

11.9 Drilling waste management steps

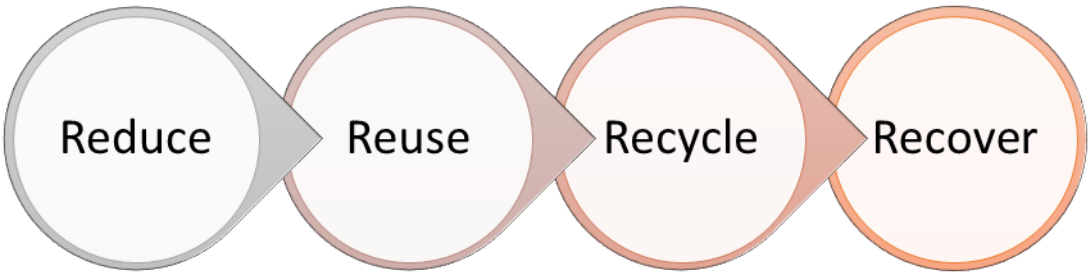


Figure 11.1 _ Drilling waste management procedure.

11.10 Calculation

11.10.1 Waste water treatment and disposal

Dewatering System

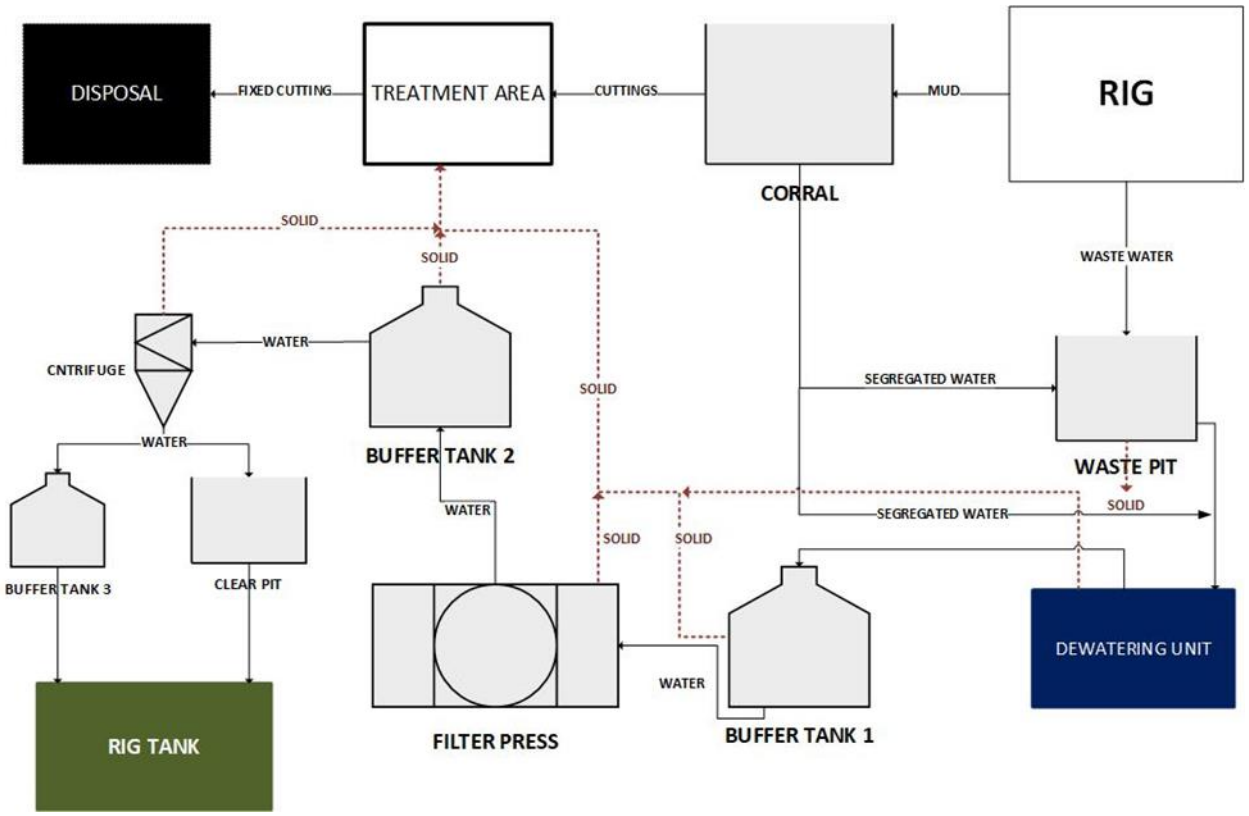


Figure 11.2_ Dewatering system flowchart.

Estimated Values

- Assumed 400 bbls in mud tank.
- The produced daily waste water of 50 m^3 and 60% mud are roughly estimated.

	unit					Total
Hole Size	Inch	**26**	**17.5**	**12.25**	**8.5**	-
Casing Size	Inch	18 5/8	13 3/8	9 5/8	7	-

Depth	Meter	152	1492	3986	4200	-
Mud system	-	-	-	Salt saturated KCl, etc. mud	Salt saturated KCl, etc. mud	-
Mud weight	PCF	-	-	75-85	100-106	-
Interval	Meter	152	1340	2493	214	-
Operating duration	Day(s)	0	0	53	44	**97**
Estimaded mud volume	M^3	-	-	987	533	**1520**
	bbl	-	-	6120	3302	**9422**
Estimated Treating waste water	M^3	-	-	3157	2449	**5605**
	bbl	-	-	19572	15181	**34753**

Table 11.1 _ Estimated treating waste water.

Dewatering Chemicals

	Application	Required concentration
Poly Aluminum chloride	Coagulant	0.5~1.5 Kg/m^3
Flocculent polymer	Flocculant	0.1~0.3 Kg/m^3
Hydrated lime	Coagulant	0.25~0.5 Kg/m^3
Soda-ash	Harness control	0.5 ~ 0.6 Kg/m^3 decreasig 1000 mg/Lit of Ca, at average
Caustic soda	pH control	0.5 ~ 0.6 Kg/m^3 Contion depended
Biocide	Microvial activity control	20 ml/ m^3

Table 11.2 _ Dewatering chemical.

Treated Water Properties

Item	Amount
pH	7~9
Total hardness	< 800 mg/litr
Turbidity	< 50 NTU

Nitrate	< 50 mg/litr
Aluminum	< 3 mg/litr
TSS	< 85 mg/litr
Sulfate	< 50 mg/litr
Ca^{2+}	< 600 mg/litr

Table 11.3 _ Treated water properties.

Material Consumption Estimation

This estimation, does not eliminate the responsibility of the test operator.

	unit					Total
Hole Size	Inch	26	17.5	12.25	8.5	-
Casing Size	Inch	18 5/8	13 3/8	9 5/8	7	-
Depth	Meter	152	1492	3986	4200	-
Interval	Meter	152	1340	2493	214	-
Operating duration	Day(s)	0	0	53	44	97
Poly alum	Dosage of kg/M^3			0.5~1.5		-
	25 kg sxs	-	-	126	98	224
Poly electrolite	Dosage of kg/M^3			0.1~0.3		
	25 kg sxs	-	-	25	20	45
Soda ash	Dosage of kg/M^3			1		
	25 kg sxs	-	-	126	98	224
Caustic soda	Dosage of kg/M^3			0.5~0.6		
	25 kg sxs	-	-	95	73	168
Biocide	Dosage of ml/M^3			20		
	25 kg sxs	-	-	631	490	1121
Lime	Dosage of kg/M^3			0.5~1		
	25 kg sxs	-	-	126	98	224
Citric acid	Dosage of kg/M^3			0.3~0.5		
	25 kg sxs	-	-	38	29	67

Table 11.4 _ Material consumption.

11.11 Cutting treatment

Cutting Removal Process



Figure 11.3 _ Cutting management process.

Estimated Cutting Volume

OBM mud is re-used in another location.

	unit					Total
Hole Size	Inch	26	17.5	12.25	8.5	-
Casing Size	Inch	18 5/8	13 3/8	9 5/8	7	-
Depth	Meter	152	1492	3986	4200	-
Interval	Meter	152	1340	2493	214	-
Operating duration	Day(s)	0	0	53	44	97
Mud system	-	-	-	Salt saturated KCl, etc. mud	Salt saturated KCl, etc. mud	-
Mud weight	PCF	-	-	75-85	100-106	-
Theorical hole cuttings	M^3	-	-	192	8	200
	bbl	-	-	1192	49	1242
Real hole cuttings	M^3	-	-	664	26	690
	bbl	-	-	4114	163	4277
Real mud solid	M^3	-	-	395	213	608
	bbl	-	-	2448	1321	3769
Total waste	M^3	-	-	1058	239	1298
	bbl	-	-	6562	1483	8045

Table 11.5 _ Estimated cutting volume.

Estimated Fixation Volume

	unit					Total
Hole Size	Inch	26	17.5	12.25	8.5	-
Casing Size	Inch	18 5/8	13 3/8	9 5/8	7	-
Depth	Meter	152	1492	3986	4200	-
Interval	Meter	152	1340	2493	214	-
Operating duration	Day(s)	0	0	53	44	97
Mud system	-	-	-	Salt saturated KCl, etc. mud	Salt saturated KCl, etc. mud	-
Mud weight	PCF	-	-	75-85	100-106	-
Total waste	M^3	-	-	1058	239	1298
	bbl	-	-	6562	1483	8045
Cement	Kg/M^3	50	50	50	-	-
	bbl	-	-	1058	239	1298
Silica	Kg/M^3	3	3	3	-	
	bbl	-	-	15	3	18

Table 11.6.Estimated fixation volume.

11.12 Waste Management Equipment's

Flocculation unit

This unit, has the bellow specifications:

- Fully automated system,
- Full 316 stainless steel material,
- Eqipped with high pressure level and pH sensors,
- Equipped with digital flow meter and totalizer to monitor flow rate, and
- Touch panel technology to control prose.



Figure 11.4 _ Flocculation unit.

400^{bbl} Dewatering unit

This tank provides more than 400 bbls capacity to store contaminated water for using in dewatering process and ability to adding chemicals such lime from hopper. Two Mixer 25 hp and Jet System give ability to holding any type of mud for hours without any stealing.



Figure 11.5 _ Conditional tank.

400^{bbl} Storage tank

Each tank provides more than 400 bbls capacity to store clear water for recycling.

High Performance Filter Unit

Filtration plays an important role in the natural treatment of waste water as it percolates through the soil. It is also a major part of most water treatment. Groundwater that has been softened, or treated through iron and manganese oxidation, requires filtration to remove floc created by coagulation or oxidation processes. Since surface water is subject to run-off and does not undergo natural filtration.
it must be filtered to remove particles and impurities. Filtration can be compared to a sieve or micro- strainer that traps suspended material between the grains of filter media. However, since most suspended particles can easily pass through the spaces between grains of the filter media, straining is the least important process in filtration. Filtration primarily depends on a combination of complex physical and chemical mechanisms, the most important being adsorption. Adsorption is the process of particles sticking onto the surface of the individual filter grains or onto the previously deposited materials. Forces that attract and hold particles to the grains are the same as those that work in coagulation and flocculation. In fact, coagulation and flocculation may occur in the filter bed, especially if coagulation and flocculation before filtration was not properly controlled. Incomplete coagulation can cause serious problems in filter operation.



Figure 11.6 _ High Performance Filter Unit.

Centrifugal pump

This pump can be utilized in different situation. 3 x 4 pump can be used for transferring processed water back to rig water tanks for reuse.



Figure 11.7 _ Centrifugal pump.

Submersible pump

These pumps designed to clear and waste water from water pits and tanks to designated areas or tanks. They are very easy to handle and can be very useful in case of transferring from bottom of tanks or pits.



Figure 11.8 _ Submersible pump.

200 bar wash down guns

These wash guns minimized water consumption and can be used on the shakers to unplug screens or washing al the area on the rig from rig floor to mud pits.



Figure 11.9_ 200 bar wash down guns.

Fluid transfer diaphragm pump

This pump works with air pressure and has the responsibility of transferring mud and contaminated water.
Three inches diaphragm pump transfers mud and waste from cellar to coral to keep cellar empty.



Figure 11.10_ 3” Diaphragm pump.

Two inches diaphragm pumps transfer waste water and mud between different spots on location. They also can be utilizing to transfer mud from mud tanks dead volume to treatment tanks.



Figure 11.11_ Diaphragm pump

These pumps designed to transfer clear and waste water from water pits and tanks to designated areas or tanks. They are very easy to handle and can be very useful in case of transferring from bottom of tanks or pits.

DWM laboratory cabin

Lab equipment for DWM is consisting of:

- Retort kit,
- API Filtrate test kit,
- Testing reagents and equipment for Chloride, Calcium, Hardness and PH, and
- Hatch Water Test kit





Figure 11.12_ Hatch Water Test kit

This kits analys all conserned parameters, including aluminum, iron, nitrat, nitrite, zinc, phosphate, sulfate, sulfide, copper and cyanide concentration.

Storage container



Figure 11.13 _ Storage container.

Loader



Figure 11.14 _ Loader.

12. Health, Safety and Environment

12.1 General Quality, Health, Safety & Environmental Requirements

- PPL/PEDEC's committed to conducting operations in a manner which minimizes the risks to the health and safety of those involved in, and affected by, such operations. PPL/PEDEC's also committed to conducting its operations in a manner which minimizes the impact on natural and 'built' environments in the area of operations.
- This section sets out minimum Quality, Health, Safety and Environmental (QHSE) requirements for the WORK that MSA and subcontractors will be required to comply with whilst conducting operations for PPL/PEDEC.
- PPL/PEDEC may perform audits and assessments of MSA's and its subcontractors' performance under the CONTRACT. PPL/PEDEC reserves the right without prior notice to conduct quality, health, safety, security and environmental audits of the operations.
- MSA shall at all times be responsible for performing the WORK in a manner so as to protect against loss of life, injury to people, damage to property and the environment. The detailed manner means and methods of performing the WORK shall be under the control and direction of MSA.
- Prior to the start of Operations, certain QHSE processes must be completed. These QHSE processes include developing the final QHSE plan format and frequency of any QHSE audits, permitting and licensing and PEDEC's approvals. MSA shall ensure that sufficient resources are dedicated to this task to ensure completion prior to mobilization.
- PPL/PEDEC may at any time stop the WORK or any portion of the WORK without penalty to PPL/PEDE Cif the WORK is not conducted in accordance with the Quality, Health, Safety or Environmental guidelines specified in this CONTRACT or by the laws of the appropriate jurisdiction. PPL/PEDEC reserves the right to require MSA to remove any person from the operations in the case of non-compliance with QHSE procedures. PPL/PEDEC may also prohibit the use of equipment and plant in cases where that equipment or plant does not meet the required standards or is deemed unsafe.
- MSA shall ensure that activities associated with this CONTRACT are conducted in compliance with applicable national and local HSE laws and

regulations, MSA's standards and PEDEC's requirements and IADC documents.

- Limits of Authority
- PEDEC's Representative(s) do not have the authority to relax health, safety or environment standards. If they judge the operation to be inadequate with regards to any aspect of quality, health, safety, security or the environment, then the operation may be immediately suspended.
- MSA shall be required to develop a project-specific QHSE plan to manage the WORK and associated risks. This plan shall, as a minimum, address topics included in the list detailed in Item 7. Where existing MSA QHSE publications detail the topics listed, it is sufficient to either include them verbatim, or provide reference to where they may be found in MSA's QHSE Manuals. The final project-specific QHSE plan shall satisfy MSA standards, PPL/PEDEC requirements and applicable HSE laws and regulations.
- MSA should establish an effective HSE system and implement it strictly.
- MSA must adhere to the standards and regulations of Health, Safe and Environment constituted by the local government.
- MSA must strictly adhere to the procedures / standards and regulations of Health, Safety and Environment promulgated by NIOC.
- MSA should comply with all applicable local government regulations including guidelines and recommended practices and any other Iranian authorities having jurisdiction.
- All personnel should respect the local culture, traditions and social values of the indigenous population of the Iran and South Azadegan region.
- MSA shall hire qualified and experienced worker and familiar his personnel with hazards especially with H2S and Mine and UXO (unexploded ordinance) according to related procedure.
- MSA shall conduct proper training about UXO (unexploded ordinance) and mine awareness to all personnel

12.2 Health & Safety Management

As per national/local and client (PPL/PEDEC) standards and MSA/OEOC HSE plan. Following are minimum requirement for the AZNS project:

- MSA/OEOC shall take all necessary safety measures in relation to the WORK to be provided and shall conduct itself and its work force in such a way as to comply at all times with the provisions of the national and local health and safety regulations pertinent to the contracted Scope of WORK.
- MSA/OEOC shall also comply with all and any of PPL/PEDEC'S safety standards and requirements as particularly applicable to the Operating Area and/or as may be imposed by PPL/PEDEC from time to time.
- MSA/OEOC and its subcontractors shall take all necessary steps to provide a safe and healthy working environment for employees, PPL/PEDEC's personnel and third parties in the performance of the WORK.
- MSA/OEOC shall operate a management system that ensures:
- MSA/OEOC's and sub contractors' personnel operate safe systems of work at all times for the performance of the WORK.
- Hazards associated with the work are identified and appropriate actions are taken to prevent accidents.
- Risks to safety and health of personnel involved in the WORK are identified and assessed, and any actions and measurements are implemented to either eliminate or reduce these risks to a level that is considered as low as reasonably practicable, given the nature of the WORK.
- Personnel are aware of the identified hazards and risks to their safety and health and to the safety and health of others. The detailed safe working procedures established by MSA/OEOC to ensure that the WORK will be conducted with due regard to the QHSE aspects, shall be detailed or referenced in the QHSE plan. Copies of the source documents shall be provided to PPL/PEDEC if the procedures are referenced only.

QHSE Representative:

- MSA/OEOC shall provide a dedicated, on-crew QHSE Representative. The primary responsibility of the QHSE Representative shall include, but not be limited to, the maintenance and monitoring of the implementation of HSE guidelines and procedures and where required provide HSE awareness training for the crew. Such activities shall be approved and actively supported by MSA/OEOC at corporate management level.

Training and Competence:

- MSA/OEOC shall ensure that its employees and those of its subcontractors are trained and competent to perform the WORK in a safe and environmentally responsible manner. Such training and demonstration of

competence will include but not be limited to vocational, safety, emergency, environmental and other training required by regulatory agencies and IADC/OGP guidelines. MSA/OEOC's senior staff shall have received training in safety management.

Accident/Incident Reporting:

- MSA/OEOC shall report all incidents. Any hazardous incident/near miss involving PPL/PEDEC, MSA/OEOC's, or any third-party personnel, plant or equipment, shall be reported to PPL/PEDEC, whether or not injury to personnel or damage/loss to plant or equipment resulted. The following incidents shall be verbally reported immediately to the on-site PPL/PEDEC Representative(s): damage to property or equipment belonging to PPL/PEDEC, third parties, MSA/OEOC or its subcontractors; injury to personnel of PPL/PEDEC, third parties, subcontractors or MSA/OEOC; near misses which may have had serious potential consequences; environmental damage; media attention. An initial written report, containing factual information only, shall be prepared by MSA/OEOC and delivered to the PPL/PEDEC Representative within twenty-four (24) hours of the incident.
- PPL/PEDEC reserves the right to participate in the investigation of any incident arising out of performance of the WORK.
- MSA/OEOC and all subcontractors shall have an accident and incident reporting system, details of which shall be included in the QHSE plan.
- MSA/OEOC shall maintain and forward to PPL/PEDEC, a monthly summary of its safety performance.
- MSA/OEOC shall prepare and forward to PPL/PEDEC, a weekly report stating all incidents and near misses that occurred during that week.
- All MSA/OEOC and subcontractor personnel and PPL/PEDEC Representative shall attend safety meetings on a regular basis. PPL/PEDEC Representative shall be given a copy of the minutes of the meetings.

Safety Equipment & Personal Protective Equipment (PPE) requirements:

- MSA/OEOC shall ensure that appropriate PPE compliant to national standards is supplied to MSA/OEOC's personnel involved in the WORK. PPL/PEDEC intends to use water-based mud and hence MSA/OEOC's Personnel that may be exposed to this must be provided with appropriate PPE.
- MSA/OEOC shall be solely liable for all costs in equipping its own and all other personnel on contract to, or under the control of, MSA/OEOC with the required safety equipment. MSA/OEOC will provide to all MSA/OEOC's

Personnel appropriate PPE throughout the WORK. Any defective PPE will be replaced without delay. Personnel without the correct PPE will not be allowed to continue work, until replacement PPE has been issued. If any person is disallowed to work owing to inadequate PPE, then PPL/PEDEC will deduct his day- rate from MSA/OEOC's invoice.

- MSA/OEOC shall detail in the QHSE plan the PPE proposed for issue to the crew. This proposal should include the arrangements for replacement (due to wear and tear and damage) of this equipment.
- This well is committed to develop for oil in a responsible manner that promotes employees' and local inhabitants' health.
- A trained Paramedic and proper MEDIVAC equipment and ambulance shall be stationed on the drilling rig during the drilling campaign.
- An emergency medical evacuation plan shall be coordinated, and shall keep in touch with emergency response centers manned by qualified doctor 24-hours per day in Tehran or Ahwaz city.
- Medical emergency report and investigation shall be performed as soon as possible in accordance with Medical Emergency Procedures.
- MSA/OEOC shall, as a minimum, conduct the WORK and provide personnel, equipment, facilities and services consistent with good oil field practices and OGP Guidelines. An assessment of the Contract Area with regard to "Environmental and Occupational Health Risks" and "Local Medical Support" shall be provided by MSA/OEOC in the QHSE plan. This assessment and the resulting "Health Risk Assessment Score" will be discussed with PPL/PEDEC, and a final assessment will be agreed.
- To ensure the fitness of those personnel engaged to undertake the WORK, MSA/OEOC shall conduct periodic medical examinations of its staff and ensure that its subcontractors' personnel are also subject to periodic medical examination. As a minimum this shall include pre-hire medical examinations of temporary local personnel and bi-annual medical examinations of its permanent and expatriate staff. Evidence should be made available to PPL/PEDEC to verify an appropriate program of immunizations consistent with the recommendations of the World Health Organization for the area of operations.
- MSA/OEOC shall be responsible for the cost of medical examinations as required to provide PPL/PEDEC with medical certificates which confirm the fitness of MSA/OEOC's assigned personnel for the WORK. Each drilling rig shall be equipped with a hospital, doctor and an ambulance type-2 level.

Drug Abuse and Control:

- MSA/OEOC warrants that it shall obtain valid written consent of all Personnel to a search of property of such Personnel when PPL/PEDEC suspects that such Personnel have in their possession drugs and/or other prohibited substances or any other item likely to prejudice safety or security. PPL/PEDE will provide a representative to attend the search.
- MSA/OEOC warrants that it shall obtain valid written consent of all Personnel to medical testing for the presence of drugs or other prohibited substances.
- MSA/OEOC further agrees that any Personnel who refuse to provide such consent or consents or withdraws such consent or consents shall at PPL/PEDEC's option be removed from the WORK and shall not, at PPL/PEDEC's option, be assigned to any future services to be provided by MSA/OEOC for PPL/PEDEC.
- MSA/OEOC further agrees that it shall immediately notify PPL/PEDEC in the event that MSA/OEOC becomes aware that any of its Personnel has been found in possession of drugs and/or any other prohibited substance. MSA/OEOC is required to notify PPL/PEDEC in the event that any Personnel provided for the WORK have previously been found in possession, or have abused drugs and/or other substances prohibited at Location. The acceptance of such Personnel for the WORK shall be at the sole discretion of PPL/PEDEC.

Camp Standards:

- MSA/OEOC shall abide by all statutory Iranian governmental health and safety regulations, and maintain acceptable standards for cleanliness and hygiene for quarters, camps and the Operating Area(s) in general. MSA/OEOC shall detail the standards and facilities proposed for its camp in the QHSE plan.

Emergency Preparedness:

- MSA/OEOC and its subcontractors shall have emergency preparedness and response plans plus procedures available at all times throughout the duration of the CONTRACT.
- MSA/OEOC shall take all necessary measures to ensure that:

I. Potential emergency situations have been identified and assessed;
II. An emergency response plan, including clear responsibilities, is established;
III. MSA/OEOC's management, employees and third parties are aware of response procedures and responsibilities;

IV. Regular drills and exercises are carried out to test the emergency response organization and procedures and that any resultant issues are properly addressed.

MSA/OEOC shall notify PPL/PEDEC immediately with respect to any emergency situation and shall provide regular updates on emergency planning procedures, tests etc.

MSA/OEOC shall detail in the QHSE plan a list of potential contingencies for which emergency response procedures are planned. The QHSE plan should also detail a schedule of proposed exercises and drills to verify the effectiveness of the proposed arrangements in place. Development and implementation of the above plan shall incorporate any PPL/PEDEC assigned personnel to the project.

12.3 Environmental Management

- COMPANY shall comply with any and all laws, rules and regulations of governmental agencies having jurisdiction, which now exist or may be promulgated during the term of the CONTRACT, relating to the control and prevention of damage to the environment.
- Without prejudice to above, COMPANY shall ensure that all personnel use the highest prevailing standards of care and diligence to prevent unlawful disposal of all waste oil, water and other waste materials or fluids which may accumulate or be caused by COMPANY's performance of the CONTRACT. COMPANY shall not dispose of any objects, scrap, waste, rubbish, equipment, fluids or fuel into the surrounding area or allow anything to be stored loose where it can be driven into surrounding areas by wind. COMPANY shall take all reasonable precautions to prevent pollution of any nature or kind from its performance of the WORK. COMPANY shall have contingency plans and procedures in place for such events.
- In the event of any disposal prohibited by Item 3.2 above, whether by dumping, spilling, emission or otherwise arising from COMPANY's performance or non-performance of the CONTRACT, COMPANY shall be fully liable for and bear the costs of removal of any such item(s) dumped, spilled, emitted or otherwise disposed regardless of the sole, concurrent or contributory negligence or breach of duty (statutory or otherwise) of PPL/PEDEC. In the event of any such disposal, COMPANY shall immediately undertake all reasonable remedial and protective measures required in such circumstances.
- COMPANY shall notify PPL/PEDEC immediately with respect to any pollution, loss, damage, claim or demand (or occurrence which may give rise to the same) resulting from the CONTRACT. COMPANY shall report to

PPL/PEDEC any incidents of non-compliance with legislative and regulatory environmental requirements that occur during the performance of the WORK. Without relieving COMPANY of any of its obligations under Item 3.2, if COMPANY does not promptly proceed to perform its obligations under this Item 3, PPL/PEDEC may take part to any degree it deems necessary in the control and removal of any pollution, dumping, spillage or contamination which is the responsibility of COMPANY under this Item 3.2, or which is due to any act or omission of COMPANY. COMPANY shall reimburse PPL/PEDEC for costs arising from such actions of PPL/PEDEC upon receipt of invoice from PPL/PEDEC. PPL/PEDEC shall have the right to deduct such costs from any sum due or becoming due to COMPANY under the CONTRACT.

- COMPANY shall operate a management system that is able to demonstrate compliance with environmental laws, rules and regulations applicable to its activities, products and services.
- MSA/OEOC and its subcontractors shall take all necessary steps to ensure that the environmental impacts of their work are responsibly managed and shall take all necessary measures to ensure that:
- COMPANY's personnel are aware of the environmental impact of the COMPANY's operations and are sufficiently trained and competent to perform the WORK.
- COMPANY's personnel operate systems, which will ensure that significant environmental impacts of WORK are identified and managed in accordance with a commitment to legal compliance and continual improvement.
- The environmental risks of operations associated with the WORK are evaluated and appropriate actions are taken to prevent accidents and reduce pollution, and that contingency plans are in place in the event of an incident.
- COMPANY will have adequate processes for monitoring the environmental performance of their WORK.
- COMPANY will, as a minimum, conduct the WORK and provide personnel, equipment, facilities and services consistent with IADC Environmental Guidelines. COMPANY's proposals for environmental protection should be detailed in the QHSE Plan. This should include, but not be limited to:

- procedures and precautions to minimize damage to crops and vegetation, the local ecology and disturbance of wildlife and farm animals;
- recycling and waste disposal;
- Clean-up and restoration of seismic lines, Rig sites access roads and trails, fixed sites occupied during the survey.

- An ecological team shall be set up by COMPANY, with the purpose of collecting all waste, rubbish, food scraps, office waste and other discarded material from camps, created by the WORK. Such activities shall be performed on a daily basis. Such material shall be appropriately disposed of, according to PPL/PEDEC approved methods or existing methods established by COMPANY, which are proven to be acceptable to PPL/PEDEC. Well location and camp site shall be delivered to PPL/PEDEC without any waste, rubbish,

food scraps, office waste and other discarded material. PPL/PEDEC has the right to hold Completion certificate till cleaning the well location and camp site by COMPANY.

- The access road and well location construction shall be accorded with construction program of well location. It minimizes any impacts on the local residents, prevent all soil contamination, prevent all ground water contamination, and prevent all surface water contamination. Cutting and waste water pit shall be lining and drainage system shall be good and to avoid farm and season canal contamination.
- Management regulation for transportation truck, drilling operation machine and tools shall be carried out strictly to avoid damage farm and trees, etc.
- Well site shall be flat and clean. All the equipment, materials, drilling tools shall be put in order.
- Control the rig noise to decrease the influence to surroundings while drilling at the inhabitation area.
- COMPANY shall perform on site segregation of wastes prior to transfer to the transfer area and/or delivery to the transportation COMPANY in accordance with the following waste hazard classification.
- Non-Hazardous Wastes (Re-usable, Recyclable and Recoverable Wastes)
- Ferrous and non-ferrous metals, plastics, and wood wastes, PETs and potentially paper and glass
- Materials shall be separately segregated and provision may be made for size reduction and containerization equipment.
- The COMPANY/Generators are encouraged to seek for recycle/re-use alternatives based on the Local available facilities, for final disposal of these materials.
- It should be noted that landfill disposal of metal waste materials and wood wastes shall be prohibited. For wood wastes re-use of and/or salvage of commercial timber, or deliver/release to
- local community are recommended.

Hazardous Wastes:

- Hazardous Waste, such as clinic waste, oil waste, waste produced by painting, insulation works shall be duly segregated.

Facilities of environment protection supplied to well site:

- Mud purifying circulation system
- Cooling water spraying and circulating system for drilling pump
- Special recovery tank for deposed oil
- Reserve house for valuable mud chemicals
- Automatic mud weighting supply system
- BOP stack is equipped according to drilling design

- Define a separation area for gathering all waste according to waste management procedure.
- Requirement of environment protection on location Sewage pit Area of the sewage pit: 1500 m2
- Prior to spud mud pit and sewage pit must comply with the standard. Install cooling water spraying and circulating system for drilling pump and sewage circulating system for cleaning rig floor and equipment.
- Install BOP stack and pressure test shall be operated according to the drilling design to avoid blowout.
- Install oil supply lines and check whether there is any leakage.

Management of environment protection on location:

- In drilling operation keep mud and sewage from overflowing. If it happens, clean up timely.
- Mud chemicals shall be stored according to the standard and shall not be scattered around well site. If mud chemicals scatter while loading or unloading, clean it up at once. The sack of mud chemicals shall be recovered. Do not dispose it on site.
- Enhance water management at well location and do not wash equipment with fresh water. Stop water flowing for long time. Overhead tanks shall be utilized.
- Oil management shall be carefully at the well site. No oil dirt on site, no dripping and leaking at oil tank and manifold system area.
- After the well is completed, reinforce the sewage pit， solidify the disposed mud and have environment procedure fulfilled.
- Install fuel supply lines and keep them in good condition.
- All the drilling fluid materials shall be recovered and shall not be left at the well site after completion.
- The environmental protection personnel from COMPANY and operator side shall check and accept the well site environment according to the regulation after completion. Only well site environment is qualified, the well can be delivered.
- Transition Area for Hazardous Waste shall also be provided by each COMPANY in his allocated area,
- PPL/PEDEC shall approve each area before use as well. The transition area of hazardous waste shall be:

- Paved, fenced, marked and illuminated
- Equipped with appropriate Fire Extinguisher and other emergency response equipment (e.g., Eye-Wash and Emergency Shower, appropriate First Aid kits, neutralizer substances etc.).
- In compliance with the requirements of handling, transportation and storage of hazardous waste (based on usage chemical MSDS).
- Configured to accommodate a safe physical separation of incompatible waste materials (e.g., acids, and alkalis, organic materials and strong oxidizing agents).

- Capable enough to provide storage for the entire range of hazardous wastes generated during the work.
- The area shall be provided with Spill Containment and means of preventing run off to and from the area.
- All waste containers to be stored shall be appropriately labeled.
- Hazardous Waste shall be transferred in a timely manner, within a maximum period of 48 hours, for any period in excess of this permission shall be granted from PPL/PEDEC.

  - Standard of utilization and discharging of wastes on location
  - Conduct drilling operation according to the design, drilling mud is needed before spud.
  - Sewage that cannot be recovered shall be put into the pit, do not discharge
  - Make full use of waste water. Water in mud pump and water brake must be utilized by circulating. Water that is used to wash drill floor, shall be recovered after deposition. The recovery rate of waste water shall reach 85%.
  - Disposed oil shall be recovered entirely. Recovery rate shall be 100%. Dumping of waste oil shall be prohibited.
  - In drilling operation, non-chrome thinner high temperature mud system shall be used often if engineering is qualified.
  - While drilling in special area adopt mud creation inhabitation techniques to decrease the waste mud creation.
  - After completion, all the mud chemicals, bentonite, clays shall be recovered and shall not be left at the well site. Mud in purifying tank shall also be recovered after completion.
  - In case of any flammable or noxious gas encountered, steps shall be taken at once to prevent kicks and blowout. Gas shall be guided into ignition equipment to burn it away.
  - Burning material that may result in serious smog or horrible smell at the well site is not allowed. For cement or barite that may result in granular dust pollution, operation shall be performed in a closed system.
  - For abandoned well, filled with cement plugs to seal the oil, gas or water formation.
  - COMPANY/Generators are required to transport all generated wastes to the location/facilities designated, and licensed by authorities and/or approved/verified by the PPL/PEDEC (as appropriate).
  - All wastes shall be transferred by approved Sub- COMPANY (which could be a 3rd party Sub-COMPANY of the COMPANY/Generator itself). The Waste Transporter and any other waste management service COMPANY shall be licensed in accordance with local regulations and are in compliance with such regulations. PPL/PEDEC shall verify and approve the Waste Transportation COMPANY by checking licenses and qualifications, audit and inspections, verification of human resources and received trainings etc.

- The transportation COMPANY shall ensure that all wastes are transported in a safe and responsible manner.

12.4 Security

As per PPL/PEDEC as main client and MSA/OEOC agreed plan.

- Security for the WORK shall be arranged and administered by the MSA/OEOC. MSA/OEOC shall be fully advised in writing covering all aspects of security upon. Company shall liaise with PPL/PEDEC QHSE to develop operational and contingency plans prior to Mobilization.
- The Security guards shall be appropriately housed and provided for. Transport for the Security guards to perform field activities shall be made available according to standard Iranian Oilfield procedures.
- All MSA/OEOC's personnel involved in the WORK shall possess the necessary security clearance documentation required by the Iranian authorities to work within the Operating Area.

12.5 Quality

1. Ensure that relevant experiences from the appraisal and production wells (if any) are captured and included in the operation programs /procedures.
2. Check lists to be prepared for all equipment, available for PPL/PEDEC prior to start of operations.
3. Procedures planned for all operations, available for PPL/PEDEC prior to start operations.
4. High focus and follow up of KPIs (Key Performance Indicators) by NIOC procedures.
5. Drilling risk assessment.
6. All goods and equipment shall be checked as per engineering design and inspection certificates, PPL/PEDEC approval shall be obtained before running any goods in hole.
7. Routine reports shall be submitted to PPL/PEDEC and any commented shall be corrected and informed engineering team.
   - Rig Specification shall be submitted to PPL/PEDEC any commented shall be corrected and informed engineering team.
   - Rig Layout shall be submitted to PPL/PEDEC any commented shall be corrected and informed engineering team.
   - Rig certification (COF) and record shall be submitted to PPL/PEDEC any commented shall be corrected and informed engineering team.
   - Apply PPL/PEDEC inspection work instructions.

- Drilling String inspection must be done before and during operation based on client request and its certificates and record shall be submitted to PPL/PEDEC.
- Pre-Spud check list must be verified by PPL/PEDEC before operation.
- NDT certificates and record shall be submitted to PPL/PEDEC.

12.6 Meetings

HSE meetings shall be convened on a monthly basis with an agenda to discuss the general issues about HSE, any change in the contractor, in the nature and risk of the activities, HSE training program, safety analysis of the activities, as well as investigation and analysis of the accidents. The Contractor's site manager and HSE Representative shall attend such meetings and report to the Client's site manager and HSE representative.

Weekly HSE meeting shall be conducted with the presence of MSA/OEOC HSE & Rig Contractor HSE Supervisor.

12.7 List of Minimum Required Contents

As per below tables:

Table 12-1: Minimum Onshore Rig HSE Requirements

Minimum Onshore Rig HSE Requirements	
ITEM/DESCRIPTION	**Specification**
General Safety Equipment	
Muster station sign board	2
Warning Sign (H2S Danger, Danger zone...)	As needed.
MSDS For all Chemicals	As needed.
T- Card Board	1 Panel in entrance of Rig & 1 Panel in Camp

Wind Socks	3
Fall Arrestor	Minimum 6 sets with different length Install in (Monkey board, 2 for Monkey Board ladders, Crown, In mast for hanging block, Under Rig floor)
Bug Blower (Explosion proof)	Minimum 3 no. Set in (Rig Floor, Cellar area, Spare)
Explosion Proof Torches	Minimum 6 (Company man, Tool pusher, HSE, Medic, Maintenance crew, Mud Crew)
Rescue Equipment	As needed.
Security Well Marker	By MSA/OEOC HSE Concern
Derrick man's escape system I.e. REM chute	By MSA/OEOC HSE Concern
Emergency shower with eye wash to be situated at assigned positions	Minimum 3 sets (Rig floor, chemical matters storage and handling areas)
Portable eye wash set	Minimum 5 sets
Life Ring	2
Spill Kit	3
EX Radio	At least 7
First Aid Kits	Rig Floor, Medic room, Camp Boss, Client Office, Store, Camp Galley
Rig Site Training Materials	Required
Personnel Protective Equipment	
Coverall+ Safety Helmet + Safety boots for all personnel on location and for visitors.	As needed.
Safety Eye protection & Ear Protections (Ear muff & Ear Plug)	As needed.
Anti-Acid (Apron, Gloves, Boot, Face Shield, chemical suits) for all operational personnel	As needed.
Fire Fighting Equipment	
Fire extinguishers: "Powder-50 kg"	7(In Rig site) 4(In Camp site)
Fire extinguishers: "Powder-12 kg" for all fire potential areas.	By MSA/OEOC HSE Concern

Fire extinguishers "CO2- 9 KG" For all electrical fire potential areas.	By MSA/OEOC HSE Concern
Fire extinguishers "Foam-45 Liter" For fire potential areas where required.	4 No. wheeled (Set in Diesel tanks, Engine generators in Rig site and Camp site)
Fire extinguishers "ABC 20 lbs" for all fire potential areas where required.	By MSA/OEOC HSE Concern
Mobile Foam Package (Fiberglass Body-Z4 Inductor) + 1000 Liter FFFP 3% Foam + S4 Low expansion Nuzzle (B.S)	By MSA/OEOC HSE Concern
Fire hoses with (B.S) couplings placed around the location as per firefighting plan.	By MSA/OEOC HSE Concern
Fire blankets for galley, Welding & Cutting operations	As needed.
Fire Fighting Suits	Minimum 4 sets in each Rig site and Camp site
Water pump diesel, capable of charging the fire main in case of deep well failure.	1(Rig site)1(Camp site)
Medical Equipment	
Minimum requirements for clinic and ambulance and relative medical equipment in accordance with fallowing requirements: 1. IOGP publication, Standards for local medical support, Report No: 6.44/222 2. Managing health for field operations in oil and gas activities, IOGP Report Number 343 (Revision – Version 1/ October 2011)	Required
Stretcher	Scoop Stretcher (1No.) Basket Stretcher with lifting belts and hooks) (1No.) Long Back Board with spider straps and immobilizer) (1No.) Army Stretcher(2No.)
Ambulance	1No. (B Type)

Table 12-2: Minimum Rig-less HSE Requirements for South Azadegan Project

Minimum Rig-less HSE Requirements for South Azadegan Project	
ITEM/DESCRIPTION	**Specification**
General Safety Equipment	
Eye wash & safety Shower to be situated at assigned position	(1) chemical storage and handling area
Muster station Sign board	2No.
Warning Sign (H2S Danger, Danger zone, ...)	Required
MSDS For all Chemicals	Required
T- Card Board	Required
First Aid Kits	Hospital, All offices
Wind Socks	2No.
Air Fans (Explosion proof)	1No.
Explosion Proof Torches	Required
Rescue Equipment	Required
Spill kit	3
EX Radio	At least 6
Security Well Marker	Required
Rig Site Training Materials	Required
Personnel Protective Equipment	
Coverall + Safety Helmet + boots for all personnel on location and for visitors.	Required
Safety Eye protection & Ear Protections (Ear Plug)	Required
Anti-Acid (Apron, Gloves, Boot, Face Shield, chemical suits) for all operational personnel	Required
Fire Fighting Equipment	
fire extinguishers: "Powder-50 kg"	At least 3No.
Fire extinguishers: "Powder-12 kg" for all fire potential areas.	Required
Fire extinguishers "CO2- 9 KG" For all electrical fire potential areas.	Required

Fire extinguishers "ABC 20 lbs" for all fire potential areas where required.	Required
Fire hoses with (B.S) couplings placed around the location as per firefighting plan.	Required
Fire blankets for galley, Welding & Cutting operations	Required
Fire truck (10000 Liter Water-1000 Liter Foam) Or (Firefighting systems)	1No.
Fire Fighting Suits	As per fire fighters' number
***The following equipment recommended meanwhile the risk can be retained in case that the company risk assessment considering the risk is so far as reasonably foreseeable. The measures and precautions also need to be as low as reasonably practicable.**	
Medical Equipment	
Minimum requirements for clinic and ambulance and relative medical equipment in accordance with fallowing requirements: 1. IOGP publication, Standards for local medical support, Report No: 6.44/222 2. Managing health for field operations in oil and gas activities, IOGP Report Number 343 (Revision – Version 1/ October 2011)	Required
Stretcher	Scoop Stretcher (1No.) Back Board (1No.) Army Stretcher (2No.)
Ambulance	1(B Type)

12.8 Well Control

1. **Installation Requirements of BOP Stack and Control System:**

- The distance between BOP control lines and relief lines shall be not less than 1m. Steel Gap Bridge should be placed in such a way to protect lines from damage by vehicle crossing. Sundries shall not be packed on the control line shield, on which welding operations are forbidden.
- The air supply shall be separated from that of driller's console and directly connected to the main air supply system, and shall be equipped with liquid/gas separators. Excessive bending and pinching on air hoses are forbidden.
- Electric power to BOP control system shall be directly from the main switch of the power board, and controlled by independent switches.
- Accumulators shall be in good condition, charged at its rated value, and ready for work at any time.

2. **Installation Requirements of Choke Lines and Kill Lines, and Relief Lines:**

a) Beside the drilling spool, one manual and one HCR gate valves on the side of chock manifold and two manual gate valves on the side of kill manifold shall be equipped, and the manual gate valve close to the spool shall be always open.

b) The high-pressure co-flex hose shall be installed as straight as possible and fixed by anchor after installation.

c) Relief lines (burning line and bleed line):

- No field welding is permitted on relief lines and the connecting style of relief lines must be flanged.
- The arrangement of relief lines shall take wind directions, dwellings, roads, oil tank areas, power lines and other facilities into consideration.
- While the two lines go in the same trend, the distance between the lines shall be at least 0.3m, and the two lines shall be fixed independently.
- The outlets of bleed line connecting with choke manifold shall be in a safe area which is at least 300m. The bleed line on the kill manifold side connecting with drilling spool shall be prepared and connected when necessary (about 70m from the wellhead).
- At every 10m~15m, turning points, and outlets, the lines shall be supported and fixed by cement foundations with anchor bolts or ground anchors. Hanging

portions shall be supported firmly. If crossing the stumbling blocks such as brooks or ponds is needed, metal bridging supports are necessary.

- Pre-buried anchor bolts for cement foundations shall not be less than 20mm or 3/4" in diameter, and greater than 0.5m in length.

3. Pressure test value and requirements:

a) A pressure test shall be conducted successfully where there is no pressure decline in 5 minutes for a low-pressure test and a minimum of 10 minutes for a high-pressure test.

b) Prior to each high-pressure test, a low-pressure test with 200-300 psi shall be carried out to verify sealing capability at the circumstance of low pressure and avoid washing out sealing elements.

c) High pressure test should be to the rated working pressure of ram preventers or the rated working pressure of the wellhead, whichever is the lowest, or to the rating required in DRILLING ROGRAM. Variable bore rams should be initially pressure tested on the largest and smallest OD pipe sizes that may be used during the OPERATION. Annular preventers, with a joint of drill pipe installed, should be tested to 70% of their working pressure. The high test for Kelly cocks, kill lines and choke lines should be to their rated working pressures, each part in the choke manifold is tested respectively to its rated working pressures, the relief lines be pressure tested to no less than 1,450psi (10Mpa).

d) When casing cup tester is used, the above test pressure should not exceed 80% of the internal yield pressure (burst pressure) of the exposed casing.

e) The control system of BOP shall be pressure tested to 3,000psi (21Mpa) for its reliability before putting into use.

f) Except that the BOP control system is tested with hydraulic oil, other well control equipment shall be pressure tested by clear water.

g) Testing tools

- All required sizes of the plug type testers, and cup type testers should be available for pressure testing purpose.
- When the cup type tester(s) is used for pressure test, the connection of the cup testers has to be checked, to ensure it will stand the load during the high pressures test. For surface casing, the cup tester is to be set in middle of the first

casing joint in the top, so the first casing connection (between wellhead and casing head housing) below the well head is to be tested too while testing BOP and wellhead.

4. Pressure test frequency:

All ram type preventers, annular BOP, valves, choke and kill lines, choke manifold, Kelly valves and safety valves should be pressure tested at the following frequencies:

- Prior to installation where possible.
- After installation of wellhead and BOP assembly and prior to drilling.
- After the disconnection or repair of any pressure containment seal in the BOP stack, choke line, or choke manifold, but limited to the affected component.
- Prior to drilling into a confirmed or suspected high-pressure zone.
- At any time requested by the COMPANY REPRESENTATIVE.
- In any case, every one month or during the first trip after the one-month interval with a maximum interval of 45 days. This frequency is not required for shear rams. As a minimum, the shear rams are to be tested prior to resume drilling operation after 9-5/8" casing string has been set and adequate WOC.

5. Work Requirement of Well Control Equipment:

- With shut-in casing pressure lower than 1000psi, while snubbing with annular preventer, drill tools shall have 18-deg shoulder tool joints, and the snubbing velocity shall not be greater than 0.2m/s.
- After shut off the annular preventer or ram preventer, with the shut-in casing pressure not greater than 2000psi, drill tools are allowed to pull up and down with the velocity less than 0.2m/s, however, rotating or tool joint passing through are forbidden.
- While drilling through pay zone and thereafter, BOP function (close-open) test for the BOPs shall be performed regularly.
- Kill manifolds shall not be used for drilling fluid fill-in.
- The maximum allowed shut-in casing pressure shall be remarkably indicated on choke manifolds.

- All gate valves of well control manifolds shall be tagged with serial numbers, and be indicated with on/Off.
- In case shear rams were used, after shearing the drill pipe with shear rams, retrieve the upper sheared drill pipe, lower a drill pipe joint with TIW valve installed into the hole and close pipe rams on it to prevent the possibility of shear ram's failure.

6. Management of Well Control Equipment:

- Well control equipment used by the drilling crew shall be managed and operated by designated personnel with clear post responsibilities.
- Well site shall have specified component storeroom and rubber element storeroom with air conditioners respectively. The temperature of storerooms shall meet storage requirements of components and rubber elements.
- At least 1 set is available as back-up for every type ram assembly in the well site. Ram preventer side door seals and ram rubber, every type gasket and bolt etc. should be enough for backup.

7. Killing Method and Shut-in Method:

- Driller's method is recommended for well killing.
- The hard Shut-in method is recommended for shutting in the well in Fahliyan reservoir.

8. Well Control during drilling:

a) Shut in Procedure in While on Bottom

- Position Kelly (if in use) and tool joint so that tool joints are clear of sealing elements.
- Shut off pumps, check for flow.
- If flow is noted, close the annular preventer and open remote-controlled choke line valve (HCR).
- If the well has been closed using the annular preventer, check space out (pipe rams vs. toll joint relative position) and close pipe rams and ram locks
- Bleed off pressure between pipe rams and annular preventer (if possible).
- Record shut-in drill pipe and casing pressures.
- Record volume gain and mark pits.

- Weigh mud in suction pits.
- Check blowout preventer and manifold for leaks.
- Check flowline and choke exhaust for flow.
- Check accumulator pressure.

b) Shut in Procedure in While Tripping

- If a flow is detected, install full opening safety valve and close same.
- Position tool joints clear of the sealing elements and close annular preventer and open remote-controlled choke line valve (HCR).
- Make up grey valve (dart type inside BOP on full open safety valve). Then Make up Kelly or top drive (insert a pup joint or single between safety valve and top drive) and open safety valve.
- Record casing pressure.
- Check blowout preventer, stack, and manifold for leaks.
- Check valves for correct position.
- Check accumulator pressure.
- Begin stripping to bottom.

c) Shut in Procedure for Empty Wells

- Try to RIH with drill pipe as more as possible if hole condition permits, and the drill pipe must be installed a Kelly cock or safety valve.
- Close the annular preventer and open remote-controlled choke line valve (HCR).
- If the well has been closed using the annular preventer, check space out (pipe rams vs. tool joint relative position) and close pipe rams and ram locks
- Bleed off pressure between pipe rams and annular preventer (jf possible).
- Record casing pressure

9. Well Control During Running Casing:

The well control while running casing is important and critical, best practices shall be followed, including but not limited to:

- Change out pipe rams to casing rams prior to running casing into hydrocarbon bearing formation, if possible.
- The crossover between casing and drill pipe has to be available, and good practice is to make it up with a joint of DP and keep on V-door.

10. Shut in Procedure During Logging:

Cable shall be pull out of the hole immediately when overflow is found during logging. If pit gain exceeds the specified amount, the cable shall be cut immediately and treat the overflow in accordance with the shut-in procedure for empty wells. Continuing to pull the cable through the closed annular preventer is not recommended.

11. Well Control During Testing:

- No difference to while DRILLING OPERATION, since either there is testing string in the well (same as Shut-in procedure while on bottom), or tripping the string. (Same as shut-in procedure while tripping).
- Likely there is some crude in the testing string, and precaution has to be taken while circulating the crude out of hole during killing operations to ensure it is properly handled without causing environmental impact.
- To minimize the formation damage, the killing fluids used shall be weighted brine if possible, however in well control case when it occurs, killing well safely will take priority rather than protecting formation from damage on selection of killing fluids.

12. Well Control During Stimulation:

- Similar to during testing.
- For any residual or alive acid to be circulated out during killing operations, measures have to be taken to treat the returned acid properly to avoid personnel from getting hurt, and without causing any environmental damage.

13. Well Control Drills:

Drills shall be conducted only when hole conditions and other rig activities permit. Drills should be scheduled so that crews can practice drills while on bottom, while tripping in and while out of the hole. Each operating shift shall perform at least one BOP drill under different situations per month.

14. Basic Requirement of Well Control:

- For 13-5/8”, 10K BOP, BOP should be schemed as follows: upper rams 5”, middle rams are shear rams, lower rams is variable rams (2-7/8” to 5”). For easily connecting, armored fire-resistant 10000psi and ID 4-1/16" flexible hoses are recommended as kill/choke lines which connected with BOP directly.
- The BOP stack and choke manifold, kill line and choke line, HCR and manual gate valves, inside BOP (including kelly cock) have to be H2S trimmed and stand the working temperature from -10 degrees C (in case of doing N2 for

DST) to the maximum up to 100 deg C (estimated highest temperature condition).

- The BOP equipment shall be visually inspected, function tested, and hydrostatically pressure tested during commissioning and Acceptance of Drilling Unit prior to Commencement Date, then upon installation on a wellhead, prior to drilling any hole section with a potential hydrocarbon bearing reservoir, after breaking any BOP connection (such as after installation of an additional wellhead section after running casing), and at least every 14-21 days while the BOP is installed on the Well.
- CONTRACTOR shall submit the current well control procedures manual of CONTRACTOR to COMPANY for review and approval by COMPANY. Valid IWCF Well Control Certificates (or IADC equivalent) must be held by all tool pushers, Drillers (optional) at Drilling Supervisor level and Drillers and Assistant Drillers at Driller level, as further specified in CONTRACTOR Personnel HSE Requirements.
- The accumulator must be of sufficient capacity so that it is possible to close one (1) ram and the bag, open a ram and the bag, re-close a ram and the bag and retain 1200psi accumulator pressure.
- The “OPEN” and “CLOSED” operating positions of BOP activators of the rams, annular, and remote actuated valves must be clearly identified on the BOP master control panel and remote panels, in English in addition to the native language of the majority of the senior drilling crew.
- Shear-ram BOP activator must have a flap-cover to ensure only pre-meditated and deliberated activation and to prevent accidental WORK.
- All lines and connections in the choke and kill manifold, standpipe manifold, choke and kill lines and high-pressure mud lines should be welded hard-piping with weld-on FMC Weco “hammer-lug” 1502 unions (or Western equivalent, such as Anson) or weld-on API flanges. The well control system should not include any threaded screw-on hammer-lug unions, flanges, or other threaded screw-on connections.
- All adapter or spacer spools between the wellhead and BOP must be manufactured in accordance with API 6A, and must be complete with all required studs, nuts, and ring gasket seals for both lower and upper connections; minimum spare parts should include two (2) OEM spare studs complete with nuts and two spare OEM ring gasket seals for both bottom and top connections.
- There should be a wide-bore gas vent line from the “poor-boy” degasser; the line and line outlet must be securely anchored, the line outlet must be a safe distance from the well cellar, rig floor, choke manifold, and mud logging unit.

- Well Control Equipment Welding: All welds in the well control system must be NDT inspected and records maintained of all welding and weld inspections, including choke and kill manifold, standpipe manifold, mud pump discharge manifold, choke and kill lines.
- Well Control Rig Floor Information: The following must be posted on the rig floor (in the Driller's shack / "dog-house"): Well Shut-In Procedure, Well Information Sheet (showing casing and hole sizes, depths, and such like), Kill Sheet (including most recent slow circulation rate pressures, formation integrity leak-off test information and MAASP, present mud weight, present drill string and well volumes, and such like).
- The derrick substructure clear height (rotary table to ground level) needs to be enough for fitting in above BOP configuration (including a tubing head and a bell nipple). This need to be double checked in rig tender process. According to the calculation of height of BOP stack, the substructure clear height is at least 9 m.
- In the case if drilling 6-1/8" hole becomes necessary, or using 3-1/2" drill pipe for the cleanup run (including drill 7" shoe track), then taper drilling string (5" DP + 3-1/2" DP) is to be run, so a pair variable ram be installed in the single ram preventer. The defect installing variable ram in single ram preventer is that the ram may be damaged and the string fall in hole bottom due to the ram hanging up too much weight when the drill pipe be cut. If the pipe ram be installed in the single ram preventer and the variable ram be installed in the double ram preventer, it would be very dangerous due to the variable ram failure in frequently using.

12.9 H_2S Monitoring and Prevention Measure

As per PPL/PEDEC SOW South Azadegan oilfield contains H_2S bearing formations, precautions shall be taken in place especially in 12-1/4" & 8-1/2" hole section (for typical Fahliyan vertical well) while drilling, completion and well testing operation.
H_2S can be released any time while drilling, completion and well testing operation, therefore:

- The Rig shall be prepared for H_2S (Training, procedures and protective gear).
- Make sure everyone on the Rig should be made aware of the hazard presented by H_2S during their induction when they first arrive on the rig.
- The mud shall be treated to protect the drill pipe and the casing.
- All training program shall be recorded and reported.

- H_2S emergency drill must be conducted frequently.
- Adequate sensors and alarms must be installed for the scope of monitoring the H_2S releases.

13. H_2S Monitoring & Control

NOTE: Final plan for H2S Monitoring and Control will be issued in due time by Service company instruction and Rig Contractor standards based on Oilfield safety and NIOC standards.
Minimum H2S Requirements for Rig site will be as following:

Table 13-1: Minimum Rig H2S Requirements for South Azadegan Project

Minimum Rig H_2S Requirements for South Azadegan Project	
Gas Detection Equipment	
H_2S Service cabin	1 No.
H_2S Unit location point	Comply with MSA/OEOC HSE
Main Monitoring Panel	1 No. - 8 channels
Max. number of channels	8 No.
Number of H_2S channels	4 No.
Measuring range	0 - 100
Adjustable alarm siren	Minimum 4 set (Min 120db voice load)
Measuring range	0-25 PPM
Adjustable alarm setting levels	2 No. (low 10 PPM and high 15 PPM)
Gas concentration reading displays	As sensors head number
Electrical Classification	Required
Certification	Required
Monitor panel installation site	H2S Unit
Automatic switch-on back up battery (min. 12 hr. power supply)	1
SO_2 Equipment	
Portable Single Detector	2 No.
H_2S Sensors	

H_2S Sensor installation points	Comply with MSA/OEOC HSE
Electrical Classification	For hazardous area
Certification	Required
Spare sensor head	2 No.
Response gas concentration	10 ppm low & 15 ppm High
LEL Sensors	
LEL Sensor installation points	Comply with MSA/OEOC HSE
Electrical Classification	For hazardous area
Certification	Required
Response gas concentration	20 % low & 40% High
Spare sensor head	Required
LEL Sensor installation points	Comply with MSA/OEOC HSE
Electrical Cables (connecting sensors with monitor panels)	
Suitable for hazardous area	as much as required
Electrical Classification	as much as required
Visual-Acoustic H_2S Alarms	
Low level visual alarms	Yellow flashing lights
High level visual alarms	Steady red lights
Low level acoustic alarms	Intermittent horn
High level acoustic alarms	Continuous horn
Above items suitable for hazardous area	Required
Number of alarm stations	5 No.
Alarm points location	comply with MSA/OEOC HSE
Calibration kit	At least 2 No.
PORTABLE H_2S MONITORING SYSTEM	
Hand pump colorimetric tubes H_2S detectors	1 No.
H_2S colorimetric tubes ranges	Req., comply with MSA/OEOC HSE

H_2S colorimetric tubes quantity	Req., comply with MSA/OEOC HSE
Personal electronic H2S detectors	Min. 6 No
Multi Gas Portable Detectors	
Quantity	4 (two main two spare)
Measuring range	0 – 100 ppm
Alarm setting levels	two
Certification	Required
Calibration kit	Required
Electrical Classification	For hazardous area
Certification	Required
Respiratory Protection Equipment	
Air Compressors	
Quantity	2 No. (Electrical- Diesel)
Approved for air breathing	Required
Gross weight	Required
Flow rate capacity	480 lit./min each
Charging pressure	3200
Filtration and purifier device	Specify approval
Certification	As per EN-132
Alarm for contaminating gases	Required
Auto shut down for contaminating gases	Required
Air compressor driving system	Electrical
Suitable for refilling bottles of air breathing apparatuses	Required
Air test kit	At least 1 No.
FIXED CASCADE AIR BREATHING SYSTEM	
The fixed cascade air breathing system complies with code type	Specify
Certification details	Required

No. of racks	At least 3*8
Dimensions (LxWxH) each rack	required
Bottles certified as pressure vessel	Required
Total air storage capacity	50 Liter
At max. working pressure of	3300 Psi
Air Pressure regulator	Req., victory Type
Safety/relief valve	Required
Low pressure alarm device	Required
Stainless steel piping/lines	Required
Air Breathing Delivery Stations	
Number of stations	At least 7 No.
Stations locations	Comply with MSA/OEOC HSE
Each Fixed Station Consisting of:	
Outlet manifold	4-man manifold 3 No & 6-man manifold 2 No
Connection outlets	Required
Hose lines	20(25 feet)20(15 feet)
PORTABLE AIR BREATHING APPARATUSES	
Self-Contained Breathing apparatuses (SCBA 45 min)	25
EEBA (20 min.)	20
Mask type	Full face (positive pressure)
Compliance with specifications	Comply w/ EN 137
Approval Authority	required
Air bottle working pressure	required
Air bottle certificate	required
Air regulator	required
Low Pressure audible alarm	Required

Suitable to be quick connected to cascade outlet manifold	Required
Stowage suitcase	Required
SCBA Radio Communication	Required
Spare Bottles for SCBA	At least 3
BREATHING APPARATUSES location points	comply with MSA/OEOC HSE

14. Requirement

14.1 12-1/4” Hole Section

Table 14-1: 12-1/4" Hole Section Requirement

Item	Item No.	Supplier	12-1/4” Hole Section Requirement Description	Quantity	Unit
BHA	1	MSA/OEOC	12-1/4" TCI Bit, 6-5/8" Reg Pin, IADC Code: 537	1	Ea.
	2	MSA/OEOC	12-1/4” PDC Bit, 6-5/8 Reg Pin, Matrix Body, 6-7 Blades, IADC Code: M323, M423, M433	3	Ea.
	3	MSA/OEOC	Nozzle (15/32”) W/ O-Ring & Snap Ring for Tri-Cone Bit	1	Set
	4	MSA/OEOC	Nozzle (16/32”) W/ O-Ring & Snap Ring for Tri-Cone Bit	1	Set
	5	MSA/OEOC	Nozzle (13/32”) for PDC Bit W/ O-Ring	2	Set
	6	MSA/OEOC	Nozzle (14/32”) for PDC Bit W/ O-Ring	2	Set
	7	Rig Cont.	Nozzle Installation & Removal Tool for Tri-Cone Bit	1	Ea.
	8	MSA/OEOC	Nozzle Wrench for PDC Bit	2	Ea.
	9	Rig Cont.	12-1/4" Roller Cone Bit Breaker	1	Ea.
	10	MSA/OEOC	12-1/4" PDC Bit Breaker	3	Ea.
	11	Rig Cont.	Pipe screen for inside 5" DP	3	Ea.
	12	Rig Cont.	12-1/4" Ring Gauge	1	Ea.
	13	Rig Cont.	12-1/4” Junk Sub, 6-5/8” Reg Pin-Box	2	Ea.
	14	Rig Cont.	8-1/2" Bit Sub, 6-5/8” Reg Box-Box	3	Ea.

Item	Item No.	Supplier	12-1/4” Hole Section Requirement Description	Quantity	Unit
	15	Rig Cont.	8-1/2” Spiral Drill Collar, 6-5/8” Reg Pin-Box	12	Jt.
	16	Rig Cont.	12-1/4” String Stabilizer (IBS, Spiral)	1	Ea.
	17	Rig Cont.	8” Drilling Jar, 6-5/8” Reg Pin-Box	3	Ea.
	18	MSA/OEOC	Cement, Class "G", 110 lb/sxs	As Req.	As Req.
	19	MSA/OEOC	Cement Additives	As Req.	As Req.
	20	MSA/OEOC	XOS 9-5/8" BTC Pin * 4-1/2" IF BOX	1	Ea.
	21	Rig Cont.	XOS, 6-5/8” Reg Pin x 4-1/2” IF Box	2	Ea.
	22	Rig Cont.	5” HWDP, 49.7 PPF, 4-1/2” IF Pin-Box	9	Jt.
	23	Rig Cont.	5” DP, G-105/S-135, 19.5 PPF, 4-1/2” IF Pin-Box	As Required/1000	Mt.
	24	Rig Cont.	Fishing Tools (Complete Set for This Hole Section)	As Required	Set
	25	Rig Cont.	Float Valve	2	Ea.
	26	MSA/OEOC	20-3/4” x 13-5/8” CHS Wear Bushing w/ extra O-Ring	1	Set
	27	MSA/OEOC	20-3/4” x 13-5/8” CHS Plug Tester w/ extra O-Ring	1	Set
	28	MSA/OEOC	13-3/8” Cup Tester (Suitable for 68-72# Grade)	1	Ea.
Casing & Accessories	1	MSA/OEOC	9-5/8” Float Shoe, 110ss, 53.5#, Premium Gas Tight Con., PDC Drillable (Special Drift for 8-1/2" Bit)	1	Ea.
	2	MSA/OEOC	9-5/8” Float Collar, 110ss, 53.5#, Premium Gas Tight Con., PDC Drillable (Special Drift for 8-1/2" Bit)	1	Ea.

Item	Item No.	Supplier	12-1/4” Hole Section Requirement Description	Quantity	Unit
	3	MSA/OEOC	9-5/8" CSG, 110ss, 53.5#, Premium Gas Tight Con., (Special Drift for 8-1/2" Bit)	4420	Mt
	4	MSA/OEOC	9-5/8" CSG Pup Joint, 110ss, 53.5#, Premium Gas Tight Con., (Special Drift for 8-1/2" Bit), Length: 1.5m	1	Jt.
	5	MSA/OEOC	9-5/8" CSG Pup Joint, 110ss, 53.5#, Premium Gas Tight Con., (Special Drift for 8-1/2" Bit), Length: 3m	1	Jt.
	6	MSA/OEOC	9-5/8" CSG Coupling, 110ss, 53.5#, Premium Gas Tight Con., (Special Drift for 8-1/2" Bit)	2	Ea.
	7	MSA/OEOC	API Drift for 9-5/8" CSG, 110ss, 53.5#, Premium Gas Tight Con., (Special Drift for 8-1/2" Bit)	1	Ea.
	8	MSA/OEOC	Top Cement Plug, Non-Rotating Type, Suitable for Running in 9-5/8" CSG (53.5# SD), Flexible Rubber Elements, HNBR, PDC Drillable	1	Ea.
	9	MSA/OEOC	Bottom Cement Plug, Non-Rotating Type, Suitable for Running in 9-5/8" CSG (53.5# SD), Flexible Rubber Elements, HNBR, Diaphragm Burst Pressure: 220-400 psi	1	Ea.
	10	MSA/OEOC	12-1/4” x 9-5/8” CSG Bow Type Centralizer c/w Nails, Non-Welded	30	Ea.
	11	MSA/OEOC	13-3/8” x 9-5/8” CSG Rigid Type Centralizer c/w Nails, Non-Welded	40	Ea.
	12	MSA/OEOC	9-5/8” CSG Stop Collar c/w Spiral Nails	150	Ea.
	13	MSA/OEOC	API CSG Thread Lock Compound (Baker Lock)	1	can
	14	MSA/OEOC	API CSG Thread Sealing Compound (Dope)	2	Dr.
Well Head	1	MSA/OEOC	TBG Hanger: 11" BOWL*4-1/2" 18.9 ppf BOX VAM TOP BOTTOM & 4-1/2" 4-2G ACME BOX	1	No.

Item	Item No.	Supplier	12-1/4” Hole Section Requirement Description	Quantity	Unit
Well Head	2	MSA/OEOC	Metal Seal + S-Seal	1	Set
	3	MSA/OEOC	BX-158 Ring Gasket (11" * 10000 psi)	1	Ea.
	4	MSA/OEOC	1-3/4" * 381 mm stud & nut (11" 10000 psi)	16	Ea.
	5	MSA/OEOC	Slip & Seal Assembly 13-5/8" * 9-5/8" OD casing Automatic (Non-Auto. & Auto.)	1+1	Ea.
	6	MSA/OEOC	2-1/16" Instrument Flange, WP: 10000 psi	1	Ea.
	7	MSA/OEOC	2-1/16" * 10000 psi Side outlet Valves	4	Ea.
	8	MSA/OEOC	BX-152 Ring Gasket (2-1/16" * 10000)	7	Ea.

Item	Item No.	Supplier	12-1/4” Hole Section Requirement Description	Quantity	Unit
	9	MSA/OEOC	3/4" * 139 mm stud & nut (2-1/16" * 10000)	48	Ea.
	10	MSA/OEOC	2-1/16" * 10000 psi Companion Flange + 2" Bull Plugs + 1/2" Pipe Plug	2+2+2	Ea.
	11	MSA/OEOC	1-1/2" VR-Plug	2	Ea.
	12	MSA/OEOC	Pressure Gauge, 10000 psi Range	2	Ea.
	13	MSA/OEOC	Angle Needle Valve,WP:10000 psi	2	Ea
	14	MSA/OEOC	11" Wear Bushing to Fit with T.H.S W/O-Ring	1	Set
	15	MSA/OEOC	11" Plug Tester to Fit with T.H.S W/ O-Ring	1	Set

Item	Item No.	Supplier	12-1/4" Hole Section Requirement Description	Quantity	Unit
	16	MSA/OEOC	Tubing Head Spool (11", 10K WP Flange Top x 13-5/8", 5K WP Flange Bottom)	1	Set
Well Services	1	Service Co.	Drilling Fluid Service (Material, Lab. & Engineering)	As Required	
	2	Service Co.	Mud Logging Service	As Required	
	3	Service Co.	Waste Management Service	As Required	
	4	Service Co.	Fishing service	As Required	
	5	Service Co.	H2S Service	As Required	
	6	Service Co.	Casing Running Service	As Required	
	7	Service Co.	Cementing Service (Material, Pump Truck, Silo, Engineering)	As Required	
	8	Service Co.	Well Head Service	As Required	
	9	Service Co.	Wire Line Logging Service	As Required	

Note: Qty. and specification to be confirmed with contracts SOW. In case of any shortage or discrepancy, the Contracts SOW will be reference.

14.2 8-1/2” Hole Section

Table 14-2: 8-1/2" Hole Section Requirement

Item	Item No.	Supplier	8-1/2” Hole Section Requirement Description	Quantity	Unit
BHA	1	MSA/OEOC	8-1/2" Milled Tooth Bit, 4-1/2" Reg Pin, IADC Code: 217	1	Ea.
	2	MSA/OEOC	8-1/2” PDC Bit, 4-1/2” Reg Pin, Matrix Body, 6 Blades, IADC Code: M433	1	Ea.
	3	MSA/OEOC	5-7/8” Milled Tooth Bit, 3-1/2” Reg Pin, IADC Code: 217	1	Ea
	4	MSA/OEOC	5-7/8” Mill (Junk Mill or E.Mill), 3-1/2” Reg Pin,	1	Ea
	5	MSA/OEOC	Nozzle (14/32”) W/ O-Ring & Snap Ring for 8-1/2” Tri-Cone Bit	1	Set
	6	MSA/OEOC	Nozzle (15/32”) W/ O-Ring & Snap Ring for 8-1/2” Tri-Cone Bit	1	Set
	7	MSA/OEOC	Nozzle (13/32”) for 8-1/2” PDC Bit W/ O-Ring	1	Set
	18	MSA/OEOC	Nozzle (14/32”) for 8-1/2” PDC Bit W/ O-Ring	1	Set
	9	MSA/OEOC	Nozzle (15/32”) W/ O-Ring & Snap Ring for 5-7/8” Tri-Cone Bit	1	Set
	10	MSA/OEOC	Nozzle (16/32”) W/ O-Ring & Snap Ring for 5-7/8” Tri-Cone Bit	1	Set
	11	Rig Cont.	Nozzle Installation & Removal Tool for Tri-Cone Bit	1	Ea.
	12	MSA/OEOC	Nozzle Wrench for 8-1/2” PDC Bit	1	Ea.
	13	Rig Cont.	8-1/2” Tri-Cone Bit Breaker	1	Ea.
	14	MSA/OEOC	8-1/2” PDC Bit Breaker	1	Ea.
	15	Rig Cont.	5-7/8” Tri-Cone Bit Breaker	1	Ea.

Item	Item No.	Supplier	8-1/2” Hole Section Requirement Description	Quantity	Unit
	16	Rig Cont.	8-1/2” Ring Gauge	1	Ea.
	17	Rig Cont.	5-7/8” Ring Gauge	1	Ea.
	18	Rig Cont.	8-1/2” Junk Sub, 4-1/2” Reg Pin-Box	1	Ea.
	19	Rig Cont.	5-7/8” Junk Sub, 3-1/2” Reg Pin-Box	1	Ea.
	20	Rig Cont.	6-1/2" Bit Sub, 4-1/2” Reg Box-Box	1	Ea.
	21	Rig Cont.	4-3/4” Bit Sub, 3-1/2” Reg Box-Box	1	Ea.
	22	Rig Cont.	6-1/2" Spiral Drill Collar, 4-1/2” Reg Pin-Box	12	Jt.
	23	Rig Cont.	8-1/2" String Stabilizer (IBS, Spiral) 4-1/2” Reg Pin-Box	1	Ea.
	24	Rig Cont.	6-1/2” Drilling Jar, 4-1/2” Reg Pin-Box	1	Ea.
	25	Rig Cont.	XOS, 4-1/2” Reg Pin x 4-1/2” IF Box	1	Ea.
	26	Rig Cont.	4-3/4” Spiral Drill Collar, 3-1/2” IF Pin-Box	18	Jt.
	27	Rig Cont.	4-3/4” Drilling Jar, 3-1/2” IF Pin-Box	1	Ea.
	28	Rig Cont.	XOS, 3-1/2” IF Pin x 4-1/2” IF Box	1	Ea.
	29	Rig Cont.	5” HWDP, 49.7 PPF, 4-1/2” IF Pin-Box	9	Jt.
	30	Rig Cont.	5” DP, G-105/S-135, 19.5 PPF, 4-1/2” IF Pin-Box	As Required/1000	Mt
	31	Rig Cont.	3-1/2” HWDP, 23.2 PPF, 3-1/2” IF Pin-Box	6	Jt.

Item	Item No.	Supplier	8-1/2" Hole Section Requirement Description	Quantity	Unit
	32	Rig Cont.	3-1/2" DP, G-105, 13.3 PPF, 3-1/2" IF Pin-Box	± 400	Mt
	33	Rig Cont.	9-5/8" CSG Scraper (43.5-53.5#), 4-1/2" Reg Pin-Box	1	Ea.
	34	Rig Cont.	7" CSG Scraper (32-35#), 3-1/2" Reg Pin-Box	1	Ea.
	35	Rig Cont.	Float Valve for 6-1/2" Bit Sub	2	Ea.
	36	Rig Cont.	Float Valve for 4-3/4" Bit Sub	2	Ea.
	37	Rig Cont.	Pipe Screen for inside 5" DP	3	Ea.
	38	MSA/OEOC	X.O.S 7" , 32 ppf, 110SS, Premium Gas Tight Connection pin * 4-1/2" IF Box	1	Ea.
	39	MSA/OEOC	Cement "G", 110 lb./sx.	As Req.	
	40	MSA/OEOC	Cement Additives	As Req.	
Casing Accessories	1	MSA/OEOC	7" Float Shoe, 110ss, 32#, Premium Gas Tight Con.	1	Ea.
	2	MSA/OEOC	7" Float Collar, 110ss, 32#, Premium Gas Tight Con.	1	Ea.
	3	MSA/OEOC	7" LDC, 110ss, 32#, Premium Gas Tight Con.	1	Ea.
	4	MSA/OEOC	7" Liner, 110ss, 32#, Premium Gas Tight Con.	416	Mt
	5	MSA/OEOC	Hydraulic Set, Rotating Liner Hanger, 7", 29-32 ppf,110ss, Premium Gas Tight Conn., Tandem Cone, with Integral/Split Weight Set Top Packer, 10 ft PBR, Load Capacity 2,000 KN, Packer Seal Capacity 10K, Temp. 150 ºC, with Liner Setting Sleeve c/w DPOB and Setting Tool thread, to Set Inside 9-5/8", 47-53.5 ppf Casing.	1	Set
	6	Service Co.	7" Liner Hanger Running Tool	1	Ea.
	7	MSA/OEOC	Liner Wiper Plug for 7" Liner, 110ss, 32#, Premium Gas Tight Con.	1	Ea.

Item	Item No.	Supplier	8-1/2" Hole Section Requirement Description	Quantity	Unit
	9	Service Co.	Polish Mill for 7" Liner, 110ss, 32#, Premium Gas Tight Con. PBR	1	Ea.
	10	MSA/OEOC	5" DP, G-105/S-135, 19.5 PPF Pump Down Plug (Dart)	1	Ea.
	11	MSA/OEOC	7" Coupling, 110ss, 32#, Premium Gas Tight Con.	3	Ea.
	12	MSA/OEOC	API Drift for 7" Liner, 110ss, 32#, Premium Gas Tight Con.	1	Ea.
	13	MSA/OEOC	8-1/2" x 7" Bow Type Centralizer	9	Ea.
	14	MSA/OEOC	8-1/2" x 7" Rotating Type Centralizer C/W Nail and Pins	9	30 Ea.
	15		S.N. Packer 7" 32 ppf, 110SS	1	Set as Req.
	16	MSA/OEOC	9-5/8" x 7" Rigid Type Centralizer c/w Nails, Non-Welded	3	Ea.
	17	MSA/OEOC	7" Stop Collar c/w Spiral Nails	80	Ea.
	18	MSA/OEOC	API CSG Thread Lock Compound (Baker Lock)	1	Can.
	19	MSA/OEOC	API CSG Thread Sealing Compound (Dope)	2	DR.
Well Head	1	MSA/OEOC	THS Adaptor 11", 10K x 4-1/16", 10K Integrated w/ one 4-1/16", 10K Gate Valve & X.M.T Cross T-Type Solid Block 4-1/16", 10K	1	Set
	2	MSA/OEOC	4-1/16" Positive Choke Valve W/ 10000 psi WP W/Bean Sizes for Positive Choke valve (8,12,16,20,24,28,32,36,40,44,48,52,56,60,64,128/64 inch)	1	Ea.

Item	Item No.	Supplier	8-1/2” Hole Section Requirement Description	Quantity	Unit
	3	MSA/OEOC	Companion Flange , 4-1/16",10000 psi WP	1	Ea.
	4	MSA/OEOC	Blind Flange, 4-1/16", 10000 psi API WP, C/W 1/2" NPT Female Thread Body.	1	Ea.
	5	MSA/OEOC	Gasket Self Sealing Ring BX-155, 4-1/16" 10000 psi Material : SS316	10	Ea.
	6	MSA/OEOC	Instrument Flange, 4-1/16",10000 psi API BX155, with four ports 1/2" NPT ,Aligned Including 4 Pieces od 1/2" NPT Plug, M/C:FF-0.5 T/C:U PSL:3G PR:2 Forged Body	1	Ea.
	7	MSA/OEOC	Pressure Gauge, 1/2" NPT Male Connection, 0-10000 psi, 4" Face Diameter, Stainless Steel for Cover and Internal Parts	2	Ea.
	8	MSA/OEOC	X-Mass Tree CAP, 4-1/16" API 5000 psi WP, Flanged Bottom (4-1/16",Bore) 11-1/2" 4TPI ACME Thread Union Type, Tapped with 1/2" NPT Port M/C:FF-0.5 T/C:U PSL:3G PR:2 Forged Body	1	Set.
	9	MSA/OEOC	2" Bull Plug & 1/2" Pipe Plug NPT Thread, Forged Body, Stainless Steel AISI 410, Hexagonal 10000 psi WP, M/C:FF0.5 T/C:U PSL:3G PR:2	2+2	Ea.

Item	Item No.	Supplier	8-1/2” Hole Section Requirement Description	Quantity	Unit
	10	MSA/OEOC	STUD & NUT 1 1/8" * 203 MM Long ASTM A193 B7/A194 2H Stud/Nuts PTFE Coated for 4 1/16" 10000 Flange	56	Ea.
	11	MSA/OEOC	Angle Needle Valve 1/2" NPT Male*Female 10000 psi WP M/C FF-0.5 T/C:U PSL:3G	2	Ea.
	12	MSA/OEOC	Manual Gate Valve API 4 1/16" 10000 WP Flanged Ends and Full Bore M/C:FF-0.5 T/C:U PSL:3G	2	Ea.
	13	MSA/OEOC	4" Back Pressure Valve , One Way , WP:10k , w/Redress kit	1	Set
	14	MSA/OEOC	4" Back Pressure Valve , One Way , WP:10k , w/Redress kit	1	Set
	15	MSA/OEOC	Welding Flange, 4-1/16",10000 API Flange* Welding Neck, for Pipe Size 4"Material:410SS,M/C:FF-0.5 T/C:U PSL:3 Forged Body	1	Ea.
Well Services	1	Service Co.	Drilling Fluid Service (Material, Lab. & Engineering)	As Required	
	2	Service Co.	Mud Logging Service	As Required	
	3	Service Co.	Waste Management Service	As Required	

Item	Item No.	Supplier	8-1/2" Hole Section Requirement Description	Quantity	Unit
	4	Service Co.	Directional Drilling Service (Only MWD unit, MWD Tools, MWD Engineer)	As Required	
	5	Service Co.	Fishing service	As Required	
	6	Service Co.	H2S Service	As Required	
	7	Service Co.	Casing Running Service	As Required	
	7	Service Co.	Cementing Service (Material, Pump Truck, Silo & Engineering)	As Required	
	8	Service Co.	Well Head Service	As Required	
	10	Service Co.	Liner Hanger Service	As Required	
	11	Service Co.	Wireline Logging Service (Logging and Perforation)	As Required	
	12	Service Co	Dry Testing & DST/TCP Service	As Required	
	13	Service Co	Coiled Tubing Service	As Required	
	14	Service Co	Stimulation & N2 Lifting Services	As Required	
	15	Service Co	Completion Service	As Required	
	16	Service Co	Slick line Service	As Required	

Note: Quantity and specification to be conformed to Contact's SOW. In case of any shortage or discrepancy, the Contract's SOW will be reference.

15. Past Revision Corrections

Drilling Engineering Department

Table 15-1: Past Revision Correction

Drilling Engineering							
1	• برنامه مورد تایید می باشد.						

Contact List

Table 15-2: Contact List

Contractor				
Contact	**Position**	**Phone**	**Ext.**	**Mobile**
MOSTAFA KHEDRI	Head of Drilling Engineering	+982140880435	210	+989121933919
PIRAN FARZANEH	Head of Drilling Operation	+982140880435	211	+989173042055
AMIN FATTAHI	Head of Drilling Fluids	+982140880435	-	+989133089336
DAVOOD DEHDARI	Head of Geology Operation	+982140880435	-	+989177052300